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Conduct a study to formulate an updated program for the progressive replacement of firewood as an energy source in the residential sector in Colombia, with the necessary components for its implementation.

UPME Contract C-031-2019

Final Report

Bogotá, December 9, 2019





Sustainable Rural Strategy Consortium

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Executive Summary

The UPME Contract C-031-2019 was developed in three stages, culminating in the results presented in this final report. The following section describes the main achievements of each stage, corresponding to each of the three deliverables.

In Stage I, the state of the art in energy use in the residential sector was presented, both nationally and internationally, and lessons learned from these experiences were analyzed in the implementation of solutions for replacing solid fuels in low- and lower-income families. The target population for the short-, medium-, and long-term strategies for the progressive replacement of firewood in the residential sector consists primarily of families living in dispersed rural dwellings, some without access to public services, and generally with limited financial resources, who must rely on solid fuels to meet their cooking and heating needs.

The first step in designing the substitution strategy is to understand and analyze the efforts made so far to propose solid fuel substitution programs in Colombia and internationally, and the lessons learned from such experiences.

Therefore, in Deliverable I, the Efficient Stoves programs developed to date and Biogas were analyzed, for which a broad description of international and national experiences is presented.

The State of the Art section included other experiences such as the use of induction cooktops, which encompasses not only the electric stove but also the appropriate cookware and kitchen equipment. Finally, national and international experience in the use of propane gas or LPG was presented, including programs to replace firewood use implemented in the mid-1990s and during the first half of the 2000-2010 period.

These programs constituted the first massive effort to bring the LPG solution to a significant number of low-income families who consumed firewood.
Among other international experiences, the LPG replacement program in India was also presented, which is currently in full implementation and constitutes a world reference for its design and results achieved to date, as explained in the Report.

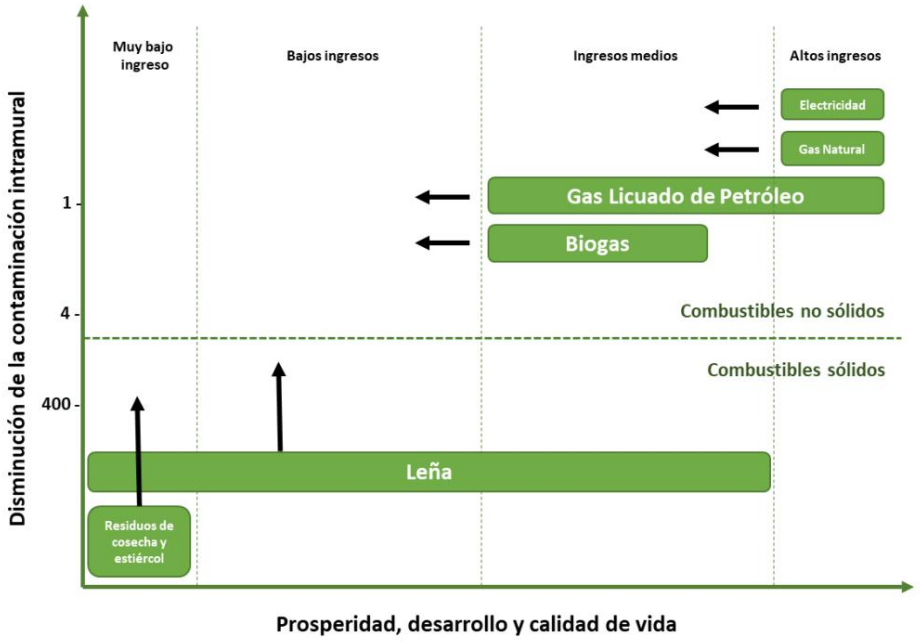
The common denominator of all the experiences consulted for the construction of chapter 3 of this Final Report is the impact that the consumption of firewood has on the health of families, particularly women and children, which raises issues of gender equality and child protection, which defined the path of the analyses that were developed below.

Health is the message, meaning that strategies and solutions to firewood consumption will prioritize addressing the public health problem. To this end, the analysis will focus on four energy sources or technologies that form the backbone of the Progressive Firewood Substitution Plan: Improved Stoves, Biogas, Electricity, and LPG. Investment in these substitution programs has additional benefits, such as increased well-being and prosperity for the beneficiaries. This is known as the “energy ladder.”

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which correlates the evolution in energy use in homes with improvements in well-being, as shown in the following illustration presented in Deliverable I.

Illustration 1. Energy ladder



Fuente: Adaptada de Smith, K; Sagar, A. Making the clean available: Escaping India's Chulha Trap, 2014

In Deliverable II, and during the period from September 26 to October 17, 2019, four *Regional Workshops* were held in the cities of Montería, Pasto, Medellín, and Sogamoso, and on October 24, the *Expert Workshop* took place in Bogotá. These meetings with communities, academia, the private sector, and the public sector provided a valuable opportunity for the Team to understand the challenges of firewood use in these areas and to hear firsthand the perceptions of users regarding the impacts of firewood smoke on issues such as family health, particularly that of women, who, for cultural reasons, have traditionally been responsible for firewood collection and cooking in kitchens without adequate ventilation. The women who attended the regional workshops expressed their concern about the deterioration in their health and quality of life, as well as the impacts on the child population, given that minors are the ones who remain in the family home exposed to smoke from zero years old until well into childhood, seeing their quality of life deteriorate due to the permanent inhalation of the fumes from the combustion of wood.

During the workshops, the consulting team had the opportunity to exchange views with local communities and academics, which are detailed in the corresponding reports. Similarly, the Expert Workshop included panel discussions on each of the energy sources/technologies.

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Identified as alternatives to firewood are: Improved Stoves, Biogas, LPG, and Electricity. The final panel of the *Expert Workshop* analyzed the issues of Health and Culture, concluding that it is necessary to design firewood replacement solutions that consider cultural and gender aspects. Experts reaffirmed the negative health effects of firewood consumption on people exposed to smoke due to inadequate smoke extraction and poor ventilation in the spaces where firewood is burned. Deliverable II presents a development of the cultural, ethnic, and gender issues, the effective treatment of which is essential in the development stage of the Replacement Plan.

During the second part of the Study, two themes were defined that are necessary for defining the substitution strategies in the final part of the Study. These themes are: the scenarios and the costs of the technological options for the substitution process.

Starting with the costs of technological options for replacing firewood, the
The study conducted a detailed analysis of all cost components for each option, which are detailed below:

- Costs of cooking equipment by energy and technology.
- Costs of the energy used in each case. For firewood, it was decided to include the collection costs, assigning a value to the work of the people, particularly women, who dedicate hours of their time to these tasks. Although this work is not actually paid, the use of time should have an associated cost.

- Environmental costs of CO2e emissions . For this purpose, the emission factors for each energy source are defined in the technical literature.
- Health costs that allow measuring the impacts per family using the national mortality and morbidity costs developed by the National Planning Department. This element of the total cost seeks to highlight a component of the cost of firewood that has not been included in the analyses until now.

The scheme described above aims to make visible and quantifiable all the costs associated with using firewood for cooking and heating food. It concludes that firewood is not a low-cost energy source; on the contrary, considering collection costs and health impacts, it is the most expensive of all. The table below, the first version of which was included in Deliverable II and revised for the Final Deliverable, shows the following results:

Table 1. Cooking costs according to technology

Energy used	Type of stove	Monthly cost of initial stove investment	Energy cost	Healthcare costs	Environmental cost of emissions of CO2	Total monthly cost
Firewood only	Stove	0	160.000	184.976	8.935	353.912
Firewood stove only	Stove with chimney	16.667	160.000	25.897	6.255	208.818
Firewood and LPG	Stove with chimney	20.833	112.500	12.948	4.834	151.116

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GLP	LPG stove	4.167	65.000	0	732	69.899
Biogas	Biogas greenhouse	12.500	160.000	0	0	172.500
Conventional electricity. Conventional stove.		2.500	66.000	0	891	69.391
Induction electricity	Induction stove	12.500	59.400	0	802	72.702

Source: Prepared by the consulting team

It is observed that LPG and electricity are the most economical options of all four analyzed. Deliverable III also presents the calculations from the analysis, which do not consider the time spent on firewood collection. Although the wood-burning stove remains the most expensive, the improved stove is more economical than electricity, and LPG and biogas are the least expensive of all. However, for the consultancy's analyses, the time spent on firewood collection per household and on biogas operation tasks is included to reflect the work performed, particularly by women.

The strategy presented in the final report, which will be explained below, does not focus solely on the most economical options, but proposes an inclusive program in which Improved Stoves, Biogas, Electricity, and LPG each have their place according to the population group resulting from the following analysis, which defines two scenarios as follows:

Scenario A: Defined based on the DANE 2018 Quality of Life Survey, which presents the result to the following question: “What energy or fuel do you mainly use for cooking?” A total of 1,664,229 households responded as follows: “firewood, wood”.

Scenario B: Households that answered the previous question were asked the following: “Additionally, do you use any other fuel for cooking?”

Approximately 30% answered the second question positively, and of that percentage, 93% indicated that their alternative fuel is LPG. The total number of users in this category is 499,809, leaving a total of 1,164,421 who exclusively use firewood without any other energy source.

Furthermore, according to information from the DANE's own Quality of Life Survey, approximately 252,765 families lack electricity service because they live in Non-Interconnected Zones (ZNI) or in areas isolated from the National Interconnected System. These families are considered to be among the group that relies solely on firewood (70%) due to a lack of access to any other energy source, or because, if LPG is available, transportation costs are prohibitively high.

With the previous analysis and information, the Study focused on the topics defined in the Contract C-031 of 2019 with UPME with the results explained in the report, of which the following will be noted for this summary:

a. The Strategy:

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From the scenario analysis, 3 User Groups are then configured based on the ECV figures from DANE 2018:

- Group I. Households that cook only with firewood and have access to the service of electrical energy. (911.656)
- Group II: Households that cook only with firewood and do not currently have Access to electricity service. (252,765).
- Group III: Households that cook with firewood and diversify with another energy source, mainly LPG. (499,809)

This analysis provides us with the elements of judgment for defining the strategy:

- Group 1: Being located in the area of influence of the National Interconnected System, it is expected that there will be transportation facilities that allow them to have access to LPG and in general to all available substitution options.

- Group 2: Currently located in non-interconnected zones or isolated areas within the interconnected system, options like LPG are more expensive and suggest the advantage of having fixed replacement options, such as improved cookstoves or biogas programs. Even with access to electricity, access difficulties persist, so the recommendation to use fixed options remains valid. • Group 3: Presents an energy consumption pattern in which LPG is used for one or at most two meals a day, leaving the rest for cooking with firewood. With LPG-based cooking equipment, the preferred option is to maintain and increase consumption of this energy source in order to continue the firewood replacement process.

Based on the above definitions, the strategy comprises actions to be taken in each of the three periods indicated—short, medium, and long term—for the three family groups defined in the previous section, considering the four energy sources identified as options in the Substitution Plan. In other words, the strategy comprises three main areas:

- Implementation Timeframe
- Population group targeted by the Plan
- Substitute energy sources.

The details of the figures included in the strategy are in Deliverable III and its main results are as follows:

Table 2. Groups identified for the substitution strategy

Group	Number of households	Features
G1	911,656	Firewood, does not diversify and has electricity
G2	252,765	Firewood and NO electricity
G3	499.809	Firewood and LPG

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	1.664.230

Group	**Phase**			**Total**
	2020-2022	**2023-2030**	**2030-2050**	
G1	100.000	511.656	300.000	
G2	30.000	150.000	72.765	
G3	30,000	200,000	160,000	269.809
Total Phase	650,000	Source: Prepared	854.230	***1.664.230***

by the consulting team

Deliverable III presents the details of the distribution of the above values by Departments and grouped into Regions as requested in the terms of reference.

b. The Environmental Impact of the Program.

The environmental balance in terms of net CO2 emissions is obtained by calculating the total emissions generated in the period 2020-2050, if current cooking practices with firewood continue for all families identified in the DANE Quality of Life Survey, minus the emissions avoided by implementing the substitution strategy mentioned above.

The net emissions result is shown in the table below:

Table 3. Net CO2 emissions by phase

	Tons of reduced CO2	**Tons of CO2 accumulated**
Fase I (2020-2022)	1.538.370	1.538.370
Phase II (2023-2030)	18.540.197	20.078.567
Phase III (2030-2050)	108,682,341	128,760,908

Source: Prepared by the consulting team

The above translates into a consolidated total of 128.76 million net tons of CO2 avoided by the replacement program during the Plan's horizon compared to the baseline of maintaining the consumption of firewood for cooking according to the current situation.

c. Costs and Subsidies

The analysis conducted previously leads us to the final part of the Study, in which the program costs, subsidies, and potential sources of funding are determined. The policy proposal is to subsidize the installation costs of the various proposed energy solutions. Regarding consumption, it is proposed to subsidize 50% of the monthly electricity and LPG costs, considering that the cost analyses of the four technological options resulted in very similar final electricity and LPG costs. Subsidies are not necessary for Improved Cookstoves and Biogas.

consumption.

This situation is summarized in the table below:

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Table 4. Costs and subsidies by technology

Technology	Technology	Operational Subsidy
Biogas	1.500.000	
Improved stoves	2.000.000	
Electrical energy	1.500.000	33.000
GLP	500.000	30.000

Source: Prepared by the consulting team

With the above information and following the strategy definitions of the Plan Substitution, the total costs turn out to be the following:

Table 5. Total implementation costs by technology and phases, option A

Technologies
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Biogas	2.250,0	22.500,0	150.000,0	174.750,0
Improved stoves	58.500,0	300.000,0	365.530,0	724.030,0
Electrical energy	43.875,0	202.500,0	202.500,0	448.875,0
GLP	50.000,0	175.000,0	218.232,5	443.232,5
	154.625,0	**700.000,0**	**936.262,5**	**1.790.887,5**

Subsidies
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Electrical energy	34.749,0	520.344,0	2.370.060,0	2.925.153,0
GLP	108.000,0	1.296.000,0	6.382.548,0	7.786.548,0
	142.749,0	**1.816.344,0**	**8.752.608,0**	**10.711.701,0**

Implementation
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Months	36	96	240	
Monthly Cost	100	100	100	
	3.600	**9.600**	**24.000**	**37.200**

Total
Millions of Pesos

Phase	I (2020 - 2022)	II (2023 - 2030)	III (2030 - 2050)	
Total Cost	300.974,0	2.525.944,0	9.712.870,5	**12.539.788,5**

Source: Prepared by the consulting team

The total cost of the program for the period 2020-2050 amounts to 12.5 trillion pesos, of which 14.3% corresponds to the installation costs of the technological options and 85.4% is the subsidy that must be granted for electricity and LPG consumption according to the defined Strategic Plan. The remaining 0.3% would correspond to the estimated budget of the Implementation Agency, which, as its name indicates, is the entity that will be in charge of developing the program.

This initial estimate assumes that the subsidy becomes permanent throughout the entire execution period of the Program, which means that a user who switches from using firewood to electricity or LPG would have a permanent subsidy for the remainder of the program period, until 2050.

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Based on the above elements, a second cost scenario is proposed, which consists of limiting the subsidy period to a maximum of 10 years. This scenario assumes that, after that time, if a family has successfully completed the firewood substitution process, it is unlikely they will return to using firewood for cooking. Furthermore, it is expected that the quality of life will improve with the use of an alternative energy source, in accordance with the Energy Ladder concept presented in Deliverable I, such that the substitution after 10 years can be considered permanent. Under this new scenario, the cost of the Program would be as follows:

Table 6. Total cost of program implementation by technology and phases, option B

Technologies
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Biogas	2.250,0	22.500,0	150.000,0	174.750,0
Improved stoves	58.500,0	300.000,0	365.530,0	724.030,0
Electrical energy	43.875,0	202.500,0	202.500,0	448.875,0
GLP	50.000,0	175.000,0	218.232,5	443.232,5
	154.625,0	**700.000,0**	**936.262,5**	**1.790.887,5**

Subsidies
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Electrical energy	34.749,0	508.761,0	641.520,0	1.185.030,0
GLP	108.000,0	1.260.000,0	1.823.274,0	3.191.274,0
	142.749,0	**1.768.761,0**	**2.464.794,0**	**4.376.304,0**

Implementation
Millions of Pesos

Technology	2020 - 2022	2023 - 2030	2030 - 2050	Total Cost
Months	36	96	240	
Monthly Cost	100	100	100	
	3.600	**9.600**	**24.000**	**37.200**

Total
Millions of Pesos

Phase	I (2020 - 2022)	II (2023 - 2030)	III (2030 - 2050)	
Total Cost	300.974,0	2.478.361,0	3.425.056,5	**6.204.391,5**

Source: Prepared by the consulting team

The total cost of the program for the period 2020-2050 has been reduced from 12.5 trillion to 6.2 trillion pesos, a decrease of 50%. Of the new figure, 28.9% corresponds to the installation costs of the technological options, and 70.5% is the subsidy that must be granted for electricity and LPG consumption. The remaining 0.3% would correspond to the estimated budget of the Implementation Agency.

From the above, it can be concluded that the subsidy period could even be adjusted during the implementation stage according to market and user perspectives, which could lower the total budget of the Replacement Plan without any detriment to the coverage goals.

d. Funding of the Program

One of the fundamental issues is the financing of the Program. To this end, Deliverable III, in its regulatory chapter, shows how each of the technological options defined for firewood substitution has a defined regulatory framework for service provision. Furthermore, it outlines subsidy schemes for energy supply and demand, for LPG distributed in cylinders, as well as royalty resources and subsidies for non-residential areas.

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Interconnected power grids and other potential sources of funding, such as the Fenoge tax and the Carbon Tax established in Law 1819 of 2016, can provide financial support for the implementation of the Fuel Substitution Plan. In addition to these potential local funding sources, it is necessary to explore, in a subsequent phase, securing external resources from international organizations and NGOs whose mission is to protect nature, promote energy efficiency, and improve the living conditions of impoverished families.

In the section on Available Funding Sources, an analysis is carried out that considers sources such as the Fund for Non-Conventional Energies and Efficient Energy Management -FENOGE-, created by Article 10 of Law 1715 of 2014; the Programs for the Benefit of Communities, PBC - Agreement 02 of 2017 of the National Hydrocarbons Agency - ANH; International Cooperation; the General Royalties System -SGR; the national carbon tax and the request for non-accrual due to carbon neutrality, Decree 926 of 2017 of the Ministry of Environment; and Other private initiatives.

In order to channel the above efforts, the Implementing Agency of the Firewood Substitution Plan, which will be discussed below, would lead a one-stop shop, which, as envisioned by FENOGE, would achieve the interaction of various actors (public sector, private sector, academia, trade associations, local banks, international banks, multilateral banks and international organizations) in terms of financing, cooperation, technical assistance and contribution of resources, among others.

In conclusion, FENOGE is proposed as the program's financial base, along with resources from the Program for the Benefit of Communities (PBC). Once the Implementing Agency is operational, it will be responsible for securing other funding sources such as the General Royalties System.

SGR (General Royalties System) and the National Carbon Tax, along with the request for carbon neutrality exemption, Decree 926 of 2017 from the Ministry of Environment, are key components of the program. The Agency would be responsible for engaging with exploration and production area operators, who are responsible for managing the resources allocated under Article 80 of ANH Agreement 02. Similarly, it would present the program to various international cooperation agencies to secure support resources, technical assistance, access to lessons learned, and other resources. This work would take place during the first year of operation and would guarantee funding for implementing the solutions in this phase by combining these three sources of financing.

e. The Implementation Strategy

The implementation strategy considers that the Plan will impact more than 5 million people, not only by changing their food cooking and home heating habits, but, most importantly, by improving their health and the quality of life of their family environment.

The implementation scheme must also consider that the Plan covers the period of the current Government and several more four-year periods and requires for its successful implementation a demanding inter-institutional coordination effort so that each actor in the program carries out its task according to the established objectives.

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For the reasons stated above, the Study recommends that the Firewood Replacement Plan be elevated to the category of public policy and therefore be considered, evaluated and approved at the highest levels of Government.

Therefore, the body that should review the Firewood Substitution Plan and propose its approval as a Public Policy is CONPES. The document that the DNP submits to CONPES for consideration must establish the mechanisms for implementing the Substitution Plan, including an allocation of responsibilities and definitions of the resources that must be budgeted for its execution in the different stages, beginning with the first short-term phase from 2020 to 2022.

As mentioned previously, the Study proposes hiring an Implementing Agency with specific functions to handle the implementation of the program
initially in the short term 2020-2022.

This process does not involve the creation of a new state entity but rather the hiring of a Team with a fixed period and the following main tasks:

a) Define in detail the portfolio of energy products that will be part of the Substitution Program during that period and the subsidy schemes that will be applied to the process.
b) Define the coverage in terms of geographical areas, number of families to be replaced and the criteria for identifying end users.
c) Define the Replacement Program Budget and schedules.
d) Propose the area allocation scheme, which could be: concessions or exclusive service areas, leveraging the experience of previous programs such as the Gas Plan, and other models that are evaluated for the implementation stage. e) The Implementing Agency shall open the bidding processes to select the agents or entities that will deliver the energy to the beneficiary families, as stipulated in the replacement plan.

f) Develop the promotion of the Program before technical cooperation entities and international support funds.
g) Design outreach programs with the communities and ethnic groups that are will benefit from the Replacement Program.

Finally, at the close of Deliverable III, the proposed text for the First Debate before the Fifth Committee of the Senate of the Republic was released for House Bill 044 of 2019, combined with another Bill No. 213 of 2019: “through which the supply, continuity, reliability, and coverage of combustible gas in the country are promoted.” Article 7 of the Bill is titled: “Program for the substitution of firewood, charcoal, and waste with combustible gas,” which aims to: “ensure access to combustible gas service for those families who continue to cook with firewood, charcoal, and waste.” Furthermore, Article 9 proposes the Declaration of National and Strategic Interest for: “the widespread use of natural gas, vehicular liquefied petroleum gas, and marine liquefied petroleum gas, given their multiple environmental, health, and economic and social competitiveness benefits.”

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Consequently, it will be necessary to monitor the progress of the aforementioned Bill in the coming weeks and months and determine the impact of said process on the Firewood Replacement Plan in the short, medium and long term.

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1. Objectives

1.1. Project Objective

Conduct a study to formulate an updated program for the progressive replacement of firewood as an energy source in the residential sector in Colombia, with the necessary components for its implementation.

1.2. Specific objectives

1. Formulate an updated program for the progressive replacement of firewood in the sector residential with its strategy and roadmap.
2. Evaluate the lessons learned from previous programs implemented in Colombia to replace or use firewood more efficiently and cleanly in the residential sector; what has worked and what has not worked; lessons learned.
3. Evaluate international experiences in programs to replace firewood with other energy sources and/or technologies (both commercially available and those under research or not yet in the commercial phase in Colombia) that can guide the development of the program.
4. Evaluate the legal and regulatory aspects required to enable the implementation of the progressive replacement program for firewood in the residential sector in Colombia.
5. Evaluate the cultural aspects related to the use of firewood in the prioritized regions of Colombia where firewood is still used as an energy source.
6. Analyze gender and ethnic aspects related to the use of firewood as energy in these issues
7. Analyze existing information and figures to establish benchmarks and standardize data. Specifically, analyze DANE data on households that use firewood and consider at least two firewood consumption scenarios.
8. Identify and prioritize, in a participatory manner with relevant actors, the cleanest and most efficient available energy sources that could replace firewood as an energy source.
9. Estimate the CO2e reduction associated with the implementation of the program to replace firewood as an energy source in the residential sector of Colombia.
10. Make an assessment of the positive environmental and social externalities resulting from the replacement of firewood with other cleaner, modern energy sources.
11. Conduct a cost-benefit analysis of the various cleaner and more efficient energy sources and technologies available in Colombia that are in a commercial phase, to replace firewood as an energy source.
12. Feasibility analysis for the installation of the most viable solutions that are in a commercial phase in Colombia to progressively replace firewood in the residential sector.
13. Evaluate and propose the available sources of financing to implement a program for the progressive replacement of firewood as an energy source; include an analysis of the subsidies and incentives that the program would require for its implementation;

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estimate the generation of carbon credits that could be obtained by replacing firewood with combustible gas or another cleaner energy source.

14. Feasibility analysis for the installation of the most viable solutions that are in a commercial phase in Colombia to progressively replace firewood in the residential sector.

15. Establish the estimated costs to implement the program.

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2. Introduction

In Colombia, firewood has traditionally been one of the most widely used energy sources for cooking in homes. According to the 2018 Quality of Life Survey (ECV), 1,664,229 households use firewood as their primary cooking fuel. To address the problems identified in the country stemming from firewood consumption in Colombian households, the Mining and Energy Planning Unit (UPME) has commissioned a consultancy to *"formulate, based on secondary data, an updated program for the progressive replacement of firewood as an energy source in the residential sector in Colombia. This program will include a strategy and roadmap, an assessment of the current state of knowledge on the issue, an evaluation of what has worked and what hasn't, and lessons learned. It will also analyze the technological options available for its progressive replacement in UPME's expansion plans, thereby facilitating the reduction of energy poverty and the health and environmental impacts of firewood use."*

[1]

In the development of this consultancy, the analysis of the state of the art of national and international experiences and lessons learned from programs implemented to replace or use firewood more efficiently and cleanly in the residential sector has been presented previously in deliverable 1, with the objective of guiding the development of a program for the progressive replacement of firewood as an energy source for cooking in Colombia.
This deliverable served as input for a preliminary identification of the technologies and some of the financing mechanisms used, as well as aspects that had not been considered in the planning process and that have led to implementation difficulties or unexpected results. Once the state-of-the-art analysis was completed, four regional workshops and one national expert workshop were held, which yielded new input for the development of the plan for the progressive substitution of firewood in Colombia. Both the proceedings of these workshops and the plan as a whole will be presented as part of deliverable 3, the final report of the project.

Deliverable 2 presented the analysis of the legal and regulatory aspects that must be considered to enable the implementation of a progressive replacement program for firewood in the residential sector in Colombia, for each of the energy sources considered as a replacement alternative, and of some current funds and subsidies.
Similarly, an assessment was conducted of the cultural aspects related to firewood use in the prioritized regions of Colombia where firewood is still used as an energy source, along with an analysis of the gender and ethnic aspects related to firewood use. This deliverable, in addition to addressing the problem from a social perspective, presents some observations on health issues affecting people exposed to smoke from firewood combustion, especially women and children. Furthermore, an analysis of data is presented that aims to contribute to the standardization of information on firewood consumption.

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1 Taken from the section “Background and Justification of the Present Process” of the Final Terms of Reference of the Public Request for Proposals process No. 002-2019 of the UMPE.

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current trends in Colombia, and proposed two consumption scenarios based on various sources, mainly data from the DANE Quality of Life Survey.

This final report presents in Chapter 77 the estimate of the CO2e emission reduction associated with the implementation of the progressive replacement program of firewood as an energy source in the residential sector of Colombia, an assessment of positive environmental and social externalities derived from the progressive replacement of firewood with other cleaner modern energy sources (Chapter 88), the cost-benefit analysis of the various sources and technologies available to progressively replace firewood as an energy source in the residential sector (Chapter 99), the feasibility analysis for the installation of viable solutions (Chapter 1010), the institutions and stakeholder maps (Chapter 1111), as well as the strategy proposed by the consulting team for the development of a Progressive Massification Plan for firewood (Chapter 1212), along with the estimated costs for its implementation (Chapter 1313), and the sources of financing available to make it possible (Chapter 1414).

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3. State of the art by identified technology

3.1. Improved Stoves

The following describes the main progress made both nationally and internationally regarding the implementation of improved stove projects in different territories.

3.1.1. Improved Stoves in Colombia

Guidelines for a National Program of Efficient Wood-Burning Stoves.

The preparation of this document arises from the need of the national government and the Ministry of Environment and Sustainable Development to present the guidelines of a national program of improved wood-burning stoves, aimed at the sustainable and rational use of this energy source, in such a way as to contribute to improving the quality of life of the population, alleviating the energy poverty in which they live, while reducing the negative impacts on health and the environment.

According to the Economic Commission for Latin America and the Caribbean , *"most families still cannot access modern fuels for cooking food, and when they do, they pay a disproportionate proportion of their income for it, which accentuates social inequality in the region. Likewise, in areas where higher per capita firewood consumption is observed, low human development indices are generally recorded"* (ECLAC, 2009).

[2]

Problems associated with the consumption of firewood for cooking

Under the scenario presented by the National Administrative Department of Statistics -
According to the DANE (National Administrative Department of Statistics) Quality of Life Survey (ECV, 2013), 1.6 million families in Colombia use firewood daily for cooking, of which 1.4 million are rural families and the remaining 200,000 are urban families. It is estimated that more than 50% of the country's rural population uses fossil fuels or firewood for cooking (approximately 13.6% of the total population), demonstrating that this is **not** an isolated energy source in the country. As part of the efforts to mitigate this situation, it is necessary to promote the implementation of cooking systems with more efficient combustion and improved systems for the evacuation and control of emissions and ash, in order to reduce polluting emissions and protect the health of the affected population.

Impact on health (risk factors for respiratory diseases in the rural population)

"Identifying territorial gaps is especially relevant considering that the lack of access to electricity – mainly in rural areas – has negative impacts on the health of the inhabitants, affecting women to a greater extent.

2 ECLAC. Contribution of energy services to the achievement of the Millennium Development Goals and the mitigation of poverty in Latin America and the Caribbean, 2009.

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those who currently dedicate more time to food preparation (DANE-ENUT, 2017). Although, between 2011 and 2017, the proportion of people who cook with firewood, wood or charcoal decreased by half, 29% of rural households still use this type of fuel for cooking3

As a result of the assessment of environmental health costs in Colombia carried out by the World Bank in 2012, and as part of the technical assistance and analytical support program to the Ministry of Environment and Sustainable Development, it was estimated that the average annual cost of health impacts from indoor air pollution associated with the use of traditional fuels (mainly firewood) in rural areas of Colombia for that year was 1.13 trillion pesos (0.22% of GDP in 2009).

Infant mortality accounts for 6% of the costs; female mortality accounts for approximately 78% of the costs. Acute respiratory infection (ARI) in children and adult women and COPD morbidity in adult women account for 16% of the costs . ⁴

Subsequently, in 2017, the National Planning Department (DNP) presented an updated study on the costs of deaths and illnesses associated with environmental degradation, which included the costs associated with indoor air pollution. This pollution is attributed to 2,286 deaths and 1.2 million illnesses, with costs for premature mortality and disease treatment exceeding 3 trillion pesos, equivalent to 0.38% of GDP in 2015 .

The following conclusions were drawn from studies by the World Bank in 2012 and the DNP in 2016:

- Among the diseases attributed to indoor air pollution, three health effects were identified with strong evidence corresponding to acute lower respiratory tract infections (ARI) in children under 5 years of age and in women over 44 years of age; chronic obstructive pulmonary diseases (COPD) in women and lung cancer in women exposed to coal smoke6 7
- as well as hypertensive diseases, ischemic heart diseases, cerebrovascular diseases and lower respiratory tract diseases, all of these in people over 44 years of age8 .
- Exposure to indoor air pollution, resulting from incomplete and inefficient biomass combustion, is one of the most significant risk factors in the proliferation of respiratory diseases.

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[3] DANE, Quality of Life Survey, 2017

4 Banco Mundial, Environmental Health Costs in Colombia: Changes from 2002 to 2010.

[5] DNP, The health costs associated with environmental degradation in Colombia amount to $20.7 billions, 2018

6 Torres Duque C, Maldonado D, Pérez Padilla R, Ezzati M and Viegi G. Biomass Fuels and Respiratory Diseases, on behalf of the Forum of International Respiratory Societies Report, 2008.

[7] DNP, The health costs associated with environmental degradation in Colombia amount to $20.7 billions, 2018

[8] DNP, The health costs associated with environmental degradation in Colombia amount to $20.7 billions, 2018

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especially in children and adult women, since they spend more time at home and therefore have a higher degree of exposure.

The level of evidence for COPD and lung cancer in men exposed to coal smoke is considered moderate; while the association between this same exposure and the occurrence of lung cancer and asthma in children under 5 years of age as well as the presence of tuberculosis in adults are still considered weak9 .

It is estimated that in 2009 there were around 1000 premature deaths among women and children, attributed to the use of solid fuels for cooking.
Around 200 of those deaths were children under five years old and 800 were adult women. COPD is the leading cause of death among women. These deaths represent 11,635 disability-adjusted life years (DALYs) lost annually, of which about 60% are among children under five . ¹⁰

The highest number of cases of mortality attributed to indoor air pollution occurs in the Eastern region of the country with an age-adjusted case ratio of 2.9 deaths per 10,000 inhabitants11 .

Regarding morbidity, it is estimated that nearly 1,000,000 children and women became ill due to the use of solid fuels for cooking in Colombia in 2009; of these, 60% were children under 5 years of age12 .

" The *morbidity rate from indoor air pollution is 104 per 1,000 inhabitants and primarily affects children under five (78%) and women over 44 (22%). In terms of mortality, women are the main victims, with a rate of 32 per 100,000 women over 44. By region, the Caribbean coast and the* east *have the highest levels of impact (DNP, 2018)13."*

Environmental impact

Similarly, as a result of the negative environmental impact of inefficient firewood use for cooking, the World Bank published a report in 2011 which concluded that:

Inadequate cooking practices using firewood in developing countries generate short-lived pollutants (black carbon, organic carbon) that affect the climate. Therefore, an efficient stove can reduce these pollutants.

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9 Torres Duque C, Maldonado D, Pérez Padilla R, Ezzati M and Viegi G. Biomass Fuels and Respiratory Diseases, on behalf of the Forum of International Respiratory Societies Report, 2008.

10 Banco Mundial, Environmental Health Costs in Colombia: Changes from 2002 to 2010.

11 DNP, Health costs associated with environmental degradation in Colombia amount to $20.7 trillion, 2018

12 Banco Mundial, Environmental Health Costs in Colombia: Changes from 2002 to 2010.

13 DNP, Health costs associated with environmental degradation in Colombia amount to $20.7 trillion, 2018

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Up to 66% of CO2 or GHG equivalent emissions are linked to improper food cooking practices at the household level14

The generation of greenhouse gases (GHG), indoor air pollution, and the degradation of forests due to the use of firewood are associated with the inefficient use of firewood for cooking in homes15 .

On the other hand, according to the Intergovernmental Panel on Climate Change (IPCC)16, the use of fuelwood for cooking accounts for 2% of global emissions, making it a significant source of greenhouse gas emissions and one of the causes of climate change. Deforestation and forest degradation activities in tropical areas produce around 20% of global CO2 emissions and, in most cases, constitute the bulk of emissions from developing countries. In this way, emissions could be reduced from 1.45 tons CO2eq/ year to 0.37 tons CO2eq/year, with a biomass renewability factor of approximately 80%.

According to Johnson's study, "emission reductions can vary depending on four main variables: type of fuel, efficiency of the equipment used for cooking or heating, non-renewable biomass, and cooking practices" (Johnson, 2008)17

Roadmap for efficient and improved cookstove programs in Colombia

In 2011, the Faculty of Public Health of the University of Antioquia carried out isokinetic sampling in the chimneys of efficient stoves and traditional hearths, in homes benefiting from the *Huellas prototype,* located in municipalities of eastern Antioquia, jurisdiction of Cornare, whose results showed that with the use of *Huellas stoves*
94.2% of particulate matter is reduced and the reduction of other pollutants (CO2, CO, NOx, SO2) has the same trend.

Based on an analysis of the country's experience in the development and implementation of improved stove projects, it was found that:

With the implementation of improved stoves in Colombia, only 2% of the potential of 1.6 million families that use firewood has been covered (ECV, 2013)

According to the 2018 Quality of Life Survey (ECV), the Caribbean Region has the highest number of households that cook with firewood (31%), followed by the Pacific Region (20%). The Pacific Region, despite having one of the highest percentages of firewood consumption for cooking in rural households, has only 5% of stoves installed .¹⁸

[14] http://climatechange.worldbank. org/sites/default/files/documents/ Household%20Cookstoves-web.pdf

[15] http://climatechange.worldbank. org/sites/default/files/documents/ Household%20Cookstoves-web.pdf

[16] http://www.ipcc.ch/pdf/assessmentreport/ar4/syr/ar4_syr_sp.pdf

[17] Johnson, M., Edwards, R., Frenk, C. A. & Masera, O. (2008). In-field greenhouse gas emissions from cookstoves in rural Mexican households. Atmospheric Environment, 42(6). 1206-22.DOI: 10.1016/ j.atmosenv.2007.10.034.

[18] DANE, Quality of Life Survey, 2018

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Stoves have been subsidized in all cases, only in some cases with a small in-kind contribution from users.

The evaluation of pollutant emissions from traditional household stoves and efficient cookstoves has been poorly monitored, resulting in limited information on the efficiency levels of each prototype. In other words, there are no efficiency tests or ex-post evaluations of emission reductions from improved cookstove prototypes.

o It is evident that the prototype of the *footprint* stove is the most widespread in the country and the difference in costs of the *footprint* stove prototype between regions is significant.

o The monitoring and evaluation mechanisms for improved stove projects in the country are currently under construction.

Lessons Learned

Within this category, it's important to highlight that the successful implementation of efficient stoves in rural areas of the country is not guaranteed. Many evaluations of the subsidized projects have shown that most households have reverted to using traditional stoves for cooking.

The following is a compilation of the main lessons learned internationally by the GIZ HERA Cooking Energy Compendium19 and those obtained from the development of guidelines for a national program of efficient wood-burning stoves in Colombia:

- There must be comprehensive and participatory planning of the projects (poverty reduction, gender issues, cost-benefit, quality of life, energy efficiency, environmental impact), which guarantees their success in the long term and involves the communities, raising awareness of the importance of acquiring new cooking technologies.

Stoves must meet minimum quality standards for proper operation, as their lifespan depends on the quality of the basic materials and the maintenance they receive. The international ISO standard is still under development, so it must be adapted to the specific conditions of each country to ensure that installed stoves are easy to use and maintain. Likewise, stoves that reach the end of their useful life must be replaced promptly.

On the other hand, it is important to provide an advisory and technical assistance approach for stove maintenance to ensure the successful installation in the home. It is recommended that the stoves be built by people from the area where the measures are being implemented, as this

19 GIZ HERA, Cooking Energy Compendium, consulted at: http://CookinEnergycompedium.www.energypedia.info/wiki/GIZ

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It will allow and guarantee that there is trained personnel to repair and perform the necessary maintenance on the new stoves.

To ensure long-term sustainability, a purely commercial approach is necessary, meaning the creation of a market (unless specific local conditions warrant a different approach) that enables the development of microfinance programs to improve access to cookstoves or the use of carbon finance. Offering microcredit opportunities and long repayment terms is more advantageous than building stoves for free.

- It must be clear that it takes time to change habits and establish new local structures within households; therefore, projects should be thought of in the long term, a minimum of 5 years, and ideally 10 years.

- It is important to have impact indicators that allow monitoring of the reduction of GHG emissions and other criteria pollutants, as well as short-lived climate pollutants (SLCPs), and the occurrence of possible co-benefits resulting from the implementation of the new technology.

Efficient wood stoves as a contribution to improving the quality of life, the efficient use of energy and the reduction of greenhouse gas emissions in rural areas of Antioquia and Santander - IICA Cornare, Corantioquia and Fundación Natura in 2014.

According to the Mining and Energy Balance 1975-2012, published by UPME, *"the consumption of Firewood went from covering 19% of national energy needs to only 5.3%."* [20]
It should be noted that, although it is a reduction of 13.7%, according to the Food and Agriculture Organization of the United Nations - FAO, heating and cooking food with firewood will prevail in the next 20 years[21] .

This is why the need arises to analyze the impact that the use of this energy source as the main fuel has on the quality of life of rural households in the country, and in turn the impact of its use on the environment.

In this way, some government, private and NGO entities have been promoting measures to adopt more efficient wood stoves with the intention of reducing the social, environmental and health impact in Colombian homes.

Regional autonomous corporations, such as the Regional Autonomous Corporation of the Negro and Nare Rivers (CORNARE), have been working for 20 years to build 23,000 wood-burning stoves in their jurisdiction (eastern Antioquia Department). As a result of this effort, the *HUELLAS* program (Clean and Environmentally Sustainable Gardens and Wood-Burning Stoves) was created, which links the replacement of traditional open fires to the replacement of traditional cooking stoves.

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20 UPME. Mining and Energy Balance 1975 – 2012. Bogotá UPME – SIMEC, 2012. Excel file.
21 FAO. Forests and energy: key issues. Rome: FAO, 2008. p. 12

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Efficient stoves are being used, and the creation of orchards for planting trees is being encouraged as a backup measure for the availability of energy in the medium and long term.

Similarly, the Regional Autonomous Corporation of Central Antioquia (CORANTIOQUIA), in its jurisdiction for the year 2015, began a process of design and evaluation of the SCE stove (Efficient Cooking System),22 of which it has built 22 units and is in the process of validation to be able to be disseminated in the coming years.

On the other hand, the Natura Foundation (FN) has championed this issue since 2008 with a pilot project addressing the problems in the Guantiva-La Rusia conservation corridor. This project focused on protecting the remaining oak forests traditionally used as a source of firewood by surrounding communities, leading to deforestation and ecosystem loss. With the aim of reducing inefficient firewood consumption, Natura developed and built 50 FN stoves as a pilot project. Mass distribution of the model began in mid-2014, and by the end of that year, nearly 2,000 stoves had been built .²³

However, to achieve widespread adoption of efficient stove initiatives aimed at improving the quality of life for rural households in the country, a financing strategy is needed to ensure the sustainability of these projects in the medium and long term. Therefore, and by combining efforts, the potential of leveraging carbon financing was identified. This financing can help stimulate the large-scale production and dissemination of efficient stoves, reducing their purchase price and making them more affordable for users . ²⁴

Objective of the aforementioned project, specific case of Antioquia and Santander

With the goal of improving quality of life and reducing environmental health problems associated with indoor air pollution in rural homes in Antioquia and Santander, the Natura Foundation in 2014 sought to address this environmental and health issue through the use of *improved FN wood-burning stoves* for cooking. Among the motivations behind the project's design were the conservation of the forests in the Chicamocha River basin and the improvement of the quality of life for rural families.

In this way, the implementation of 2,000 efficient FN model stoves was carried out in the Chicamocha river basin and the department of Antioquia.

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22 Fundación Natura. Efficient wood-burning stoves as a contribution to quality of life, efficient energy use and the reduction of GHG emissions in rural areas of Antioquia and Santander, Colombia: Systematization document and technical supplement / Gómez, Roberto León; Aristizábal, Javier Darío and Cárdenas, Luis Mario. Bogotá, DC Colombia, Fundación Natura. 2015. 130 p

23 Fundación Natura. Efficient wood-burning stoves as a contribution to quality of life, energy efficiency and the reduction of GHG emissions in rural areas of Antioquia and Santander, Colombia: Systematization document and technical supplement / Gómez, Roberto León; Aristizábal, Javier Darío and Cárdenas, Luis Mario. Bogotá, DC Colombia, Fundación Natura. 2015. 130 p

[24] BLUNCK, M. et al. Carbon markets for improved cooking stoves. 4th Edition. Eschborn (Germany): GIZ-HERA, 2011. p. 5.

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As for the department of Antioquia, during the first phase stoves were only distributed in the jurisdictions of CORNARE and CORANTIOQUIA, while in Santander stoves were only built in the provinces of García Rovira and Guanentá.

The main partners in the project included the municipalities of San Andrés, Mogalavita, Macaravita, San José de Miranda, Málaga, Capitanejo and Enciso (Santander).
In Antioquia, CORNARE and the municipalities of Concepción, Marinilla, El Santuario, La Unión, San Vicente de Ferrer, Cocorná, El Peñol, Granada, La Ceja, Sonsón, Rionegro and Guarne. Just like each of the project's beneficiaries.

Problems associated with the consumption of firewood for cooking25

Colombia is still a country with significant rural characteristics. As a developing nation, access to energy is one of the main limitations to improving living conditions for the population. According to the 2013 National Quality of Life Survey, 50.5% of rural families rely on firewood as their primary cooking fuel, and within this group, 38% combine it with LPG .

On the other hand, in 2014, DANE reported that 4,900 women had died from COPD and at least a third of them lived in rural areas27, raising the question : *How can rural households access more efficient and cleaner stoves?*

One of the alternatives mentioned in previous paragraphs is to implement a carbon finance mechanism to leverage resources directed toward efficient stove projects.28 In this way, using an efficient stove would imply less firewood use and, therefore, a reduction in the amount of GHG emissions that can be traded in regulated or voluntary markets at prices that may vary depending on the quality of the project and its collateral benefits. This would guarantee that the emissions are offset by the implementation of improved stoves. According to the World LPG Association (WLPGA), it is estimated that a person consumes 400 kg of firewood annually for cooking,29 the lowest of the calculations presented so far and assuming a minimum firewood consumption scenario. Under this premise, and considering that in Colombia families currently consist of 3.4 people,30 firewood consumption per household amounts to 1,360 kg of firewood per year. This value, multiplied by the number of households that use firewood as their main energy source according to the ECV (Environmental and Climate Survey) for

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25https://sostenibilidad.semana.com/medio-ambiente/articulo/estufas-eficientes-a-cocinar-aire-limpio/32770
26 DANE. National Quality of Life Survey 2013. Op. Cit.
27 DANE. Non-fetal deaths 2014 – Preliminary. Bogotá: DANE, 2015. Table 5. Excel file.
28 Given that it will not be easy to do without the subsidy scheme because it is not always possible for the user to acquire a new stove
29 WLPGA. (2018). SUBSTITUTING LPG FOR WOOD: CARBON AND DEFORESTATION IMPACTS. Neuilly-sur-Seine, France: WLPGA.
30 Quality of Life Survey, 2018. Excel document

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2018 (1.6 million) gives us an estimated value of 2,176,000 tons of firewood per year. This translates to emissions of $_2$ is equivalent to 3,699,200 tons of $_2$ per year.

Based on the above, synergies have been generated with regional and local entities to ensure proper coordination among the different project stakeholders. The Alliance for Energy and Environment with the Andean Region (AEA) program has been presented to the Ministry of Foreign Affairs of Finland and the Inter-American Institute for Cooperation on Agriculture (IICA). The systematization of experience is underway to develop and register a national initiative for improved cookstoves in the carbon market, from which the country can benefit and which can strengthen its sustainability and replication in the medium and long term. This will be achieved through the formulation, validation, and implementation of a program of activities (PoA) for emissions reduction in the voluntary market, which will facilitate the development of improved cookstove projects . ³¹

The PoA estimates that greenhouse gas emissions will be reduced through the production and dissemination of wood-burning stoves that have been field-tested by local corporations, in collaboration with the Natura Foundation, and have proven to be more efficient than traditional cooking methods.32 The PoA seeks to ensure long-term viability and continuity after the first phase through the commercialization of “The Gold Standard- Voluntary emission reductions” (GS-VERs).

In light of the above, the efficient wood-burning stove project, aimed at improving quality of life, promoting energy efficiency, and reducing greenhouse gas emissions in rural areas of Antioquia and Santander (Colombia), seeks to benefit from this mechanism so that future efficient stove initiatives in the country can be registered. The project's implementation formally began in April 2013 with the development of the Annual Operating Plan (PoA).

Initially, the implementation of the AMS.II-G33 methodology was proposed; however, it was found that this protocol requires technologies with a minimum thermal efficiency of 20%, and none of the stove models existing in the country had this requirement.
This data. On the other hand, when efficiency measurements were taken, the stoves showed efficiencies of less than 20%, so it was not possible to use this methodology.

Subsequently, the Technologies and Practices to Displace Decentralized Thermal Energy Consumption (TPDDTEC) methodology was used, in which firewood savings are measured by applying the Cooker Performance Test (KPT). This test evaluates the situation before and after the introduction of the new stove to determine the amount of fuel saved per family. For the case study, a

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31 IICA. Project monitoring and evaluation guide. s. l: IICA – AEA – Ministry of Foreign Affairs of Finland, 2012. p. 11.
32 In developing the PoA and its components, it was possible to estimate the existing firewood savings for the FN, Huellas, and SCE stoves. With this information, South Pole Carbon was able to finalize the PoA-DD and its two components (VPA for the Robles corridor and CORNARE).
33 The small-scale methodology AMS.II-G corresponds to a protocol for estimating the reduction of emissions from efficient stove projects called “Energy efficiency measures in thermal applications of non-renewable biomass” which has been developed for the Clean Development Mechanism.

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A significant number of families dropped out after the initial measurements were completed, forcing the search for new families and a complete restart. Similarly, the enactment of Law 996 of 2005 suspended contracting for six months, halting stove construction and delaying the publication of the findings.

In the scheme used by this project, an agreement was reached with the beneficiaries of the stoves to transfer the rights to the VERs (Energy Verification Resources) as consideration for their construction. In this way, the beneficiaries become co-financiers of the asset they acquire and strengthen their sense of ownership over something in which they "invested," which ultimately benefits them by increasing the overall adoption rate of the project.

Among the results obtained through the measurement process is the reduction in individual emissions from each stove, resulting from the reduction in firewood consumption, which was measured for the different types of efficient stoves used; this reduction is between 1.5 and 2.0 tCO2e per year, which means that each project instance could comprise between 5,000 and 6,667 stoves.

With the implementation of the PoA, and despite the limitations, it was possible to determine the firewood savings for the FN, Huellas and SCE stoves:

Table 7. WBT results for the Huellas, FN and SCE stoves

STOVE	PHASE	METRIC UNITS		AVERAGE OF 5 TESTS
FOOTPRINTS	Cold start	Thermal efficiency	%	12,2
	Hot start	Thermal efficiency	%	13,4
	Moderate heat	Thermal efficiency	%	6,6
FN	Cold start	Thermal efficiency	%	1,30
	Hot start	Thermal efficiency	%	1,14
	Moderate heat	Thermal efficiency	%	0,45
SCE	Cold start	Thermal efficiency	%	1,48
	Hot start	Thermal efficiency	%	2,07

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	Moderate heat	Thermal efficiency	%	2,12

Source: Prepared from: Fundación Natura. Efficient wood-burning stoves as a contribution to quality of life, efficient energy use and the reduction of GHG emissions in rural areas of Antioquia and Santander, Colombia: Systematization document and technical supplement / Gómez, Roberto León; Aristizábal

It is hoped that the foundations of this project will allow for the creation of a carbon market to reduce firewood consumption in Colombian households.

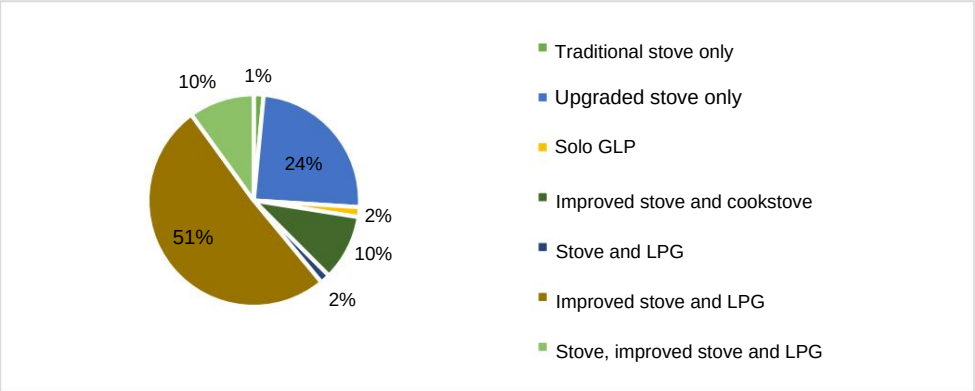
Results

In 2018, the Natura Foundation published the study *"Adoption levels and impact of an improved wood stove in rural communities of the department of Santander, Colombia"* which aimed to evaluate the level of adoption and impact of an improved stove model (FN type) in 70 families benefiting from the aforementioned carbon financing project34 The methodology was based on the implementation of surveys, from which the following results were obtained:

- It was found that 77% of the families evaluated do not use the traditional wood stove/hearth but the improved one, while the remaining 23% admitted that they continued to use it as the only cooking "technology", either inside the same dwelling (21%) or in a covered space but separate from it (2%).

More than half of the respondents stated that they used both the improved cookstove and the gas (LPG) stove for their cooking activities (51%), while 24.5% indicated that they used the efficient cookstove exclusively. It was evident that the traditional open fire remains important for 13% of the households surveyed, of which 1.5% stated that they use it exclusively and 1.5% use it in conjunction with other cooking methods.

Figure 1. Multiple use of cooking technologies



34 Made in 2014 by Natura in partnership with CorNare and CorAntioquia.

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Source: Adapted from "Adoption levels and impact of an improved wood stove in rural communities of the department of Santander, Colombia, 2018" [35]

On the other hand, the efficient stove is used an average of 6.17 days per week, while the traditional wood-burning stove is used 2.5 days per week, depending on what is being prepared. In homes that use LPG, the average use of this fuel is 6 days a week, noting that the gas is mainly used in the mornings to prepare breakfast or to reheat food that was previously cooked on the wood-burning stove, thus optimizing cooking time.

Regarding health perceptions, 55% of surveyed users stated they had no health problems related to exposure to wood smoke, either before or after acquiring the efficient stove. Similarly, 33% of the population reported noticing at least one change in their health after installing the improved stove, and only 11.5% reported noticing two changes.

On the other hand, regarding the perception of firewood savings, 67.5% of respondents stated that they perceived savings in firewood consumption with the FN stove, 11.5% stated that they perceived a subtle decrease in firewood use, and 3% reported substantial reductions.

- Based on the question of whether the stoves would be delivered in the future under the same mechanism, that is, without any monetary compensation, the people surveyed responded positively; on the contrary, if they had to put up their own money to acquire a new stove, the result shows that there is no market associated with improving stoves to reduce firewood consumption in homes.

Lessons Learned

The existence of a pooled carbon project is an effective incentive for implementing efficient wood-burning stove projects, with an emphasis on climate change mitigation, guaranteeing clear additional social and environmental benefits. It is also proposed that the benefits of carbon financing be invested in activities that promote the expansion of the program, such as consumption subsidies, training of local partners, social marketing, and improvement of distribution channels.

The participation of local partners in the project, along with the active involvement of regional corporations, can generate management policies that, if implemented in a timely manner, will guarantee the long-term sustainability of the actions. This implies that sustainability strategies must be developed to ensure that the projects are enduring in the medium and long term.

- In rural areas, it has been observed that the arrival of a new fuel does not displace the other, but rather both are used concurrently36 in such a way that the

[35] Aristizabal, Javier. Adoption levels and impact of an improved wood-burning stove in communities rural areas of the department of Santander, Colombia, 2018

36 TRONCOSO, Karin. Factors associated with better adoption of clean stoves: a gender perspective. Bucaramanga, 2014. p. 29.

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The wood stove is used for preparing large quantities of food, and the gas stove for quicker and simpler preparations.

Cornare Footprints Case Study

As mentioned in the previous chapter, the Regional Autonomous Corporation of the Negro and Nare Rivers (CORNARE) has been working for 20 years on the construction of 23,000 stoves in its jurisdiction (eastern Antioquia Department). As a result of this effort, the HUELLAS program (Clean and Environmentally Sustainable Wood-Burning Gardens and Stoves) was created, which links the replacement of traditional hearths with efficient stoves and encourages the creation of gardens for planting trees. In this way, "the aim is to reduce the accelerated deforestation of native or remaining forests due to the extraction of firewood for cooking, as well as to decrease the emission of particulate matter and greenhouse gases generated in traditional farmers' kitchens, thereby reducing mortality rates from diseases associated with the inhalation of smoke and ash, improving the living conditions of homes, and generally improving the quality of life of farming families."37 By 2010, 6,000 efficient stoves had been built and the construction of another 3,070 was underway, each with its respective woodshed.

The program selected the families according to the following parameters:

- Families with a peasant economy living in the rural area.
- That they are registered in levels 0, 1, 2 and 3 of the SISBEN.
- That they use firewood as the main energy source in the cooking processes of food.
- That they possess a minimum area of 500 square meters for the implementation of the woodshed. (municipality's criteria)
- That they are owners or regular possessors of the land where the project will be implemented.
- Give priority to women and/or men who are heads of household.
- Select users in rural communities to facilitate access and technical support.
- Give priority to those rural areas that do not have any other energy service.

Cornare, in collaboration with the local government and entities such as the Finnish government, Isagen, EPM, Argos, and Viva, has worked to provide efficient stoves that range in price from $800,000 to $2 million pesos each. Since these costs are beyond the reach of end users, the stoves have been provided free of charge.

The households receive prior training on the construction and maintenance of the project after the stoves are installed. The provision of [other resources] is also being studied.

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[37] Regional Autonomous Corporation of the Negro and Nare River Basins (Cornare). MINUTES OF THE MANAGEMENT REPORT SUBMITTED TO THE COMPTROLLER GENERAL OF THE REPUBLIC January 1, 2001 to June 20, 2010

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Firewood crops for fuel production are estimated to take approximately three years to grow. This measure would have a medium-term impact.

Lessons Learned

- Firewood consumption was reduced by 55%, compared to that consumed in the traditional stoves.
- CO_2 emissions were reduced by 86%, although it was found that some people do not like to clean the chimney, so smoke emissions spread inside the kitchen.
- Pressure on natural forests was reduced with savings in firewood consumption of 70,200 tons per year.
- Fewer hours spent collecting firewood, cooking, and improvements in quality of life.

Ecopetrol – Cormacarena program developed in the department of Meta.

During the period 2002-2007, the department of Meta reached a total of 2,093.36 km² of deforested land, with an Average Annual Deforestation Rate (AADR) of 342.4 km²/year, making it the second department in the country with this persistent problem, surpassed only by the department of Caquetá.38 Among the affected municipalities is Mapiripán, a municipality that had no energy source other than firewood for cooking, which meant clearing forests to access this resource. As a consequence, there was a loss of forest cover, damage to water sources, and a loss of biodiversity.

Thus, in 2014, the project's objective was set as *"Implementation of eco-efficient stoves and the establishment of Yopo39 dendroenergy banks for 70 peasant families in rural areas of the prioritized municipalities of the Meta department, for the case of Mapiripán"* developed by the Temporary Union of Meta, in the search to reduce anthropogenic pressure on native forests through the establishment of commercial crops of Yopo for the provision of energy sources.

The functional evaluation concluded that the wood-saving systems implemented by the entity during 2013-2014 showed adequate functioning in 90% of the installed stoves, meeting the expectations of the users, since they presented multiple benefits associated with the reduction of smoke inside the house, reduction in food cooking times and a marked decrease in the amount of firewood.

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[38] Murcia, GUG; Huertas, MC; Rodríguez, JM; Castellanos, HO 2010. Multitemporal changes in forests and other land cover in the Colombian Amazon, at a scale of 1:100,000, from 2002 to 2007. SINCHI Amazonian Institute for Scientific Research. Bogotá, DC, available online: http://siatac.co/c/document_library/get_file?uuid=a0ec40be-0617-4ee8-8f27-1b66890bfcdc&groupId=762

39 The Yopo is a species that has been used culturally and traditionally as an energy source. It is a fast-growing tree and is a strategy to minimize the deterioration of natural forests.

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used and times for its collection. With the implementation of these dendroenergy banks, an increase of approximately 35 hectares of vegetation cover was contributed to.

As an added benefit, five training sessions were held on the maintenance and use of the wood-burning fuel banks, stoves, and stove construction. These sessions were aimed at master builders and the community at large, with the goal of standardizing stove construction and providing knowledge opportunities for the community. It is estimated that the sustainable livestock management and development strategy promotes the ongoing participation of landowners benefiting from the project and the environmental authority, ensuring the long-term viability of these fuel banks. This would prevent the emission of approximately 2,801.21 tons of CO2 into the atmosphere, based on the results of two similar projects in neighboring municipalities (Puerto Rico and Puerto Concordia).

The use of the stove by gender was evident, resulting in: women with 80%, men with 9% and in 11% the figure of shared use (men and women) is presented.

Similarly, stoves, due to their design characteristics and construction materials, consume less firewood than traditional stoves, thanks to the fact that the wood burns much better, in a more efficient combustion process, resulting in less energy loss, better heat retention, and less time spent cooking food.

Similarly, the efficient stove contributes to reducing the incidence of respiratory illnesses in rural families, promotes the proper use of firewood as fuel, and encourages the establishment of forestry plantations as the primary input for eco-efficient cookstoves. Therefore, another major benefit is preventing farmers from further health problems caused by smoke inhalation, which increases the risk of respiratory and eye diseases.

This project has already been developed in Mapiripán, Puerto Rico, Puerto Gaitán and Puerto Concordia, benefiting 402 farming families since 2011.

Limitations

There is no information about the success of the dendroenergy banks in Yopo, so it cannot be guaranteed that the implementation of the banks has contributed positively to mitigating the impact of environmental damage or whether it is good to associate dendroenergy crops with the implementation of improved stove projects.

Regarding this issue, a 2003 study by the Jakarta Forest Research Center stands out. It argues that, to facilitate access to firewood as an energy source for families, the Food and Agriculture Organization of the United Nations implemented a tropical forest action plan in 1985. Under this action plan, tree planting projects proliferated worldwide, notably in India. Unfortunately, *“these plantations did little to*

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"Increasing fuelwood supplies for rural users" (Arnold et al., 2003)40 and changes in domestic fuelwood use were not occurring on a large scale. Thus, development policy during the last decade of this century has shifted away from planting trees or woodlots, to give way to two new objectives: replacing fuelwood as a cooking energy source and/or improving stoves, regardless of the fuel they use, in terms of fuel efficiency, emissions, durability, and safety4142, in order to find alternatives that meet the needs of households and their geographical, environmental, economic, social, and cultural conditions.

Censat Agua Viva Project – ASOMUCALE (Association of Women of Lebrija) developed in the department of Santander in 2015.

In recent years, the government has expressed its intention to improve the situation in rural areas and implement programs that promote non-conventional energy sources. For this reason, a regulatory framework emphasizing this approach has been under development. The Program for the Rational and Efficient Use of Energy and Non-Conventional Sources (PROURE), which in turn defined an Indicative Action Plan for 2017-2022, aims to gradually implement subprograms and actions to ensure that the entire energy chain consistently meets minimum energy efficiency standards, without prejudice to current environmental and renewable natural resource regulations. Therefore, the National Government must promote the sector and establish priority lines of action for the promotion of non-conventional energy sources, focusing on education, promotion, institutional strengthening, consumer protection, and the management and monitoring of goals, among other areas.

There is also Law 697 of 2001 and Regulatory Decree 2688 of 2008 which establish guidelines, directives and functions for public and private entities, giving the greatest responsibility to the Ministry of Mines and Energy in relation to the promotion, organization, assurance of development and monitoring of programs and the design of the Program for the Rational and Efficient Use of Energy and other forms of non-conventional energy.

In Colombia, the final consumption of biomass (firewood and bagasse) as a primary energy source is significant. According to figures reported by the National Mining and Energy Planning Unit (UPME), approximately 11% of total energy consumption comes from these sources, 18% from coal, and 15% from natural gas. For rural residential consumption, firewood was the most important source (82%), followed by electricity (11%) and then LPG (5%), despite the

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[40] Arnold, M., Köhlin, G., Persson, R., Shepherd, G., 2003. Fuelwood revisited: What has changed in the last decade? Center for International Forestry Reserach, Jakarta

[41] Dept for International Development, 2002. Energy for the poor: Underpinning the millennium development goals. UK Dept. for International Development.

[42] ISO, 2012. Guidelines for evaluating cookstove performance. IWA 112012, ISO Int. Work. Agreem

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efforts by the central government and companies to replace firewood consumption with gas in rural and urban Colombia."43

Under a need to prioritize the mitigation of respiratory diseases and the reduction of pressure on forests, a project is being forged with Censat Agua Viva, located in the provinces of Soto (municipalities of Lebrija and Matanza) and García Rovira (municipalities of Concepción and Cerrito) in the department of Santander, Colombia, near a peasant and community reserve of natural forest, guarded by the locals (also known as ICCA).

One of the problems identified in the region is the exploitation of resources such as coal. Therefore, the communities have rejected the government and companies' proposal to mine the anthracite coal in the area, as this would jeopardize the supply of drinking water for their households. For this reason, the communities have organized themselves and implemented alternative economic activities to remain in the region and reclaim their rural identity.

As Soler and Rankin explain, "The United Nations World Health Organization, along with other organizations, has been gradually discrediting this cultural practice, denouncing that the use of firewood as fuel affects the health of infants and adults due to air pollution and carbon monoxide emissions, and that, in turn, it increases deforestation as a result of firewood extraction in forests. However, this discourse, rather than addressing the situation of the communities, responds to the guidelines of the World Bank, which has asked governments and companies to achieve greater coverage of rural areas with modern energy resources, such as natural gas or electricity, which prioritizes the advancement of oil companies and those who trade their derivatives without taking into account the real needs of the different communities" (Soler, 2015)44 .

In this way, the use of firewood as fuel is seen as a cultural appropriation that can be enhanced and thus strengthen identity processes, which must be accompanied by other practices that make this activity more sustainable, such as the development of dendroenergy crops, the implementation of efficient stoves, the maximum use of combustion gases, etc.

In the 1990s, they began agroecological processes on their farms and plots, educational and training programs in straw and wool crafts and the making of cakes and pastries; but at the same time they wanted to give greater added value to their products by building efficient wood-saving stoves, which allows them to increase the energy efficiency of their products, reducing the consumption of wood and therefore the pressure on local forests, which allows them to improve the air quality inside the homes and in turn increase the profitability of their processes.

43https://censat.org/es/analisis/utilizacion-de-lena-para-energia-en-granjas-campesinas-en-santander-colombia-4068 Accessed: August 29, 2019
44https://censat.org/es/analisis/utilizacion-de-lena-para-energia-en-granjas-campesinas-en-santander-colombia-4068 Accessed: August 29, 2019

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"These types of stoves allow for a 55 percent savings in firewood, which directly reduces the emission of carbon monoxide and carbon dioxide, greenhouse gases, and reduces respiratory and eye diseases by decreasing smoke inside homes by 99.5 percent. Installing an oven in the flue allows the heat from the combustion gases to be harnessed, maintaining stable temperatures of around 250°C for baking cakes and pastries."45

Lessons Learned

Some communities must travel long distances to obtain firewood due to human intervention in natural ecosystems. Therefore, it can be said that there is an ancestral and cultural practice, deeply rooted in the use of firewood as a fuel source for cooking, heating, producing panela (unrefined cane sugar), making bricks, and other purposes.

These experiences contribute to complementing the processes of geographic, territorial, and cultural resistance that began decades ago, related to culinary arts, with sustainable energy practices. This is because it helps communities achieve independence in fuel production for food preparation and cooking, reduces pressure on local resources and waterways, and promotes the use of mingas (communal work parties),46 which allows for knowledge sharing on the one hand and strengthens bonds of unity among local families on the other. Participating in this process have been Censat Agua Viva, the Lebrija Women's Association, the Collective of Peasant and Forest Reserves, the Agroecological Producers Association - Agrovida, and the Colombian Movement in Defense of Territories and Those Affected by Dams "Ríos Vivos" (Living Rivers).

The development of improved wood-burning stoves in Colombia

Since the 1980s, projects to install improved cookstoves have been implemented in the country, but it wasn't until 2009 that it was estimated that approximately 30,000 efficient stove prototypes had been installed. According to the national stove survey conducted by the Ministry of Environment and Sustainable Development in July 2014, *"28,238 efficient stoves have been installed throughout the country."* [47]

The following are the improved wood-burning stoves currently implemented in Colombia, where a total of 14 prototypes designed for 2014 are identified, with the Huellas prototypes being the most prevalent.

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45 https://censat.org/es/analisis/utilizacion-de-lena-para-energia-en-granjas-campesinas-en-santander-colombia-4068 Accessed: August 29, 2019

46 Its motto is "walking the word" which for indigenous communities means reaching agreements through dialogue, speaking and through the word recognizing the other and their truth.

47 Ministry of Environment and Sustainable Development. National Stove Survey 2014

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Table 8. Development of improved wood-burning stoves in Colombia

Region	Number of stoves installed	Installed stove prototypes	Unit cost
Atlantic	624	Footprints	721.500
		Corantioquia	530.000
Central	2.030	Footprints (UNDP)	1.350.000
		Lorena (Natural Heritage)	1.230.000
		Two positions (Heritage Natural)	1.200.000
		Rocket (Natural Heritage)	380.000
Pacific	1.336	Footprints (Natural Heritage)	1.230.000
		Footprints (Corponariño)	1.200.000
		Triangular hybrid (Patrimonio Natural)	462.300
		Triangular block (Heritage Natural)	932.840
		Linear hybrid with two burners (Natural Heritage)	491.950
Oriental	3.069	Footprints (Corpoguavio)	718.398
		Footprints (Car)	1.720.840
		Natura Foundation	700.000
		ICA-1791 (Corporate)	1.720.840
Antioquia	21.179	Footprints (Corpocaldas)	900.000
		Footprints (CAM)	931.060
		Footprints (Carder)	1.333.700
Total 28,238 14 Prototypes			

Source: National Survey of Improved Stoves, Ministry of Environment and Sustainable Development, 2014

It is important to highlight that this prototype exhibits significant price variability at the regional level, ranging from $718,398 to $1,720,840. This phenomenon can be explained primarily by the costs associated with the availability of materials and their transportation to the specific site of the prototype's installation. These factors play a significant role when conducting a cost-benefit analysis of the identified sources and technologies, as well as an analysis of the feasibility of implementing viable and sustainable solutions within the country.

According to the national survey of improved stoves, only in a few cases have field efficiency tests been carried out such as the *controlled cooking test* (CCT) which has allowed comparison of the combustion efficiency of the traditional hearth with the improved stove, but does not respond to the application of the standardized test and protocol48 .

48 *Guidelines for a national program of efficient wood-burning stoves* / Texts: Concha, María Cecilia; Pabón, Giovanni; Cerón Viviana. -- Bogotá, DC: Colombia. Ministry of Environment and Sustainable Development, 2015.

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The Natura Foundation was in charge of evaluating the Huellas FN and Corantioquia prototypes within the framework of the implementation of projects financed by Ecopetrol and the Alliance in Energy and Environment program with the Andean Region, whose results were presented in section 3.1.2.

Lessons Learned

Based on the analysis of the development of improved stoves in the country, it is important to highlight that:

According to the latest version of the Quality of Life Survey, in 2018 approximately 1,664,000 households prepared food using firewood or wood. This means that about 11% of households that cook do so with firewood or wood. This use of firewood for cooking is more prevalent in households in the Pacific region (30% of households), the Orinoquía and Amazon regions (19% of households), and the Caribbean region (18% of households). However, the largest number of households cooking with firewood is found in the Caribbean region, with 516,000 households.

- The most widespread prototype of efficient stoves in the country is the Huellas, which It presents significant cost differences between regions.
- Most of the stoves installed have been subsidized except in some cases where there is a small in-kind contribution from the users.
- There are no efficiency evaluations or tests available for most of the stoves that would allow for a comparison of their wood consumption with traditional open-fire stoves. Of those for which efficiency data is available, the Huellas stove achieves 13.4%, a still very low percentage of impact. Similarly, the stoves are not patented.

Exposure to indoor air pollution, resulting from incomplete and inefficient biomass combustion, is one of the most significant risk factors in the proliferation of respiratory diseases, especially in children and adult women, since they spend more time at home and therefore have a higher degree of exposure.

The adoption rate achieved by improved cookstove projects somewhat contradicts the idea that the proportion of abandoned/disused units tends to be high in projects where prototypes are given away for free, because it is assumed that people do not develop a sense of ownership over things that are given away. This confirms the findings of Bensch & Peters in 2014, who, after distributing free improved cookstoves in Senegal, found that almost all of them were still in use three years after delivery. Therefore, the idea of distributing fully or partially subsidized cookstoves should not be definitively dismissed (Bensch & Peters, 2014)49

The effectiveness of improved stove projects can only be assessed to the extent that they are adequately monitored and evaluated through

[49] Bensch, G & Peters, J. (2014). The intensive margin of technology adoption. Experimental evidence of improved cooking stove in rural Senegal. Essen, Germany: rwi.

Machine Translated by Google

Qualitative or quantitative techniques currently available or that may be developed in the future. This will allow for the necessary feedback, which will not only serve to make the necessary adjustments to improve the financing and execution processes by the implementing entities, but will also facilitate addressing the technical and performance aspects that are vital to improving the technology and thus adapting it to the end user (Aristizabal, 2018)50 .

3.1.2. International Experiences

This section aims to present results of programs introducing improved wood-burning stoves carried out in other Latin American countries.

Concern about firewood consumption and the problems arising from its misuse has led to the development of projects that seek to replace this fuel with others, primarily modern fuels, or to reduce firewood consumption by improving combustion efficiency and reducing users' exposure to combustion products (gases such as CO2, CO, particulate matter, and others). These measures have manifested in the development of projects specifically aimed at reducing or mitigating the aforementioned impacts.

Peru

According to the National Institute of Statistics and Informatics (INEI), one of the main fuels used for cooking food in rural areas of the country is firewood (41.4% of the population)51. It was estimated that by 2009, 2.7 million households cooked on inefficient traditional open fires52. Given its inefficient use and the health and environmental problems associated with the use of firewood as the main energy source, in 1980 measures began to be promoted for the use of improved cookstoves53 .

Among the high-impact programs developed in Peru was “Sembrando,” whose objective was to combat chronic child malnutrition. This included the mass implementation of improved cookstoves, reaching 30,000 beneficiaries by the year 2000. This initiative was linked to the Qori Q'oncha program and the carbon credit markets for the certification of improved cookstove projects in Peru . ⁵⁴

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50 Aristizábal, 2018, “Adoption Levels and Impact of an Improved Wood Stove in Rural Communities of the Department of Santander, Colombia”

51 According to INEI (2016)

52 Moreno, A. 2018. Webinar: Advances in clean cooking. Latin American and Caribbean Network of Clean Kitchens

53 GACC a (Global Alliance for Clean Cookstoves, US). 2012. Peru Market Assessment: Sector mapping (online). Accessed September 2, 2019.

54 Laurent, A. 2014. Standardism as government: voluntary carbon certification and the Peruvian cookstove sector. Tesis doctoral, Sciences-Po Paris.

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The Qori Q'oncha program (meaning "golden stove" in Quechua) is a nationwide project implemented in 2008. It was the first project registered in the voluntary carbon market using the Gold Standard methodology, which promotes improved cookstoves through carbon credit financing. The project involved the installation of over 100,000 cookstoves, resulting in emissions reductions of over 800,000 tons . The structure of the Activities Program allows for the inclusion of additional projects under the same program in the future.

The project is supervised by the Swiss-based NGO Myclimate and the French social enterprise Microsol, based in Peru.55 Within the ex-post evaluation of this program, it was found that 89.60% of households that consume firewood collect it, while 10.40% buy it.56

As of 2018, they provide services in three districts of the country, each leading a participatory local project. Nearly 30,000 improved cookstoves were distributed between 2008 and 2010, achieving an emissions reduction of 33,000 tCO2e/year by 2017.⁵⁷ The inclusion of financing through carbon credits has increased the distribution of improved biomass cookstoves and contributed to the development of new initiatives in other parts of the country. “Consequently, the advance revenues resulting from carbon credit financing serve to sustain market growth and improvements in cooking technology.”⁵⁸

In 2001, a national campaign was launched through Emergency Decree No. 069-2009, called “Half a Million Improved Cookstoves, for a Smoke-Free Peru,” supported by the Pan American Health Organization (PAHO) and the German Development Cooperation (GIZ), which authorized regional and local governments to use the equivalent of 5% of mining royalties, over-royalties, and other mining revenues for the implementation of improved cookstove projects combined with safe water and fecal matter management.59 The most widely implemented model was Inkawasi, which cost approximately USD 30.

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[55] Myclimate y Microsol, 2010; Microsol, 2011, Simon et al, 2012

56 Microsol. 2018, “Report on the relationship between short-lived climate pollutants and rural cooking in the residential sector - Prospective study applied to Colombia, Mexico and Peru”

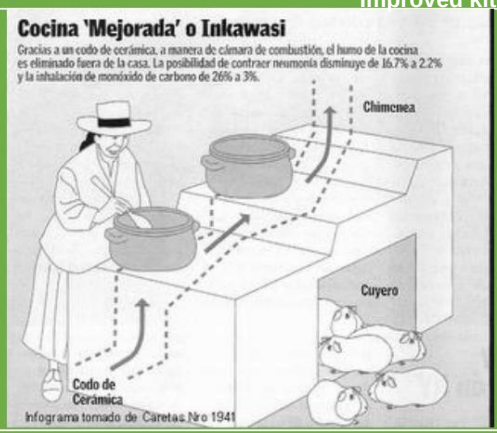
[57] Myclimate y Microsol, 2010; Microsol, 2011, Simon et al, 2012

[58] Myclimate y Microsol, 2010; Microsol, 2011, Simon et al, 2012

59 Emergency Decree No. 069-2009, Peru

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Table 9. Images of the Inkawasi Improved Cookstove

Improved kitchen Inkawasi	
	
Improved kitchen prototype	Sechura Province

Source: Developed by the Pan American Health Organization and German Cooperation

Furthermore, in 2012, as part of a regional project called Carbon 2012, the UNDP provided technical assistance to access carbon markets, thereby ensuring the sustainability of previously implemented projects. International funds and financial resources will be invested in the maintenance of existing efficient cookstoves, as well as in training and awareness-raising about the benefits of cooking with efficient stoves for Peruvian families.

Cultural aspects

- In rural areas where firewood is used as the main energy source for cooking food, there are different cooking habits and differences in food preparation, depending on the region.

Preparing lunch takes approximately half an hour to two hours.
hours and is carried out mostly by women

- The kitchen is located inside the homes and it is necessary to use at least two burners or stoves where you can cook with different sizes of pots.

- Regarding the requirements for the stove to be used, it is noted that at least two burners are necessary where cooking with different sized pots can be done.

Lessons Learned

Although this type of project has experienced significant growth and adoption throughout the country, one of the limitations in implementing projects financed by carbon credits is the lack of reliable information for estimating emission reductions and monitoring the installed cookstoves. This challenge is a constant in the regions where these solutions have been implemented with the aim of mitigating the impact associated with cooking food.

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Mexico

According to Masera's estimates in 2005, the population in Mexico that used firewood (approximately 25 million people, or 5 million households) used it for cooking or heating water; of these, 19 million used firewood as their sole fuel and 9 million used firewood and LPG. The consumption of this energy source is most frequent in rural areas, where 89% of the population uses it as their primary energy source for cooking .

Among the limitations to the implementation of improved stove projects that allow for a better quality of life for rural households in the country is the fact that the exact number of stoves installed in the country is unknown; it is estimated that up to 2017 900,000 improved stoves had been installed, the vast majority through public tenders.

Between 2006 and 2012, under the Special Climate Change Program (PCC), Mexico designed a national program for improved cookstoves to distribute 600,000 improved biomass cookstoves, under the responsibility of the Ministry of Social Development (SEDESOL), which would distribute 500,000, and the Ministry of Agriculture, Livestock, Rural Development, Fisheries and Food (SAGARPA)62, which would distribute 100,000. However, no ex-post evaluations have been carried out to determine the effectiveness of the installation of the cookstoves.

[60] Masera, O; Díaz, R; Berrueta, V. 2005. From cookstoves to cooking systems: The integrated program on sustainable households energy use in México. Energy for Sustainable Development: Special Issue on Clean Cookstoves 9 (1): 25-36.

61 Díaz-Jiménez. 2000. Firewood Consumption in the Residential Sector of Mexico. Historical Evolution and CO2 Emissions. Mag Sc Thesis. Mexico City, MX, UNAM. P 113.

[62] Diaz, R., Berrueta, V., Masera, O. 2011. Thematic Notebooks No. 3 on Bioenergy: Stoves of Firewood. Mexico City

Machine Translated by Google

Table 10. Improved Biomass Stoves in Mexico

Improved stoves	
	
Santa Ana Zegache, Oaxaca, Nov. 15, 2014, 63	The awareness campaign on the use of the stove is intensified64

Source: Own elaboration, Corpoema 2019

As Microsol explains, *"From the private sector, the Red Patsari initiative is noteworthy, which seeks to implement Patsari kitchens nationwide by training local operators.*
Regarding financing, the vast majority of projects in Mexico have been operated under a scheme of total or partial subsidies from the private sector, civil society and the State." 65

In November 2015, a standardization process for wood-burning stoves was established in Mexico with the participation of various stakeholders from both the public and private sectors. The goal was to create a draft Mexican Standard that would allow for the evaluation of efficiency, measurement of indoor emissions, and assessment of safety, functionality, adaptability, and durability of stoves nationwide. While there is a focus on consolidating information and implementing measures to mitigate the problem, Mexico has few laboratories that meet international standards and guidelines.

Cultural aspects

Cooking habits vary by region and are influenced by the climate.
the area, the geography, and the ethnic differences.
Their staple food is the tortilla, and its preparation requires a lot of effort.
quantities of firewood.

Lessons Learned

Since there is no database of the number of technologies installed, nor the models of stoves implemented according to the conditions of the area, nor methods

63 Photo taken from: https://oaxacaentrelineas.com/entrega-sedesol-mas-de-mil-500-estufas-ecologicas-a-cuatro-municipios-de-valles-centrales-de-oaxaca/
Photo 64 taken from: http://www.mipuntodevista.com.mx/entregadas-mas-de-5-mil-estufas-ecologicas-con-chimenea/

65 Microsol. 2018. Report on the relationship between short-lived climate pollutants and rural cooking in the residential sector. Prospective study applied to Colombia, Mexico and Peru

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Without assessment tools that allow measuring the impact of their implementation, it is not possible to design a sectoral support strategy that guarantees the sustainability of the projects.

Guatemala

The Productive Uses of Renewable Energy Program (PURE) aims *to*
To eliminate barriers to the adoption of renewable energy technologies (RETs), primarily mini/micro hydroelectric plants (MHPs), by promoting productive energy uses in one of the poorest areas of Guatemala: the departments of Alta Verapaz, Baja Verapaz, Quiché, Huehuetenango, and San Marcos.66 The project would create new local jobs and sources of income, while directly mitigating 1.75 million tons and indirectly mitigating 5.25 million tons of CO2 equivalent over 20 years. This would be achieved by promoting RETs, which are linked to income generation and improved productivity, adding value to local agricultural crops. The sustainability of productive uses would be strengthened through the application of adaptation mechanisms that were aimed at the management of natural resources and the problems of vulnerability where the projects were developed, providing the necessary structural support for the formulation of relevant policies and regulations, and through the support of national and local dialogue among the multiple stakeholders for long-term collaboration.

In the context of water resource conservation, efficient stoves were considered a means to reduce firewood consumption and improve user health conditions by replacing traditional stoves with these improved models. The use of biogas plants was also promoted.

The total budget required for this undertaking, which is complementary to the ongoing efforts of the Government of Guatemala for poverty reduction, rural development and the promotion of watershed management activities, was estimated at US$12.67 million, with US$2.55 million required from the Global Environment Facility (GEF) to cover incremental costs.

The *overall environmental objective* (project goal) was *to reduce Guatemala's greenhouse gas emissions by promoting the productive use of renewable energy, with significant rural development benefits.* The project began in August 2007 and ended in March 2012 (approximately 4.5 years). It was implemented by the NGO FUNSOLAR, supported by the United Nations Development Programme (UNDP), and funded by the Government of Guatemala and the Global Environment Facility (GEF).

The project's focal area is the northeast and northwest regions of the country (the departments of Alta Verapaz, Baja Verapaz, El Quiché, Huehuetenango, and San Marcos), which were ravaged by armed conflict and where 29% of the population is located.67 The population of this area (approximately 32,000 km2) is mainly of

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[66] Rodriguez, H. Productive Uses of Renewable Energy in Guatemala – PURE. Solar Foundation (2013) Guatemala City.

67 Population Census 2000. National Institute of Statistics (2002) Guatemala City.

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of Mayan origin, including the following ethnic groups: Q'anjob'al, Jakalteko, Chuj, Mam, Ixil, Q'eqchi', Poqomam, K'iche', and Kaqchikel, all of which speak languages other than Spanish. These populations had suffered the consequences of the war and the impact of the problems that arose with the restoration of peace. The efforts proposed in the PURE are linked to the commitments of the Guatemalan Peace Accords to activate income-generating programs and projects that take into account both cultural and environmental aspects.

The *expected situation regarding the barriers at the end of the project* was that by removing them through the project's implementation, a nationwide effect would be created to establish a market for renewable energy sources (TERs), developing both in rural and urban areas, benefiting rural sectors through productive uses of energy, promoting the conservation of renewable natural resources, and allowing for a reduction in greenhouse gas emissions produced by energy supply in rural areas.

The project *diagnosis* considered different types of barriers: Socio-institutional, political-regulatory, Market and Finance, and technical.

The project designed four outcomes to overcome these barriers. *Outcome 3, Sustainable Management of Natural and Energy Resources in River Basins,* ***considered the use of improved cookstoves and biogas plants in rural areas.***
Subsequently, rural areas were identified where firewood was a commercial fuel and at a cost that allowed the adoption of improved stoves.

Improved stoves

The stove shown in Table 11 is one of ***430 stoves*** installed by PURE II, using resources from the Guatemalan National Electric Energy Commission (CNEE), which in turn came from a fund of the Inter-American Development Bank (IDB) and CNEE energy efficiency program. The CNEE's contribution was US$25,000, of which US$20,000 was used for the revolving fund. PURE II, using its own resources and the remaining US$5,000 from the CNEE, carried out all the outreach, promotion, and information work (including transporting and setting up the promotional team).

According to average user data, they consumed 60 bundles of firewood per month. Once the efficient stove was installed, consumption was reduced to 20 bundles per month. One bundle is approximately 25 pounds of firewood. Therefore, the efficient stove saves 40 bundles per month, which corresponds to 2/3 of the monthly consumption, or, in other words, the improved stove consumes 1/3 of the previous consumption. In economic terms, this means a savings of 40 bundles per month x 5 Quetzales/bundle = 200 Quetzales per month. ***Considering that the stove cost each user 960 Quetzales, the payback period with the savings is 5 months.*** The purchase was financed by the revolving fund, with a repayment term that varied between 6 and 10 months.

68 Quetzales is a currency of Guatemala. The exchange rate was approximately 7.75 Quetzales/US Dollar.

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The improved stoves were sold not only to households but also to people who operated small food production businesses (small restaurants) and the success was equal to that of the stoves installed in homes.

Advantages of an efficient stove

According to user interviews, the main advantages of the efficient stove compared to the conventional open stove are69:

- Burns are avoided when making tortillas
- Users are not getting sick from combustion products as they used to.
- The efficient stove operates two burners simultaneously (previously they had open-fire stoves), with separate fires in each position.
- It uses less firewood
- Less dirt in the environment, less smoke, less pollution in the kitchen.
- You don't need to buy comal70 which cost Q15/month.

[69] Gabriel Tzul Sam, owner of the stove (Cell 40559538)

70 Refers to a traditional cooking vessel used as a griddle for cooking.
It is used in Mexico and Central America.

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Table 11. Images of Efficient Stoves Cobán, Alta Verapaz, Guatemala

Location: Cobán, Alta Verapaz, Guatemala	
	
Efficient stove brand Onil	combustion fume extractor pipe
	
Combustion chamber	A bundle of firewood

Source: Prepared by consultants, Corpoema

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Disadvantages of the efficient stove

One disadvantage is that since there is no smoke in the kitchen, there are more mosquitoes.

Other aspects

This efficient stove is manufactured in Guatemala by Onil71. The design is an adaptation of the Rocket stove, known for its efficiency. All materials used are sourced domestically. Onil71 sells this stove and ships it anywhere in the country. The company provides manuals and videos for installation and maintenance.
It also offers technicians for installation.

Conclusions

The PURE project achieved its objective of providing efficient technology for productive purposes, reducing costs and GHG emissions. The efficient stove proved successful because the savings quickly recouped the investment, there were no direct subsidies for users, emissions were reduced, the impact of these emissions on users' health was lessened, and CO2 emissions were reduced. This stove competed favorably against other commercial energy sources in the area (electricity and LPG) and maintained the users' traditions.

3.1.1. El Salvador

The fundamental idea of the project Protection and Recovery of the La Barra Forest and establishment of an agroforestry system in the hamlet of La Barra, Tecomapa, Metapán72 , was to reduce the pressure on the remaining forests, using efficient stoves that reduced the consumption of firewood, and producing the required firewood from the energy grove.
This program was developed by the IDB, the UNDP (GEF Small Grants Programme), and the government of El Salvador. The community had approximately 150 families at the time of the project (2004), who were exerting considerable pressure on the remaining forests along the shores of Lake Güija. The project consisted of building efficient cookstoves for users who had a plot of land with a minimum area of 100 x 100 m² . A nursery was established with madrecacao, a fast-growing, woody species endemic to the region. When the seedlings reached 30 to 50 cm in height, they were planted in plots arranged in a 2 m x 2 m grid.

A visit in 2006 established that there were groves in the vicinity of the houses, but because these were in a village, there was pressure on these groves to
to build new housing. No sources were found online that would allow us to investigate the existence of an ex-post evaluation of this project.

71 http://aspravsoluciones.blogspot.com/
72 GEF – ADESCOLAB – UNDP. Adescolab's experience with the project: Protection and recovery of the La Barra forest and establishment of an Agroforestry System. (2004) Colonia La Barra, Metapán, El Salvador.

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3.2. GLP

3.2.1. LPG in Colombia

Gas for the Countryside Program

Introduction

By the mid-1990s, ECOPETROL had regulated the LPG market through imports of the product and the release of quotas, which allowed it to reach full supply of demand without supply restrictions.

The expansion of supply, thanks to the commissioning of the Catalytic Cracking Unit with a production capacity of 4,000 barrels per day at the Barrancabermeja Industrial and Petrochemical Complex, created the need to explore market options and stimulate demand for LPG through a program called *"Gas for the Countryside."* It is worth noting that ECOPETROL had recently established a Community Relations Department, foreshadowing the concept of Corporate Social Responsibility that would gain significant traction in industrial activities a few years later.
Furthermore, the development of these programs was well suited to the character of a State Enterprise, which at that time directed the actions of ECOPETROL and allowed it to enter into projects of social and community interest.

Diagnostic elements[73]

The Gas for the Countryside program was designed in 1996. At that time, the Gas Massification Plan was in full implementation and a reduction in national LPG consumption was expected, considering that the purpose of natural gas was to reach the main cities and population centers, displacing significant volumes of LPG whose commercial destination is the periphery of cities and rural areas of the country.

Given a surplus due to the presence of new supply, it was necessary to explore market opportunities that also had a social and environmental component.

The main elements that ECOPETROL took into account for the formulation of
The Gas Program for Field I is found in the document cited in the reference, from which the following information on the characteristics and motivations of the program is taken:

Replacing firewood with LPG has profound implications for the family life of farmers: for their health, their lifestyle, and perhaps most profoundly, for the dignity of rural women. It is no secret that the task of

[73] Gas for the Countryside Program - Vice Presidency of International Trade and Gas. Final Report. May 1999.

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The task of gathering and collecting firewood is almost always performed by women. The daily work of lighting the hearth, especially critical in winter, and keeping the fire burning constantly, means having to breathe the fumes produced by combustion. The family, and especially young children, are forced to remain in these unhealthy environments, developing severe lung problems in many cases.

cases.

The combustion of firewood produces volumes of CO, CO2, N2O, NOx, VOCs, and particulate matter, and depending on the type of wood used, even heavy metals. The constant inhalation of combustion fumes is the cause of a wide variety of respiratory and pulmonary diseases in rural areas, a situation that becomes especially critical in cold climates, where kitchens are located in enclosed spaces and families are constantly exposed to breathing air contaminated by combustion byproducts.

Firewood is not always the cheapest energy resource for farmers. There are completely deforested regions where firewood is such a scarce resource that it must be imported at a much higher cost than other fuels. In the long term, it is expected that replacing firewood with LPG will result in less pressure on natural resources. It has been estimated that a farming family of four to five people consumes an average of 12 to 15 kilos of firewood per day, equivalent to between 4.4 and 5.5 tons per year.

Program development [74]

Based on the diagnosis developed in the Gas Massification Plan regarding the high consumption of firewood in the energy mix, an approximation was made to determine the areas of the country with high deforestation rates, resulting in the Departments of Nariño, Santander, and Tolima. Considering the presence of the oil industry in Casanare, it was decided to include this Department in the Program.

Overall, the program proposed to provide LPG to approximately 750,000 rural families, most of whom consume firewood and would never have access to natural gas massification programs due to the distance of these communities from natural gas distribution networks.

The requirements for users participating in the program were determined as follows:

a. Preferably firewood consumers

b. Living in areas not currently or in the future covered by gas programs natural.

c. That if they have LPG, they are not currently using it. d. That they are committed to using LPG permanently.

74 Memorandum GAS-PES-079, from the Special Programs Area to the Vice Presidency of International Trade and Gas. December 6, 1997

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Considering the low purchasing power of the beneficiary communities, it was decided to deliver the complete kitchen kit free of charge, consisting of the two-burner stove, the 40-pound cylinder, which was the best-selling size at the time, the hose and the regulator, plus 3 coupons to refill the contents of the cylinder.

In the first phase, the four departments were included with the following coverage in terms of users and municipalities:

Table 12. Coverage Beneficiaries

Department Municipalities Users		
Casanare 13,405	19	
Nariño 33,064	45	
Santander 22.570	86	
Tolima 23,599	47	
TOTAL 92.638	197	

Source: From “Memorandum GAS-PES-079, from the Special Programs Area to the Vice Presidency of International Trade and Gas. December 6, 1997”

In the end, the Program reached 90,894 families out of the 750,000 initially planned, distributed across 198 municipalities, reaching nearly half a million people. Total investments amounted to $9.2 billion in cooking equipment, LPG purchases, and payments to distributors and supervisors.

Final Comments

The Gas for Rural Areas Program I did not undergo a formal and detailed ex-post evaluation to determine its effectiveness; however, distributors in the beneficiary areas reported that the rate of orders for new LPG refills, after the initial three coupons were used, decreased significantly. Some distributors told ECOPETROL that in some cases families kept the cooking kit even though they could not afford to purchase the gas. In other cases, the stoves and cylinders ended up in pawn shops in the towns.

However, several significant aspects of this first experience with a large-scale firewood substitution program should be highlighted. First, it demonstrated that with proper organization, it is possible to implement a program of this nature, whose objective is to replace firewood with an energy source like LPG, with significant health and environmental benefits. Furthermore, the motivations and the diagnosis carried out by ECOPETROL are no different from what we are seeing more than 20 years after the first iteration of the Gas for Rural Areas Program. The problem assessment focused on the environmental issues of deforestation and also, and with greater emphasis, on the public health problems of women and children due to the constant inhalation of gases and particulate matter produced by burning firewood in traditional stoves.

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In internal reflections by the Team that directed and implemented the Program, it was suggested that for a subsequent phase it was advisable to introduce some changes to the program as follows:

- To improve the likelihood of users remaining in the Program, suggest smaller cylinder sizes that fit the users' payment capabilities.

- To propose a joint effort between the entity responsible for the Program and the beneficiary families so that they contribute a percentage of the costs with the idea that what represents direct effort is valued better.

Gas for the Countryside Program (Rural LPG)

Introduction

In 2004, ECOPETROL launched a new version of the Gas for Rural Areas Program, or *"Rural LPG"* as it was called. This chapter presents the program's definition, the evaluation of the Pilot Plan, and the subsequent evaluation of the Full Plan, conducted by a consortium of consulting firms.

The program

The main features are as follows:

- Number of families benefited: 198,465, distributed as follows:

Table 13. Families Benefiting from the Gas for the Countryside Program II

Regions	Number of Families Benefited
Andean Region[75]	93.952
Caribe[76]	88.602
On Occidental[77]	15.911
Total	198.465

Source: Final Report University of the Andes agreement no. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 2006.

- Families receive: a two-burner stove, a 33-pound gas cylinder, a regulator, a hose, and two vouchers that entitle them to a gas refill.

[75] It includes the departments of Antioquia, Santander (Central), Norte de Santander, Meta, Caldas, Quindío, Risaralda and Boyacá
[76] It includes the departments of Atlántico, Magdalena, (Sierra Nevada) and other municipalities, Cesar, Sucre, Cordoba and Bolivar
[77] It includes the departments of Chocó, Valle, Nariño and Cauca.

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each one. Additionally, a booklet is provided with all the recommendations and explanations for the proper use of the kitchen "kit".

- The Pilot Program was developed in the period December 2004 to September 2005 aimed at rural households to use LPG in cooking and water heating processes.

Other program requirements that are worth highlighting are the following:

or That users are preferably currently consuming firewood.
or That they are not current or future consumers of natural gas.
or that they are not consuming LPG nor have cooking equipment, whether stoves or cylinders.

or That users have the financial resources to continue paying for LPG, once the redemption of the two vouchers that are given as part of the program is finished.

or Other requirements such as the user's home being their permanent residence so that the other conditions that must be met can be verified.

The stated requirements aim to focus the program on firewood users and also to build a permanent base of LPG users in the medium and long term with the ability to pay to sustain consumption, once the two vouchers that are part of the Program have been used up.

The first phase of the plan was developed as a pilot program, reaching 3,465 families in 29 municipalities of the Department of Cauca, specifically in the Colombian Cauca Massif, where the Magdalena, Cauca, Caquetá, and Patía rivers originate. This region contains strategic ecosystems that are fundamental to the country.

The number of users in the Pilot Plan is included in the general scope of 198,465 and the same Program requirements outlined above must be met.

The operational part of the Pilot Plan—that is, the user registration process, equipment delivery, and redemption of consumption vouchers—was carried out through five distribution companies in the area. The distributors were to charge $25,000 per family for user registration, equipment delivery, voucher redemption, and training.

The selection of municipalities for the Pilot Plan did not include areas with current or potential natural gas coverage, and efforts were made to avoid areas with public order difficulties.
The 29 municipalities of the Department of Cauca comprise a total of 618,747 inhabitants with 74% in rural areas.

The Pilot Plan presents two important differences with respect to the Program version
Gas for Field I:

- The number of vouchers given to users to replenish the initial cylinder charge is reduced from 3 to 2.

Families pay distributors $25,000 for their services, which negates the program's permanent free nature. This is important insofar as

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that the families involved understand that they have to pay, even if it's a small sum, for the benefits received.

Program Evaluation78

The evaluation of the pilot program has four aspects: environmental, social and economic, barriers, and solutions for implementing the program in other strategic ecosystems. This evaluation was carried out through surveys in the Municipality of Totoró, with a total of 80 beneficiaries, of whom 35% were surveyed. The main conclusions of the evaluation of the LPG Pilot Program are presented below.

- Income is the variable that determines whether a family leaves the program or not. Those with higher incomes remain in the program, while the opposite occurs as income decreases. In the most extreme cases, LPG consumption is abandoned, and only firewood is consumed. In 2005, family income ranged from $100,000 to $450,000, with an average of $235,307 (61.7% of the minimum monthly wage).
- Including households located in sparsely populated rural areas, that is, far from population centers, presents difficulties for distributors due to the distances that must be covered. It should be noted that there are greater incentives for distributors to reach population centers with several houses than to reach dispersed dwellings.

- Regarding the distributors, they believe the LPG Program has very positive effects. The Program was well received by the beneficiaries, which facilitates future expansions.

- The design of strategies to expand gas access could include incentives that allow families greater access to LPG. For example, instead of paying for a full cylinder, which can last several weeks, daily payment plans are being designed recently to make it easier for families with limited cash flow to purchase the fuel, thus improving retention rates.

Distributors point out that the $25,000 charge in 2005 (6.6% of the 2005 minimum wage) could be a significant barrier. However, as mentioned earlier, charging some amount makes families appreciate the benefits they receive.

- Distributors believe that greater participation from communities and, in general, local actors is necessary in the design of programs.
  They also propose greater diversity in the size of the cylinders, which in turn would allow beneficiaries to commit less money to the purchase of the product and the expense to be better adjusted to their cash availability.

78 Final Report University of the Andes Agreement No. 88 of 2005. Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 2006

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Benefits

Fuel Consumption

The following table presents fuel consumption before and after the pilot program.

Table 14. Energy consumption, before and after79

Fuel type	Before	After
Firewood only	93%	11%
Only GLP	7%	25%
LPG and Firewood	0%	64%

Source: Final Report University of the Andes agreement no. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 2006.

The above confirms that households with the arrival of LPG do not completely replace firewood, but rather migrate towards various energy sources as income increases in what could be called an energy "portfolio" or also known as "fuel stacking".

Regarding firewood consumption in households, the information from the Study is as follows:

Table 15. Firewood Consumption in Households

Distribution of Firewood Consumption Unit
Among households that only use firewood : 9.81 Kg/day (294 Kg/month)
Combined use of firewood and LPG : 170.71 kg/month
Occasional use of firewood and LPG : 251.62 kg
LPG and occasional firewood 41.26 Kg.

Source: Final Report University of the Andes agreement no. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 2006.

Deforestation

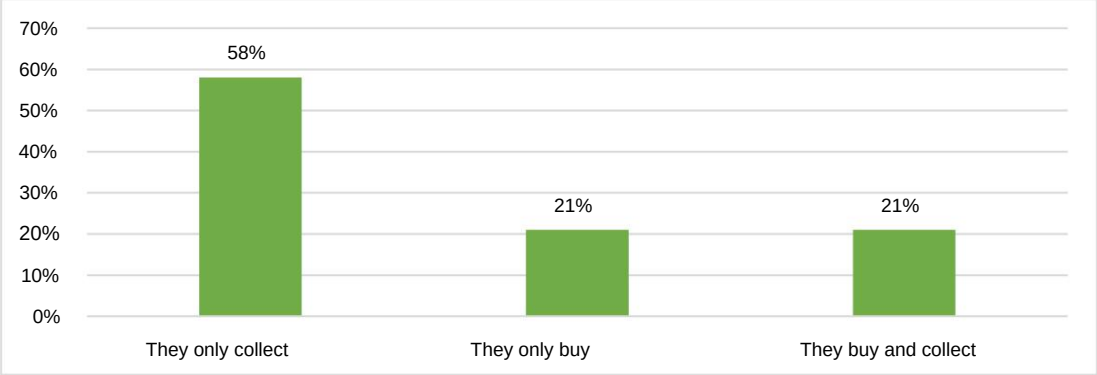
Deforestation is primarily caused by the expansion of the agricultural frontier and commercial development in forested areas. Families no longer obtain firewood directly from natural forests but from other sources such as stubble and secondary forest. This was the case for families in the municipality of Totoró, Cauca, who consumed 262 tons of firewood before receiving stoves.

From the above, it can be concluded that the relationship between firewood consumption and deforestation is not direct. Information on firewood consumption can be found in the following graphs:

79 **FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005** ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006

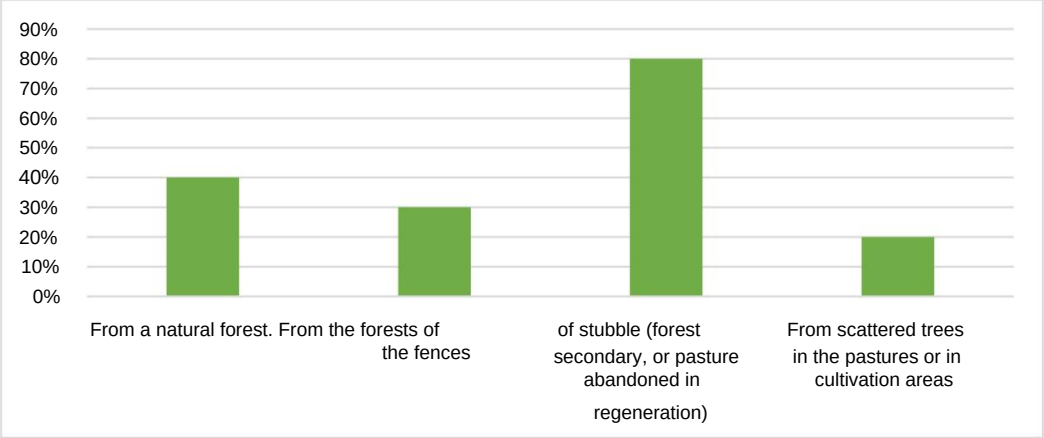
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Chart 2. For households that consume firewood: do they buy it or collect it?



Source: Final Report University of the Andes agreement no. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 2006.80

Chart 3. Origin of Firewood



Source: Final Report University of the Andes agreement no. 88 of 2005 Ecopetrol – Ministry of Environment, Housing and Territorial Development – University of the Andes May 200681

It is concluded that 80% collect firewood from stubble and secondary forest, and although 40% of the firewood collected comes from natural forest, it was found that these same households extract firewood from other sources.

In health

The greatest impact of solid fuel use is indoor air pollution, which particularly affects women and children.

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80 FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005 ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006.

81 FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005 ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006

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The survey reveals that 97% of women are in charge of cooking duties, while only 3% of men do so occasionally.

The health risk is greater when stoves lack chimneys for venting polluting gases. In the survey's target population, 58% of beneficiary families who were using firewood at the time did not have chimney systems to vent combustion gases, and 53% had some type of ventilation system that was in unacceptable condition.

In this sense, it is very notable that in Totoró, Cauca, a mainly rural municipality where most of its inhabitants cook with firewood, before the implementation of the pilot program, respiratory diseases were the main cause of morbidity and corresponded to 20% of the cases of medical consultation.

According to the results of the study:

- or *"women in the Municipality of Totoró who cook with firewood have a 1.9 times greater risk of developing lung cancer and a 3.2 times greater risk of developing chronic obstructive pulmonary diseases."82*
- With a fuel like LPG, families will see an improvement in their health situation thanks to the fact that combustion is cleaner compared to conventional firewood.

Cooking Fuel Expenditure

According to the results of the survey, the monetary cost of consuming firewood is lower than that of consuming gas, even if the firewood is purchased.
In 2005, for families that only consume and purchase firewood, the monthly monetary cost was $12,616 (3.3% of the SMMLV), which is compared to the monthly cost for families that only consume LPG, whose value was $20,000 (5.2% of the SMMLV).

For families who collect firewood (do not buy it) and also purchase LPG, the cost is $15,000 (3.9% of the SMMLV).

Opportunity Costs

At this point, reference is made to the hidden or non-monetary costs incurred in obtaining and consuming firewood.

To this end, it was determined that families: "spend 14 hours per month to obtain firewood, while those that combine LPG and firewood spend 8.14 hours per month."83

In addition to the above, the time spent lighting the wood-burning stove, which can reach up to half an hour daily, must be included. According to the survey, among households that

82 FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005 ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006.
83 FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005 ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006.

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They combine LPG and firewood; in 100% of cases, LPG is used for breakfast.
and drinks, which saves time to dedicate to other activities.

Benefits of adopting LPG

The survey shows that families using LPG take less time to cook; the energy source is faster and easier to use than firewood.

This has allowed families, particularly mothers, to dedicate more time to other household activities, and although this greater availability of time is not reflected in monetary terms - at least not at the time of the survey - it is clear that levels of well-being are higher under the new situation.

Ultimately, switching energy sources for cooking and heating translates into greater well-being for the whole family in terms of improved health and more free time for other activities. The price for these increased levels of well-being is the difference a family pays when switching from using firewood to a combination of firewood and LPG, or LPG alone.

The barriers of the program

Economic

The analyses carried out with the information from the Survey allow us to conclude that family income is the variable that determines whether families consume LPG or not, that is, whether a stable demand is established or whether the program is abandoned.

In agricultural communities, income levels are low, a situation exacerbated by the seasonality of harvests, which is why firewood will always be the most economical and preferred resource for the poorest communities.

Cultural

factors : The survey indicates that: “many rural communities have diets that include corn, rice, potatoes, lentils, or beans. All of these foods require significant amounts of energy to cook and are preferably cooked over low heat.”84

Under these conditions, firewood is the preferred energy source, reinforced by the belief that food is better cooked over wood. Furthermore, families who use firewood indicate that it facilitates cooking food on large stoves to meet the family's needs.

However, it is also to be expected that as families understand and appreciate the benefits of LPG, firewood will begin to be used only occasionally and in special circumstances.

84 FINAL REPORT UNIVERSITY OF THE ANDES AGREEMENT NO. 88 OF 2005 ECOPETROL – MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT – UNIVERSITY OF THE ANDES MAY 2006.

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The cultural aspect must be taken into account, especially in indigenous communities, for which it is necessary to understand their customs and the appropriate way to make changes in the energy portfolio.

Educational

Related to the above, understanding educational realities is a key factor in promoting the mass adoption of LPG.

Households with higher levels of education understand and accept the processes of change and better value the free time available when switching from firewood to other fuels, such as LPG in this case.

Trust

Any change in family customs, such as changes in food cooking methods, generates concerns and mistrust.

The educational work of distributors is key to facilitating the change process.

Access

Facilitating the change process with adequate information is fundamental in aspects such as distribution frequencies and delivery points.

Comments

It can be said that the LPG Program has been well received by the beneficiary population. It wasn't imposed upon us, and its advantages and benefits are widely recognized. Thanks to LPG, cooking habits have improved due to faster cooking times and a greater perception among users of an improved quality of life. At the same time, families are opting for a diversified energy portfolio that includes both LPG and firewood. It is expected that as families gain confidence in the new product, LPG usage will increase.

Although health issues may require longer evaluation periods, it is also true that families may notice signs of improvement in the short term. Regarding environmental issues, as previously mentioned, there will be no significant changes in terms of deforestation reduction since natural forests would not be subject to biomass loss due to firewood consumption.

There are things that need improvement, for example, better marketing to promote the environmental benefits in schools and communities. It would be advisable to consider smaller cylinders and lower prices to make them more accessible to families.

It is also necessary to take into account the characteristics of the communities to which the program is directed, in order to ensure its acceptability, continuity and permanence in the medium and long term.

And, as the distributors themselves acknowledge, distribution systems need to be improved to bring LPG to the communities.

The recommendations could be summarized as follows:

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- Better dissemination of the program through education and information.
- Greater participation of communities in the design of programs.
- Specific solutions that take into account the differences between communities.
- Diversify the size of the cylinders to serve communities with younger children income.
- Refine the selection criteria for beneficiaries to minimize the possibility of including families that do not require LPG or that do not have the financial capacity to continue with the program.

- Improve the selection procedures for distributors, in order to have agents that adapt to the rural and dispersed environment.

The Rural LPG Program: Evaluation

Introduction

In May 2007, the IG-Itansuca-Geoengineering Consortium presented the Evaluation of the "Rural LPG Program" carried out by ECOPETROL in the years 2005-2006. This program is the same one for which the Pilot Project was developed in the Department of Cauca, which was evaluated through Agreement No. 88 Ecopetrol. Ministry of Environment, Housing and Territorial Development-University of the Andes, delivered in May 2006.

The Consortium document mentioned above carries out the comprehensive evaluation of the Program, the main results of which are presented below.

Results

The scope of the program is as follows:

- It comprises a total of 463 municipalities and 204,678 households. The areas to be covered are the following:

Table 16. Scope of the rural LPG program

Region	Municipalities Households
Andean	293 110.525
Caribbean	84 79.992
On Western	86 15.161
Total	463 204.678

Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-2006

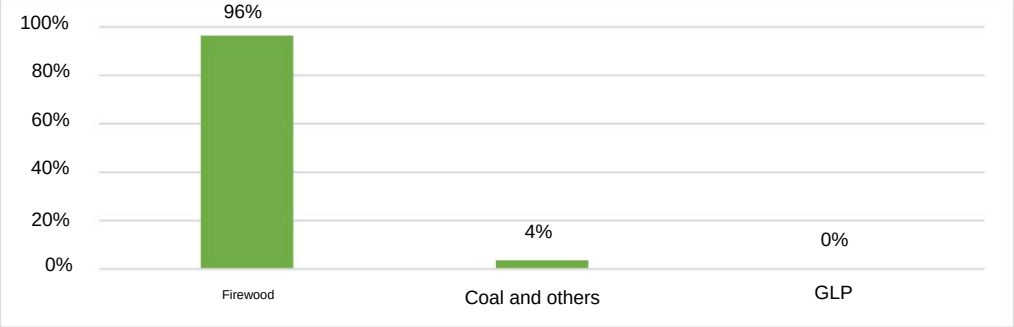
The program as originally planned covered the same regions and departments, reaching 198,465 households. It should be noted that this total includes the pilot program with 3,465 teams in the Department of Cauca.
- A cooking stove, cylinder, regulator and three LPG refills are provided with a cost of $185,000 (48% of the SMMLV in 2005) per unit.

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- Retail distributors took over the implementation of the program.
- The beneficiary population comprises approximately 1,052,000 inhabitants and the Total investments reached $41.732 billion.
- Of the total of 463 municipalities and 204,678 households, for the purposes of the evaluation, surveyed 3,336 households in 134 municipalities.

Before the implementation of the Rural LPG Program, the distribution of energy consumption in the surveyed population was as follows:

Figure 4. Distribution of energy use before the program



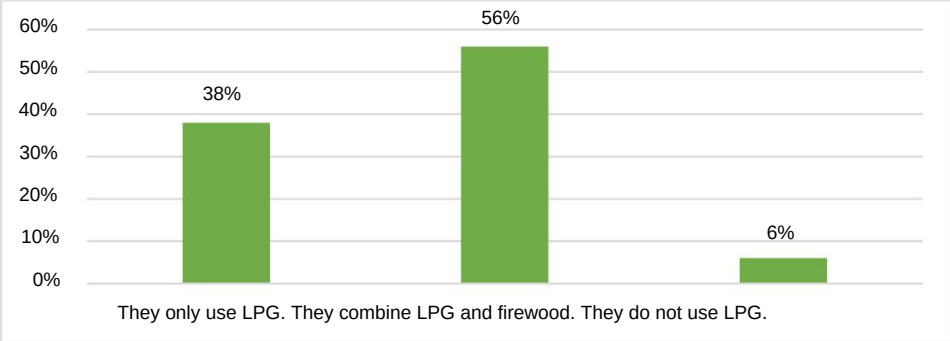
Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-200685

Of all firewood users, 86% collected or cut it and 10.4% bought it.

The main aspects of the program evaluation are summarized below.

1. The percentage distribution of energy for cooking is as follows:

Figure 5. Distribution of energy sources as a result of the LPG Program



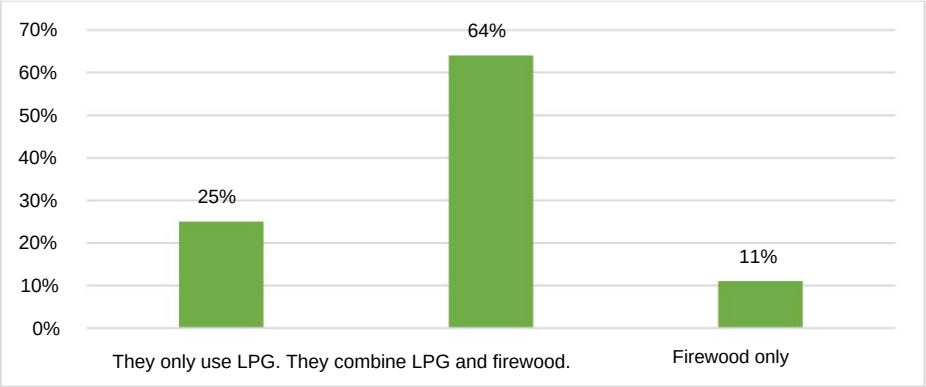
Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-200686

[85] Rural LPG Program-ECOPETROL. IG-Itansuca-Geoengineering Consortium. May 2007
86 Same Note 65

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In the Evaluation of the Pilot Program in Totoró, Cauca, these percentages were as follows:

Figure 6. Distribution of energy sources as a result of the LPG Pilot Program

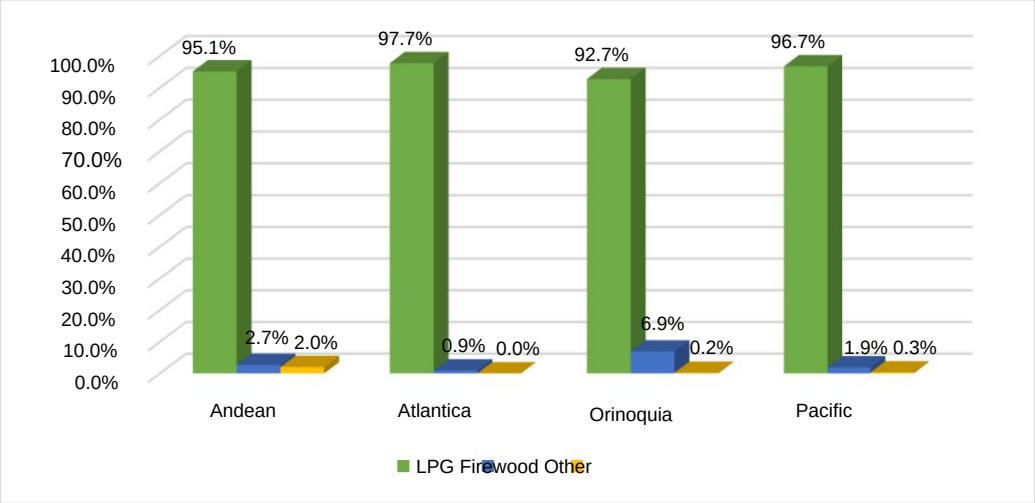


Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-2006

There is a significant difference between the Pilot Project survey and the Final Project evaluation, with a lower percentage of users consuming only LPG and a higher percentage consuming only firewood, while the percentage of households consuming both is higher.

2. Regarding LPG preferences for cooking, the graph below reveals the following:

Chart 7. Preferred energy sources by region



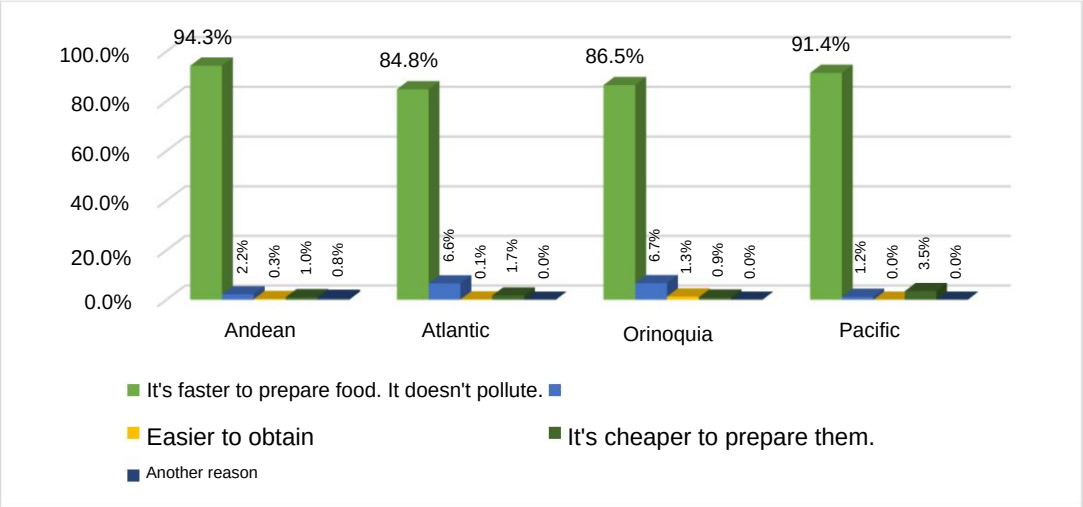
Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-200687

87 Same Note 65

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3. On average, 95.6% of beneficiaries prefer to prepare their food with LPG, while only 3.1% prefer to use firewood.

Chart 8. Beneficiary preferences



Source: From the “Rural LPG Program” advanced by ECOPETROL 2005-2006 [88]

On the other hand, in the evaluation of the Final Program, approximately 90% consider that it is faster to cook with LPG, which is quite close to the result of the Pilot Study, in which 92% consider that it is easier to cook with LPG.

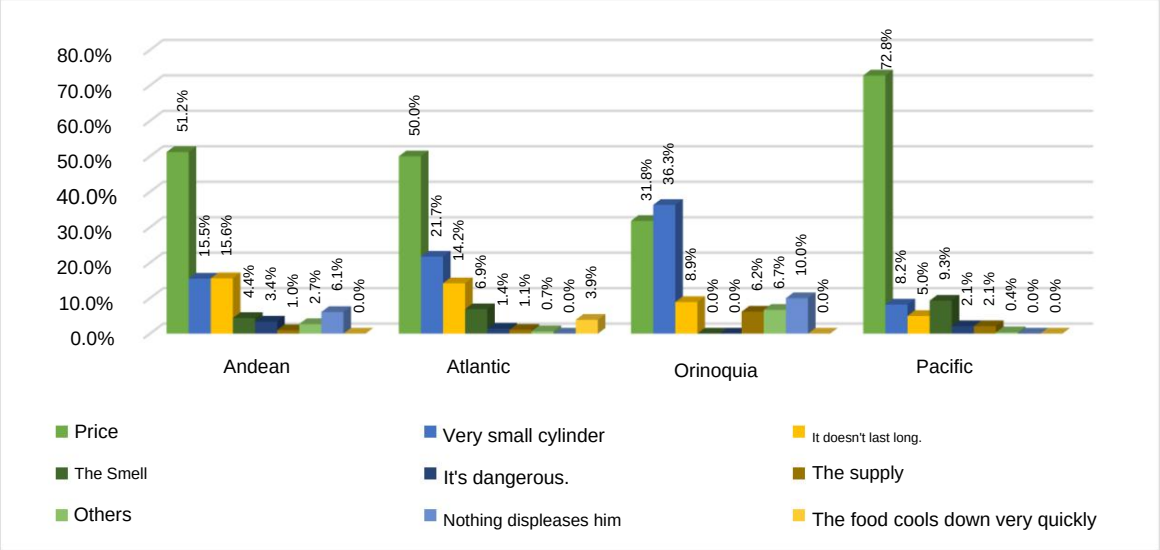
It is interesting to note that, in the Final Project survey, the perception of LPG as a non-polluting energy source is below 7%, and in some cases, such as the Andean region, it is as low as 2.2%, and in the Pacific region, as low as 1.2%. In other words, the beneficiary families do not perceive a change in terms of pollution, which can be understood considering that these effects manifest themselves over the longer term. Furthermore, regarding the question of whether LPG is easier to obtain, the response rates are very low, between 0% and 1.3%. That is, families do not assign points to LPG distribution compared to the difficulty of obtaining firewood for their consumption. This result, which is surprising due to its low scores, may be the result of ineffective work by LPG distributors, who may not adequately communicate delivery schedules and locations.

4. To the question “what do you dislike about LPG”, the following graph shows the evaluation carried out on the Project:

88 Same Note 65

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Figure 9. Assessment of negative aspects of GLP



Source: From the "Rural LPG Program" advanced by ECOPETROL 2005-200689

As expected, price is the least valued factor for new users in three of the four development zones of the Project. However, in the Orinoquía region, beneficiary families are more concerned about the cylinder size than the price of LPG. This can be explained by issues of access to the region and the fuel's durability. Another factor worth highlighting is the cylinder's lifespan, with respondents indicating that it doesn't last long, with percentages ranging from 5% to 14%.

5. According to the survey, 91.2% of respondents have family incomes below one minimum wage, which in 2007, the year of the Program evaluation, was $443,700.90, while for 44.3% of respondents, family incomes are below $200,000 (45% of the minimum wage). For the Pilot Project Survey, monthly household expenditures, which are an approximation of income level, range from $100,000 to $450,000, with an average of $235,307 (53% of the minimum wage). These figures are significant because high-income families remain in the program, while lower-income families lack the resources to continue participating by purchasing LPG supplies.

The above is also shown in the following table, where income is related to the recharge period. In other words, higher income means higher consumption and a shorter recharge period.

89 Same Note 65

[90] http://www.banrep.gov.co/es/estadisticas/salarios-t

Machine Translated by Google

Table 17. Recharge periods according to income

Income	LPG refill (days)
Under $200,000 From	37
$200,000 to $400,000	36
Over $400,000	30

Source: From the “Rural LPG Program” advanced by ECOPETROL 2005-200691

Other results, such as the reduction in firewood consumption per household and the number of hectares equivalent to deforestation, were calculated based on the results of the Pilot Program conducted in the municipality of Totoró, Cauca. Similarly, the reduction in mortality and morbidity was calculated using the results of a 2005 World Bank study in Guatemala. This study found that replacing firewood with LPG contributes to a decrease in morbidity and mortality rates caused primarily by acute respiratory infections in children, chronic lung diseases, and lung cancer, given the significant reduction in indoor air pollution.

Since cooking food is a cultural aspect, other entities could complement marketing efforts by providing support to users in the process of adapting to the new product.

Subsidies for the Distribution of LPG in Networks and Cylinders

Introduction

In Colombia, there is a subsidy scheme to promote the provision of piped gas service to the lowest socioeconomic strata as part of a general policy of public utility subsidies.

This chapter will analyze the subsidies for LPG, whether distributed through pipelines or cylinders, established in Colombian regulations. This analysis will utilize public reference sources and documents from GASNOVA, the trade association representing LPG distribution and marketing companies with majority Chilean investment groups as their capital.

The final section presents 1) the results of an audit report on the Subsidy Program for Users in socioeconomic strata 1 and 2 for LPG consumption in cylinders, commissioned by the Ministry of Mines and Energy. Although this audit was conducted three years ago, and there are no new auditing processes, the results are interesting in terms of the satisfaction of the beneficiary communities and set the standard for establishing improvements in future phases of the Program. The communities' acceptance of LPG and the

91 Same Note 65

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1) Perception of improvement in health conditions and quality of life due to the replacement of firewood, 2) What is reflected in article 295 of the 2018-2020 Development Plan regarding LPG subsidies.

Solidarity Fund for Subsidies and Income Redistribution92

The National Solidarity Fund for Subsidies and Income Redistribution -
The Ministry of Mines and Energy was established by the Public Services Law (142 of 1994) in Article 89 and in Article 4 of Law 632 of 2000. This is a special fund for the management of public resources subject to the legal and procedural rules established in the Organic Budget Statute.

These regulations incorporate subsidies for each of the public electricity and piped natural gas services. As of the first quarter of 2019, natural gas coverage in the country reached 9,419,079 users, while piped LPG coverage was 103,937 users.

Subsidies for the consumption of LPG distributed in cylinders

Decree 2195 of 2013 established the granting of subsidies for the consumption of LPG distributed in cylinders. The main characteristics of the program are as follows:

- The departments that can access this subsidy are: Caquetá, Cauca, Nariño, Putumayo and San Andrés, Providencia and Santa Catalina.
- The beneficiary population that qualifies for the subsidy is identified through: "…the information system of the Subdirectorate of Social Promotion and Quality of Life of the National Planning Department, called SISBEN, and when it comes to indigenous communities, the indigenous census administered by the Ministry of the Interior is used."93

- "The maximum amount to be subsidized per user is a percentage of the cost of basic or subsistence consumption defined by the UPME, established at 14.6 kg equivalent to 32.18 pounds and which may not exceed 50% for stratum 1 and 40% for stratum 2." 94

The subsidy program for gas cylinders began with the LPG pilot program, which is still ongoing. The subsidy amount for 2018 was set at $47 billion, based on the budget allocated to the five departments participating in the program.

92 GUIDELINES FOR THE PROJECT: REPLACING THE CONSUMPTION OF FIREWOOD, COAL AND WASTE WITH LPG. COLOMBIAN LPG ASSOCIATION – GASNOVA. 2018
93 Same Note 65.
94 Same Note 65

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According to GASNOVA figures from 2017, a total of 342,464 households have used the LPG consumption subsidy. With this Pilot Plan, the percentage of families using LPG increased from 52% to 84.58%, reducing firewood use from 25% to 1.14% and coal use from 4% to 0.04% .95

LPG Subsidy via Networks

Resolution 90032 of 2014 establishes: *"the requirements for the presentation of projects aimed at providing the public service of combustible gas through the development of liquefied petroleum gas infrastructure – LPG by pipeline network at the national level, the procedure applicable to the evaluation of the same is adopted and the methodology for calculating the prioritization index is adopted."*

Some of the most relevant topics of the cited standard are the following:

- Resources will be allocated primarily to municipalities and rural areas with the highest rates of unmet basic needs and in areas not served by main gas pipelines. These resources will also be used to co-finance connection fees for low-income users.

- The Archipelago of San Andrés, Providencia and Santa Catalina is included within the eligible areas.

LPG pipeline coverage has been gaining momentum in recent years, and although it's still far from what natural gas has achieved, partly due to the large number of users connected in cities, its increasing presence in rural municipalities makes it an attractive option for expanding coverage. The table below illustrates this situation:

Table 18. GLP by Networks

NATIONAL LPG COVERAGE VIA NETWORKS				
YEAR	QUARTER	CADASTRAL POPULATION	RINGED RESIDENTIAL COMPLEXES	CONNECTED RESIDENTIALS
2015	1Q	205.422	65.530	46.538
2016	1Q	221.098	82.349	61.680
2016	4Q	289.119	92.059	67.443
2017	1Q	304.710	98.104	75.206
2017	4Q	350.366	113.511	85.895
2018	1Q	471.217	116.099	89.367
2018	4Q	449.080	124.668	97.471
2019	1Q	428.260	131.482	103.937

95 GUIDELINES FOR THE PROJECT: REPLACING THE CONSUMPTION OF FIREWOOD, COAL AND WASTE WITH LPG. COLOMBIAN LPG ASSOCIATION – GASNOVA. 2018

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Source: From the “Rural LPG Program” advanced by ECOPETROL 2005-200696

Results of the Audit of the Subsidy Program for Users in Strata 1 and 2 [97]

In 2016, the Ministry of Mines and Energy contracted a comprehensive audit with the firm Sistemas 2000 Consultores for the review and analysis of four (4) components of the subsidy program for the consumption of liquefied petroleum gas (LPG) in cylinders for strata 1, 2 of the SISBEN database and the indigenous people of the departments of Nariño, Putumayo, Caquetá and the Archipelago of San Andrés, Providencia and Santa Catalina.

The general idea is to understand how the pilot program of providing subsidies for gas consumption in cylinders by the Ministry of Mines and Energy is operating.

The audit considered legal-regulatory, technological, financial and accounting aspects. The fourth component, social and operational, which is of most interest to the UPME Firewood Substitution Study, is based on fieldwork with a defined sample and the application of a survey. This survey was used to analyze the granting of the subsidy and measure environmental variables, quality of life, subsidy coverage, and other issues. All of this was done to determine the level of satisfaction among the population benefiting from the program and to examine whether the process of allocating subsidies to the beneficiaries was carried out correctly.

As indicated above, the maximum amount to be subsidized per user was a percentage of the cost of basic or subsistence consumption defined by the Mining and Energy Planning Unit, UPME, which may not exceed 50% for stratum 1 and 40% for stratum 2.

The Ministry of Mines and Energy, through Resolution 9 0855 of 2013, defined that the procedure to be applied to assign LPG subsidies for cylinders would be a Pilot Program to users of strata 1 and 2 in 15 municipalities, located in the departments of Caquetá, Nariño and Putumayo.

The aim is to provide the poorest and most vulnerable populations in these areas with access to piped natural gas, facilitating access to LPG. This seeks to achieve several objectives:

- Improve the health of the beneficiary communities by providing a cleaner-burning energy source.
- Minimize the use of energy sources that can compromise the health of the people, particularly women and children.

96 Source: Ministry of Mines and Energy and Consultant's calculations

[97] Ministry of Mines and Energy, Comprehensive Audit of the Subsidy Program for Users of Strata 1 and 2 for the Consumption of Liquefied Petroleum Gas, LPG, distributed through cylinders in the Departments of Caquetá, Nariño, Putumayo and Archipelago of San Andrés, Providencia and Santa Catalina Contract 245 of 2016.

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- The pilot program was expanded through Resolution 9 0580 of 2014 to all users in strata 1 and 2 in the departments of Caquetá, Nariño, Putumayo and the Archipelago of San Andrés, Providencia and Santa Catalina.

The pilot program was conducted between December 23, 2013 and February 28, 2014.

The following requirements were established for the design of the Program:

- "Departments with a high reported sales of LPG to users in socioeconomic levels 1 and 2 (information consolidated based on reports submitted by LPG marketers, excluding municipalities that had piped gas service).

- Potential beneficiaries of the LPG subsidy; that is, those citizens who at the time of the review of the municipalities by the Ministry of Mines and Energy did not buy LPG.

- Municipalities that do not have piped natural gas service, excluding, for example, the municipality of Túquerres in the department of Nariño, *"because the piped natural gas service is about to become operational in its urban center."* 98

In general, the communities have received the Cylinder Plan favorably for the following reasons:

- With the subsidies for LPG, which translates into lower rates, a positive impact has been generated among users, who have been able to access the energy at lower costs.

- Replacing other energy sources with LPG cylinders has resulted in less deforestation, with a decrease in the purchase of wood, coal and gasoline as cooking fuels.

- Households perceive an improvement in their quality of life. By reducing smoke from cooking food, respiratory illnesses decrease.
- The LPG consumption subsidy allowed many households to access this service and *"free up resources for other activities or the purchase of various goods,"* which translates into a better quality of life.
- "With the issuance of Resolution 4 0720 of 2016, the Ministry of Mines and Energy expanded the beneficiaries of the subsidy program to indigenous communities and users in strata 1 and 2 of the rural areas of the municipalities of the department of Cauca, which are part of the protected areas of the National Parks system with jurisdiction in the Colombian Massif."99

98 Same Note 65
99 Same Note 65

Machine Translated by Google

The main results of the study, taken directly from the Audit Report, are the following100:

- Of the 477,846 potentially eligible households that can access the pilot program, 342,464 households have made use of the benefit of the subsidy for gas consumption in cylinders in these departments.

- The energy consumption pattern in the households surveyed in the four departments before the program's implementation was as follows: 1. Gas cylinder, 52%. 2. Firewood, 25%. 3. Electricity, 16%. 4. Gasoline, 2%. 5. Natural gas, 1%. 6. Coal, 4%. Once the subsidy program was implemented, 84.58% of the households surveyed switched to LPG, replacing fuels such as gasoline and reducing firewood use to 1.14% and coal use to 0.04%.

- 19% of households surveyed and with access to the subsidy believe that respiratory illnesses in homes were reduced thanks to the replacement of firewood, gasoline and coal with LPG in cylinders.

- 92.53% feel satisfied or very satisfied with the subsidy, and only 6.86% feel somewhat or not at all satisfied.

- 34% of the households surveyed that use the subsidy indicated that they receive it very quickly, followed by 34% who receive it quickly and 20% who receive it moderately quickly.

- Of the beneficiaries with access to the program, some do not actually take it for a variety of reasons: they are natural gas users, or there is no LPG coverage in their places of residence, or they do not have sufficient resources to acquire it.

- In the case of indigenous communities in the 4 departments, it was found that 7.7 out of every 10 indigenous families use cylinder gas as fuel for cooking, and 2.3 out of every 10 families use other types of fuel such as natural gas and firewood.

Overall, the results show that users are satisfied with the LPG consumption subsidy implemented by the Ministry of Mines and Energy, and also state that it is necessary to contribute to improving the quality of life of the households that use said subsidy.

While public health impact studies require more long-term evidence regarding the benefits of GLP use, it should be noted that the perception among the beneficiary population is one of well-being and improved living conditions. Indeed, those surveyed indicated the following:

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100 “Comprehensive Audit of the Subsidy Program for Users of Strata 1 and 2 for the Consumption of Liquefied Petroleum Gas, LPG, distributed through cylinders in the Departments of Caquetá, Nariño, Putumayo and Archipelago of San Andrés, Providencia and Santa Catalina Contract 245 of 2016. Ministry of Mines and Energy – Sistemas 2000 Consultores

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- Purchases of firewood, coal, gasoline, and tree felling have decreased due to the use of LPG in cooking food.
- It is believed that respiratory illnesses in households have decreased. beneficiaries of the subsidy.
- The living conditions of families have improved.

In conclusion, families are better off with the LPG subsidy, which has enabled them to change their energy consumption habits. The perception is one of improved health and greater well-being for families, which is the central objective of any public policy. The challenge now is to make these changes sustainable and permanent, and that will be the challenge for energy policy going forward.

National Development Plan 2018-2022

According to Article 295 of the 2018-2022 National Development Plan, "Nationwide subsidies for liquid fuels, biofuels, and combustible gas will be reviewed to establish an efficient methodology that guarantees competitive pricing and the best consumption signal for users, fiscal sustainability, and the appropriateness of their gradual adjustment, without overlapping subsidies. Likewise, the Fuel Price Stabilization Fund (FEPC) will be reviewed to establish a methodology that enables a long-term fiscally sustainable structure. The Ministry of Mines and Energy will review transportation subsidies for liquid fuels, combustible gas, and crude oil, as well as transportation subsidies in special zones and marine diesel."

In addition to the above, the National Government had established, through Decree 2195 of 2013 and another regulation, Resolution Minminas 9-0855 of the same year, a program of subsidies for LPG covering some municipalities in Cauca, Nariño, Putumayo, Caquetá, and San Andrés. Later, Amazonas and Vaupés were included. According to information from GASNOVA, the 2020 National Budget included budget allocations to complete coverage for approximately 342,000 families, who receive a subsidy of $16,000 per month per 33-pound cylinder, representing approximately 29% of the retail price.

Environmental and Public Health Considerations

Introduction

The consumption of firewood in Colombia has been diagnosed on several occasions in the National Energy Plans and in Public Policy documents.

Thus, the CONPES document number 2646,101, which provided the guidelines for the Gas Massification Plan, indicated that 62% of energy consumption in the residential sector was met with firewood, warning that due to its low performance

101 Document DNP UINF – DIREN – MINMINAS. December 18, 1991.

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As an energy source, it required 5 times more energy than that required with gas, which has negative environmental consequences.

The aforementioned CONPES document established that although during the 20 years prior to the definition of the Gas Plan, the supply of firewood had decreased with the increase in the supply of electricity, from the point of view of supply costs, this situation was unsustainable in the future considering that: "… *it is necessary to invest $7.4 of electricity to achieve the boiling of 1 liter of water, while it would only cost $2.4 with propane or $1.0 with natural gas"* 102

The objective of the long-term Gas Plan outlined in the referenced document is to achieve a more balanced distribution of final energy consumption nationwide, resulting in a more equitable energy mix. To this end, four objectives were established, as follows:

- To expand the use of propane gas in most major cities and introduce its use in rural areas to combat deforestation. It was proposed that once natural gas began its widespread adoption, LPG should focus on meeting the needs of urban peripheries and rural areas.

- To encourage private participation in the production and marketing of energy.
- To bring energy prices closer to the real costs of production and service delivery of the services.
- Optimize the use of natural gas reserves.

In the development of point 1, it was determined that in order to facilitate the import of propane gas, the port and storage infrastructure in Cartagena would be adapted.
Similarly, domestic LPG production would increase with the commissioning of the new catalytic cracking unit in Barrancabermeja. With sufficient supply, it was hoped that the LPG market would normalize.

This account of the events that led to the promotion of LPG as part of the gas plan gave this energy source the necessary prominence to participate in the programs for replacing firewood.

Firewood Consumption, Deforestation ***103*** *and the Impact on Health*

Firewood consumption in Colombia raises two major concerns from a public policy perspective: deforestation and its impact on user health. This analysis is fundamental to defining the study's focus regarding the proposed solutions.

102 Same Note 65
103 The article is based, among other sources, on: "Final Report University of the Andes" – ECOPETROL-MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT- UNIVERSITY OF THE ANDES – May 2006. Bogotá

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Starting with deforestation, as will be discussed in this report, the causal relationship between fuelwood consumption and deforestation is unclear. This uncertainty stems, among other reasons, from the fact that consumption patterns vary considerably between regions and individuals, which can only be documented through surveys that assess user behavior, particularly regarding fuelwood and biomass consumption patterns, whether acquired through informal trade or directly by users. 104

Through surveys it is possible to identify the sources of firewood supply, which can be very varied: forests, living fences, fields, waste from industries where wood is processed, among others.

The sources consulted indicate that the most likely cause of deforestation is the expansion of the agricultural frontier, along with forest clearing resulting from formal exploitation such as sawmills and road construction. In other words, the traditional view that the use of firewood in homes causes deforestation assumes that this firewood comes from natural forests and jungles, when in reality, homes may obtain firewood from their own properties and along roads. This point is highly significant because it means that firewood collection costs tend to increase as longer distances must be traveled to obtain it, which implies higher costs in terms of travel time or, potentially, the need to acquire biomass from informal markets.

Poor households that cannot obtain firewood near their homes or do not have the resources to acquire it will have to use another resource such as animal biomass or manure to be able to cook.

However, the Energy Balance figures reveal the need to find explanations for the high values of firewood consumption in the residential sector found in the 2017 Energy Balance of the UPME, which indicates the following: [105]

Table 19. Consumption in Terajoules per energy source

Energy Consumption	
Firewood consumption	103,357 Terajoules
Natural gas consumption	49,061 Terajoules
LPG consumption	14,511 Terajoules

Source: BECO figures, UPME Energy Balance, 2017

From the above, it can be concluded that, despite efforts to increase the use of natural gas and LPG, the figures show a high share of firewood in the residential sector, which is concluded from the following items:

104 ECP-MAVDT-2006

[105] http://www1.upme.gov.co/InformacionCifras/Paginas/BECOCONSULTA.aspx

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- The consumption of firewood energy for the residential sector is more than double the equivalent consumption of natural gas energy in the same sector, even though the ratio between the population that consumes natural gas and firewood is 6 to 1, according to information from the Ministry of Mines and Energy for the first quarter of 2019 and the Quality of Life Survey of 2018.

- The energy consumption of firewood for the residential sector is more than seven times higher than the equivalent energy consumption of LPG, even though the ratio of LPG users to those who consume firewood is around 2 to 1.

public health impact of solid fuel use in Colombia, it has a 106, 107, 108, Indoor Pollution. Regarding the significant impact on the most vulnerable and lowest-income populations. Some figures are as follows:

- Air pollution from the use of solid fuels in Colombia caused 2.89 deaths per 100,000 inhabitants in children under 5 years of age, according to data reported by the Global Burden of Disease Study for 2013.

- Costs for premature mortality and disease care exceed $3 trillions of pesos, equivalent to 0.38% of the GDP of 2015.
- 42% of cases of Chronic Obstructive Pulmonary Disease (COPD) are related to environmental factors.

- The average exposure for women is 25 times the maximum level recommended by the WHO and 21 times in the case of children under 5 years of age.

Intramural contamination is associated with:

- Acute respiratory infections in children, being the leading cause of child mortality in developing countries.
- Chronic obstructive pulmonary disease, COPD.
- Lung cancer.

In addition to the direct health impacts mentioned above, the references cited in the document relate to other risks derived from high firewood consumption, such as the long distances that poor users must travel in search of firewood, as well as burns and fires and the hours that need to be spent collecting this energy source.

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106 "Papers on Health" – Edition No. 11 – April 2017 Ministry of Health

107 "Cooking for life – Elements for the formulation of an Energy Use Policy for Cooking in Rural Areas of Colombia". Luis Augusto Yepes G. International LPG Congress – GASNOVA August 2018.

108 Same Note 86

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Another document prepared by the National Planning Department analyzes the problem from another perspective, through the economic valuation of environmental degradation109. In the conclusions phase of the document, the following can be read:

> *"The economic value of environmental degradation in 2015, estimated in this document, is 16.6 trillion pesos, equivalent to 2.08% of GDP. Poor urban air quality has the largest contribution to this result (1.5% of GDP), followed by indoor air pollution (0.38% of GDP) and inadequate access to drinking water and basic sanitation (0.15% of GDP). In the first two components, mortality accounts for a large part of the value, while in the third, it is morbidity. The results described are consistent with Colombia's income level. While indoor air pollution and inadequate access to drinking water and basic sanitation are major problems for low-income countries, urban air pollution is a relatively common problem in upper-middle-income countries, such as Colombia."*

The conclusion is that there is a correlation between firewood consumption and public health, which is also quantified in the Colombian case, according to the previous documents.

The tables below show the emissions from different types of energy sources. It can be concluded that LPG has very low health impacts compared to other energy resources, with the exception of biogas, although the latter has a higher particulate matter content. The figures are as follows:

110 111

Table 20. Pollutant emissions related to LPG112

	Biogas	LPG	Kerosene	Firewood	Roots	Crop waste	Manure
CO	0.1	1	3	19	22	60	64
Hydrocarbons	0.3	1	4,2	18	18	32	115
PM	2.5	1	1,3	30	30	124	63

Source: (Smith, 2005) Contaminants that affect health per unit of energy supplied. The values (on a log scale) express the amount in grams of the contaminant per one million Joules of energy supplied to a cooking vessel.

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109 Source: "Economic Valuation of Environmental Degradation - 2015". National Planning Department 110 Source: "Final Report University of the Andes" – ECOPETROL-MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT- UNIVERSITY OF THE ANDES – May 2006. Bogotá

111 Fuente: "The Mega Conversion Program from Kerosene to LPG in Indonesia: Lessons learned and recommendations for future clean cooking energy expansion". K.Thoday, P. Benjamin, M.Gan, E. Puzzolo. Energy for Sustainable Development 46 (2018) 71–81

112 Source: "Final Report University of the Andes" – ECOPETROL-MINISTRY OF ENVIRONMENT, HOUSING AND TERRITORIAL DEVELOPMENT- UNIVERSITY OF THE ANDES – May 2006. Bogotá

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Table 21. Relative emissions of various fuels with respect to LPG113

	Biogas	LPG	Kerosene	Wood	Crop residues
CO	0,1	[1]	3	19	60
Particulate matter (PM)	2,5	[1]	1,3	26	124

Fuente: Relative health- damaging polluting emissions of various fuels relative to LPG. (Smith, 2005)

In the previous comparison between fuels, natural gas is missing, whose emission level is lower than that of LPG due to its molecular structure, having only one carbon atom (C1), unlike LPG which has C3 and C4. However, since the origins of the Gas Massification Plan, it has been clear that this energy source is intended to provide service in large population centers and not in isolated areas.

Finally, while it is acknowledged that there are other risks associated with the use of LPG, such as leaks and potential explosions due to its density, it is also clear that this sector has increasingly better performance indicators, which reduces the possibility of accidents.

Conclusions

- It has traditionally been thought that lower firewood consumption tends to reduce the impact on deforestation. However, based on observations to date, this correlation between firewood consumption and deforestation is not direct.
- On the contrary, the relationship between firewood consumption and public health is direct, it has an impact on people's quality of life and has been quantified in various studies presented in the Chapter.

This suggests that the solutions proposed for replacing firewood, which is the objective of the Study, should focus on technologies that reduce the health impact on families who currently lack access to clean energy resources and must make do with what they can obtain from nature.

113 Fuente: “The Mega Conversion Program from Kerosene to LPG in Indonesia: Lessons learned and recommendations for future clean cooking energy expansion”. K.Thoday, P. Benjamin, M.Gan, E. Puzzolo. Energy for Sustainable Development 46 (2018) 71–81

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3.2.2. International Experiences

India

Introduction

India has designed one of the most successful programs for replacing firewood with LPG, which is now a benchmark case in the world.

This program is a response to the problem of indoor air pollution in homes caused by cooking with firewood, coal, manure, and other biomass resources.
This is one of the greatest environmental health risks worldwide, and it likely affects more people today than at any other time in human history. India is one of the countries most severely affected by this problem, which has led its authorities to develop a large-scale strategy to address it.

The LPG Replacement Scheme has been led by the Government of India and in particular by Prime Minister Narendra Modi, who suffered from this situation as a child, and has led the country on the path to solving one of the most serious health problems of the population of that country.

This chapter presents information on the situation in India and the circumstances that motivated the structuring of the Program, its characteristics and results to date.
Likewise, a look will be given at the Ujjwala 2.0 Program, which contains the steps to follow in the next stage of the process.

The “Cooking for life” campaign

This initiative by the Government of India is part of the "Cooking for Life" campaign, sponsored by the World LPG Association (WLPGA). This campaign aims to facilitate the transition by 2030 of 780 million people who currently cook with traditional and solid fuels (as well as other dirty and hazardous fuels) to cleaner combustion, such as that generated by LPG. It also seeks to prevent 900,000 premature deaths annually, primarily among women and children, due to exposure to indoor air pollution, and to save 2.65 million hectares of forest—an area equivalent to 51% of net annual global deforestation—for every 268 million households converted to LPG.

It is a global implementation initiative that seeks to demonstrate the benefits of LPG (in health, economic advancement, women's empowerment and education, and the environment) to policymakers and opinion leaders; to enable WLPGA member companies, governments, and national development agencies to serve as channels to inform consumers about the benefits of LPG; and to promote the expansion of demand for and use of LPG in developing markets, working hand in hand with governments and development agencies.

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According to WLPGA, India has a disease burden from indoor pollution that amounts to between 900,000 and one million premature deaths per year, of which 100,000 are children.

This is how the Indian government decided to act, implementing programs to provide subsidies for access to LPG to those who need it most. The most relevant elements of the strategy are explained below.

Program Elements

Three major national initiatives of the Government of India (called GLP Joint Implementation Measures, Catalyst for Social Change114) since 2014 aim to provide India with access to GLP by March 2019, with a rapid-growth strategy to reach 50 million low-income households, or approximately 300 million people. These initiatives are:

The PAHAL is a subsidy delivery scheme called direct benefit transfer for the LPG subsidy, where the consumer receives the subsidy in their bank account to help pay for the product. It's important to note that the end user pays the non-subsidized portion of the price, while the government covers the remaining cost.

In addition to introducing the sale of LPG at a single market price, it also eliminates the intermediary between the government and the consumer, thus preventing any part of the effort from being lost along the way. In this way, the end user remains the focus of the process. This financing scheme has been declared *"the world's largest cash bank transfer"* (GDP, 2018)115. "PAHAL, together with another government scheme, Jan Dhan Yojana, ensures both the financial inclusion of poor households and the transfer of subsidies directly into the bank accounts of poor households" (Tripathi, 2019)116. It is concluded that one of the keys to the Program's success in India has been the high degree of "banking penetration," which allows the user to see the government subsidy deposited into their account as soon as the product is received and the unsubsidized portion of the LPG is paid.

However, one of the limitations of the Program occurs when the non-subsidized component of the LPG price is high, since poor households find it difficult to pay this amount, thus creating a limitation on using the product.

Thus, while the government guarantees LPG accessibility to poor households by providing them with free connections, ensuring the constant use of LPG by

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114 https://www.wlpga.org/cfl-mediaroom/the-story-of-the-national-lpg-program-of-india/

115 PIB, 2018, "Cabinet approves enhancement of target under Pradhan Mantri Ujjwala Yojana", Press Information Bureau, overnment of India, Cabinet Committee on Economic Affairs (CCEA), http://pib.nic.in/newsite/PrintRelease.aspx?relid=176351, accessed on 24th June 2018.

116 Tripathi A, Sagar A, 2019, Ujjwala, V2.0: What should be done next?, Policy Brief, CCAPC/2019/02, Collaborative Clean Air Policy Centre, New Delhi

Machine Translated by Google

These households depend on the price they pay for the product. If the price of unsubsidized LPG increases, families' ability to pay decreases, making it more difficult for them to use this energy source for cooking.

Give it Up: This is a campaign launched in April 2015 that asks wealthy people in India to voluntarily give up their LPG subsidy and raise awareness among themselves about the benefits for poor households of switching their energy to LPG. This scheme has been very successful, and with extensive media coverage, by March 2019, around 1.04 million middle-class households had given up their subsidy to provide free connection costs for poor families (MOPNG, 2019)117. This scheme makes a collective appeal to those who have more to give up a subsidy they don't need, in favor of the poorest. Although it has been a successful policy, it is not enough to expand LPG coverage to the tens of millions of additional connections needed for poor households.

These elements are part of the ***Prandhan Mantri Ujjwala Yojana Plan,*** launched by the Central Government on May 1, 2016, considered one of the most comprehensive worldwide, to provide access to clean cooking energy for low-income families in the Indian region. “This amounts to suddenly doubling the ‘natural’ growth rate of LPG from 5-6%, which was only keeping up with the population, to a rate that can lead India to meet the UN goal of clean domestic fuels for all by 2030” (WLPGA, 2016)118 .

One of the plan's objectives is to bring joy to the faces of women from impoverished families in India, empowering them on this issue while simultaneously protecting their health by reducing the serious risks associated with cooking over open fires, primarily due to the accumulation of particles and smoke in the air. It also aims to prevent young children from suffering acute respiratory illnesses caused by indoor air pollution.

Under this access scheme, the Ujjwala project provided financial assistance of Rs. 1,600 (approximately US$22) to each eligible household on the Socioeconomic and Caste Census (SECC) list to subscribe to an LPG connection with one of the Petroleum Marketing Companies. An initial budget allocation of Rs. 8,000, equivalent to US$112, was made to cover 50 million households under the scheme. Once this initial target was reached, the government raised the goal to 80 million households by 2020 .

Indian women can obtain free GLP connections through their BLP (Citizens' Identification Card). The application process involves completing a scheme application form with name, address, Aadhaar number, and Jan Dhan/Bank account number, and an application form indicating whether they wish to receive free GLP connections.

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117 MOPNG website, 2019, http://petroleum.nic.in/, accessedon 23rd March 2019.

118 WLPGA website, 2019, https://www.wlpga.org/cfl-mediaroom/the-story-of-the-national-lpg-program-of-india/ , accessedon 27rd August 2019.

119 Same Note 95

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If you wish to obtain a 14.2kg or 5kg gas cylinder, once the information has been completed, it must be sent to the nearest supplier's application center and you must wait for the service to be approved.

As previously mentioned, to date, the national government has committed over one billion dollars, in addition to the personnel resources of its ministries. Furthermore, middle-class LPG users are contributing approximately 250 million dollars annually to the effort by voluntarily forgoing their LPG subsidy. India's largest company (Indian Oil) and the fifth and sixth largest (Bharat Petroleum and Hindustan Petroleum) are implementing the program on the ground through their network of 18,000 distributors. Similarly, 10,000 new distributors are being recruited to work in underserved areas, such as Agricultural Cooperative Societies and women's self-help groups. In addition, oil companies are doubling the LPG pipeline network, creating at least two new import terminals and several large new bottling plants.

The program is being implemented using modern IT technology, which in India is called JAM. This refers to the combination of electronic bank accounts, Aadhaar digital ID cards, and mobile phones, along with integrated national databases from all stakeholders. First implemented for the GLP, the implications of the expanded use of JAM are enormous for the country: the Prime Minister's chief economist stated at the meeting that he believed it would improve GDP growth by 0.5%.

The results of applying Ujjwala showed that:

- As of 2018, 9 out of 10 Indian households had access to LPG for cooking as a clean fuel, compared to approximately 4 out of 5 households in previous years. These results have had a profound impact on people's health, air quality, and emissions in the region.

- The widespread and sophisticated application of social marketing has been successfully implemented, including digital media (Twitter, SMS, interactive websites, etc.) and hundreds of educational fairs to promote understanding of the health benefits of GLP.

According to findings from the Asia GLP120 summit held in early 2019, the Indian government has been a pioneer in increasing LPG penetration in the country through policy interventions, aiming to provide LPG connections to 80 million households by 2020, which offers hope for the LPG market in India. Furthermore, LPG consumption in India has increased considerably to 23 million tons annually.

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120 Taken from the website:
https://currenthunt.com/2019/02/%E0%A4%8F%E0%A4%B6%E0%A4%BF%E0%A4%AF%E0%A4%BE-%E0%A4%8F%E0%A4%B2%E0%A4%AA%E0%A5%80%E0%A4%9C%E0%A5%80-%E0%A4%B6%E0%A4%BF%E0%A4%96%E0%A4%B0-%E0%A4%B8%E0%A4%AE%E0%A5%8D%E0%A4%AE%E0%A5%87%E0%A4%B2/

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which makes India the second largest consumer of LPG in the domestic sector.

- The Indian case presents a challenge for other countries in terms of what they can do to match its reach.

The first step to improving people's living conditions is to raise awareness and analyze the effect on health and the environment of not implementing measures to reduce the risks of indoor air pollution.

In the case of India, the program was very well received given its focus on empowering women, who are primarily responsible for household chores and directly exposed to pollution from cooking on open stoves or wood-burning cookstoves. Thus, health is the driving force behind the Ujjwala initiative, and the situation of women is a key factor in the new approach, as all new connections and subsidies are now exclusively for women, with the goal of achieving greater participation.

The Ujjwala Project 20121

The project identified three main barriers to LPG access for low-income households: accessibility facilitated by LPG distribution networks, affordability in terms of high initial cost and high refill cost, and awareness of the impact of LPG use.

Ujjwala focuses primarily on providing and facilitating access to energy and covering initial and installation costs. While this is a significant step in policy implementation, ensuring the sustained use of LPG in homes is crucial. This need has led to the creation of the **"Ujjwala 2.0" program.**

One of the challenges regarding the sustainability of LPG in households is the government's limited ability to provide subsidies to the poor to ensure access to this new energy source. Therefore, increasing coverage would mean reducing the percentage of LPG subsidized for all beneficiaries of the project. Indeed, as the program is currently designed, the Indian government is navigating a delicate balance between providing LPG at a price that poor families can afford and the normal budgetary constraints that governments must contend with.

To maintain and grow the dynamics of the Program, the following actions are proposed in Ujjwala 2.0:

- That subsidies have a differential rate for two levels of households: LPG at subsidized prices for the poorest households and a decrease in the subsidized price for consumers with the ability to pay.

121 Same Note 95

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- That the subsidized price for the poorest families be based on their willingness and ability to pay. In this regard, Tripathi and Sagar address the issue, stating that: “Our analysis of the National Sample Survey data suggests that households should be willing to use LPG as their primary cooking fuel if fuel costs are within 4% of their total monthly expenditure” (Tripathi, 2019)122 ; thus, “the amount of subsidized LPG could be restricted to 126 kg, or 9 cylinders per year. The delivery of the subsidy would continue to be through conditional cash transfers (known as the PAHAL scheme), minimizing the risks of subsidy diversion to unintended beneficiaries” (Tripathi, 2019)123 .

This approach to Ujjwala 2.0 has the potential to take the Ujjwala program to the next level and help the poor reap the benefits of the clean cooking energy transition. Thus, while the implementation of Ujjwala 2.0 is still under discussion, the guidelines proposed by these two authors provide a basis for studies on LPG substitution in other countries, incorporating concepts such as the valuation of morbidity costs, the statistical assessment of human lives lost, and the improvement in economic productivity due to time saved in firewood collection.

Final Considerations

It is crucial to emphasize that the LPG subsidy should not be viewed as a financial burden, as providing LPG leads to improved health outcomes for poor households, particularly women and children. LPG also enhances household economic productivity by reducing cooking time. Government spending on healthcare is further reduced due to the decrease in illnesses attributed to household air pollution. Therefore, providing subsidies to the poor and ensuring the regular use of LPG for cooking should be seen as a social investment by the government.

- To increase awareness among rich and poor about the health impacts, under Ujjwala 2.0, the Government should include intensive education campaigns that sensitize the general public about the need to direct the subsidy only to the poor and debunk myths about the taste of food cooked with LPG compared to firewood.

- Finally, it is worth highlighting that there is full support from all the main agencies involved, including frequent mention in the speeches of the First
Minister Modi and the firm commitment and, to date, a spirit of strong collaboration between the heads of the oil companies and a directive

122 Same Note 95
123 Same Note 95

Machine Translated by Google

stating that this is a social development program, not one focused on profits.

Although the Government of India has been successful in its ambitious effort to provide LPG connections to the poor, the inability of beneficiaries to afford the sustained use of LPG cylinders remains a concern. Therefore, ideally, a tiered pricing mechanism based on household income should be implemented to benefit the most vulnerable population. This would gradually address the limitations of PAHAL and Give it Up.

Mexico

Currently, a public-private partnership is being considered to expand access to LPG for people who use traditional wood-burning stoves. The program is in the pilot phase and is expected to benefit approximately 13,400 households. To implement this strategy, SEDESOL has contacted companies in the gas industry and Distribuidora e Impulsadora Comercial Conasupo (DICONSA), which operates the Rural Supply Program through affiliated stores.124 DICONSA distributes LPG to beneficiary families who are eligible because they live no more than 2.5 km from a DICONSA branch and receive cash transfers of $1,000 to $3,000 MXN to guarantee the purchase of a 10 kg cylinder.

Peru

As an initiative by the Peruvian government to improve the national production matrix, the Energy Social Inclusion Fund (FISE) was created. This entity is responsible for promoting the use of LPG in urban and rural areas of the country through a subsidy for the purchase of gas cylinders. If a family does not have a gas stove, the State provides one. However, maintaining the subsidy has proven difficult, given that the cost of the fuel is tied to the international price and the cost of importing it. Furthermore, families who use firewood for cooking obtain the biomass free of charge, which discourages them from accessing other energy sources for cooking.

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124 Microsol. 2018. Report on the relationship between short-lived climate pollutants and rural cooking in the residential sector. Prospective study applied to Colombia, Mexico and Peru
125 The FISE was created by Law 28853, which creates the Energy Security System in Hydrocarbons, and the Energy Social Inclusion Fund

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3.3. Biogas

3.3.1. Biogas in Colombia

A specific case126

In Colombia, the main sectors that have implemented anaerobic digestion processes are swine, poultry, dairy, the palm oil industry, and some small farmers. These sectors use part of the biogas for thermal energy generation, having to burn the remainder in many cases due to the lack of refining systems.127

Biogas is not a new topic in Colombia; the first plants were built as part of wastewater treatment facilities in Cali, Medellín, and Bogotá. In the first two cities, biogas is used for power generation, while in Bogotá it is burned in a chimney. Studies on biogas recovery and utilization have also been conducted at some municipal solid waste landfills, such as Doña Juana in Bogotá and Curva de Rodas in Medellín. In the agricultural sector, biogas has been viewed primarily from an environmental perspective, and the Ministry of Environment and Sustainable Development (MADS) has some regulations in this regard, especially for pig farms.

Colombia has made national and international commitments, such as the one made at COP 21, where it pledged to reduce greenhouse gas emissions by 20% compared to projected emissions for 2030. With the OECD group of countries, noteworthy aspects include the national policy for Integrated Solid Waste Management, corresponding to CONPES 3874 of 2016. Likewise, strategies such as Green Businesses, the carbon tax, and a gaining momentum initiative—the promotion of non-conventional renewable energy projects such as wind and solar power—have been developed. Biogas is an ideal complement to these, as it is a strong renewable energy source that provides firm energy to the aforementioned sources.

On the other hand, Colombia has joined different global strategies related to strengthening the environment, such as “Tropical Forest Alliance 2020 (TFA)”, a public-private alliance that aims to address deforestation, contributing to the reduction of greenhouse gas emissions, improving the quality of life, and conserving biodiversity, among other factors.

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126 This section was prepared by consultant José María Rincón Martínez in collaboration with Diana Marcela Durán Hernández; the findings and contributions contained herein are their own and are under review for publication.

127 Mario Enrique Velásquez Lozano *et al.,* “ESTIMATION OF THE POTENTIAL FOR CONVERSION TO BIOGAS OF BIOMASS IN COLOMBIA AND ITS USE FINAL REPORT INTERDISCIPLINARY TEAM MADE UP OF THE COLCIENCIAS RESEARCH GROUPS”, Bogotá, 2017.

128 C. Serrano Camacho, “Alternatives for the Use of Biogas in Sanitary Landfills in Colombia,” *Rds.Org.Co,* p. 93, 2006.

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In its domestic commitments, the country has established two important lines of work in its National Development Plan (PND): adaptation to the impacts of climate change and mitigation of greenhouse gases. These are achieved through three fundamental objectives: Objective 1, *"Moving towards sustainable and low-carbon growth";* Objective 2 , *"Protecting and ensuring the sustainable use of natural capital and improving environmental quality and governance";* and Objective 3 , *"Achieving resilient growth and reducing vulnerability to disaster risks and climate change"* (National Planning Department, 2014). Biogas is an ideal alternative that contributes to achieving these objectives.

At the beginning of 2019, Colombia inaugurated a regasification plant in the city of Cartagena, the first liquefied natural gas import plant located in the port society of El Cayao.

With Colombia becoming an importer of natural gas, the plans to meet the commitments made above may fail; therefore, adjustments must be made such as changing fossil fuels (coal, oil and natural gas) for biogas or biomethane, classified as renewable fuels and more efficient in terms of saving CO2 to the environment.129

Colombia has a long history in the development and dissemination of biodigesters among small and medium-sized agricultural producers. In 1986, the first manual for the installation of plastic biodigesters was published by Dr. Thomas Preston, based on previous small-scale projects in Ethiopia and Australia.130

This manual initiated the dissemination of this appropriate technology throughout the continent. The CIPAV Research Center promoted the democratization of tubular plastic biodigesters in the 1990s, but at the beginning of the new millennium, it was the Foundation for Sustainable Tropical Agricultural Production (Fundación UTA) that took the initiative in implementation, training of installers, and developed research on tubular plastic biodigesters.

Olade comments that this foundation convened in 2012 the meeting between actors for the formation of the Colombian Biomass Energy Network of Colombia (REDBIOCOL), made up today of 55 organizations of different types, NGOs, grassroots organizations, universities, social movements, associations of farmers and indigenous people, urban collectives.

The advantage of the tubular plastic biodigester is its low cost and the ease with which local technicians can be trained to install them. This is why this technology spread so rapidly throughout the country (and Latin America). ASPROINCA's experience is noteworthy, as it has over 300 systems installed through its own unique process, which allows its members to access tubular plastic biodigesters and the necessary financing via a revolving fund, and provides them with their own network of installers.

129 Oxford, "Biogas Contribution decarbonising gas markets," no. June 2017, 2017.
130 OLADE, "Available Technologies for Implementing Rural Biogas Programs in Latin America," no. 11

Machine Translated by Google

Another example is EL COMUN, an association of peasant and popular organizations of Colombia, who in the last two years have trained promoters through rural schools, disseminating the technology and obtaining support from the Ministry of Agriculture and international organizations for the installation of more than 100 biodigesters in the department of Santander131

Colombia demonstrates its own process, from the bottom up, managed by the local actors themselves, coordinated through a network of exchange of experiences and mutual support. In the REDBIOCOL network, biodigesters are yet another tool for strengthening the sovereignty of small and medium-sized producers, also taking into account other aspects such as the social and solidarity economy and the formation of new youth networks. Given the wide variety of institutions participating in REDBIOCOL, there is also a range of biodigester types available, primarily based on tubular plastic and geomembrane biodigesters. Despite the lack of specific national policies for the development of a sustainable sector, stringent environmental regulations for small and medium-sized producers, and substantial international cooperation funding, Colombia has managed to connect and consolidate a biodigester sector.

The development of biogas as one of the country's sustainable development strategies creates There is a demand for qualified personnel in the design and calculation of biodigesters, in the construction of the equipment, and in the operation of the plants. It also generates employment in the field, such as in the collection and transport of agricultural waste, and in maintenance. The plants, distribution networks, and energy crop production used in agriculture support the seasonality of agricultural waste, ensuring a year-round supply of biogas. This creates a demand for rural workers, preventing migration to cities and eliminating the need for permanent workforce training. It's important to note that the technology developed using this model is easy to learn and readily adaptable to Colombian agriculture.

Lessons Learned

An estimated 125 million people worldwide used biogas for cooking in 2017, primarily in China (111 million) and India (9 million). That year, approximately 15.5 billion cubic meters of biogas were produced for cooking, with around 13.1 billion cubic meters produced in China and 1.7 billion cubic meters in India.

Taking into account the experiences in the installation of biodigesters, mainly from Georgia132 , the following lessons learned are presented.

The use of biodigesters improves living conditions for households. People in rural areas, particularly women, dedicate a significant amount of time to biodigesters.

131 Idem Note 109
132 GM Karanja and EM Kiruiro, “Biogas production 107,” *Statistics (Eng.),* pp. 132-1 107–208 ,

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Time spent collecting and storing firewood. Using biogas frees up a lot of time and reduces the need for hard physical labor (felling and storing firewood).

- Biodigesters provide the expected biogas production under the expected conditions real.
- The costs of bioreactors of all types, especially thermophilic ones, They are still high.
- The amount of biogas used to maintain the substrate temperature within the limits of 50-55 °C (thermophilic bioreactors), in most cases does not exceed 25% of the biogas produced; Georgia is a clear example.
- Visual aids (brochures, television broadcasts) play a role important in the promotion of biogas technologies.
- Interest in developed and adapted biogas technologies is growing extending.

Farmers' knowledge of biogas and access to this information is very limited. They generally show little interest in the initial stages of biogas development.

- The more farmers who participate in the project, the longer they will keep the biodigesters in working condition.
- Farmers' interest in biogas is growing, as a result of pilot implementation and information campaigns.

Despite increased interest, most farmers lack the financial resources to install biogas production plants. Subsidies are essential.

- The lack of a strategy to finance biogas projects and the absence of credit lines for farmers are preventing the take-off of biogas production.
- Biodigester technologies must be adapted to local climatic conditions; in any case, it is possible to implement biogas production plants in hot and cold climates.

- It is important to have a minimum amount of manure of at least 20-30 kg/day, is approximately equivalent to 3 bovines.
- It is important to have a stable that ensures the collection of manure.
- It is necessary to ensure that there are local building materials available for the installation of biodigesters.
- Low incomes of farmers make it difficult for them to build their own biodigesters.
- High construction costs require support with subsidies, especially in the early stages of a national biodigester program for the replacement of firewood at the rural level.

- One factor that facilitates the installation of biodigesters is an average annual temperature above 20°C.
- The need for chemical-free fertilizers promotes the use of the byproduct from the anaerobic digestion process, which can be used as a biofertilizer.
- Within the interest of farmers in energy efficiency and environmental protection, anaerobic digestion is a process that contributes to both sectors, energy and environment.

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Biogas is considered a fully renewable source; the production of biogas from biomass waste avoids methane emissions into the environment due to the natural processes of decomposition of organic matter; it also avoids the formation of nitrogen oxides.

Implementing processes that combine adaptation and mitigation of greenhouse gases, such as anaerobic digestion, can create rapid and systematic transitions in rural and urban areas, knowing that the benefits outweigh the drawbacks of achieving sustainable development (particularly for Sustainable Development Goals (SDGs) numbers 3, health and 7, affordable and clean energy).

Advantages and Disadvantages of Using Biogas

Advantages

- It is an environmentally friendly process, since it disposes of organic waste that causes bad odors during decomposition, prevents the proliferation of pests, and destroys most pathogenic microorganisms.
- It prevents greenhouse gas emissions and helps the country meet its commitments made at COP 21 by reducing methane emissions, which have a global warming potential (GWP) 21 times greater than that of CO2. It also reduces NOx emissions from nitrogen in organic waste, which are 300 times greater than that of CO2, and those avoided by replacing fossil fuels such as natural gas, liquefied petroleum gas, kerosene, and coal.

- By replacing firewood, it prevents deforestation of the countryside.
- It improves the living conditions of rural women and children by giving them more time to use in creating new businesses, studying, and participating in local sports.

- It eliminates morbidity and lung diseases caused by breathing carbon monoxide, unburned and carcinogenic products from the incomplete combustion of solid fuels such as wood and coal.
- In terms of finances, it avoids hospital expenses for illnesses pulmonary.
- It improves the quality of life of the peasant and therefore helps to regulate the movement of the peasant to the slums of the big cities.
- Using digestate as a biofertilizer improves the quality of agricultural soils and crops due to nutrient recirculation.
- Avoid the use of chemical fertilizers, especially nitrogenous products produced by chemical synthesis.
- It prevents GHG emissions caused during fertilizer synthesis chemicals.
- It helps create businesses and promote green growth by guaranteeing organic fertilization and ensuring sales to the international community.
- Its energy efficiency is greater. Comparing the combustion of biogas to firewood, which is 50% less efficient, it can be turned off or shut down once its use is finished.

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Whereas, with firewood, the fire goes out when the solid fuel runs out, so some heat is lost.

Disadvantages

- Financial support is needed to establish a National Biogas and Renewable Energy Program for the rural sector, construct demonstration plant sites, and develop cost-effective and efficient digesters. CONPES 3924 establishes green growth, competitiveness, and social inclusion as key drivers for its development.

- The initial investment in biodigesters is expensive for the family's means.
  peasant woman.
- It is important to establish a learning process to carry out the transfer
  Knowledge for the farmer. Training is needed.

3.3.2. International Experiences

The following describes, at an international level and subsequently in Latin America, the experiences related to the incorporation of biodigester technology and the progress made to progressively replace firewood with biogas in the residential sector.

China and India are pioneers in the installation of small-scale biodigesters in rural areas; therefore, this section presents their experience, which serves as an example for other regions of the world, with or without programs to replace firewood with biogas. Germany, for its part, is a leader in the installation of larger industrial biodigesters and is presented as an example of developed countries where biogas is purified into biomethane, which is used as a substitute for natural gas in all its applications.

China133

At the beginning of the 20th century, Luo Guorui created the biogas storage method and opened China's first biogas company in Shantou City in 1929. Since the late 1970s and early 1980s, the Central Government has emphasized that "the development of biogas technology is an effective way to rationally utilize natural resources."

Subsequently, the Biogas Research Institute of the Ministry of Agriculture (BIOMA) was established, an institute of great international prestige, which in 2014 was designated by the Food and Agriculture Organization (FAO) as the FAO Reference Centre for Biogas Research and Training.

The implementation of biodigesters began with conventional models to provide cooking fuel in rural areas of China. The most commonly used substrates were cattle manure, food waste, and energy crops such as straw. In China,

133 Idem Note 105

Machine Translated by Google

They identified two types of models: digesters for domestic use, with a fixed dome design and volumes of 6, 8 and 10 m3 And there are biogas projects with small volumes, less than 150 m3 Medium, less than 5,000 m3 and large, above 5,000 m3 134

Due to the Arab countries' oil embargo on Western countries in retaliation for the Yom Kippur War, oil prices increased, spurring research into alternative energies, including biogas bioenergy. During this period, the Chinese government promoted the use of biogas in all rural households, and more than seven million digesters were installed.135 From the second half of the 1980s onward, biogas technology found further applications, particularly in the treatment of industrial waste. Since the beginning of this century, the number of installed plants has increased, and by 2007 there were 26.5 million biogas plants,136 mostly in Chinese homes, with capacities ranging from 6 to 10 m3 .

Since 2003, the “national debt aid” program has financed biogas projects, particularly for agricultural plants. Following the passage of China’s Renewable Energy Law in 2016, there was an increase in biogas engineering projects from one billion Chinese yuan (CNY) in 2003–2005 to 2.5 billion CNY in 2006–
2007.

Support for biogas reached 5 billion CNY in 2010, leading to an increase in biogas users in the first decade of the 21st century from 11 million in 2003 to 43 million in 2013. In turn, there was an increase in the number of biogas engineering projects from 2,300 in 2003 to almost 10,000 in 2013. ¹³⁷ Among China's policies is the goal of achieving, by 2020, 30 million kilowatts (kW) of power generation from biomass, primarily agricultural and forestry waste and the organic fraction of municipal solid waste (OFMSW). Biogas production for that same year was projected to be 44 billion m³ .
138

The Chinese-designed digester, known as a fixed-dome biodigester, consists of a gas chamber constructed of brick, stone, or concrete. The top and bottom are hemispherical and joined by straight sides (see Figure 2). The inlet pipe is straight with level ends. The biogas is stored under the dome with pressures of 1 to 1.5 meters of water. This creates high structural forces that are mitigated by the hemispherical shape of the top and bottom . ¹³⁹

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134 Sun Liying, “Status and prospect of biomass energy development in China´s rural areas,” 2016.

135 T. Bond and M. R. Templeton, “History and future of domestic biogas plants in the developing world,” *Energy Sustain. Dev.,* vol. 15, no. 4, pp. 347–354, 2011.

136 T. Bond and M. R. Templeton, “History and future of domestic biogas plants in the developing world,” *Energy Sustain. Dev.,* vol. 15, no. 4, pp. 347–354, 2011.

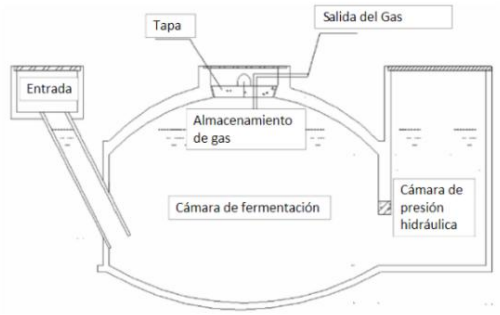
137 World Biogas Association, “What does the future hold for biogas in China?,” 2019. [Online]. Available: http://www.worldbiogasassociation.org/what-does-the-future-hold-for-biogas-in-china/. [Accessed: 14-Aug-2019].

138 Sun Liying, “Status and prospect of biomass energy development in China´s rural areas,” 2016.

139 “Biogas Manual,” 2011.

Machine Translated by Google

Illustration 2. Model of a Chinese-type digester



Source: "Biogas Manual," 2011.

India140

India began implementing anaerobic digesters to meet its rural biogas needs by installing small-scale biogas plants and further developed this technology with the launch of the National Project on Biogas Development (NPBD)141. India has installed approximately 4.6 million biogas plants, primarily using cattle manure; this number is expected to exceed 6.5 million by 2022. The most commonly used digesters are fixed-dome and floating-drum types, which are considered artisanal due to their simple design. The floating-drum digester is considered the Indian model in the literature (See Illustration **3).** Originally made of steel, it was prohibitively expensive and was subsequently replaced with fiberglass-reinforced plastic, a more corrosion-resistant material. The digester walls and bottom are constructed of brick and reinforced with concrete to prevent leaks. The biogas is stored under a floating drum that moves up and down on a central guide. The digester is fed semi-continuously through the inlet pipe.

The Indian Government has financed the biogas plants; the beneficiary only has to pay about 400 rupees, out of 20,000, which is the estimated cost of each plant142 .

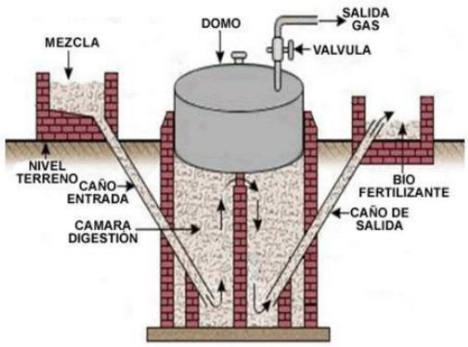
140 Same Note 105

141 R. J. Patinvoh, J. V Arulappan, F. Johansson, and M. J. Taherzadeh, "Biogas digestersÿ: from plastics and bricks to textile bioreactor — A review," *J. Energy Environ. Sustain.,* vol. 4, pp. 31–35, 2017.

142 Econoticias, "In India, biogas plants are an excellent solution to climate change," 2018. [Online]. Available: https://www.ecoticias.com/energias-renovables/181705/India-plantas-biogas-excelente-solucion-frente-cambio-climatico. [Accessed: 14-Aug-2019].

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Illustration 3. Diagram of an Indian-type biodigester



Source: “Biogas Manual,” 2011.

Germany143

The first biogas production plant in Germany was built for wastewater treatment and began operation in 1920; while the first plant fueled by agricultural waste began operating in 1950. Germany has largely been the main promoter of biogas in Europe; its development of biogas plants began in the 1990s and grew rapidly between 2006 and 2013144 .

However, a new renewable energy law in Germany, effective from 1 August 2014, reduced the use of energy crops and redirected industry towards raw materials derived from waste.145 Around 1,000 new plants per year were added in Germany between 2009 and 2011, which fell to just 150 in 2014.

The German government has set a target of feeding 6 billion cubic meters of biomethane into the natural gas grid by 2020. This represents almost 7% of Germany's current natural gas consumption. The target for 2030 is 10 billion cubic meters .

In Germany, according to data compiled by the International Energy Agency (IEA), there were 10,000 biogas-fired power plants, totaling 28,000 GWh/year in 2015.146

143 Idem Note 105

144 Oxford, “Biogas Contribution decarbonising gas markets,” no. June 2017, 2017.

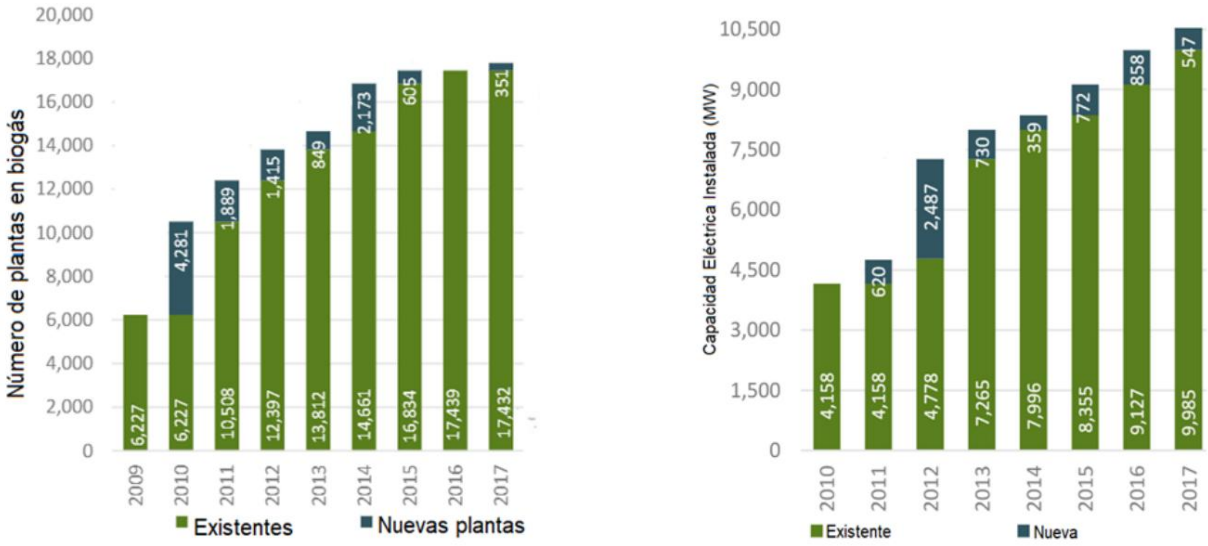
145 Lozano, M et al., “ESTIMATION OF THE POTENTIAL FOR CONVERSION TO BIOGAS OF BIOMASS IN COLOMBIA AND ITS USE FINAL REPORT INTERDISCIPLINARY TEAM MADE UP OF THE COLCIENCIAS RESEARCH GROUPS”, Bogotá, 2017.

146 J. Stern, The Future of Gas in Decarbonising European Energy Markets: the need for a new approach, no. January. 2017.

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Graph **10 (Error! Reference source not found)** presents the number of biogas plants installed in Europe for the period 2009-2017, and the increase in plants installed in the period 2010-2017, along with their installed electrical capacity.

Figure 10. Biogas plants installed in Europe (left), installed electrical capacity in Europe (right)



Source: Del EB Association, “EBA Statistical Report,” *Eur. Biogas Assoc.,* 2017.

Georgia147

Georgia is a mountainous country in the Caucasus, with a regime of extreme temperatures. Its experience with extreme temperatures is important for its application in the high mountains of Colombia. The information presented below is based on the report “Biogas Production”148

The development of biogas technologies in Georgia began in 1993-1994 with the German Agency for Technical Cooperation (GTZ). The technical support provided by GTZ enabled Georgian experts and engineers to study advanced designs and adapt technologies to Georgia's climatic and economic conditions. Georgia now has several engineering companies, research institutes, and professionals with experience in the field of biogas production.

In the 1990s, the Georgian company Bioenergy Ltd. developed small-scale, fixed-dome and floating-drum biogas reactors for mesophilic conditions. These systems are easy to operate but less effective in terms of biogas production.

147 Idem Note 105

148 GM Karanja and EM Kiruiro, “Biogas production 107,” *Statistics (Eng)* . Rev. 107–208 ,

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Taking local conditions into account, these reactors represent the most attractive technologies for most rural households with 1-2 cattle. Subsequently, more efficient mesophilic biogas reactors with a volume of 6 m^3 were developed. , for families with at least 4 cattle149

In 1994, the first bioreactor was built in Sasireti, Kaspi. That same year, Bioenergy Ltd. received a patent and distributed its booklet, "Construction and Maintenance of Biogas Plants." Between 1994 and 1996, bioreactors were installed in Gurjaani, Dedoplistskaro, Gardabani, Tsalka, and Chakvi, some with assistance from the United States Agency for International Development (USAID). The publication of the booklet resulted in approximately 60 bioreactors being installed by farmers, primarily using their own resources.¹⁵⁰

In 1999, with financial support from the Coordination Centre for Agricultural Project Development, Bioenergy Ltd. built four small-scale bioreactors in the Terjola region, two of them of the floating drum type, with thermal insulation, equipped with a solar collector, and the other two of the fixed horizontal dome type.

Three different types of biogas plants have been tested in Georgia, with support from the World Bank project "Reducing Pollution in the Agricultural Sector." In 2002, the project installed 12 bioreactors in the Khobi, Chkhorotsku, and Tsalenjikha regions. Eight bioreactors were of the floating dome type, two were of the Chinese fixed dome type, and the other two were locally improved versions of the fixed dome type. In 2003, the Coordinating Center announced a tender for the construction of an additional 45 biodigesters.

In parallel with the work of the Coordination Centre, Bioenergy built a 30 m3 bioreactor in Sachkhere, with financial support from the United Methodist Relief Committee (UMCOR) and 9 bioreactors in Akhaldaba under the community mobilization program.

In Georgia, only 0.1–0.2% of households use biogas. Most bioreactors are mesophilic, and only a few are thermophilic, primarily because the latter is more expensive and requires more biomass resources.

Based on these experiences, no technical barriers to the successful operation of biogas reactors in Georgia are perceived. Different types have been tested, showing good results. Some are easier to operate but less effective in biogas production. Others require more attention to operate but produce significantly more biogas. These different types address different needs.

On the other hand, the construction of biogas reactors has not taken off. Most efforts are funded by international organizations, suggesting a problem with scaling up. This is because many farmers are still unaware of this technology and, more importantly, would have serious difficulty finding the $500 needed for even the cheapest bioreactor.

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149 Same Note 127
150 Idem Note 127

Machine Translated by Google

Nicaragua151

Nicaragua is the only country that has developed a National Biogas Program (NBP) based on experiences from Asia and Africa. The NBP-Nicaragua began in 2012, following the positive results of a feasibility study. The feasibility study identified a technical potential of 55,000 biodigestion systems, demonstrating the viability of a biogas program.152 The country has one of the lowest electrification rates in Latin America. Approximately one-quarter of the population lacks access to electricity. Additionally, a large proportion of rural households use firewood as an energy source, which leads to a decline in the health of those who use this type of technology.153,154

The Biogas Nicaragua program, which was implemented by the Dutch development cooperation service (SNV), under the supervision of FOMIN (an entity attached to the IDB), was proposed in order to address the aforementioned problems.

The strategy employed was to strengthen biogas production systems: facilitating access to credit and incentivizing the first 1,000 families with US$480 for tubular systems and US$970 for fixed-dome systems. This program was implemented between 2012 and 2017, with an investment of US$2,082,750 from FOMIN and US$1,976,550 from the NDF. The main beneficiaries were small-scale farmers who were able to establish energy production systems on their farms by replacing their wood-burning stoves. By 2014, 76 families had biogas systems for domestic use.

Nicaragua's socio-economic conditions, with 27% of the population lacking electricity (a figure that rises to 64% in rural areas), place the country close to the circumstances of other African countries where biodigesters have been successfully implemented. The program was financed by the IDB-FOMIN, the Nordic Fund, and HIVOS for the first two years. Its initial goal was to install 6,000 biodigesters by 2018, a target that was reduced to 1,500 in a mid-term evaluation. By the end of 2017, 1,200 biodigesters had been installed . ^{155,156}

The PNB-Nicaragua has faced several achievements and challenges that explain the low number of installations after 6 years compared to other Asian and African programs; however, it is the program that has managed to install the most biodigesters in Latin America.

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151 Same Note 105

152 OLADE, “Available Technologies for Implementing Rural Biogas Programs in Latin America,” no. 11,

153 Biogas Nicaragua, “Nicaragua Biogas Program,” 2018. [Online]. Available: http://programabiogasnicaragua.org.

154 FOMIN (IDB) and SNV, “DEL MARCADO DE BIOGAS EN NICARAGUA (ATN / ME-13067-NI Y ATN / NV-13169-NI)”, 2012.

155 Idem Note 133

156 Idem Note 131

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Olade notes that forming a team to coordinate the various components was difficult due to a lack of prior experience with market-oriented biodigester projects. Africa was the first continent to receive the successful experiences of the National Biodiversity Program (NBP) from Asia. Experienced personnel were relocated from Asia to Africa, facilitating knowledge transfer to local staff. In Nicaragua, however, the knowledge transfer was limited to the technical aspects—design and construction of fixed-dome biodigesters—and did not address implementation strategies. Consequently, the initial years were characterized by self-learning, delaying the program's execution.
Drawing on Asian and African experience, the fixed-dome model was chosen. The program allowed the import of tubular biodigesters made of polyethylene geomembrane, distributed by the Nicaraguan company Tecnosol. The construction cost of the fixed domes in Nicaragua was three times higher than in other countries, while the tubular geomembrane biodigesters also proved expensive and did not achieve widespread distribution.
This entailed serious difficulties in promoting the technology in the early years.
Subsequently, adjustments and adaptations were made to the fixed dome model, in a joint effort between staff with experience in Asia and local staff, which seriously lowered costs to as low as $1300 USD for 6 m3 biodigesters .
Subsequently, the use of various technologies was promoted, allowing companies supplying tubular biodigesters with PVC and polyethylene geomembranes to enter the National Biodiversity Program (NBP), with similar prices.157 As a result, better promotion of the technology was ensured, and most of the biodigesters installed are tubular geomembrane models. Thus, Nicaragua's NBP has been a pioneer in consolidating technological diversity (fixed dome and tubular biodigesters) compared to other NBPs.

The bureaucratic level of management has been very high due to requirements from the funders, an aspect that has limited the execution capacity of the companies supplying the technologies involved.

The high cost of the selected technology has necessitated a 62% subsidy on the construction price of the smallest fixed-dome biodigester (4 m^3). The beneficiary contributes US$300 and the National Productive Group (NPG) contributes US$480. This subsidy represents 37% of the installed cost of the fixed-dome biodigester.
In other national production systems, access to credit has been the key to overcoming the barrier of limited repayment capacity among rural producers. However, in Nicaragua, approximately 80% of producers have paid in cash, and many have not installed a biodigester because they are unwilling to take out a loan. The interest rate on loans to purchase a biodigester is around 20%, making it less accessible. Lower interest rates can be obtained if the user takes out a larger loan (up to US$5,000) to make other investments, a factor that also discourages many potential users. Another issue is the collateral required for many of these loans; putting up land ownership to access a biodigester also discourages potential users from taking out a loan. The high cost of biodigesters (compared to other national production systems) and the difficulty producers face in accessing attractive credit options have limited the achievement of targets for the number of systems installed.

157 Idem Note 131

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Another pioneering contribution of Nicaragua's PNB has been the development of a whole new market (systems, technology providers, etc.), focused on the productive uses of biogas from much larger biodigesters (biogas-powered systems such as mechanical milking machines, milk cooling, grass cutters, mills, boilers, electric generators, water pumps, etc.). Other National Biodiversity Programs (NBPs) focus on domestic biodigesters, leaving the productive uses of biogas for specific, demonstrative cases (since employing this type of machinery requires much larger digesters that, in turn, require significantly more feedstock: manure or other organic matter). In this way, Nicaragua's NBP has managed to make the productive uses of biogas accessible to producers through a market-driven approach. Developing this complementary market has required adapting technologies to biogas and validating the different systems considered, as well as generating interest from suppliers to showcase and sell them. This has made the technology attractive to medium-sized agricultural producers who also have greater purchasing power and access to credit.

Other aspects to highlight are the successful development of the process of obtaining carbon credit registration for emission reductions under the Clean Development Mechanism (CDM) and the Gold Standard, the dissemination of biodigester technology in the country, which has overcome the previous lack of knowledge or bad reputation of the technology, and how to monitor and guarantee the functionality of the biodigesters installed in the program.

Costa Rica158

Since 2014, the ICE Group (Costa Rica's public companies group), in accordance with its social duties and responsibilities, has promoted the biogas program in Costa Rica, which aims to provide technical assistance to the agricultural sector in generating biogas from its waste. In this way, the productive sector alleviates the burden on the energy system and contributes to mitigating the environmental impact generated by the use of conventional technologies.¹⁵⁹

The program has a broad portfolio of projects. For example, one of these is Sermide, which has an installed capacity of 60 kW. It is capable of supplying electricity to a privately owned farm through biogas production.

Among the most recent projects, a project called **"Implementation of an Electricity Generation System from Biogas Obtained Through the Treatment of Pig Manure Generated by the Agricultural Farm (Pig Farm)"** stands out .
"Cerdos El Cerro" is located in Cañas, Guanacaste province and focuses on

158 Same Note 105

159ICE, "Grupo *Programa Biogas*", 2017. http://www.grupoice.com/wps/wcm/connect/669c79e9-84c9-4682-b63b-136238ebc7e2/tarifas+ice+gaceta+alcance+106+junio+27+2016.pdf?MOD=AJPERES. [Online]. Available:

Machine Translated by Google

Biogas production in an anaerobic manner using pig manure from the farm as raw material. It consists of a generating plant, with an installed capacity of 70 KW, which is producing around 135,000 kWh/year160

Thus, the aim is for the agricultural sector to be able to cover a large part of the energy demand, and even supply rural areas within Costa Rica.

Olade notes that Costa Rica, after Colombia, is another leader in the dissemination of tubular plastic biodigesters, thanks to Raúl Botero, a professor at EARTH University who, in 1986, along with Thomas Preston, published the first manual on this technology. EARTH University, which has several operational tubular biodigesters on its premises, trains 100 students from Latin America and Africa each year, equipping them with knowledge of these systems. This has made the university a focal point for the dissemination of the technology at a regional level. Several Costa Rican graduates from EARTH have launched businesses related to tubular biodigesters, replacing greenhouse plastic with geomembrane, usually made of PVC, thus increasing the system's durability, as seen in companies like VIOGAZ and Biosinergia Alternativa.161 These companies have become Latin American leaders in this type of system, developing their activities with a market-driven approach and without support from international cooperation. The country's government's intensive environmental policies have incentivized the adoption of biodigesters as waste treatment systems, which also produce other benefits such as biogas and biofertilizer. The Ministry of Agriculture, the National Learning Institute, and the state-owned electricity company (ICE) have worked to promote biodigesters, providing various types of funding for pilot projects. Many of these initiatives are driven by EARTH University graduates, as well as other stakeholders and enthusiasts. Thus, even without a specific policy for developing a biodigester sector in the country, environmental policies are catalyzing this market-based process.

Guatemala

In Guatemala, several projects have been developed around the implementation of biogas as an energy source. Among these, those promoted by the company Agrogeneradora SA (a member of the Central Agricultural Group) stand out. The Agrogeneradora SA plant, which received an investment of US$4.7 million over five years starting in 2014, has a capacity of 1 MW, of which only ten percent is used for self-sufficiency, while the remainder is injected into the electrical grid, thus mitigating the environmental impact.¹⁶²

Similarly, Aqualimpia SA built a chicken manure processing system for biogas production at the Victoria poultry farm. Four 3,500 m³ biodigesters were constructed with the capacity to process 160 tons of chicken manure per day.

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160 Idem Note 131

161 Same Note 131

162 Grupo Central Agrícola, “Agrogeneradora SA,” 2018. [Online]. Available: http://www.centralagricola.com/nuestrosnegocios#energia.

Machine Translated by Google

The power plant has a capacity of 1 MW. The energy produced is sold to the Guatemalan power grid.¹⁶³

The implementation of small biogas systems in rural areas has also been promoted as a replacement for conventional technologies such as biomass burning for energy production. Mr. Daniel Buchbinder, as part of Alterna (an organization that promotes entrepreneurship in Central America), designed a two-compartment system that stores biomass and decomposes it to produce biogas for household use. Although it was successfully implemented, barriers were encountered regarding the expense that users must incur to implement the design.164

Biogas plant

The family business consists of buying two-day-old chicks and raising, slaughtering, and selling them. The community engagement process involved all the necessary work to get the families to adopt the biogas plants. Six potential plants were identified, but only four were built.

Mr. Manuel Xol *adopted biogas technology because of its advantages:* greater chicken growth due to disease control, lower chicken mortality, fertilizer production, flaming of the breeding facility, water heating for the slaughtering process, **cooking in the kitchen,** lighting, and final disposal of excrement and blood.

The plant has a volume of 12 m³ and a production capacity of 4.5 m³ /day. 2 m³ of biogas corresponds to the calorific value of 1 lb of LPG. One lb of LPG costs 5.5 Q.
Therefore, the equivalent value of biogas is 12.40 Q/day or US$1.60/day or US$48/month in LPG.

The gas produced saves 3 m3 of firewood per month, with a monthly value of 100 (Q/month of US$12.8/month) and uses a fuel with a higher calorific value as calculated above.

The plant cost Q10,000 (US$1,280). It was 100% financed by PURE II and the owner contributed in kind.

Conclusion

The advantages of this plant are appreciated by users.

In terms of project funding, **it was fully subsidized by the PURE program.**

Table 22. Chicken farm biogas plant

Location: San Juan Chamelco, Cobán, Alta Verapaz	

[163] Aqualimpia, “Aqualimpia Engineering,” *Aqualimpia engineering,* 2018. [Online]. Available: https://www.aqualimpia.de/antorchas-biodigestores/.
[164] Alterna, “Alterna,” 2019. [Online]. Available: https://www.alterna.pro/.

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One-week-old chicks | Biogas plant

Biogas flaming to sterilize the soil | Single-burner stove burning biogas

Source: Prepared by consultants, Corpoema

2. Biogas plant at an ecotourism center

The business is an ecotourism center that includes birdwatching, nature walks, and a restaurant. The socialization process consisted of all the work necessary for the owner to adopt the technology.

FUNSOLAR hired a biogas plant expert who carried out the following activities: induction, supply, component installation, assembly, support, and training. The plant is a fixed-bell type, has a volume of 50 m^3 , and a production capacity of 20 to 25 m^3 /day of biogas, requiring a feed of 400 lb/day of organic matter and 1 m^3 /day of water.

The cost of the project was Q 54,158, of which the owner contributed 25% and PURE II contributed 75%. Construction began in June 2012 and was in the commissioning phase by the end of the year.

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Conclusion

In this case, the program co-financed 75% of the project, partly because the project is a showcase for the technology. The project's benefits include the elimination of firewood consumption, the production of compost, and the reduction of CO2 emissions .

Table 23. Biogas plant at the Cuevas de Ma'xivan eco-tourism center

Location: Cobán, Alta Verapaz	
	
Tourist Center	Biogas plant
	
Exterior view of the biogas plant	Two-burner stove burning biogas

Source: Prepared by consultants, Corpoema

Lessons learned

- One of the essential aspects for the success of projects with TER is to design and consider different management schemes that lead to the sustainability of the projects.

- The introduction of renewable energy technologies (RES) in rural areas is costly for the income level of the beneficiaries, therefore they must receive support from the state (economic, technical and know-how resources).

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The high cost of various renewable energy technologies (RETs) continues to be a barrier to rural development. The appropriate selection of technologies and their adoption by stakeholders in the country are essential.

The private sector's perception of high risks and weak market signals, without regulatory and tariff signals to encourage private sector projects, will continue to limit private sector participation in the development of renewable energy sources.

Mexico165

Mexico has high potential for biogas production due to the large populations of its major cities and the high volume of organic waste generated. Initially, the IRRI-Mexico Institute, in partnership with the company Sistema BIOBolsa, promoted a national biogas program. This program focused on manufacturing biogas systems (biobags) in Mexico capable of producing one cubic meter of biogas per day, equivalent to 2.2 kWh of energy. The program also conducted research, promotion, education, quality control, and monitoring of biogas production projects. It is worth noting that the program targeted rural and/or marginalized communities in Mexico, where rudimentary energy production methods were in use. Furthermore, these communities lacked a culture of proper disposal and utilization of the organic waste they generated . ¹⁶⁶

The Ministry of Environment and Natural Resources (SEMARNAT) and the Ministry of Energy (SENER), in collaboration with the German Federal Ministry for Economic Cooperation and Development (BMZ), through the German Agency for International Cooperation (GIZ), promoted a project called “Energy Recovery from Urban Waste in Mexico,” which aimed at generating biogas and valorizing urban waste by training the population in the use of biogas production technologies and their utilization. The project was scheduled to run for four years (2014-2018) .[1]ÿÿ

Olade notes that Mexico has achieved similar impacts to those of a national biogas program, but without one, as in the case of Costa Rica, to incentivize the use of biodigesters as a waste treatment system for small and medium-sized producers. The NGO (IRRI) and the social enterprise (BIOBOLSA) are obtaining results similar to national programs in Africa and Asia, with more than 1,000 tubular polyethylene biodigesters installed per year. Partial support has been obtained from the government (Ministry of Environment, Livestock, Rural Development, Fisheries and Food) for the implementation of pilot projects. Subsidies have been secured for the

165 Same Note 105

166 GMI, “Global Methane Initiative,” 2014. [Online]. Available: https://www.globalmethane.org/.

167 SEMARNAT, “Energy use of urban waste (EnRes),” 2018. [Online]. Available: https://www.semarnat.gob.mx/.

Machine Translated by Google

The installation of biodigesters for small and medium-sized agricultural producers is facilitated through support from existing social networks (such as agricultural cooperatives, livestock associations, rural fairs, and demonstration events), access to financing (0% interest microloans), and the ability to attract large international cooperation budgets. Less than 30% of the biodigesters installed in Mexico have received subsidies (national or international), and when subsidies have been provided, they have been as high as 80%, and more frequently around 40-50%. This NGO-business consortium is expanding into other countries in the Americas (Nicaragua, Haiti, Colombia), Africa (Ghana, Ethiopia), and Asia (India), demonstrating a replicable business model . ¹⁶⁸

Chile169

The Chilean government, through its Energy 2050 policy, has promoted the development of renewable energy, with German cooperation. The roadmap was presented in 2014, and it is expected that by 2035, 60% of Chile's energy will be produced from renewable sources, while by 2050, this figure will reach 70%. This year, the aim is to ensure that the environmental regulations and standards for energy projects are consistent with international guidelines and the interests of society . ¹⁷⁰

The 4e program was also developed, which, through the NAMA project, seeks to promote the application of technologies for energy production using renewable resources, in this case for self-consumption (including biogas). These energy advancement policies are especially important for the development of communities that want to improve the use of their own resources.171172 Some initiative and promotion programs presented by the GEF granted approximately US$1.7 million to the government to implement biogas production technologies in the dairy sector.

The aforementioned programs aim to strengthen the biogas production sector in the country. Some of the most important projects include those carried out in regions such as Valparaíso, Maule, Metropolitan, O'Higgins, Los Lagos, and Biobío, which are of medium capacity (1 MW on average). Currently, only 4% of the energy generated comes from biomass. Within the country, 104 biogas plants have been identified, of which 60% are operational . ¹⁷³

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168 Idem Note 131
169 Idem Note 105
170 Ministry of Energy of Chile, “Description of the Energy 2050 process,” 2018. [Online]. Available: https://www.opia.cl/static/website/601/articles-76887_archivo_01.pdf.
171 Idem Note 149
172 4e Chile, “NAMA-Renewable Energies for Self-Consumption,” 2018. [Online]. Available: http://4echile.cl/4echile/wp-content/uploads/2018/05/NAMA-Chile-Factsheet-esp-2018.pdf.
173 Ministry of Energy of Chile, “Description of the Energy 2050 process,” 2018. [Online]. Available: https://www.opia.cl/static/website/601/articles-76887_archivo_01.pdf.

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Brasil174

Brazil is the country that has made the most progress in biogas production and its applications in Latin America. The National Electric Energy Agency (ANEEL) reported in 2014 that electricity production from biomass accounted for 8.75% of Brazil's total electricity production, representing an installed capacity of 12,303 MW. This includes thermal generation using both biomass and biogas. Three new biogas plants were installed in 2014, bringing the total number of biogas plants connected to the electrical grid to 25. That year saw a 12% reduction in biogas generation, attributed to the decay curve of landfill biogas. Most biogas plants are located on agricultural properties for waste processing and at landfills . ¹⁷⁵

In Brazil, most biogas is used to produce electricity and heat (cogeneration), while its use as a vehicle fuel is uncommon. However, a project developed by ITAIPU Binacional, the Itaipu Technological Park Foundation, Scania, Haacke Farm, and the International Center for Renewable Energies-Biogas/CIBiogás-ER demonstrated the feasibility of using biomethane as a vehicle fuel, generated from municipal solid waste, wastewater, and pruning waste from the ITAIPU educational and industrial center.

The financial support systems shown below were used to stimulate biogas in Brazil:

- The program is financed with funds from BNDES, Rural Savings Booklet (MCR 6-4) and Constitutional funds, which are obtained through taxes on public resources.
  The target group is farmers and their cooperatives, including transfers to members, with a credit limit of one million US dollars per beneficiary per crop per year. This loan can be received independently of other credit to which the producer or cooperative is entitled and is based on controlled rural credit resources. The interest rate is 5.5% per year.

- PRONAF ECO: Credit for projects using renewable energy technology in wastewater and effluent treatment plants. The target group is farmers enrolled in PRONAF. The credit limit is R$ 50,000 per beneficiary. The interest rate is 2% per year. Term: 10 years with up to 5 years grace period.

Bolivia176

In 2007, Bolivia began a pilot project to promote biodigesters, sponsored by the German Development Cooperation's (GIZ) Endev-Bolivia program and coordinated by CIMNE. This project, which ran for five years, was updated every six months based on the results obtained. It concluded in 2012.

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174 Ibid. Note

175 Mario Enrique Velásquez Lozano *et al.,* “ESTIMATION OF THE POTENTIAL FOR CONVERSION TO BIOGAS OF BIOMASS IN COLOMBIA AND ITS USE FINAL REPORT INTERDISCIPLINARY TEAM FORMED BY THE COLCIENCIAS RESEARCH GROUPS,” Bogotá, 2017

176 Idem Note 105

Machine Translated by Google

With 750 domestic plastic tubular biodigesters installed, the project's nature—marked by multiple extensions and a small team—prevented the realization of a long-term vision for developing a biodigester sector in the country. Among its achievements was the development and validation of adapting tubular biodigesters to cold climates through a passive solar heating design, thus expanding the operating range of these low-cost systems. Positioning plastic tubular biodigesters as a cheaper and more accessible technology, aligned with the economic realities of the poorest producers, was another contribution to the international landscape. Another key aspect was the involvement of universities and NGOs in research and development under real-world operating conditions, including biodigester evaluation, biogas and biofertilizer design and applications. In 2011, the Biodigester, Biogas, and Biofertilizer Research Center (CIB3) was founded in the Bolivian Altiplano. Because Bolivia has a gas subsidy that makes it very accessible to families, biodigesters lose some of their appeal (biogas) and it was decided to make a strong commitment to the productive uses of biogas (although without specific results, unlike the case of Nicaragua) and uses of biol (fertilizer).

This led to a large number of producers being interested in biodigesters, not for the biogas or their waste treatment capacity, but for the fertilizer they produced. Without specific government policies to promote the use of biodigesters, but with support from international cooperation, the project was always focused on the poorest, smallest, and most vulnerable producers, with subsidies covering 33% of the biodigester's cost. In 2012-2013, feasibility studies and a plan for a national biogas program were carried out, but the necessary financing for its development could not be secured. The case of Bolivia represents an intermediate, indeterminate point, lacking business ventures supported by environmental policies or international cooperation, a strong social base organized through a network, and a long-term national biogas program. Impacts were achieved only during the project's duration, with little subsequent replication.

Argentina177

The Argentine government, through its RenovAr program, promoted the development and implementation of renewable energy. The program, which began in 2016 and is expected to conclude in 2025, is being conducted through a call for bids by CAMMESA, representing the Argentine government and stakeholders in the wholesale electricity market. The goal is for energy sector companies to enter the market and fulfill the total power allocation required (15 MW for biogas).

Although Argentina has broad energy system coverage in major cities and provinces, several rural areas still rely on technologies such as burning firewood and kerosene. For this reason, Argentina was one of the first countries in Latin America interested in implementing policies that would promote the development of renewable energy technologies.

177 Idem Note 105

Machine Translated by Google

renewable energy production. This can be observed through the application of Law 26190 of 2006, which establishes that 8% of energy production must be done through renewable energy sources by 2016. In addition, Law 27191 of 2016 (which replaces the previous one), under regulated conditions, aims for 20% by 2020. It should be noted that these laws and incentives received criticism for showing low requirements compared to policies used in other countries.178

Alongside the implementation of the RenovAr program, the creation of the renewable energy development fund, the World Bank's backing, and the key incentives provided for in Law 27191 of 2015, Argentina is positioning itself as a leader in the transition from conventional to renewable energy. These benefits include: early VAT refunds and the option of accelerated depreciation of assets, in addition to the fact that these assets will not be considered when calculating income tax.

Furthermore, biofuel producers will not have to contribute to the water infrastructure fund or the diesel tax, depending on whether the biofuels are pure or a mixture179 .

Across Argentina, the Renew project has awarded 37 biogas generation projects (64.9 MW total), which have a weighted average price of USD 159.7/MWh. These projects have a capacity ranging from 0.72 to 3.00 MW.

3.4. Pellets, Briquettes and Charcoal 180

3.4.1. Pellets, Briquettes and Charcoal in Colombia

A specific case

In Colombia, according to Law 1715 of 2014, non-conventional renewable energy production systems were included, within which are those that operate through the use of biomass (pellets, briquettes, among others).
This policy was considered to be of social interest and a pillar for the development of the country181 .

Furthermore, the presidency has also launched programs such as “Colombia E2,” which aims to replace conventional energy systems with renewable energy sources through entrepreneurial ventures. This program subsidizes companies with two billion pesos . ¹⁸²

178 P. Schaube, W. Ortiz, and M. Recalde, “Status and future dynamics of decentralised renewable energy niche building processes in Argentina,” *Energy Res. Soc. Sci.,* vol. 35, no. November 2017, pp. 57–67, 2018.

179 Argentine Congress, “Law 27191 of 2016,” no. 67, p. 1235213, 2015.

180 Idem Note 105

181 Congress of Colombia, “LAW 1715 OF 2014,” *law,* vol. 151, no. 4. pp. 1–46, 2014.

182 Presidency of the Republic of Colombia, “President Duque presented the 'Colombia E2_ Entrepreneurship x Energy' program, to strengthen the renewable energy industry,” 2019.

Machine Translated by Google

In 2015, the Colombian government launched a program aimed at replacing wood-burning heating systems with more efficient ones, fully subsidized.183 The guidelines defined a clean and efficient stove as one that met three conditions: a thermal efficiency greater than or equal to 25%, and the production of less than 0.49 g/min of CO and less than 8 g/min of particulate matter (for indoor emissions), and ÿ 9 g/MJ of CO and ÿ 168 mg/MJ of PM 2.5 particulate matter (for overall emissions). Accordingly, the new heating systems were implemented, many of which used the same type of biomass as the previous ones (firewood), but in a more efficient manner. In fact, the program guidelines state that pollution can be reduced by up to 94.2%, based on the fact that only the design has been improved, but more efficient raw materials such as biomass pellets or chips have not been used.

The previous project shows as lessons learned the fact that it is important to change cultural habits in the populations to which the project is directed and the fact that to achieve this, time is needed (a reasonable period of between 5 and 10 years).
It is also emphasized that improvements in efficiency, pollution reduction, and the improvement in people's quality of life are essential standards for project compliance.

3.4.2. International Experiences

Given the energy inefficiency of charcoal, the experiences in the use of pellets and briquettes in countries such as Spain, Russia, and Canada are presented below as an example.

Spain

The Spanish government launched the Renewable Energy Program (2011-2020), which contains the roadmap the country will follow in the second decade of the 21st century. The program's objectives are aligned with Directive 2009/28/EC of the European Parliament and of the Council of 23 April 2009 on the promotion of the use of energy from renewable sources. Additionally, it complies with the provisions of Royal Decree 661/2007, which regulates electricity generation under the special regime, and Law 2/2011 on Sustainable Economy .

Currently, most of Spain's final thermal consumption of biomass comes from the forestry sector, being used in the domestic sector through inefficient conventional systems (using firewood in outdated equipment), which generate pollution and can be harmful to human health. Biomass is also used in forestry industries for self-consumption or cogeneration, where there is an installed capacity of 533

183 AD . C. Concha, María Cecilia; Pabón, Giovanni; Cerón Viviana. -- Bogotá and 2015. Ministry of Environment and Sustainable Development, *Guidelines for a National Program of Efficient Wood-Burning Stoves.* 2015.

184 IDAE, “IDAE,” 2019. [Online]. Available: https://www.idae.es/.

Machine Translated by Google

MW which is supplied with agricultural and forestry waste. To mitigate the situation created by the use of conventional energy sources, the 2011-2020 renewable energy program aims to promote the development and application of technologies that use pellets in the residential sector. This is achieved through awareness campaigns, legislative development, and new systems of financial support, incentives, and public aid for investment. The growth in electricity production from biomass is achieved through small cogeneration plants and power plants with an installed capacity of around 15 MW, for which new financing systems and improvements to the renewable electricity remuneration system are proposed, especially for those installations with less than 2 MW .

Russia

In accordance with the Russian government's annual biofuels report ,186 biofuels represent a very small percentage of the nation's energy mix (only 1.2%). Biomass use is even lower (0.5%). However, Russia has significant potential for energy production from biomass. For example, the Federal Forestry Agency considers biomass energy production the primary development alternative, given the large quantities of wood waste (up to 8 million cubic meters) that are used for energy production during energy shortages. It is worth noting that the lack of biomass utilization policies, due to the high dependence on fossil fuels, means that only large industries are interested in entering the biomass market (especially pellets) .187

In addition, governments such as those of Japan, Korea, and some members of the European Union have increased their interest in obtaining pellets. For this reason, the demand for this raw material has increased in Russia . ¹⁸⁸

Finally, in 2016, ENBIO was created. This organization was established with the aim of developing the field of renewable energies in Russia and aligning the energy sector with global objectives such as conservation and combating climate change caused by pollution. One of its main goals is to generate a self-consumption policy in isolated locations within Russian territory . ¹⁸⁹

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185 Idae, “Summary of the Renewable Energy Plan.” p. 64, 2011

186 FAS-Staff, “Russian Federation Biofuels Annual Biofuels Sector Update.” p. 21, 2016.

187 FAS-Staff, “Russian Federation Biofuels Annual Biofuels Sector Update.” p. 21, 2016.

188 FAS-Staff, “Russian Federation Biofuels Annual Biofuels Sector Update.” p. 21, 2016.

189 ENBIO, “Russian Bioenergy Association (ENBIO) · Bioenergy Europe,” 2016. [Online]. Available: enbio.ru/.

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Canada

For some years now, Canada has become one of the largest producers of energy from biomass. According to the publication by García et al.190, Canada is one of the world's leading producers of wood pellets, reaching almost two million tons in 2014, second only to the United States and Germany.
Additionally, the Canadian Wood Pellet Association of Canada (WPAC) reports that since 2010, pellet exports have increased by 240% to date.191 According to NEB,192 Canada has a biomass energy system with an installed capacity of 2408 MW with 70 plants.

Regarding the development of technologies, the WPAC has entered into several agreements with European entities, the United States and in general at the international level, to promote research and contribution to standardization of pellet production processes193 .

In order to help improve energy production figures from biomass, the Canadian government launched the so-called "Biomass Program".
For example, in the province of Manitoba in 2016, the government, through the program, subsidized coal users with US$500,000 for transitioning to biomass-based technologies, while also providing another US$500,000 for research and development. The benefits also extended to small consumers who wanted to switch their heating systems to those using wood pellets or briquettes. This subsidy amounted to US$50,000 .

Argentina

Through the RenovAr program, Argentina promoted the implementation of renewable energy. The program's implementation period runs from 2016 to 2025.
Thanks to a call for proposals made by CAMMESA, companies in the sector are expected to participate.

190D. P. Garcia, J. C. Caraschi, G. Ventorim, and F. H. A. Vieira, “Trends and challenges of Brazilian pellets industry originated from agroforestry,” Cerne, vol. 22, no. 3, pp. 233–240, 2016.
191 Wood Pellet Association of Canada, “Government organizations contribute to our success: WPAC,” 2017. [Online]. Available: https://www.pellet.org/wpac-news/wpac-government-organizations-contribute-to-our-success.

192 NEB, “NEB - Canada’s Adoption of Renewable Power Sources - Energy Market Analysis - Biomass,” 2019. [Online]. Available: https://www.neb-one.gc.ca/nrg/sttstc/lctrct/rprt/2017cnddptnrnwblpwr/bmss-eng.html.
193 Wood Pellet Association of Canada, “Government organizations contribute to our success: WPAC,” 2017.

194 Government of Canada, “Biomass Program Continues to Support Transition to Renewable Energy in Manitoba - Canada,” 2016. [Online]. Available: https://www.canada.ca/en/agriculture-agri-food/news/2016/07/biomass-program-continues-to-support-transition-to-renewable-energy-in-manitoba.html.

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energy sources enter the market and cover the total power allocation required, between 1 and 65 MW for biomass, pellets and/or briquettes195 .

Additionally, through the implementation of Law 26190 of 2006, a work plan was established so that 8% of energy production should be done through renewable energies in 2016. Subsequently, Law 27191 of 2016 establishes that 20% must be produced in 2020.196 These laws establish certain benefits for the producer of renewable energies such as the early recovery of VAT and the option of accelerated depreciation of assets, in addition to the fact that these are not included in the concepts to be taken into account when paying income tax.197

The Renovate program has awarded 18 energy production projects using biomass (pellets and briquettes) with an installed capacity of 157.7 MW, at a weighted average price of USD 116.5/MWh. These projects have capacities ranging from 2.00 to 37.00 MW. Among these projects, one of the most important, the San Alonso Thermal Power Plant, located in the province of Corrientes, has an installed capacity of 37 MW. This energy is produced from wood chips derived from forest biomass waste.

In addition, in recent years there has been a noticeable increase in interest in producing renewable energy from biomass, on the part of non-governmental entities.
For example, the company PetroChubut implemented pellet production for use and sale in the province of Chubut, in order to encourage the use of this type of fuel by the population. In addition, 8,000-calorie stoves, which consumed between 10 and 12 kilograms of pellets, were installed at various stations in towns throughout the province, in order to reduce local energy demand . ¹⁹⁹

Another example is the "Project for the Promotion of Biomass-Derived Energy," presented by the Argentine organization PROBIOMASA200, which aims to transition from conventional energy sources to biomass. The project seeks to convert industrial boilers, dryers, and ovens to use briquettes as fuel, as well as residential stoves and boilers to use pellets.

The aim is also to encourage the marketing of these types of biofuels.

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[195] MEMyM, “Supply of Electricity from Renewable Sources through Cammesa on behalf of the Distribution Agents and Large Users of the Wholesale Electricity Market.” 2016.

196 P. Schaube, W. Ortiz, and M. Recalde, “Status and future dynamics of decentralised renewable energy niche building processes in Argentina,” Energy Res. Soc. Sci., vol. 35, no. November 2017, pp. 57–67, 2018.

[197] Argentine Congress, “Law 27191 of 2016,” no. 67, p. 1235213, 2015.

198 M. of Energy and Mining, “Ministry of Energy and Mining, Awarded Projects,” 2019. [Online].
Available: https://www.minem.gob.ar/www/833/25897/proyectos-adjudicados-del-programa-renovar

199 R. CHUBUT, “PetroChubut promotes the use of pellets as heating fuel in Argentina,” *EL CHUBUT,* 28-Jul-2018.

200 PROBIOMASA, “Project for the promotion of energy derived from biomass,” 2018. [Online].
Available: http://www.probiomasa.gob.ar.

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Chile

Through its "Energy 2050" policy, Chile has promoted the growth of renewable energy production technologies. The plan, presented in 2014, envisions that by 2035, 60% of Chile's energy will be produced from renewable sources. It also expects that by 2050, this figure will reach 70% .

The production of renewable energy was also promoted through the 4e program, which, through the NAMA project, seeks to encourage the application of technologies for energy production through self-consumption (biomass, among others). These types of energy advancement policies are particularly important for the economic, social, and industrial growth of communities that want to mitigate public spending.²⁰²²

Over the past few years, several programs have emerged focused on increasing the use of biomass as a biofuel to replace conventional energy production technologies such as wood-burning stoves. In 2012, the Chilean government, through the Ministry of the Environment, launched a program called the "Stove Replacement Program in the Residential and Industrial Sectors." It was demonstrated that air pollution in Chile was largely caused by the use of wood-burning stoves. PM2.5 at a concentration of 20 µg/m3 was present in the environment, where approximately ten million people lived. By 2012, up to 4,000 stoves in the residential sector had been replaced. By 2017, this number had tripled thanks to collaboration with local manufacturers. In 2013 pellet production was 29,000 tons per year, while by 2020 production is expected to be 190,000204 .

The program to replace wood-burning heaters, proposed by the Chilean government through the Ministry of Environment, achieved a number of 23,795,205 by 2017 .

Brazil

According to García et al.206, Brazil produced 62,000 tons of pellets in 2014. This does not position it as a strong competitor in this type of energy market when compared to other countries like the United States or Canada, which produce two million tons or more per year. However, a study by the same authors concluded that Brazil has the capacity to produce up to 200,000 tons per year, based on the pine agro-industry and waste products such as

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201 Ministry of Energy of Chile, “Description of the Energy 2050 process,” 2018. [Online]. Available: https://www.opia.cl/static/website/601/articles-76887_archivo_01.pdf.

202Congress of Colombia, “LAW 1715 OF 2014,” *law,* vol. 151, no. 4. pp. 1–46, 2014.

203 JM Rincón and E. Silva, Editors; Bioenergy. Bogotá, 2014.

[204] Bioenergy International, “Chile increases the use of pellets,” 2017. [Online]. Available: https://bioenergyinternational.es/chile-aumenta-el-uso-de-pellet/.

205 University of Chile, “Evaluation of the Wood Stove Replacement Program of the Ministry of the Environment FINAL REPORT,” Santiago, Chile, 2019

206 D. P. Garcia, J. C. Caraschi, G. Ventorim, and F. H. A. Vieira, “Trends and challenges of Brazilian pellets industry originated from agroforestry,” *Cerne,* vol. 22, no. 3, pp. 233–240, 2016.

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sugarcane bagasse. This implies that Brazil has a high potential for expansion in pellet production.

Furthermore, the Brazilian government's ten-year energy plan (PDE) to 2026 shows that in 2017, energy production capacity reached up to 9.4 GW solely from biomass generated in ethanol production processes. This market presents an opportunity to enter with other types of plant waste . ²⁰⁷

Renewable energy growth policies have provided incentives for companies wishing to enter this market. Although at the regional and residential levels, improvements to the efficiency of wood-burning stoves have been more prevalent than their replacement with biomass-fueled systems,208 the industrial-level growth in biomass use is remarkable. For example, FEPAM, a regulatory and environmental protection agency in the state of Rio Grande do Sul, granted an operating license to a 50 MW plant powered by wood pellets produced by Pellco Brasil, which, in addition to generating electricity, will be able to export wood pellets for use in countries such as Japan.209

Lessons learned

Based on the experiences described above, the following lessons learned are highlighted below.

- Pellets and briquettes are modern biomass fuels that, due to their low moisture content and high calorific value, have good combustion and low greenhouse gas emissions.

- Pellets, due to their size, can be easily controlled when feeding into the combustion chamber, thus avoiding unnecessary burning of fuel and therefore achieving efficient use, which does not occur with firewood.
- Pellet production facilitates the use of agricultural waste and pastures near the farmer's home.

- Most countries are implementing pelletizing plants for the production of domestic fuel and the generation of electricity to replace fossil fuels.

- Most developing countries subsidize the implementation of pellet production plants for domestic use and replacement of firewood.
- Charcoal has environmental advantages and can be used in apartments without the need for chimney extractor hoods, but the energy efficiency in its production is low, so it is only used for purposes to increase user comfort such as traditional barbecues and heating.

207 Ministry of Mines and Energy Brazil, “PDE BRASIL 2026.” 2018
208 Ecofogão, “Ecofogão ecological wood-burning stove.” [Online]. Available: https://ecofogao.com/
209 FEPAM, “FEPAM,” 2019. [Online]. Available: http://www.fepam.rs.gov.br/.

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Advantages and disadvantages

Table 24. Advantages and Disadvantages of Implementing Pellets, Briquettes, and Charcoal

Energetic	Advantages	Disadvantages
Pellets	- Low pollutant emissions of - Use of plant waste - Low or zero energy expenditure in carbonization, since a thermal process is not necessary - High density, therefore the combustion chamber volume is not large - High mechanical resistance - Uniform chemical composition facilitates the use of pellet in continuous processes - Easy disposal and transport of briquettes - Low pollutant	- The use of pellets is more for mass systems such as small industries, heating systems, and cannot be used to replace firewood in domestic wood stoves without first making the modifications for the stove, depending on the type of pellet on the market. - High energy costs in biomass drying and compression processes, therefore specialized equipment is needed for its production - High dependence on the particle size of the material; it is necessary to grind the biomass to a uniform size.
emissions.	of - Use of plant waste - Low or zero energy expenditure in carbonization, since a thermal process is not necessary - Ease of disposal and transport	- You may need Binders, usually clays and starches, are used - High energy costs in the biomass drying - High dependence on size of particle of the material - Variable heating power - Low density, compared to pellets - Low mechanical resistance compared to that of pellets - Large particle diameter. - Variable heating power
Charcoal	- Simple biomass pretreatment - High calorific value - High-volume production in batch systems - Controlled CO emissions, CO2, H2S y SOx	- High energy costs in biomass carbonization Difficult arrangement and transport

Source: Prepared by the authors. Corpoema Consultants

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3.5. Electrical Energy

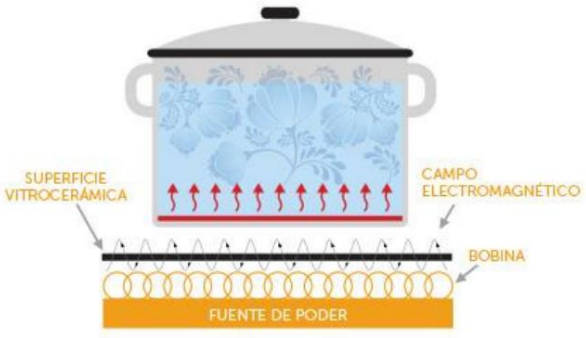
3.5.1. Induction cooktops in Colombia

Definition

An induction cooktop is a ceramic cooktop that runs on electricity and is considered highly efficient because it uses energy much more efficiently than any other type of cooker, since it heats the cookware directly by induction instead of heating the cooktop itself.

The main difference between an induction cooktop and a traditional electric cooktop is that traditional electric cooktops use heating elements to heat the surface and then the cookware, while induction cooktops generate electromagnetic induction and heat the cookware directly in a single step, making them much more efficient and economical. An illustration of how an induction cooktop works is shown below.

Figure 4. Operating principle of an induction stove



Source: Taken from the website: Ecuador Cambia. Link: http://www.ecuadorcambia.com/home.php?op=info

The way an induction cooktop works is through an electronic generator, which supplies energy to a coil that produces a high-frequency electromagnetic field. This electromagnetic field penetrates the cookware (which must be made of ferromagnetic material) and establishes an electric current that generates heat. The heat generated in the cookware is then transferred to its contents. When using cookware sets specifically designed for this type of cooktop, the field does not affect anything outside the cookware, meaning that as soon as the cookware is removed from the cooktop, the heat generation stops.

Currently, there are no programs in Colombia promoting the use of induction cooktops or cooking with electricity. Therefore, the following describes a specific case in the municipality of Mitú, Colombia, involving food cooking, and a case study of induction cooktop implementation in Ecuador, where the experience was not positive, not due to a lack of efficiency.

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not only because of the stoves themselves, but also because of the poor execution of the projects and the lack of information provided to users on how to use them.

3.5.2. Case: Electricity in Mitú

The municipality of Mitú has a population of 32,457 inhabitants for 2018 according to projections by DANE, of which 51.91% are located in urban areas, the multidimensional poverty index is 76% and 51.77% of the population has their basic needs unmet, demonstrating a lag in relation to the national level of 19.66%; it is the capital of the department of Vaupés located on the slopes of the Vaupés River belonging to the Colombian Amazon, due to its forest georeferencing, access and connectivity is difficult, limiting it to air and river routes that affect the levels of development in the area.

The provision of electricity service has been one of the factors affected by access difficulties in the municipality, a fact reflected in the unit costs. According to the Electrical Energy Service Provision Diagnosis presented in November 2018 by the Superintendency of Public Utilities, the cost per kWh is $3,117, the highest recorded in Colombian municipalities. This is explained by the application of Resolution 181891 of 2008 from the Ministry of Mines and Energy, which recognizes the real costs of generation. This value is contrasted with the rates applied in December 2017 in Mitú: 251.53 for the lowest income bracket, 599.94 for commercial customers, and 807.5 for government customers, using the local electricity provider, the Guaviare Department Electric Energy Company, as a reference (Superintendency of Public Utilities, 2018, p. 35). This leads us to affirm that the Colombian state assumes 91.93% of the costs for users in stratum one, 80.75% for commercial users, and 74.09% for official users, which becomes a significant investment in the area.

It is also necessary to consider that in 2017, 1,770 subscribers were registered. During 2018, the company GESTION ENERGETICA SA ESP (GENSA) was carrying out a project to install meters in the municipality to guarantee a sustainable energy scheme. However, the lack of social awareness among the communities prevented the implementation of these meters, making it difficult to measure user consumption levels and establish the true magnitude of the problem. Even with these characteristics, the National University news agency reported that the average monthly consumption of a residential user in Vaupés is 314 kilowatt-hours, which exceeds 200% of the national average, according to data obtained from the census conducted by the University in conjunction with GENSA (Manizales News Agency, 2018).

The absence of natural gas and the prohibition of tree felling in the municipality means that families use electricity for food preparation, excessive or improper use of air conditioning and public lighting, unconventional uses that determine the high levels of electricity consumption in the area (Manizales News Agency. 2018).

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During recent visits to Mitú, it became evident that the issue of installing water meters has generated significant resistance among the population, due to fears of increased bills in the municipality, as current consumption amounts are negligible.

Thinking about solutions to the excessive consumption of electricity in Mitú is a necessary task that involves taking several aspects into account:

- Carry out consultation processes with local authorities and communities, enabling the inhabitants of the population to recognize that it is a problem that requires a change in their daily habits and does not generate resistance that prevents the execution of the process.

- Take into account the cultural traits of the populations to be involved since, as José Iturriaga points out, "Eating is a biological act; cooking is a cultural act," also taking into account the cultural diversity present in the municipality.
- Recognize that domestic activities are mainly carried out by women, which implies identifying them as a central actor in the development of these processes of reducing electricity consumption.

3.5.3. International Experiences

Induction cooktops in Ecuador

Approximately 92% of the liquefied petroleum gas (LPG) used in Ecuador is consumed by the residential sector, of which 78% must be imported. Because the price of the fuel has remained low despite fluctuations in the international market, the state has had to assume a substantial subsidy of USD 700 million per year.²¹⁰ This situation reflects the country's dependence on imported energy, resulting in a significant outflow of foreign currency and negatively impacting, among other things, the trade balance.

With the aim of changing the energy matrix to reduce LPG use in the country, efforts were made to replace LPG-powered stoves with induction stoves that run on electricity. This would allow Ecuadorian households to utilize clean energy sources focused on locally generated energy.

Thus, in early 2014, the Efficient Cooking Program emerged, which sought to introduce approximately 3 million electric induction cookers in the country, with the intention of benefiting the same number of households and solving their dependence on LPG.
These cookers would be accompanied by their respective induction kits, consisting of a set of pots with features suitable for the new technology.

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210 Ministry of Industries and Productivity. Call for expressions of interest and qualification of induction cooker suppliers for the efficient cooking program, 2014

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The Ministry of Electricity and Renewable Energy, through the country's electricity distribution companies, is reinforcing the electrical distribution networks (including service connections, meters, and 220V internal circuits) to enable the use of new cooking technology in Ecuadorian homes. “The state will provide a soft loan to Ecuadorian electricity subscribers who request it, so they can purchase an induction cooktop. Subscribers should visit one of the appliance retail chains and choose an induction cooktop model that qualifies under the Efficient Cooking Program. Subscribers can also purchase a set of induction cookware with financing, and if they choose a two-zone induction cooktop, they can also purchase a rice cooker with financing.” 211

In this way, subscribers who have been selected and acquire the service—that is, those who meet the technical requirements for stove installation—will be immediately registered in the electric companies' databases as beneficiaries of a tariff or tariff incentive that will be applied the month following the purchase of the new stove. It is important to note that the Program's incentives were exclusively for induction cooktops and not for conventional electric stoves. The incentives for acquiring the new stove included government financing for the purchase over a period of up to 36 months, payable through the electricity bill, and up to 80 kWh of free cooking energy per month until 2018.

Results

This program, hailed as a flagship initiative by the previous government, failed to meet expectations. While it aimed to distribute 3 million induction cooktops, only 9.7% (291,413) were actually delivered. By the end of 2018, this figure had risen to 388,762.

Through the National Electricity Corporation (CNEL), the purchase of approximately one million electric stoves was promoted through three contracts with Chinese companies (Midea, Haier, and Kenmec) for a total of 216.4 million soles. Of these, only 145,000 stoves arrived in Ecuador, valued at 23 million soles, according to documents from the Public Procurement Service (Sercop) this year—three years after the contracts were signed. Two of the three contracts were unilaterally terminated by CNEL despite documented breaches of contract by the contractors.

- Similarly, the kitchen kits were not delivered on time, further limiting the use of induction stoves in Ecuadorian homes.

Although the implementation of induction stoves is a beneficial measure for countries that do not have an LPG market that self-sufficiently meets household consumption, its implementation poses a challenge to the application of effective policies.

211 Ministry of Industries and Productivity. Call for expressions of interest and qualification of induction cooker suppliers for the efficient cooking program, 2014

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The monitoring and evaluation of these measures must be effective and must guarantee their long-term sustainability, as well as having progressive subsidies.

3.6. Conclusions on the state of the art

Among the main findings on the substitution of biomass-firewood both in Colombia and internationally, it was found that:

Improved Stoves

- Efficient stove programs lack efficiency tests and expos evaluations regarding the reduction of emissions that the new prototypes would help mitigate with their implementation.

- There must be comprehensive and participatory planning of the projects, which guarantees their long-term success and involves the communities, raising awareness of the importance of acquiring new cooking technologies.

On the other hand, it is important to provide an advisory and technical assistance approach for stove maintenance to ensure the successful installation in the home. It is recommended that the stoves be built by people from the area where the measures are being implemented, as this will ensure the availability of trained personnel to repair and perform necessary maintenance on the new stoves. It should be noted that the successful implementation of efficient stoves in rural areas is not guaranteed due to external factors related to use and handling.

To ensure long-term sustainability, a purely commercial approach is necessary, meaning the creation of a market (unless specific local conditions warrant a different approach) that enables the development of microfinance programs to improve access to cookstoves or the use of carbon financing. This measure is counterproductive, given that access to firewood is primarily free and there would be no incentive to switch energy sources. Therefore, the implementation of new projects must have a commercial focus, but considering the low income levels of families, subsidy schemes will be necessary.

- In rural areas, it has been observed that the arrival of a new fuel does not displace the other, but rather both are used concurrently, such that the wood stove is used for preparing large quantities of food and the gas stove for quicker and simpler preparations.

- The use of firewood as fuel for some communities is not seen as a problem; on the contrary, they seek to enhance it and thus strengthen identity processes, which must be accompanied by other practices that make this activity more sustainable, such as the development of dendroenergy crops, the implementation of efficient stoves, the maximum use of combustion gases, etc.

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- In general, the implementation of efficient stoves has been well received by the communities, and the way the new prototypes are used is perceived positively.

GLP

It can be said that the LPG Program has been well received by the beneficiary population. It was not imposed upon them, and its advantages and benefits are recognized. Thanks to LPG, cooking habits have improved due to the speed of food preparation and the users' perception of an improved quality of life. At the same time, families are opting for a diversified fuel portfolio that includes both LPG and firewood. It is expected that as families gain confidence in the new product, LPG usage will increase.

- LPG implementation programs need better outreach through education and information, given that one of the concerns of communities is the design of the programs and how to provide solutions that take into account the differences and limitations of the communities in acquiring the energy
- Subsidies for LPG, which translate into lower rates, have had a positive impact on users who have been able to access the energy source at lower costs. Therefore, the selection criteria for beneficiaries must be refined to minimize the possibility of including families who do not require LPG or who lack the financial means to continue in a subsidy program.

While public health impact studies require more long-term evidence regarding the benefits of LPG use, it should be noted that the population benefiting from it perceives well-being and an improvement in living conditions. Reducing smoke during food preparation decreases respiratory illnesses.

It is crucial to emphasize that the LPG subsidy should not be viewed as a financial burden, as providing LPG leads to improved health outcomes for poor households, particularly women and children. LPG also enhances household economic productivity by reducing cooking time. Regular use of LPG for cooking should be seen as a social investment by the government.

In the Colombian case, the programs to replace firewood with LPG implemented in the mid-1990s and the first decade of this century have only one ex-post evaluation, conducted by ITANSUCA-GEOINGENIERIA for ECOPETROL in 2007. This evaluation identified the advantages of reducing firewood consumption in households, the benefits of the programs implemented to date, and LPG use by families. Subsequently, and years after the program's implementation phase concluded, no further evaluations have been conducted to assess the long-term sustainability of LPG use among the families who benefited from the program.

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Biogas

- Biodigesters are yet another tool to strengthen the sovereignty of small and medium-sized producers, also considering other aspects, such as the social and solidarity economy, without concrete national policies for the development of a sustainable sector, nor demanding environmental regulations for small and medium-sized producers, nor large budgets for international cooperation. Despite the above, Colombia has managed to connect and consolidate a biodigester sector.

The use of biodigesters improves the living conditions of households. People in rural areas, particularly women, dedicate a significant amount of time to collecting and storing firewood. However, with biogas, it is necessary to dedicate hours of labor to the processes of material handling and biodigester operation, which will be taken into account in the economic evaluation of

the replacement options.

Farmers' knowledge of biogas and access to this information is very limited. They generally show little interest in the initial stages of biogas development.

- The lack of a strategy to finance biogas projects and the absence of credit lines for farmers are preventing the take-off of biogas production.
- Biodigester technologies must be adapted to local climatic conditions; in any case, it is possible to implement biogas production plants in hot and cold climates.

- Biogas is considered a fully renewable source; the production of biogas from biomass waste avoids methane emissions into the environment due to natural processes of decomposition of organic matter.

It also prevents the formation of nitrogen oxides.

- Avoid the use of chemical fertilizers, especially nitrogenous products that are
They are produced by chemical synthesis.
- Financial support is needed for the establishment of a National Biogas and Renewable Energy Program for the rural sector, the construction of demonstration plant sites, and the development of economical and efficient digesters. CONPES 3924 establishes green growth, competitiveness, and social inclusion.

Pellets, Briquettes and Coal

- In the case of Pellets and Briquettes, there are low emissions of pollutants, a use of plant waste and a low or zero energy expenditure in carbonization, since a thermal process is not necessary.

The use of pellets is more suited to large-scale systems such as small industries and heating systems, and they cannot be used to replace firewood in domestic wood-burning stoves without first making the necessary modifications to the stove, depending on the type of pellet available. This is because there are high energy costs involved in drying the biomass and in the compression process, so specialized equipment is required for their production.

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On the other hand, coal has a simple biomass pretreatment which contributes to it being a high calorific value energy source, but it is difficult to obtain the resource given the high acquisition costs in the market.

It is important to change cultural habits in the populations targeted by the project, and to recognize that achieving this will take time (a reasonable timeframe of 5 to 10 years). It is also crucial to evaluate improvements in efficiency, pollution reduction, and the overall improvement in people's quality of life, as these are fundamental standards for achieving the objectives of projects using this energy source.

Electrical energy

As evidenced in the case of Mitú, the installation of water meters has generated significant resistance among the population due to fears of increased utility bills, even though the actual amounts paid for consumption are negligible and include a subsidy. Therefore, it is necessary to propose solutions and engage in dialogue with local authorities and communities to help residents recognize the need for a change in their daily habits and avoid resistance that could hinder the implementation of the process.

- In the international context, while the implementation of induction cooktops is a beneficial measure for countries that lack a self-sufficient LPG market to meet household consumption, its implementation presents a challenge in applying effective policies. The monitoring and evaluation of these measures must be effective and ensure their long-term sustainability, as well as include progressive subsidies. It is worth noting that in countries like Ecuador, the experience was not positive, and that in addition to the cooktop, kitchen utensils are also necessary.

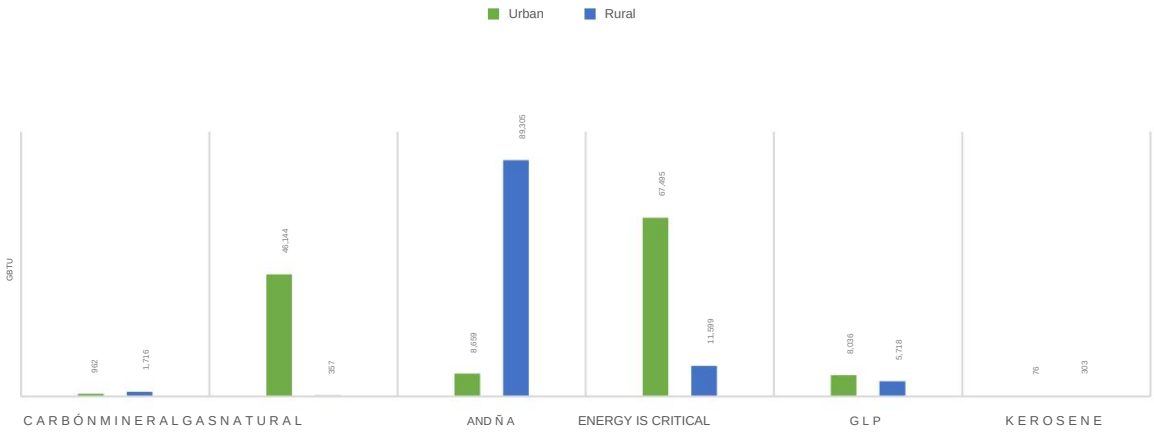
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4. Legal and regulatory aspects

4.1. Background

According to figures from the World Health Organization (WHO), approximately 3 billion low-income people depend on solid fuels (firewood, agricultural crop residues, and coal) that are consumed inefficiently to meet their cooking and heating needs, exposing this population to indoor air pollution that causes nearly 4 million deaths of children and adults annually from respiratory diseases.212 Colombia is no exception, and according to figures from the UPME (Colombian Energy Balance of 2017), firewood is the main energy source consumed in the country's rural residential sector, as shown in the following graph.

Figure 11. Final energy consumption in the residential sector



Source: Colombian Energy Balance of 2017

It can also be observed that electricity and natural gas have become widespread in urban areas; however, in rural areas, electricity is the second most used energy source, far surpassed by firewood. Natural gas still has a marginal presence in rural areas, where LPG distributed in cylinders is the most prevalent and ranks as the third most used energy source.

There are many reasons why electricity and natural gas cannot completely replace firewood consumption in rural areas, and even if they did, their reach would not eliminate it entirely; however, as will be presented later, some financing schemes have been implemented in the country that have allowed the coverage of these services to be extended beyond the possibilities or viability of the market.

212 Stroke (34%), ischemic heart disease (26%), and chronic obstructive pulmonary disease (22%). Pneumonia and lung cancer account for 12% and 6% of deaths, respectively (Source: World Health Organization).

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This section aims to put forward specific recommendations and proposals regarding legal and regulatory aspects required to enable the implementation of the progressive replacement program for firewood in the residential sector in Colombia. To this end, the operation of the supply chains of the energy sources considered as alternatives for the replacement objectives will be reviewed first.

4.2. Energy sources selected as alternatives

The energy sources that have been identified and selected as alternatives for the gradual replacement of firewood in the rural sector are:

4.2.1. LPG distributed in cylinders

Supply and demand

In terms of supply, LPG is a fuel produced in Colombian refineries and is also obtained as a byproduct of the natural gas separation process in natural gas fields within the country. Importation options are also available to meet local demand. Historically, the country has had a sufficient supply of LPG to satisfy demand; however, in recent years shortages have begun to appear, requiring imports at higher prices.

Regarding demand, it has remained stable in recent years, with consumption occurring primarily in municipalities lacking natural gas service due to the absence of pipelines. The widespread availability of piped natural gas, supported by cross-subsidies, has led to the replacement of LPG in urban markets and the subsequent shift of LPG use to rural areas.

Service delivery scheme

The delivery of LPG in cylinders is provided by private companies (some with foreign capital that have consolidated ownership of the distribution business) that face a concentrated market, particularly with a single producer (Ecopetrol). Distributors are responsible for procurement, product collection at the production site or at terminals along the main pipeline system, storage, transport to bottling plants, and subsequent distribution and sale to end users. In other words, private agents (distributors and retailers) are responsible for the commercial activity of purchasing the product, managing the logistics of transport and bottling, and competing in the end-consumer market with other energy substitutes.

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Regulatory aspects

The producer price of the product is regulated by a system focused on sales by the dominant player (Ecopetrol) and subject to monitored freedom for sales by other players. The regulated price is defined as a netback based on a point between export and import parity, thus exposing domestic demand to international price conditions, which largely depend on the price of oil.

The product price is calculated monthly and is also affected by fluctuations in the exchange rate (representative market rate). Distribution and marketing margins can be freely defined by the agents to determine, from the sum of all service components, the retail price of the cylinder, which is defined in pesos per unit of weight ($/Kg) according to each cylinder size.

The distributor owns the cylinder and must add a brand to identify their product. According to regulatory definitions, retail marketing is the activity of delivering LPG in cylinders to the end user's home or to retail outlets. This includes purchasing the bottled product through an exclusive contract with a distributor, when applicable, the freight of the product in cylinders (transport from the bottling plant to the user or consumer), the execution of public service contracts with users, and customer service.

Seeking to generate tools to control informality, the regulation defined the requirements that retail marketers must meet to carry out their activity, among which the following stand out:

- To organize itself into a Public Utility Company in accordance with the provisions of Law 142 of 1994.
- Have a sufficient fleet of cylinder transport vehicles to meet demand your market.
- For warehouses, outlets and points of sale, you must have the certifications of compliance with the Technical Regulation established by the Ministry of Mines and Energy, as well as the required insurance policies (Extra-contractual Civil Liability).
- Have an office for petitions, complaints, and appeals. Also, have a hotline for 24-hour emergency care.
- Having a supply contract for bottled LPG with a single distributor. In other words, there is a limitation on retailers being able to sell cylinders of various brands.
- Define and widely publicize in the market where it operates, the uniform terms and conditions of the contract under which it will provide the service to end users.

4.2.2. Biogas from Organic Waste

Supply and demand

Although there is no consolidated official information on the use of biogas as fuel, some experiences in the country are known (cited in CREG Document 040 of 2016):

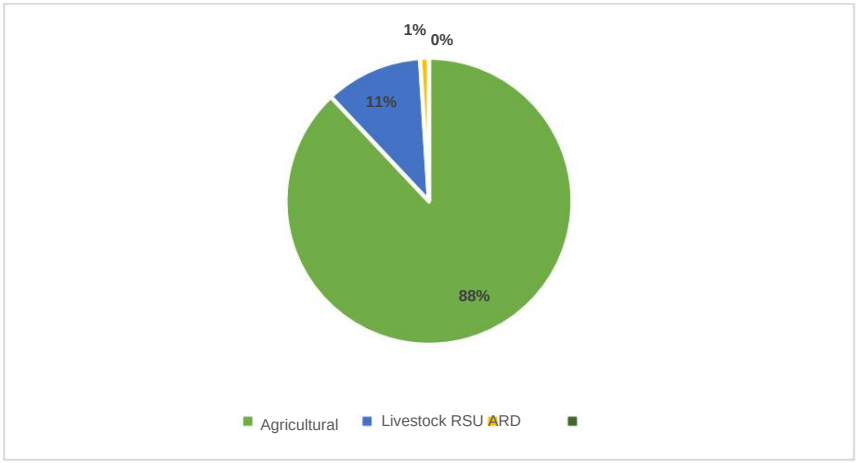
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- San Fernando wastewater treatment plant of EPM in the municipality of Itagüí.

- La Pradera landfill in the municipality of Don Matías in the department from Antioquia.
- Manuelita oil plant in the municipality of San Carlos de Guaroa in the Meta department.

In 2009, the UPME (Mining and Energy Planning Unit) identified potential sources of biogas generation in Colombia, taking into account urban waste and wastewater generation reports. Significant potential was identified in agricultural waste, primarily from sugarcane, rice, corn, and coffee crops. The document published by the UPME on its website, prepared by the National University of Colombia, estimated the biogas potential from urban landfills in major cities; however, no information was found regarding rural areas.

Figure 12. Biogas generation potential



Source: 2016 CREG Paper 040

In summary, biogas can play a significant role by becoming a fuel that can be made available for use in the rural sector.

Service delivery scheme

Private initiatives that have shown evidence of implementing biogas as a fuel have been developed as self-consumption schemes, meaning that all activities from production, treatment and distribution are carried out by the same agent.

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Regulatory aspects

According to the classification of Colombian Technical Standard 3527, biogas is part of the first family of combustible gases, which means that economic regulation is part of the competencies of the Energy and Gas Regulatory Commission (CREG).

Indeed, the CREG, in compliance with its functions, has issued regulations applicable to this energy source: i) CREG Resolution 135 of 2012; and ii) CREG Resolution 204 of 2016.
The most recent resolution is the one that applies, from which the following elements stand out.

First, current regulations allow all activities in the service delivery chain to be carried out by the same entity, meaning vertical integration is permitted in the biogas supply chain. A single company can be the producer, transporter, distributor, and marketer of this energy source, which is an exception for biogas compared to others. This applies both to isolated biogas networks and to networks interconnected to the national transportation system.

The regulation defined the public utility service based on two possible sources:

- Public household gas service with biogas, that is, that which comes from a mixture of gases that are the product of anaerobic decomposition of organic or biodegradable matter.

- Public residential service of combustible gas with biomethane, that is, that which comes from biogas that has been treated to achieve high concentrations of methane, which improve its calorific value and eliminate unwanted components, thus complying with the quality specifications required for natural gas that is transported by pipelines (contained in the Single Regulation for Natural Gas Transportation).

The provision of residential service based on this energy source is subject to the following rules:

- The provider must be a Public Service Company established under the Law 142 of 1994.

The service provider may freely define the service rates based on the monitored freedom scheme. In any case, it must inform the CREG and the Superintendency of Residential Public Services of these rates once they are defined and each time they are updated.

- Design a contract with uniform conditions in accordance with the provisions of Law 142 of 1994.

- Comply with the quality conditions established in CREG Resolution 240 of 2016 (Art. 8), regarding calorific value, Wobbe index, and the content of CH4, H2S, and carbon dioxide. To this end, the producer must install measuring equipment to verify compliance with these standards. In the case of biogas from wastewater, industrial, and urban sources, monitoring of siloxane and halogenated compound content is required. The producer must perform a semi-annual review.

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You must report the measurements of these parameters to the Superintendency of Public Utilities.

- Comply with current technical and environmental standards.

It is important to note that the regulations described apply to public utility service provision schemes, meaning when an agent intends to serve a market or group of end users with this energy source. In cases where biogas generation projects are implemented for self-consumption, these rules do not apply, as the beneficiary is the same: both the service provider and the end consumer.

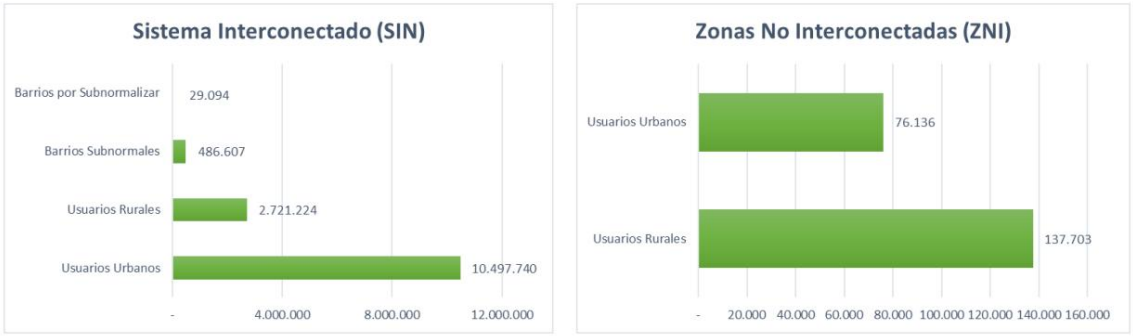
4.2.3. Electrical Energy

Supply and demand

The public electricity service is one of the services with the greatest coverage nationwide, reaching 97.02% in December 2016.213 The largest proportion of users is located in urban centers; however, in rural areas, a significant effort has been made to increase the supply of this service, with the particular help of some funds that have been created for this purpose, as will be detailed later.

forward.

Chart 13. Users of the public residential electricity service



Source: Consultant with UPME data for the year 2016

As can be seen, in isolated areas of the interconnected market, the largest proportion of users are in rural areas where the service is more expensive due to the length of the networks and the low density in their use, which does not allow the economies of scale inherent to this type of infrastructure to be reflected in the service rates, as is achieved in urban areas where the largest number of users are concentrated per unit of network length.

213 Taken from the coverage indicators for the year 2016, the most recent published by UPME in the Colombian Electrical Information System (SIEL).

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Electricity is produced locally, and the country has sufficient installed capacity to generate the energy required to meet demand. From the perspective of expanding supply, there are specific mechanisms in the current regulations (such as the reliability charge) that ensure the installation of new capacity when the country needs it.

From the demand point of view, the behavior is stable and it can be stated that the country is not intensive in the use of energy, which is concluded when analyzing the per capita consumption indicator (in Colombia a per capita consumption of 1,312 kWh/per capita was recorded, compared to 7,819 and 12,994 kWh/per capita of Japan and United States, respectively for the year 2014).

With the enactment of Law 1715 of 2014, Colombia has embraced the introduction of non-conventional sources for electricity generation, particularly solar and wind power. Specifically, the government (Congress and the Ministry of Mines and Energy) has been working on designing mechanisms and incentives to promote the deployment of these technologies, thereby reducing installation costs and increasing the competitiveness of this energy source, making it a viable alternative to meet the population's energy needs.

The Institute for Planning and Promotion of Energy Solutions for Non-Interconnected Zones (IPSE), created in 1999214 from the transformation of the Colombian Institute of Electrical Energy (ICEL), has been developing its social purpose, that is the *"identification, planning and promotion of comprehensive energy solutions, financially viable and sustainable in the long term, for the non-interconnected zones of the country".*

The role of this institution becomes relevant to the object of the present study, insofar as, as already presented, the consumption of firewood is concentrated in rural areas and presumably in the areas that are called "Non-Interconnected Zones" on which IPSE focuses its competence.

During the previous government period, IPSE achieved 75% of the established goal for the incorporation of non-conventional energy sources as a solution for the ZNI, as shown in the following table.

Table 25. Installed Capacity of Non-Conventional Energy Sources in Non-Interconnected Zones (MW)

Year	Cumulative Target	Advance		
		Cumulative % 2015-2018		
2015	0,50	0,60	120%	
2016	2,50	1,66	66%	
2017	5,00	4,53	91%	**75%**
2018	9,00	6,79	75%	
2015- 2018	9,00	**6,79**	**75%**	

214 Decree 1140 of 1999, which transforms the Colombian Institute of Electrical Energy, ICEL, at the Institute for Planning and Promotion of Energy Solutions

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Source: Energy Planning Sub-Directorate, IPSE. January 2019

Similarly, using resources from specific funds (which will be described in detail later), IPSE structured and awarded electrification projects in the ZNI that allowed 5,101 new users to access the electricity service in the period 2015-2018, as shown in the following table.

Table 26. New users with electricity service - ZNI

Indicator Name	Year	Meta Annual	Advance			
			Input Sector	% Contribution ITSELF		% Advance
New users with electricity service located in non-interconnected zones (ZNI) belonging to areas previously without coverage through public resources	2015	843	2583	311%	946	60%
	2016	1687	5409		1987	
	2017	2530	13564		1369	
	2018	3374	4649		799	
	2015- 2018	8434	26205		5101	

Source: Energy Planning Sub-Directorate, IPSE. January 2019

Service delivery scheme:

In an interconnected system, the service is provided to the end user by energy retailers. For residential users, the distributor operates the medium- and low-voltage distribution networks within cities. The retailer manages the energy supply from the wholesale market and subsequently receives the energy through the transmission system in its distribution network for later delivery to the end user.

The marketer handles all customer service, from signing the standard contract terms in accordance with the public utilities law, to meter reading, billing, and collection. This agent calculates service rates and is also responsible for applying consumption subsidies, as will be described in detail later in this document. In non-interconnected areas, different service delivery models apply:

- Free competition zones
- Operators provide service without coverage commitments
- Exclusive Service Areas designed to achieve coverage
- Continental
- Saint Andrew, Providence and Saint Catherine

Regulatory aspects:

The electricity service is regulated by the CREG, both for its provision in the National Interconnected System and in Non-Interconnected Zones. There are

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formulas and conditions for the provision of the service and there is a market structure, both at the level of vertical and horizontal integration.

There is a separation of activities, meaning each agent in the chain must be independent; however, in the case of providing the service in non-interconnected areas, there is a single integrated agent that offers the service and can operate all activities, from generation to marketing to the end user.

At the tariff level, there is a price structure defined by the CREG in most of the service components, as follows: transmission and distribution networks, as well as the intermediation margin of the marketer are regulated values; while the generation price is freely defined in a short-term market (exchange) and by mutual agreement between the agents in the case of long-term supply contracts.

4.2.4. Firewood consumed in Improved Stoves

Firewood is consumed by users in multiple ways, without official records that would allow for specific analysis. It is known that in most cases, the firewood is obtained by the users themselves, who are also responsible for transporting or collecting it from their location to the point of consumption. In many cases, no price or cost is assigned to these activities, and the opportunity cost for those who perform them may be lower than the associated cost, thus obscuring the economic value of carrying them out.

Several public initiatives have been identified to address or reduce firewood consumption for energy needs. These initiatives have focused on offering more efficient cooking appliances with less negative impact on household health. While these initiatives may have value in reducing consumption, they have been implemented in isolation, without integration with other stakeholders.

For its part, the Ministry of Environment and Sustainable Development launched in 2015 a *"comprehensive national program for the use of improved wood-burning stoves in rural and peri-urban areas, aimed at the sustainable and rational use of firewood, contributing to improving the quality of life of the population, alleviating energy poverty, and reducing negative impacts on health and the environment."* Among the objectives of this program are:

- To establish an inter-institutional working group to coordinate efforts between government entities, the private sector, non-governmental organizations, universities and other sectors of society, to promote and ensure that the populations of rural and peri-urban areas of the country, identified as those with the highest consumption of firewood for cooking food, have efficient, adaptable, durable and good quality stoves.

- Strengthen local capacity for the creation of markets for improved stove technologies that serve as a financial sustainability strategy, promoting and expanding the use of improved stoves nationwide.

- To disseminate the use of improved wood-burning stoves and woodlots through the implementation of pilot projects, in accordance with the pattern

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of the use of firewood, taking into account the economic, social, cultural and environmental aspects.

- Promote training and technology transfer in the construction and maintenance of improved stoves, based on the exchange of successful experiences with different technologies already implemented.

In this document, the Ministry cites the definitions of clean and efficient stoves, which meet the following requirements:

- Stoves / fuels that meet efficiency level 2 or higher (Efficiency thermal greater than or equal to 25%);
- Stoves / Fuels that comply at least with Level 3 for indoor emissions (ÿ 0.49 g/min of CO and ÿ 8 g/min of particulate matter)
- Stoves / Fuels that meet at least Level 3 for overall emissions (ÿ 9 g/MJ of CO215 and ÿ 168 mg/MJ of particulate matter PM 2.5)

Likewise, the diagnosis relates an estimate of the number of stoves that have been installed in Colombia from 2009 to 2014 in different regions of the country, as shown in the following table.

Table 27. Number of stoves installed in Colombia. 2009-2014

Region	Number of Greenhouses	Improved stove prototypes
Atlantic	624	Footprints (1)
		Lorena (2)
		Two positions - Natural Heritage (3)
		Rocket (4)
Central	2.030	Footprints
Pacific	1.336	Footprints
		Triangular Hybrid- Natural Heritage (5)
		Linear hybrid - Natural Heritage (6)
		Triangular with refractory blocks- Natural Heritage (7)
		Riscaleña- Fundación Mar Viva (8)
Oriental	3.069	Footprints
		Natura Foundation (9)
		ICA- 1791 (10)
Antioquia	21.179	Footprints
		Corantioquia (11)

215 CO: Carbon monoxide

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		Orchid Park - Natural Heritage (12)
		Catalan Eco-stove. EPM (13)
		Ecoestufa- EPM (14)
Total	28.238	14 Prototypes

Source: National Survey of Improved Stoves, Ministry of Environment and Development Sustainable, 2014

In designing this program, the Ministry of Environment and Sustainable Development cites some State policies that have been issued to seek solutions aimed at resolving the problem of firewood consumption for cooking, among which the following stand out:

- National Policy on Sustainable Production and Consumption of 2010.
- National Forestry Development Plan: approved by Conpes Document 3125 of 2001.
- Air Pollution Prevention and Control Policy of 2010: adopted through Conpes Document 3344 of 2005.
- Forest Policy: adopted through Conpes document 2834 of 1996 identifies the consumption of firewood as one of the main causes of deforestation in the country.
- Comprehensive Environmental Health Policy: adopted through Conpes Document 3550 of 2008.
- National Implementation Plan for the Stockholm Convention.
- Program for the Rational and Efficient Use of Energy and Other Non-Conventional Forms (PROURE): in its indicative action plan adopted by Resolution 180919 of 2010 by the Ministry of Mines and Energy. It was also included in the PROURE of 2017 (UPME Resolution 585 of 2017).

Recently, the Colombian Institute of Technical Standards (ICONTEC) issued a new standard (NTC 6358: Biomass stoves for cooking food) that defines the standards that biomass stoves for cooking food must meet, that is, what agents who manufacture this type of appliance must observe for sale in the national territory.

4.3. Current Funds and Subsidies

In the energy and natural gas sector, different subsidy schemes can be identified, both for supply and demand or consumption. Since the enactment of Law 142 of 1994 (the public utilities regime), which defined the roles of the different institutions that make up the sector and established criteria for setting tariffs that reflect the economic costs of providing the services, principles of solidarity and income redistribution were also defined. These principles were implemented through a cross-contribution scheme from socioeconomic levels 5 and 6, as well as industrial and commercial users, to finance subsidies for users in levels 1, 2, and 3.

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4.3.1. Supply-side Subsidies

The way to subsidize supply has been through earmarked funds financed by fees levied on electricity rates within the sector. In the electricity sector, the following funds stand out, as they are used to finance the expansion of service coverage. In 2018, the FAZNI fund, which is allocated to non-interconnected areas, raised $122 billion, as shown in the following table:

Table 28. Supply-side subsidies

Contribution 2014	2015	2016	2017	2018
PHASE	69.429 76.385 115.959 119.571 122.090			
FENOGES				13.538
MAYOR	79.452 87.402 107.350 124.052 134.625			
FOES	342	186	99.679 124.093 138.637	
PRONE 54.146 52.861 94.507 112.237 121.803				
TOTAL 203.369 216.834 417.495 479.953 530.693				

Source: XM SA ESP (Figures in Millions of Pesos)

PHASE:

The Financial Support Fund for the Electrification of Non-Interconnected Zones was created by Law 633 of 2000; subsequently, Law 1099 of 2006 and Law 1715 of 2014 extended its validity until December 31, 2021, and it is regulated by Decree 1124 of 2008. For each kilowatt-hour dispatched on the Wholesale Energy Exchange, the administrator of the commercial exchange system (ASIC) collects one peso ($1.00) m/cte216, which is paid by the generating agents. Article 190 of Law 1753 of 2015, regarding the Financial Support Fund for the Electrification of Non-Interconnected Zones (FAZNI), establishes:

"(…) *The Financial Support Fund for the Electrification of Non-Interconnected Zones (FAZNI), administered by the Ministry of Mines and Energy, will receive, starting January 1, 2016, the resources collected by the Administrator of the System of Commercial Exchanges (ASIC) corresponding to one peso and ninety cents ($1.90) for each kilowatt hour dispatched on the Wholesale Energy Exchange, of which forty ($0.40) will be allocated to finance the Fund for Non-Conventional Energies and Efficient Energy Management (FENOGE) referred to in Article 10 of Law 1715 of 2014.* (…)"

The FAZNI seeks to finance plans, programs and investment projects in energy infrastructure in non-interconnected zones (ZNI), for the construction and installation of new electrical infrastructure and for the replacement or rehabilitation of existing infrastructure, with the purpose of expanding coverage and ensuring the satisfaction of energy demand in the Non-Interconnected Zones.

216 In December 2014, the value of the FAZNI contribution is $1.15/kWh.

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Plans, programs, and projects eligible for FAZNI funding allocation can be submitted through the following mechanisms

- Public invitations designed by the Ministry of Mines and Energy.
- Initiative of the Territorial Entities, of IPSE, or of the companies providing the electricity service, whether belonging to the National Interconnected System – SIN, or to the Non-Interconnected Zones - ZNI.

Assets financed with FAZNI resources may be contributed to the service provider for operational purposes, in which case their value should not be included in the calculation of the rates to be charged to users. In practice, this condition may not be accounted for or reflected in the rate approval, thus hindering the verification of the transfer of the fund's benefits to end users.

PRONE

The Electrical Network Normalization Program – PRONE was created through Law 1117 of 2006, and seeks financing from the National Government for plans, programs or projects that aim to legalize users and adapt networks to current technical regulations in subnormal neighborhoods located in municipalities of the National Interconnected System – SIN.

The PRONE program is financed with up to 20% of the revenue collected from the Financial Support Fund for the Electrification of Interconnected Rural Areas (FAER), in accordance with Article 1 of Law 1117 of 2006 and the resources provided for in Article 68 of Law 1151 of 2007, with the adjustments established in Resolution CREG-003-2008 and any subsequent amendments or replacements. The System Administrator for Commercial Exchanges (ASIC) will collect the corresponding amount from the owners of the assets of the National Transmission System (STN).

-,

MAYOR

The Financial Support Fund for the Electrification of Interconnected Rural Areas (FAER) was created by Article 105 of Law 788 of 2002 and regulated by Decree 1122 of 2008. This fund allows local governments, with the support of the electricity service providers in their area of influence, to manage prioritized investment plans, programs, and projects for the construction and installation of new electrical infrastructure. The objective is to expand coverage and ensure the satisfaction of energy demand in interconnected rural areas, in accordance with the coverage expansion plans that each network operator will develop and that must be approved by the Mining and Energy Planning Unit (UPME).

For every kilowatt-hour dispatched on the wholesale energy exchange, the Commercial Interchange System Administrator (ASIC) will collect one peso ($1.00) in legal tender, which will be allocated to the Financial Support Fund for the electrification of interconnected rural areas. This fee will be paid by the owners of the STN assets. The Fund is administered by the Ministry of Mines and Energy.

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Royalties

Royalties are also a source of funding for the construction of infrastructure for the provision of electricity service. Under the new General Royalties System, the activities related to assessing the feasibility of projects to be financed with these resources are the responsibility of the Collegiate Bodies for Decision-Making and Administration (OCAD).

In December 2019, the Congress of the Republic approved a constitutional reform to the royalties scheme217, according to which the income of the general system will be distributed as follows:

- or 37% for direct allocations. Of the total resources allocated to direct allocations, a percentage equivalent to 80% will be allocated to producing entities and 20% to river and maritime ports.
- or 28% for the Regional Development Fund and the Compensation Fund Regional;
- or 8% for the Science, Technology and Innovation Fund;
- or 10% for territorial pension savings;
- or 14% for the Savings and Stabilization Fund;
- or 1% for the Monitoring, Tracking, Control and Evaluation System of the Royalties;
- or 1% for the operation of the General Royalties System;
- or 0.5% for the auditing of royalties from the exploitation of deposits and;
- or 0.5% for the riverside municipalities of the Magdalena River and Canal del Dique.

On the other hand, the approved law seeks to redesign the Collegiate Bodies for Administration and Decision (OCAD). Regional impact projects for departments, municipalities, and districts, to be financed with resources from the Regional Development and Compensation Funds, will be defined through regional planning exercises by collegiate bodies for administration and decision. These bodies will include four (4) ministers or their delegates, one (1) representative from the National Planning Agency, the respective governors or their delegates, and a representative number of mayors. It is announced that a law will be enacted to regulate the General Royalties System, defining the operational details of the OCADs. This law may also consider incentivizing the submission of projects or programs to replace the use of firewood for cooking, financed with royalty funds.

For its part, the following funds are identified in the fuel gas sector, which are intended to subsidize supply:

FECF

The Special Development Quota Fund (FECF) was created by Article 15 of Law 401 of 1997, amended by Law 887 of 2004 and Law 1151 of 2007, and is administered by the Ministry of Mines and Energy. This fund is financed with resources from

[217] Draft Legislative Act 040 of 2019

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three percent (3.0%) on the value of the tariff charged for the gas subject to transport, actually carried out, paid by all senders of the National Natural Gas Transport System.

The fund's objective is to promote and co-finance projects aimed at developing infrastructure for the use of natural gas in municipalities and rural areas, primarily within the area of influence of main gas pipelines and those with the highest Index of unmet basic needs.

In natural gas, there are also royalty resources that are allocated to the construction of infrastructure for the provision of the service, and their characteristics are the same as those described in the electricity section.

4.3.2. Demand Subsidies

The current system of subsidies and contributions implemented in Colombia (cross-subsidies) has been developed and strengthened since 1991, the year in which the Colombian Constitution permitted the participation of private companies in the provision of public utility services. Chapter Five of the Constitution also established that public utility services are a responsibility of the State, thus assigning it the duty to ensure their efficient provision to all inhabitants of the national territory. Article 386 stipulated that subsidies could be granted to enable low-income individuals to pay the rates for public utility services that cover their basic needs.

Law 142 of 1994 regulates public utility services and establishes that public utility services are the services of: water supply, sewerage, cleaning, electricity, public telephone, rural mobile telephone and distribution of combustible gas.
In particular:

> *Residential electricity service: This refers to the delivery of electricity from regional transmission networks to the end user's home, including connection and metering. This law also applies to complementary activities such as generation, marketing, transformation, interconnection, and transmission.*

> *Public residential service of combustible gas: is the set of activities ordered to the distribution of combustible gas, by pipe or other means, from a large volume storage site or from a central gas pipeline to the installation of the final consumer, including its connection and measurement.*
> *This law will also apply to complementary marketing activities from the production and transport of gas through a main pipeline, or by other means, from the generation site to the site where a secondary network is connected.*

> *Subsidy: The difference between what is paid for a good or service and its cost, when that cost is greater than the payment received.*

The main demand subsidy granted in Colombia for the provision of residential public services of electricity and gas are the so-called cross-subsidies between users, which were conceived from Law 142 of 1994 and 143

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of 1994, and Decree 3087 of December 23, 1997. In this type of subsidy, higher-income users finance the subsidy granted to lower-income users. Thus, residential users in strata 5 and 6 contribute 20%, and in the gas sector, commercial users contribute 8.9%. Industrial users, as described in Decree 4956 of 2011, were exempt from this contribution. In the case of electricity, industrial users who meet the characteristics described in Decree 2915 of 2011 are exempt from the 20% contribution.

Subsidies for strata 1 and 2 apply to both natural gas and LPG when distributed through pipeline networks in different markets; that is, they are applicable as long as the service is provided to a home, as is the case with network systems.

To manage the scheme of subsidies and contributions, the Solidarity Fund for Subsidies and Income Redistribution – FSSRI was created.

FSSRI

The Solidarity Fund for Subsidies and Income Redistribution (FSSRI) was created by Laws 142 of 1994 and 286 of 1996 as a fund to manage and distribute resources allocated by the fund and the National General Budget, intended to cover subsidies for low-income users in the electricity and gas sectors. The fund is governed by Decrees 847 of 2001 and 201 of 2004, which establish the procedures for the liquidation, collection, and management of subsidies and contributions.

FOES

The Social Energy Fund – FOES was created by article 118 of Law 812 of 2003 and extended by article 59 of Law 1151 of 2007. It is a fund financed with resources from eighty percent (80%) of the congestion rents calculated by the Administrator of the Commercial Exchange System, as a product of the exports of electricity to the countries of the Agreements of the Andean Community of Nations.

The Ministry of Mines and Energy administers the FOES as a special account system, designed to cover, starting in 2007, up to forty-six pesos ($46) per kilowatt-hour of the cost of electricity consumed by users located in hard-to-reach areas, less developed rural areas, and substandard urban areas as defined by the National Government. It does not cover unregulated users.

Law 1450 of 2011 continued this fund with the objective of covering up to forty-six pesos ($46) per kilowatt hour of the value of the electricity intended for the subsistence consumption of residential users of strata 1 and 2 of the Rural Areas of Less Development, Difficult Management Zones, and Subnormal Neighborhoods.

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GLP

With the issuance of Decree 2195 of October 2013, the Ministry of Mines and Energy established a program for granting subsidies for the consumption of LPG distributed in cylinders, indicating that the subsidies will be granted to LPG users who do not have the service of natural gas distributed by networks, and that in order to design and implement the ideal mechanism for it, the MME may carry out a pilot program, from which it would define the application period, the area of influence, the delivery procedure, among others.

Resolution MME 40720 of 2016 establishes the guidelines for granting subsidies for the consumption of LPG distributed in cylinders. The regulation stipulates that the maximum subsidy percentage corresponds to that defined in Decree 2195 of 2013; that is, a subsidy is granted up to the maximum subsistence consumption value defined by the UPME at 14.6 kg per month per household (Resolution 129 of 2007), and may not exceed 50% for socioeconomic stratum 1 and 40% for stratum 2.

The following table presents a list of users who have benefited from this subsidy scheme:

Table 29. Users Benefiting from the LPG Subsidy Scheme

Year	Average subsidy per cylinder (COP$)	Average number of users benefited	Number of times subsidized in the year	Total
2014	15.462	20.798	6	19.295
2015	15.412	22.103	12	40.946
2016	15.424	216.885	12	40.144
2017	15.305	237.260	12	43.575
Average	15.401	124.262		143.960

Source: Taken from the Ministry of Mines and Energy

The subsidy is provided to low-income users in areas of the country where natural gas service is not available through pipelines, specifically the departments of Caquetá, Nariño, Putumayo, and San Andrés, Providencia and Santa Catalina, as well as indigenous communities. Beneficiaries are identified through the SISBEN system, and in the case of indigenous users, the Indigenous Census administered by the Ministry of the Interior is used.

Article 295 of the National Development Plan (Law 1955 of 2019) establishes that subsidies at the national level for combustible gas will be reviewed in order to establish an efficient methodology that guarantees a competitive price and the best consumption signal for users, fiscal sustainability and the relevance of its gradual adjustment, without there being a concurrence of subsidies between them.

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Consumption subsidies in non-interconnected areas

With the enactment of Law 142 of 1994, and as contained in Article 99, the Congress of the Republic determined that the subsidies applicable to the Non-Interconnected Zones (ZNI) would be equivalent to those of the National Interconnected System (SIN), without differentiating the provision of the service in ZNI; and without taking into account that the generation component is based on diesel or heavy full oil (HFO) used in San Andrés Islands and Leticia (Amazonas), which makes the cost of the kWh generated much higher with respect to the SIN.

Until 2004, the provision of service in the ZNI was characterized by: i) low coverage; ii) high costs; iii) deficient management of the service by public service companies, municipalities and governorates; iv) lack of adequate inspection, surveillance and control mechanisms; v) constant assistance from the Nation, with resources for maintenance, replacement of the electrical infrastructure, supply of fossil fuels and investment for the expansion of the service; and vi) difficulty in monitoring subsidies.

With the aim of ensuring continuity in the provision of the service, in 2004 the Institute of Planning and Promotion of Energy Solutions – IPSE – initiated a strategy consisting of taking over the generation parks it owned through agreements with qualified public operators (Gensa and Cedenar), mainly in concentrated markets, such as the capitals of the departments of the ZNI (Leticia, Mitú, Puerto Carreño and Inírida) and other localities on the Pacific coast (Guapi, El Charco, Salahonda, López de Micay).

This strategy achieved in the short term the improvement of the quality of service, the reduction of the costs of its provision, the rationalization of the use of fuels used for generation, the increase of energy hours and the acquisition of technical and economic information of the ZNI, which laid the basis for a public policy for the provision of the electricity service in ZNI, which was defined in the Conpes document 3453 of 2006.

Conpes 3453 of 2006: *"Management Schemes for the Provision of Electricity Service in Non-Interconnected Zones"* determined the actions to improve the provision of the service in ZNI, including: i) ensuring that the private sector was involved in the provision of the service; ii) designing an efficient scheme for the allocation of subsidies; iii) designing an appropriate tariff framework for ZNI; iv) allowing the tariffs paid by users of ZNI to gradually converge with the average tariffs paid by users of the National Interconnected System.

In compliance with the above actions and especially point iv, the Government Nacional obtained powers for the Ministry of Mines to calculate subsidies for the ZNI, which are contained in Law 1117 of 2006 *"By which rules are issued on the standardization of electrical networks and subsidies for strata 1 and* 2"; Article 2 provided that (sic): Add a numeral to Article 99 of Law 142 of 1994, as follows:

> *99.10. Electricity sector subsidies for non-interconnected areas will be granted to users under the conditions and percentages defined by the*

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Ministry of Mines and Energy, taking into account the ability of users to pay in these areas.

The subsidies mentioned in this article may not be disbursed to service providers who have not promptly reported the requested information through the Single Information System, SUI.

With the powers granted to the Ministry of Mines and Energy by Law 1117, the
The Ministry, through Resolution 18-2128 of 2007, established the *"Procedure for granting subsidies in the electricity sector in Non-Interconnected Zones"* and Article 2 stipulated the following (sic):

Determination of the conditions for calculating subsidies. The total amount of electricity sector subsidies for Non-Interconnected Zones will be determined by referencing the socioeconomic stratification of users in the localities within these zones and the difference between the cost of providing the service approved by the CREG (Energy and Gas Regulatory Commission) for said localities, updated for December of the year immediately preceding the respective effective date, and the tariff applied in December of the immediately preceding year to residential users corresponding to the same stratum of the incumbent marketing market of the National Interconnected System (SIN) in the department where the localities are located. If the locality is located in a department that does not belong to the National Interconnected System, the tariff applicable in the capital of the department of the SIN with the 115 kV connection point closest to the capital of the department to which the locality belongs will be used as a reference.

This means that to balance energy payments by users between the ZNI and the SIN, a user in a ZNI located in Inírida pays the same as a user in San José del Guaviare; this measure began to take effect in January 2008, so for example the cost of providing service in Inírida in August 2019 corresponds to Cu = 1.294 $/kWh, a user in stratum 1 pays 227 $/kWh which corresponds to the Cu of San José del Guaviare, that is, the Nation assumes a subsidy of 82.46%.

4.4. Other sources of resources

Carbon tax

Law 1819 of 2016 (Tax Reform) created the carbon tax as an economic instrument to incentivize compliance with greenhouse gas (GHG) emission reduction targets. The tax is levied on fuel consumers based on the carbon content of each energy source (liquid fuels, gas used in refineries and petrochemicals, and LPG for industrial use are all subject to the tax).

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For its part, Decree 926 of 2017 created a carbon tax non-accrual scheme, which consists of allowing the taxpayer to cancel the tons of CO2 in their favor, which is achieved through projects that comply with:

- or Transparency in its implementation
- or Reliability in its methodological development
- or Publicizing its mitigation results

Eligible initiatives include: i) implementation of renewable energy; ii) forestry projects; iii) projects that reduce methane emissions into the atmosphere from landfills; iv) sustainable livestock projects; v) improvement of energy efficiency in boilers; among others.

As can be seen, this non-accrual mechanism currently exists that allows those liable for the carbon tax to "offset" the amount due with the
Funding for an emissions mitigation project could be explored to determine if a project to replace the use of firewood in the country could qualify under the scheme, thus providing an additional source of funding to achieve the goal of reducing firewood consumption. If the criteria are not met, a modification to the regulations could be proposed to allow for the inclusion of funding for centrally implemented national programs (by a designated entity, such as IPSE) aimed at replacing firewood consumption.

Law 1715 of 2014 and FENOGE

In 2014, Law 1715, known as the Renewable Energy Law, was enacted. This law established tax benefits for non-conventional renewable energy generation technologies and created the Fund for Non-Conventional Energy and Efficient Energy Management (FENOGE) to finance renewable energy and energy efficiency projects. This fund receives its resources from a tax levied on electricity sales on the wholesale market (0.4 $/kWh218), but it can also receive resources from public, private, or multilateral and international entities.

Article 287 of the National Development Plan Law (Law 1955 of 2019) allows FENOGE to finance energy efficiency management projects and individual self-generation systems with NCRE (non-conventional sources) in non-interconnected areas and in the National Interconnected System, including the maintenance and replacement of equipment and the transfer of ownership of assets to the beneficiaries of the respective projects.
In other words, FENOGE could be a source of resources to bring solar self-generation solutions to users who currently cook with firewood, thus facilitating the penetration of these technologies as a result of reducing their cost thanks to the contribution of resources from this fund.

[218] Regulated by Law 1753 of 2015, article 190

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4.5. Institutional Aspects

In formulating a policy to replace fuelwood, the need for an institutional framework to support its successful implementation is identified. While the institutional framework is addressed in a separate section, this section analyzes the idea of including the Institute for Planning and Promotion of Energy Solutions for Non-Interconnected Zones (IPSE) as a coordinating element for the policy's implementation.

The IPSE aims to identify, plan, and promote comprehensive, financially viable, and long-term sustainable energy solutions for the country's non-interconnected zones (ZNI). ZNIs are geographic areas where public energy service is not provided through the National Interconnected System (SIN), excluding the Department of San Andrés, Providencia, and Santa Catalina.

Another institutional aspect that can be reviewed in detail is related to the competition for the formation of exclusive service areas that can be awarded to public service providers that have the capacity to acquire commitments to meet the energy needs of delimited geographical areas with a guarantee of exclusivity.

In structuring these areas, it is possible to evaluate the grouping of rural areas with municipalities with higher urban concentrations (which may or may not be ZNI) to define public service rates (electricity and combustible gas) that average the costs of serving the different areas that are part of the exclusive area.

In the case of electricity service in Non-Interconnected Zones (ZNI), there is experience with this type of scheme based on the provisions of Article 40 of Law 142 of 1994 and Article 1151 of 2007, according to which, for reasons of social interest and with the purpose of extending the coverage of public services, the Ministry of Mines and Energy may establish Exclusive Service Areas and the CREG will verify compliance with the reasons that allow the inclusion of exclusivity in the service provision contracts, at the time:

- The configuration of the area must ensure the extension of service coverage and improve the quality of its delivery.

- The area's configuration should aim for the lowest costs by taking advantage of economies of scale and scope derived from geographical location and the endowment of natural resources.

Thus, the legal feasibility of repeating these experiences could be studied by using the powers of the Ministry of Mines and Energy and the CREG to establish exclusive service areas to provide those public services that allow meeting the energy needs of rural areas and ZNIs, with the aim of reducing or replacing the consumption of firewood for cooking in these areas.

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4.6. Public policy

The objective of this consultancy is to formulate an updated program for the progressive replacement of firewood as an energy source in the residential sector in Colombia.
Having completed the diagnostic process and conducted workshops in areas of the country where significant firewood consumption is identified, the following conditions can be highlighted:

i. Firewood is an energy source available in the rural sector with an opportunity cost that is not reflected in the economic value perceived by the user.
ii. The consumption of firewood in the rural sector has a cultural basis.
iii. Alternative substitutes cannot compete with firewood in terms of cost.
iv. Traditional market schemes that regulate alternative energy sources to firewood do not allow these sources to be provided in areas where firewood is consumed.

It is for the above reasons, among others, that for the articulation of a firewood substitution program it is necessary to consider the formulation of a public policy in which the general strategies of a scheme are reflected, which must differ from the traditional regulations and rules applied by the agencies in charge of energy sources that can be substitutes for firewood.

The National Council for Economic and Social Policy (CONPES) is "(…) *the highest national planning authority and serves as an advisory body to the Government on all aspects related to the country's economic and social development* (…)219". CONPES must approve the policies, strategies, plans, programs, and projects of the national Government, and in this case, the firewood substitution program can be viewed as a policy that can be submitted for CONPES approval, indicating the objective of this program and coordinating the participation of the different State entities that have influence and interest in the program. The following are the entities that must be involved:

- Ministry of Mines and Energy: This is the governing body for the country's energy policy. These policy decisions can impact the supply and demand of a fuel alternative to firewood, such as LPG, and also send signals to investors regarding renewable energy solutions for cooking.

- National Planning Department: The DNP is the entity that proposes the National Development Plan to the Government and the Congress of the Republic, analyzes subsidies for public services, and evaluates public investment requirements in order to determine the National Budget. It also harmonizes sectoral plans with the National Development Plan.

- Mining and Energy Planning Unit (UPME): This entity is responsible for contracting the study to define the Firewood Substitution Strategy and also participates in matters such as the National Biogas Plan. As a Planning Unit, it prepares studies and proposals for policy adoption by

219 Taken from the National Planning Department

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of the Ministry of Mines and Energy. It manages and carries out the Colombian Energy Balance (BECO), as an information tool for the design of plans and programs for the replacement of firewood.

- Ministry of Health: Given that improving the health conditions of families affected by firewood consumption is the cornerstone of the long-term firewood substitution strategy, the Ministry's participation in the subsequent stages of the program is essential. This includes raising awareness of the impacts of firewood consumption on family health, particularly for women and children. It is also proposed that the Ministry of Health, through Health Service Providers, promote prevention programs to reduce the impact of firewood consumption on family health.

- Ministry of Environment: The implementation of the strategy, in addition to prioritizing health, must be compatible with the policies established by the environmental authority. Furthermore, the Ministry will develop the National Efficient Stoves Plan, which is one of the elements to be considered in the Firewood Replacement Plan.

- Ministry of Agriculture: Biogas is another element of the long-term Firewood Replacement Plan and therefore the participation of the Ministry as leader of the National Biogas Plan is fundamental in the stages of the Firewood Replacement Plan.

- Ministry of Housing: The national urban and rural housing policy must incorporate the energy concept into its designs in matters such as efficiency and the use of clean energy by the families benefiting from the new housing programs and in general in social interest housing programs and priority interest housing.

- Ministry of National Education and Ministry of Labor – SENA: The Firewood Substitution Program must be accompanied by an educational and training exercise on the following topics: a-) Dissemination campaigns about the effects of cooking with firewood on families, particularly on women and children; b-) Development of educational programs at the primary and secondary levels on energy, efficiency, health impacts, among others; and c-) Structuring programs at SENA for the design and installation of energy solutions on topics such as Efficient Stoves, Biogas, and Renewable Energies aimed at meeting the needs of cooking food and heating homes.

It is recommended that the Ministry of Mines and Energy escalate the replacement program to the National Planning Department (DNP) so that it can be submitted to the National Council for Economic and Social Policy (CONPES) for approval and thus become a national policy. The DNP serves as the technical secretariat of CONPES and is therefore responsible for submitting the program for approval. The DNP also has the authority to coordinate with all involved entities regarding the scope of their participation and their obligations. Furthermore, CONPES can define target and monitoring frameworks for the program, which will be tracked throughout its implementation.

As part of the policy and for the program to be successful, it is necessary to level the selling prices of the energy sources that have been selected as replacement alternatives.

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As mentioned previously, the economic cost of electricity and LPG distributed in cylinders does not allow, under traditional market rules, the extension of coverage of these services to rural areas where most users who currently consume firewood for cooking are located. Consequently, there is no supply available, and if it does exist (such as in municipal capitals of the departments of Vaupés, Amazonas, Guainía, Caquetá, Putumayo, Nariño, Guaviare and Vichada), the price offered to users is unaffordable or, in other words, at prices that exceed the willingness to pay of these users, given their income level.

The most direct mechanism to level the playing field in this situation is through an explicit subsidy granted to the consumer or user, reflected in a lower energy price. While from a fiscal perspective the idea of implementing a subsidy funded by the General Budget of the Nation (PGN) may initially seem impractical due to the lack of an identified source of funding (as is the case with some of the subsidies described in this document), there is an economic justification for proposing it: an economic and social benefit that outweighs the value of the public funding.

Indeed, the estimated benefit of the subsidy would be a reduction in costs to the healthcare system in the medium and long term, stemming from fewer illnesses caused by indoor air pollution in homes where wood is used for cooking. This lower healthcare cost could be further enhanced by a reduction in deaths from respiratory illnesses caused by wood consumption for cooking. The subsidy, which would be economically viable to implement to reduce the retail price of energy sources such as LPG distributed in cylinders and even electricity (from off-grid power generation solutions using non-conventional sources, such as solar photovoltaics), would be capped at the estimated cost to the country's healthcare system (which is financed by the National General Budget) for treating respiratory illnesses caused by wood smoke used for cooking in homes.

This economic and social justification is precisely what can be presented by the DNP for CONPES approval, as part of the strategies of the firewood substitution program.

4.7. Regulatory recommendations

Regarding the alternative of massifying the installation and use of improved stoves that contribute to reducing the consumption of firewood for cooking in rural areas, the following recommendations are presented:

- Implement a monitoring scheme to verify compliance with the recently issued technical regulations for biomass stoves (NTC 6358) by existing and new users.

- Design an incentive scheme for the replacement of wood stoves, in which financial aid is given to the user who changes the stove with which he currently cooks for an improved one that complies with the new technical standards.

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It is recommended to implement a scheme within the "Works for Taxes" program, in which companies with positive net income can contribute resources to a fund designated to provide the aforementioned aid to users. For this purpose, it is suggested that IPSE be the entity responsible for receiving and distributing these funds, in accordance with the proposal in the institutional chapter.

- It is suggested to also study the possibility of using resources from the general royalties system, which, according to its new operation, can distribute resources in all departments of the country, for which it is required that the projects be presented by the territorial entities to the Collegiate Bodies of Administration and Decision - OCAD, who will be responsible for defining, evaluating, making them viable, prioritizing, approving them and designating the executor of these.
- Design programs to replace cooking firewood in urban and rural areas, with a verifiable methodology that allows measuring the benefit in reducing GHG emissions, so that those liable for the carbon tax can "offset" their payment obligation by financing these types of programs.

For electricity and combustible gas, the following is recommended:

4.7.1. Gas natural

Natural gas has limited reach in terms of achieving cost competitiveness in reaching remote and sparsely populated areas, such as rural regions where firewood, which it seeks to replace, is the primary fuel. However, the sector has the Special Development Fund (FECF), whose resources are collected from the sector itself (through the gas consumed in the country) and can be an important source of infrastructure financing for the provision of natural gas service. In this regard, the following is recommended:

- Modify the FECF regulations to prioritize the use of available resources in projects that achieve the replacement of firewood with natural gas.
- Modify the FECF regulations to allow the financing of natural gas distribution assets through virtual pipelines in areas where the replacement of firewood with natural gas is achieved.

- In the regulations, funding would be prioritized for those projects with a majority proportion of users registered in SISBEN within the groups that are classified below the poverty line.

4.7.2. GLP

This energy source is characterized by its flexibility, allowing it to adapt to the location of potential users and thus achieve greater coverage. However, due to its mobility, it is not eligible for demand-side subsidies (or cross-subsidies under Law 142 of 1994), making it less competitive compared to other energy sources delivered to homes that do receive these subsidies.

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Some recommendations to improve the competitiveness of this energy source in the rural sector include:

- Expand the coverage of the subsidy program for LPG distributed in cylinders, implemented through the Resolutions of the Ministry of Mines and Energy, to include all households in rural areas whose members are registered in the SISBEN220 system within the groups classified below the poverty line. It is proposed to utilize SISBEN IV221

  which is an improved version of this system insofar as it is based on a capabilities approach and considers the household's standard of living; that is, the new methodology includes the estimation of an individual targeting index based on the presumption of income.

- Develop regulations that enable the sale of cylinders with technological devices that allow the vaporization of LPG in the cylinder to be interrupted when the balance of money that the user has deposited in their account at the time of purchase of the cylinder is exhausted.

4.7.3. Electrical energy

It is necessary for the CREG (Energy and Gas Regulatory Commission) to define the mechanism by which the costs of generating energy with non-conventional technologies, such as solar panels, are passed on to the end user in the Non-Interconnected Zones (ZNI). This includes adjusting the tariff formula to allow the transfer of these costs when the service is provided to a group of users. This clarity will enable the use of this technology and, consequently, the definition of a tariff to which the corresponding subsidy would be applied.

It is also recommended to modify the regulations for prioritizing projects financed with FAZNI resources, so that funds can be allocated for the implementation of comprehensive energy solutions aimed at replacing the consumption of firewood for cooking, which could include solutions with non-conventional sources of energy generation, that is, self-generation solutions with solar panels, installed individually for each user.

Resources from the General Royalties System could be used, which, as previously mentioned, can be distributed to all departments in the country, provided that the territorial entities structure and submit projects. To this end, a mechanism can be sought to allow these entities to participate in the policy used to design the various crop substitution programs.

4.7.4. Biogas

In the case of biogas, it is recommended to include this technological alternative in the substitution programs that the proposed agency (or IPSE if it were the one) would design and implement.

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220 System for Identifying Potential Beneficiaries – SISBEN

221 CONPES Document 3877: Declaration of strategic importance of the System for Identifying Potential Beneficiaries (SISBEN IV)

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(in this case, according to the selected institutional arrangement). In this way, resources allocated for the purpose of replacing firewood consumption could be used to subsidize the equipment required to produce biogas as an individual solution for rural households or groups of households whose activities generate sufficient waste for biogas production. To allocate the subsidy, this implementing agency would evaluate different alternatives to determine the best option for each case, based on the user's location and consumption characteristics. If biogas is determined to be the best solution, resources would be provided as co-financing to reduce costs for the user and incentivize the technological shift from firewood to biogas.

Similarly, in the case of community solutions or those involving a larger group of users with biogas production located remotely from the user and the corresponding pipeline network installed to transport it to consumption points, these could be structured as projects assigned to companies established for their execution or to natural gas companies interested in developing this type of business. In these cases, as permitted by regulations, the same entity could carry out all activities from production to sale to the end user, including the transport and distribution networks. Public resources from the aforementioned funds could also be allocated to co-finance the development of this infrastructure (the entire chain). In these cases, service providers could apply a demand-side subsidy scheme (using resources from the Solidarity and Income Redistribution Subsidy Fund – FSSRI) to users connecting to this service, thereby reducing its cost and incentivizing the substitution of firewood.

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4.7.5. Subsidies for the demand for electricity and LPG in cylinders

It is recommended to extend the application of the subsidy scheme designed for Non-Interconnected Zones to electricity and LPG cylinder solutions, specifically the recommendations in CONPES 3453 of 2006: "(…) *Management Schemes for the Provision of Electricity Service in Non-Interconnected Zones," which determined the actions to improve service provision in these zones, including: i) ensuring that private initiatives are involved in service provision; ii) designing an efficient scheme for the allocation of subsidies; iii) designing an appropriate tariff framework for Non-Interconnected Zones; iv) allowing the tariffs paid by users in Non-Interconnected Zones to gradually converge to the average tariffs paid by users of the National Interconnected System.* (…)"

The action described in point (iv) is a recommendation that can be included in the CONPES document on firewood substitution (suggested in this study). To determine the subsidy level for LPG cylinder users in areas where firewood consumption is to be replaced, the price of the same cylinder offered in the nearest urban market with a population greater than 50,000 inhabitants should be calculated, along with the price in the location of the target user (a current firewood consumer). The difference would then be subsidized with resources from the National General Budget (PGN). This subsidy would be targeted based on the SISBEN system, meaning it would only be granted to SISBEN groups classified below the poverty line.

This same scheme could be recommended as a subsidy for electricity solutions implemented in areas where firewood is consumed, aiming to attract these users. Again, the subsidy would be targeted based on the SISBEN system and the poverty line criterion, taking into account the guidelines of CONPES 3877 of 2016, entitled *"Declaration of Strategic Importance of the System for Identifying Potential Beneficiaries (SISBEN IV)."*

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5. Cultural, gender and health aspects

This chapter will explore the main cultural factors identified surrounding firewood use in Colombia. The study population consists primarily of farmers, Indigenous people, and Afro-Colombians. This section is crucial for mitigating the risk of cultural clashes that often hinder the implementation of programs such as those aimed at replacing firewood in Colombian households. It also seeks to analyze the changes in daily habits among these populations, such as food preparation and home heating.

Furthermore, considering that, following the declaration made by UNESCO in China in May 2013, *"Culture: Key to Sustainable Development,"* it was reaffirmed *"that culture must be considered a fundamental factor of sustainability, as it is a source of meaning and energy, creativity and innovation, and a resource for responding to challenges and finding appropriate solutions. The extraordinary power of culture to foster and enable truly sustainable development becomes especially evident when a people-centered and locally context-based approach is integrated into development programs and peacebuilding initiatives."*

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Under this approach, cultural factors should not be seen as a hindrance to programs that promote the development of populations, but as factors that open the doors to a series of possibilities to improve the impacts and acceptance of programs and projects for communities.

It is important to mention that culture is a fundamental element for rural development in Colombia, considering that overcoming the use of firewood as the main energy source for cooking and heating in homes is a fundamental objective in this mission, given that indigenous and Afro-Colombian communities are mostly located in these territories.

Additionally, it is necessary to identify the significant inequality between rural and urban areas of the country, marked by their social and economic disadvantages. Studies such as the one included in this plan to replace firewood as an energy source should contribute to reducing the existing inequality gap and should be viewed within the framework of the United Nations' 2030 Agenda and the Sustainable Development Goals (SDGs), to which Colombia has committed. This is not to say that the problem of cooking with firewood is limited to rural areas, since the regional workshops conducted during the consultancy confirmed that firewood use persists in urban areas, especially in smaller municipalities.

It is worth noting that the continued use of firewood in Colombian homes has become a trap, where the limitations are mostly related to access to new alternatives or reduced family incomes to acquire a new one.

222 Unesco. Culture: Key to Sustainable Development. May 17, 2013. Page 2.

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Technology. Women in Colombia continue to use firewood as an energy source, whether for economic or cultural reasons, or due to a lack of awareness of the consequences of its use. This practice continues to make them and their families ill. This situation, which must be overcome, requires economic, social, and cultural solutions that allow us to move toward a scenario where the use of firewood is an option, not a necessity, for households.

5.1. Concept of culture

The initial approaches to the concept of culture, as presented by Grimson in anthropology, are revisited. *"'Culture' was a concept that arose to oppose 'High Culture' and the racist theories that permeated an early stage of anthropology, which sought to establish hierarchical differences between the various human groups encountered by an isolated early Europe upon its first encounter with the world."223* Recognizing its historical origins has led to the identification of pejorative relationships between cultures, which have caused conflicts and violent scenarios, sometimes affecting interventions by states.

For the present study, the concept of culture will be addressed from the perspective of the UNESCO declaration of 2013, as *"an open, evolving and solidly framed vision of culture based on rights and respect for diversity, which allows individuals who freely access it to "live and be what they wish"224,* thus reinforcing their possibilities and human capacities, promoting mutual understanding and exchanges between peoples, in such a way that it is recognized that *"culture is not only a specific domain of life, but is "constructive, constitutive and creative" of all aspects of it, including the economy and development"225; this* dynamic and changing behavior of culture contains a creative potential that could drive the change of daily habits in the population to be intervened.

5.2. Use of firewood as an energy source

Among the questions that arise when evaluating and analyzing firewood consumption in homes is: What are the main reasons that encourage households to use this energy source for cooking and preparing their food?

Revisiting some conversations developed in the regional workshops of this consultancy and conducting a literature review, at first glance the most common responses have been related to the economic difficulties of the populations in accessing other energy sources such as LPG and the difficulties of extending coverage

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223 Barrera Luna, Raul. The concept of Culture: definitions, debates and social uses. Autonomous University of Barcelona. February 15, 2013. Page 3.
224 Unesco. Culture: Key to Sustainable Development. May 17, 2013. Page 3.
225 Barrera Luna, Raul. The concept of Culture: definitions, debates and social uses. Autonomous University of Barcelona. February 15, 2013. Page 14.

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natural gas due to the distances of the communities to urban centers or main networks.

However, it is necessary to consider other reasons that motivate the use of firewood:

- Regarding the tradition of using firewood, Singer points out that *"…firewood is the oldest source of heat used by man, perhaps because it is much more accessible than other fuels since it ignites easily. This accessibility explains why it is still burned in primitive homes today according to traditional methods."* 226

Families who still cook with firewood believe that preparing food with this fuel gives it a particular flavor that they enjoy, and in many cases, it's a determining factor in their choice of dishes. In a testimony given by a rural woman who attended the regional workshop held in Boyacá by the consulting firm, she stated that "when her children visit, she prefers to cook with firewood because they like it, and since she loves them, she gives them the pleasure of preparing meals on the stove." This indicates that the use of firewood is also linked to an emotional aspect.

- It is also necessary to recognize that *"social relations, evident in peasant houses, the visit is often made in the kitchen because it is the most welcoming place"* 227, a phenomenon evidenced in the kitchens, mainly of Cundinamarca, Boyacá and Santander, since it is associated with the heating of the houses and the custom of offering some food to the guests to thank them for their visit.

These three fundamental reasons are key to understanding, in part, the persistence of using firewood in homes, which clearly disregards the fact that using firewood has repercussions on the health of those exposed to the smoke produced by burning the fuel.

From now on, three uses of firewood in Colombian homes will be characterized: The first, related to the preparation of food; the second, for heating homes and the third as a use in rituals, mainly by indigenous communities.

5.2.1. Food preparation

In general we can say that:

> *"Cuisine is a distinctive aspect present in every culture, allowing it to differentiate modalities ranging from the production of ingredients to the way they are transformed through preparation into an expression of culture. A semiotic approach to culinary arts therefore requires identifying the approaches and the*

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226 Cited by Mejía Barragán, Fabiola. Environmental implications of the use of firewood as domestic fuel in the rural area of Usme. Master's Thesis Environment and Development. National University of Colombia.

227 Cited by Sierra, F. Mejia F and Guerro C. 2011. Firewood as domestic fuel in rural areas of Usme. Bogotá. University of La Rioja. Page 31.

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An approach to the culinary cultural act. Every culinary cultural act results from the act of assigning meaning to some aspect of the transformation of food into a meal, as developed by each human group. The main aspects to be addressed stem from establishing methodological boundaries, using cuisine as the primary reference: I) cuisine as a sign or cultural unit; II) cuisine as a language; III) cuisine as a narrative; IV) cuisine as an expression of culture. Each of the above aspects implies a dimension of analysis, precise instruments, and proven approaches to understand cuisine as a space of action conducive to the exercise of semiotics from an analytical perspective . "228

To further complicate the discussion of cooking, arguments developed by anthropology were revisited. Fernández (2004) states that *"cooking, or cooking as an action, is the direct consequence of the domestication of fire";* likewise, it constitutes *"one of the great revolutionary innovations in history, not so much for the way it transforms food—there are many other ways to do so—but for the way it has transformed society."229* Thus, as humankind progressed in its process of adapting to the environment, it modified its food and ways of preparing it, which could lead us to recognize that cooking is a cultural habit that has changed throughout history, allowing us to be optimistic about abandoning the use of firewood as an element that demonstrates our progress as a society.

It is necessary to further state that cooking becomes an act that not only responds to a physiological need but is also a vision of how the subject sees himself and his context, *"the taste230 goes hand in hand with the technical dimension that leads to ingredients, which are combined in a special way, transporting the cultural sense of flavor in that particular human sphere."231,* and it is because both terms synthesize *"the essence of eating as a need of double condition among human beings: physiological and symbolic, elements necessary when transforming ways of preparing food, since not only are habits changed in relation to physiological elements but also symbolic ones that determine ways of seeing and being in the world."*

If we also consider that in both the regional and expert workshops, all participants agree that cooking with wood has a particular flavor that is more

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228 Cid Jurado, Alfredo Tenoch. Culinary semiotics and cultural heritage: Colombian cuisine. 2018. Chilean Journal of Semiotics. Page 10.

229 Cited by Rosas Pachon, Javier. The kitchen: a stage for the reconstruction of memory. Pontifical Javeriana University. Page 11.

230 According to the authors, discussing tastes involves presenting taste as the cultural imprint carried by flavors, textures, and, essentially, all food and drink, which also situates a culinary preparation differently within a human context. Furthermore, from a structuralist perspective—enhanced by the dimension of linguistics—this cultural marker is revealed as something more than what a society defines as valid. According to Lévi-Strauss, tastes are the smallest units of meaning within a particular human context, thus demonstrating their great importance.

231 Delgado Salazar, Ramiro and Delgado Giraldo, Miguel. Of Gustemas and Tecnemas. Identity Culinary Arts of Knowledge and Flavors. Page 128.

Machine Translated by Google

While it is more enjoyable to cook with other technologies, it is important to return to Delgado and Delgado (2010) who refer to the fact that taste *"is essentially related to the issue of culinary identity" of societies, and behind it lies the dimension of its profound meaning in relation to the collective and the individual ."232* With this they establish and show the connection between the formation of taste through culture and society.

On the other hand, the authors show how the technema or techniques in particular can pave the way to *"reviewing everything built by a culture in its temporal dimension to appropriate inputs from the animal, mineral, vegetable and industrial world; transform and combine them, and thus arrive at the construction of its culinary universe."233 .*
Understanding that taste is a constructed element, both individually and collectively, is important in creating awareness campaigns that aim to change this conception about the taste of wood.

Another element of analysis allows us to see the world of ritual understood from the perspective that *"the kitchen is a social phenomenon, which offers ritual spaces, mythical, aesthetic, and esthetic values234; at the same time it provides segments of history and identity which, in their union, make it possible to understand the continuity, transcendence, and evolution of a given culture.* 235

To further ground this analysis in our country, we will delve into two fundamental elements to consider when contextualizing food preparation in Colombia: The first refers to the great variety of dishes in Colombian cuisine and the second to the different types of places dedicated to food preparation or types of cuisines.

5.2.2. Colombian Cuisine

In Colombia, cuisine has been linked to cultural tradition and is as varied as the diversity of the country itself. The Ministry of Culture recognizes the great value and variety of Colombian cuisine, which is why it funds several research projects and compilations on the subject, including the Great Book of Colombian Cuisine, written by Carlos Ordoñez Caicedo. In it, he discusses the rich blend of cultures present in our typical dishes. *"Old and noble sources reveal the combination of dishes: the stews of chuguas, cubios, and hibias that thicken the Boyacá-style mazamorra have a distinct indigenous flavor. Those from the coast almost all reflect the best recipes of Caribbean cuisine and the influence of Spain."*

232 Ibid. Page 128.
233 Ibid. Page 129.
234 This word is an old expression; it refers to the faculty, capacity, or aptitude of feeling or perceiving everything in one's surroundings, characteristic of living beings. It also describes the quality or condition of being sensitive, receptive, impressionable, hyper-aesthetic, or sentimental. Finally, it refers to the particular way or manner of feeling or perceiving.

235 Cid Jurado, Alfredo Tenoch. Culinary semiotics and cultural heritage: Colombian cuisine. 2018. Chilean Journal of Semiotics. Page 10.

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He jumps here and there into the secret for a sweet syrup, the point for toasting the suckling pig or the unequivocal kinship between the Madrid stew and our classic puchero"236 .

This is how we can identify the different cultures that have influenced our country's cuisine and its diversity. Reading this section, we will surely all picture traditional kitchens and cooking with firewood. It is from this image that we must try to discourage the consumption of this energy source and design alternatives that transform and preserve the country's culinary richness, in such a way that we internalize the need to modernize technology to improve the health of our cooks and their families.

In Carlos Ordoñez's book, Colombia is divided into eight subregions: the Pacific Coast, Antioquia and Old Caldas, the Llanos and Amazon, Valle del Cauca, Cauca and Nariño, Santanderes, the Atlantic Coast, Tolima and Huila, and Boyacá and Cundinamarca. The book presents some characteristics of the typical dishes found in these regions and highlights the rich culinary heritage of our country.

Table 30. Characterization of typical Colombian food

COLOMBIAN REGIONS	DESCRIPTION OF TYPICAL FOOD.
Pacific Coast 	Its coastline, with some of the most beautiful inlets and bays in Colombia, fosters the proliferation of fish and seafood dishes that form the basis of the local diet, which they enjoy with plantains and yucca. Coconut milk is also widely used as a cooking ingredient; additionally, there is a wide variety of soups that complement their diet, and finally, they also use free-range meat and chicken, especially for barbecues and picnics. 237
Antioquia and Old Caldas	This region has been very successful in showcasing its cuisine both within and outside the country, to the point that its "mountain dish" is undoubtedly its flagship dish. It consists of white rice, beans, ground beef, chicharrón (fried pork rinds), chorizo or morcilla (blood sausage), a fried egg, avocado, slices of ripe plantain, patacones (fried green plantains), and arepa (corn cake). The paisa (a person from the Antioquia region) can never be without beans, which they might eat daily; nor can the

236 Ordoñez Caicedo, Carlos. The Great Book of Colombian Cuisine. Volume 9. Ministry of Culture of Colombia. Page 23.

237 Ibid. Page 31.

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	Arepas and the mazamorra with which they finish their meals and which can be served with corn (rice) or, simply, the clear broth accompanied by split panela or guava candy"238
Plains and Amazon 	Thinking of the plains brings to mind llanera-style beef, tamales, and yucca. Cachama fish, abundant in several areas, is also part of their diet. The Amazon region lives off hunting and fishing, which it cooks and accompanies with farinha or cassava and casabe, the latter two made from bitter cassava, the staple of its diet, and exotic fruits unknown in the rest of the country. An exotic source of highly nutritious protein is the boa, which is now raised on farms. Locals smoke its splendid loins and prepare excellent dishes . ²³⁹
Valle, Cauca and Nariño 	More a country estate than a city kitchen, it has in the sugar cane, which it cultivates with talent, the basis of a varied production of excellent sweets: its famous manjar blanco, its dragees, its delicate candies. The banana is just one more fruit in the enormous palette that colors its sales stalls, with varied and delicious fruits, such as the Guacarí loquat. Cauca, perhaps the department with the richest variety of dishes in the entire country, preserves Creole diversity. It features the old, skillful, and economical indigenous recipes, as well as the colorful cuisine of the Afro-Colombian community. And let's not forget that peanuts have...

238 Ibid. Page 74.
239 Ibid. Page 112.

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	become one of its fundamental seasonings. Among the truly native things, a very special vegetable stands out: the ulluco, with its enormous variety of preparations, all original and appetizing. Nariño is the land of the Quillacingas, loyal to the Kingdom of Quito, from which they adopted their culinary customs in their gentleness. These customs are still present today, accompanying the traditional cui (guinea pig), which requires more skill to kill than to cook. Their pastries are varied and delicious .
Santanderes  	Their main crop was corn, which they called *aba.* With it they prepared a soup, *suque* (a type of porridge), cakes called *bun* or *tajitafun* (arepas), and a drink called *chicha,* which can be translated as "drink for men." The use of *quibsa* (chili peppers) was also very important in their diet. It was these indigenous people who discovered that big-bottomed ants were not only delicious but also had aphrodisiac properties, a recipe that has been preserved to this day and which we have included in this book. The cuisine of Santander has always been very rich. Santander produces the best pineapples in the country in Lebrija and magnificent sweets in Piedecuesta. Its greatest pride is its arepas, mute (a type of stew), and cabrito con pepitoria (goat stew with pumpkin seeds), which have become popular in restaurants throughout the country. Norte de Santander is very rich in traditional dishes that are proudly preserved: Cúcuta, with magnificent desserts, where they still make currant candy and a wide variety of fruits in syrup; Pamplona, with its masato and a very distinctive cuisine; and Ocaña, with the most interesting and varied cuisine in the region241 .
Atlantic Coast	His products and influences are reflected in his cuisine, which has managed to bring together the

240 Ibid. Page 141.
241 Ibid. Page 277.

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 	a refinement of the international, the local and the African touch of its cheerful cooks. The high quality of its fish and seafood comes together in the market with the excellent meats from the savannas of Bolívar. The people of Cartagena are very fond of desserts, which they find in great quantity and excellent quality in their famous Portal de los Dulces (Sweet Portal). Their signature dish is the Cartagena-style steak. Barranquilla is a culinarily young city, whose cuisine is notable for its rice with mullet and excellent international dishes. Santa Marta, despite its antiquity, has lost some of its old traditions, but talented dishes still stand out, such as its fried mojarras with patacones. La Guajira, a region with fascinating culinary habits, naturally uses a collection of meats more typical of an adventure book than a popular restaurant menu: capybara, tortoise, agouti, turtle, armadillo, iguana and a colorful array of fish and seafood, among which its wonderful lobsters stand out at very affordable prices, allowing modest people to have them for breakfast before picking up the hoe and starting their hard daily work242 .
Tolima and Huila 	Tolima is proud of its lechona (roasted suckling pig) and tamales, dishes prepared with unique touches in every town. The variety of its cuisine is further enriched by dishes made with fish from the Magdalena River, the most famous of which is the delicious viudo (fish stew). The local flour is achira flour, which is used to prepare the famous biscuits243 .
Boyacá and Cundinamarca	Its wide gastronomic variety offers nuances of open competition in Tunja, especially on Thursdays, when popular restaurants serve their cuchuco with pork spine.

242 Ibid. Page 301.
243 Ibid. Page 341.

Machine Translated by Google

 	Pork, a talented and original dish that would delight even the most sophisticated gourmet. Villa de Leyva, with its olive groves and magnificent biscuits, Paipa with its rich traditional cuisine, Tota with its trout, and much more, make this region an unparalleled destination for gastronomic tourism. Bogotá, the great capital, cosmopolitan both within Colombia and internationally, boasts restaurants representing every region of the country and a talented local cuisine exemplified by specialties such as ajiaco and Santafereño hot chocolate, always accompanied by tamales, pericos, almojábanas, colaciones, cakes, and countless other delicacies. Girardot shares with Flandes, Tolima, the honor of producing the best viudo de pescado (fish stew), which, along with the traditional subienda (fish migration), is a true festival for lovers of good river fish . Not forgetting chicha as a traditional drink still used by farmers, especially in family celebrations.

Source: Prepared by the consulting team based on information from Caicedo C. 2012

Taking this journey through Colombian cuisine allows us to appreciate the great variety of its dishes, determined especially by the diverse climates resulting from the country's different thermal zones. This diversity has shaped the crops and variety of animal and plant species available in the various regions, which have formed the basis of Colombian diets for generations. This demonstrates how deeply rooted the preparation of traditional dishes remains in the different regions. Therefore, when designing programs to replace firewood, it is essential to consider this variety, identifying cooking levels, required caloric intake, cooking times, and other factors that will allow families to maintain their eating habits and recognize the existing cultural richness.

5.2.3. Types of kitchens

This section describes the different types of wood-burning stoves identified in the country, which we will group into two main categories: traditional stoves and improved stoves:

244 Ibid. Page 345.

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Traditional Stoves

At the start of this study, before the regional workshops, the prevailing idea was that there was only one type of hearth: the three-stone hearth. However, after fieldwork, the workshops, and a review of the literature, it became clear that there are different prototypes designed according to the realities, conditions, and needs of each region. These variations are important when considering stoves or replacement strategies, as they imply the construction of different technologies, not a single standardized one, as initially assumed. Below are some of the types of hearths identified so far:

1. Three-stone hearth: This hearth is the most common and traditional type used in homes. It is located on the ground and can be found in almost all areas of the country. Sometimes it is a permanent hearth used daily, especially in indigenous communities, and other times it is used for gatherings, barbecues, picnics, or to prepare special dishes such as tamales, chicha, and other foods.
   Undoubtedly, the communities where these stoves are still used are those located in hard-to-reach areas where it is difficult to introduce new cooking technology.

Illustration 5. Traditional three-stone hearth



Source: Photo taken in the Val Hermana Indigenous Reserve. Manaure. Manaure District. La Guajira.

2. High Fireplace: There are some fireplaces with few variations, such as those shown below, which are more permanent. They are built with readily available materials like wood, mud, wattle and daub, among others, and have table-like structures that solve the problem of having to huddle together when stoking the fire. Material is incorporated around the fire to prevent the flame from going out.

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Table 31. High Stoves in Communities

	
Pijao Community Hearth Hilarquito Indigenous Reserve. Coyaima.	Afro-Colombian Community Hearth. Valley District. Bahía Solano.

Source: Photos taken by the consulting team

3. Wood-burning stove in indigenous communities of Vaupés and the Colombian Amazon in general: used primarily for preparing casabe, a type of flatbread made from bitter cassava or manioc, a traditional accompaniment to all meals. It consists of a wattle and daub base and a circular platform. It complements the three-stone stove described earlier.

Illustration 6. Wood-burning stove in the communities of Vaupés



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Source: Photo taken by the consulting team in the community of San Marcos de Fariña. Located in the rural area of Mitú.

4. Wind-assisted stove: used primarily by women in Wayuu indigenous communities. They explained that they decide where to place the stove based on their knowledge of the winds, which is one of the two reasons why stoves are located in open or semi-open spaces—to avoid having to get close to blow on and stoke the fire. The other reason is related to the heat levels inside the houses; if the stove is located in an enclosed space, the heat becomes unbearable. These are small modifications that should be considered when thinking about replacing firewood.

Illustration 7. Fire supported by the wind



Source: Photo taken by the consulting team in the Alapale ranchería. Manaure Municipality.

5. Cylinder Stove: A carpenter in Bahía Solano designed this stove from a gas cylinder. His design was replicated by several families in the area. Appliances and other metal structures or utensils rust very quickly, as do LPG cylinders, which have a shorter lifespan, leading to them being left lying around, creating clutter and litter. This occurs because the community lives very close to the sea. They use discarded or unusable cylinders as stoves. As seen in Illustration 8, the residents cut off the top of the cylinder and attach a grate to hold the pot, and open a cavity on one side at the bottom through which they insert the firewood. Since wood is used as a building material for houses, sawdust and wood shavings in this municipality are used as biomass for cooking. This biomass is free, and given the stove's design, it can be used as an energy source.

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Illustration 8. Cylinder stove in Bahía Solano



Source: Photo taken by the housing consulting team in the Valle district, Municipality of Bahía Solano.

6. Termite Nest Stove: This is a traditional "stove" from the rural area of Montería, consisting of a wall or platform that provides height, and one or more termite nests placed on top, which are used as a firebox. In this way, the fire is concentrated in the nest, which is heat-resistant and acts as an insulator against the external temperature.

Illustration 9. Termite-proof hearth



Source: Luis Naranjo (2012). Blog: The Lord of the Jungle. Retrieved from: http://elsenordelajungla.blogspot.com/2012/

It should be noted that the stoves described above were identified during the consulting team's field visits and that different prototypes may be found in other regions of the country, according to the conditions and needs of each family group.

Therefore, it is necessary to consider creating flexible methodologies in socialization and education programs for the replacement of firewood in the country, which allow for dialogue and agreements with the populations, so that the technologies or fuels introduced are appropriate for households and truly respond to their needs and culture.

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Improved stoves

This section follows the discussion held in the city of Sogamoso, as coal stoves were identified as the most common type of stove used in the area by the rural women who attended the workshop. These stoves have been built and improved for years by local master builders. During the adaptation process, the need arose for a chimney to vent the cooking smoke outside the house, as well as modifications to the oven and water heater, due to the low temperatures in some neighboring municipalities. This type of stove, which uses coal as its fuel, is found in most rural kitchens in the Boyacá, Cundinamarca, and Santander regions.

Table 32. Improved Stove in Duitama-Boyacá

	
Traditional improved stove inside the house	Improved stove flue outside the house

Source: Photos taken by the consulting team on the trip to the Firewood Workshop - Duitama Boyacá.

Developing programs to replace firewood and charcoal in the region is crucial, given that it is one of the priority regions with the highest biomass consumption in the country (sixth, according to the 2018 quality of life survey). Based on their technological advancements in the traditional stove prototype (developed through trial and error) and the success of the measures implemented over the years, the application of new technology must align with the needs already identified by households. It should be noted that a stove in good condition, with proper maintenance and cleaning of the flue, effectively reduces emissions of substances harmful to health. However, according to the firewood laboratory conducted in Duitama, the gases released by the stove inside the home were detected by the gas analyzer; this device triggers an alarm when levels exceed the maximum permitted limits. In this measurement, the levels of formaldehyde and particulate matter 2.5 and 10 were exceeded, results that exacerbate the problem of indoor air pollution and the associated health problems.

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5.2.4. Heating in homes

In an investigation carried out by Mejía (2011), on the determination of the environmental implications of the use of firewood as domestic fuel in the rural area of Usme in the city of Bogotá, it was identified that wood stoves were used for cooking in 33%, for heating and cooking in 56% and as a means of heating in 11%245 .

Adding the last two percentages together, it can be concluded that 67% of families acknowledge using firewood for home heating.

Workshops on firewood production in Cundinamarca and Boyacá demonstrated that using wood stoves for heating increased the temperature in homes, providing a solution to the effects of the cold typical of the Cundinamarca-Boyacá highlands. Thus, heating becomes an additional benefit of using firewood and a key element to consider in the plan to replace this energy source.

5.2.5. Use of firewood in rituals

This section considers the use of firewood for fire production as a fundamental element in traditional rituals of Indigenous communities. It can be used either as a means of communication with gods or higher beings to interpret what happens in their territories, or as a means of communication with God as a form of belief, which they endorse and obey. A fire origin myth from the Huitoto Indigenous communities honors this type of ritual. It centers on the story of a fire woman who brought and showed a flame to the children of a family so they could prepare casabe bread, preventing the bitter cassava, the base of the preparation, from harming their health.246 Hence the community's deep connection to fire in their worldview and the importance of preparing casabe bread as a ritual in honor of fire.

In the wood-burning workshop held in La Guajira, for example, an elderly Wayuu woman commented that *"at night she turns off all the lights and sits contemplating the fire, which gives her ideas for color combinations and images that she can include in the bags she makes."* They recognize fire as a medium that acts as an intermediary to foster creativity in the making of their bags.

One final element worth mentioning regarding Indigenous traditions is the minga. A minga is a gathering where the inhabitants of an Indigenous community carry out a collective activity or task and afterwards share a generous meal prepared by themselves. Fire is seen as a central element, required both for food preparation and for creating ancestral spaces. In this way, conversations or rituals are performed with fire as the central focus.

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[245] Mejía Barragán, Fabiola. 2011. Environmental implications of the use of firewood as domestic fuel in the rural area of Usme. Master's thesis in environment and development. Page 97.

246 Ministry of the Interior Colombia. 2014. Origin myths of indigenous peoples of Colombia. Page 26.

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in order to ask permission from Mother Earth for the planting and preparation of crops and for the curing of seeds.

Like these three examples, there may be many more throughout Colombia, where we can continue to find meanings and symbols that are created in relation to fire and that are fundamental to ways of being and existing in the world.

communities.

For some Afro-Colombian communities, the use of fire is very important in their celebrations. One example is the fire carnival held in Tumaco a week before the start of Lent, a tradition deeply rooted in their cultural identity, in which they celebrate fire and reclaim its use in traditional rituals.

Illustration 10. Carnival of Fire in Tumaco



Source: Taken from the Tumaco Mayor's Office website.

To conclude this section, we can say that the traditional use of firewood as a symbol of fire is among the most deeply rooted customs in communities, primarily Indigenous and Afro-Colombian. When approaching and negotiating with these communities, it is essential to show respect for their beliefs and ultimately assess whether the frequency with which these rituals are performed has effects on their health and living conditions. This will help determine if intervention is truly necessary in these complex situations, which are a matter of cultural identity.

5.3. Analysis of Ethnic Aspects

5.3.1. Ethnicity

It is important to mention that, although this section will delve deeper into the category of ethnicity, the previous section already made mention of the habits of the

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Indigenous and Afro-Colombian communities. This analysis will revisit the concept of race, a concept that precedes that of ethnicity, recognizing that the origin of this category stems from the monotheistic conception that considers that all human beings come from Adam and Eve and that the colonized American and African populations lacked this quality and therefore did not possess a soul.

However , *"as a result of the horrors left by the justification of the supposed existence of races and the hatred that arose between human groups, generating disastrous phenomena such as the Nazi Holocaust and slavery, the concept of race was replaced from certain positions of social thought by the concept of ethnicity to refer to certain cultural characteristics of certain groups247."*

Since the 1991 Constitution, Colombia has been recognized as a multiethnic and pluricultural country, which has *"four ethnic sectors: the indigenous peoples, the Afro-Colombian populations, including the Raizal communities of San Andrés and Providencia and the community of San Basilio de Palenque, in the department of Bolívar and the Romani or Gypsy people248".*

Indigenous peoples

According to the 2018 National Population and Housing Census, the population that identifies as Indigenous numbers 1,905,617 people, representing 4.4% of Colombia's population, belonging to 115 native groups. 78.6% of these Indigenous people live in rural areas of the country, in legally established Indigenous reserves, Indigenous communities, or in territories not legally delimited, places where firewood is still used as an energy source. One identifiable reason for this is the lack of access to alternative energy sources. Electricity coverage in these rural areas is only 66%, compared to 96.3% nationally, and natural gas coverage is only 15.3%, compared to 66.8% nationally .²⁴⁹

The largest number of people who identify as Indigenous are concentrated in the departments of La Guajira, Cauca, Nariño, Córdoba, Sucre, and Chocó. These departments account for approximately half of the country's Indigenous population, and it can be said that they also have the highest concentration of families who still use firewood as their primary fuel.

The experience gained from the two firewood workshops developed in two Wayuu indigenous reserves allowed us to identify the complexity of the discussion and the work itself, from adverse weather conditions and access difficulties to the need for approval and agreement with authorities, the mediation of a local translator, and the conceptions about the use of firewood, starting from

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247 Curiel, Ochi. Gender, race, sexuality: contemporary debates. Conference, Universidad del Rosario. Page 16.
248 Dane. Colombia A Multicultural Nation. Its Ethnic Diversity. May 2007. Page 19.
249 DANE. Indigenous Population of Colombia. Results of the National Housing Census. Page 6.

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The very location of the hearths, based on their knowledge of the winds, gives us an idea of what it means to intervene or design strategies for working with these communities. It's also important to understand that this was merely an exploratory study and that changing habits regarding firewood use requires a more comprehensive approach that includes negotiation and intercultural dialogue to ensure that one or more alternatives to firewood are well-received and adopted by the community.

Afro-Colombians

Afro-Colombian *"refers to the group of people with African ancestry who live in the country and self-identify as such"* 250; according to the *2018 Population and Housing Census,* in accordance with conversations and agreements, three ethnic communities of self-identification are defined within this group: Raizal, Palenquero and black, mulatto Afro-descendant or Afro-Colombian; in Colombia there are 4,671,160 inhabitants, which corresponds to 9.34% of the total national population.

During a tour in Bahía Solano, which included visits to the homes of Afro-Colombian families, the women commented that an indigenous hearth is not the same as an Afro-Colombian hearth; that theirs is not on the floor and that they only light it while cooking, unlike indigenous communities who keep their hearths lit all day long. This presents us with challenges in recognizing the difference between the conceptions of these two peoples and the need to include these particularities in any crop substitution program.

According to the *2018 Population and Housing Census,* the spatial distribution of the Afro-Colombian population is predominantly urban, with 75% residing in municipal capitals and 25% in the rest of the municipalities' territory.251 It is worth noting that during visits to Afro-Colombian homes located in rural areas of the Valle district in the municipality of Bahía Solano, it was observed that even in the urban center, wood-burning stoves are still used, often combined with LPG, as fuel for cooking. This is due, among other reasons, to the economic difficulties these households face, which force them to use firewood at the end of the month because they cannot afford LPG.

Palenqueros: a black community that lives in San Basilio de Palenque, a town remembered for being the first free town in America; they have a collective title, social organization and their own language of African languages, which they mix with Spanish; according to the *2018 Population and Housing Census,* it is reported that there are 6,637 people who self-identify as Palenqueros.

Raizales: According to the Ministry of Culture, in the document *"Raizales, islanders descended from Europeans and Africans" "The Raizal people are the native population of the islands of San Andrés, Providence and Saint Catherine, to avoid confusion with the term "natives"*

250 ICBF. 2017. Differential Approach Model. Page 43.
251 DANE. Black, Afro-Colombian, Raizal and Palenquera population. Population and Housing Census 2018. Page 10.

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Given their indigenous heritage, they call themselves "raizales" (Mow, 2006) and are the product of the mixing of indigenous, Spanish, French, English, Dutch, and African cultures, with British culture predominating, as it was the one that most strongly colonized the Caribbean islands. The raizal culture has its own cultural expressions: the Baptist religion, the Creole language, and its oral tradition.252" Their population is estimated at 30,565 inhabitants, corresponding to 0.06% of Colombians.

ROM

According to the *2018 National Population and Housing Census,* there are 2,649 people belonging to the Romani-Gypsy ethnic group in Colombia, representing a 45.5% decrease compared to the 2015 Census due to greater accuracy in identification and the participation of Romani census takers. 64.1% reside in 12 municipalities: Bogotá, Girón, Cúcuta, Sampués, Pasto, Guamo, Cali, Sabanalarga, Medellín, Ibagué, San Pelayo, and Sahagún. The majority of their population (82.2%) is located in the municipal capitals; however, research reveals little information regarding their food preparation habits. They are primarily a nomadic people, although changes in a globalized world have required some families to become sedentary in order to survive, even though they remain transient. They can be sedentary for years and then become nomadic again.

In relation to their rituals, the Romani people recognize the elementals or spirits of nature, and one of them, *"The elemental of fire signifies transmutation; it is the energy that transforms the solid into the liquid and, in turn, the liquid into the gaseous; it is represented through the Sun, which provides heat and light. Fire has a purifying character, and many of the traditional knowledge of the Romani people have historically developed around its use, among which the forging of metals stands out254."*

Peasant

This category is included because, although this population group is not identified as an ethnic group, it has particularities and ways of life relevant to the present study, since the peasantry is a large part of the target population.

"In Latin America, the rural population is identified by a set of economic, social, cultural, political and environmental characteristics, among which the following stand out: the dual function of agricultural activity, the intensive use of family labor with

252 ICBF. 2017. Differential Approach Model. Page 44.
253 DANE. Gypsy or Romani Population of Colombia. Results of the 2018 National Population and Housing Census. Page 10.
254 Gómez Boas, Ana Dalila. Roma Gypsy People of Colombia. National Planning Department. 2010. Page 79.

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capital limitations and because in general production is intended for self-consumption with some variable ranges that are oriented towards the market" 255

According to the *2018 National Population and Housing Census,* there is a preliminary estimate of 48,258,494 people in Colombia, of which 15.8% are located in dispersed rural areas, *"a figure that has decreased considerably in recent years"* (Rural Mission. 2013. Page 2). Clarifying that this does not allow us to identify or characterize the rural population, and in response to a petition for protection filed by rural associations regarding the need to be characterized, the Constitutional Court resolves *"TO CALL THE ATTENTION of the Ministry of the Interior, the National Administrative Department of Statistics – DANE, the Presidency of the Republic, the Ministry of Agriculture and Rural Development and the Colombian Institute of Anthropology and History (ICANH), which formed part of the opposing party by passive means256, so that, within the framework of their competencies, they prepare studies complementary to the 2014 Agricultural Census and the 2018 Population Census, which allow for a more in-depth delimitation of the concept of "rural," count the citizens who make up that population group and also that, under the leadership of the Rural Affairs Group of the Ministry of the Interior, the current situation of the rural population be identified and support be given to the formulation and monitoring of plans, programs and public policies that allow the realization of the fundamental right to material equality that belongs to the Colombian peasantry"* 257

According to the Rural Mission Foundation, the origin of studies on the peasantry begins
In Europe, as a product of the nationalist and culturalist concerns of the 18th century, which sought to preserve and recover traditional cultures, with peasants as their prime representatives, "peasant studies" began as a form of cultural ethnography. Furthermore, it is necessary to recognize that: globally, the urban population depends on the production of food and raw materials from agriculture; however, despite the fact that most of these products are the result of peasant practices (Fanon 1968), it is estimated that 70% of global poverty is concentrated in rural areas (Golay, 2009). This allows us to acknowledge the existence of a historical debt owed to these populations . ²⁵⁸

According to the research group Studies on Identity at the University of Rosario: *The category of peasant, despite being erased from agrarian policies, remains valid as a cultural category and as a form of self-determination for a large part of the rural population and behind*

255 Rural Mission Foundation. Analysis of different theoretical conceptions of the peasant and their forms of organization. 2013. Study financed by the Ministry of Agriculture. Page 2.

256 Refers to the defendant in a legal process.

257 Supreme Court Ruling on Peasant Protection. STP2028-2018 Case No. 96414. February 13, 2018. 258

Rural Mission Foundation. Analysis of different theoretical conceptions of the peasant and their forms Organizational structure. 2013. Study funded by the Ministry of Agriculture. Page 2.

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This term has a complex heterogeneity of identity that corresponds to local, occupational-productive and social particularities259 .

Bill PL 14-19, which seeks to recognize peasants as subjects of rights, is currently before Congress: “Peasant men and women are subjects of special protection. Peasant communities have a particular relationship with the land based on food production according to the peasant economy and the protection of the environment, as well as shared traditions and customs that distinguish them from other social groups.”

However, it is key to reaffirm that “unfortunately, to date the peasant has not been recognized as a social subject by those who build public policies in the country, generating a crisis at the level of this social group, aggravated by phenomena such as forced displacement, the change in land use and the concentration of its ownership, inadequate use of productive resources and in general the conditions of poverty in the rural sector”260 .

Thus, it is necessary to consider the category of peasant in this study and the creation of firewood substitution strategies becomes fundamental, recognizing their social, political, economic and cultural conditions, in order to develop programs that recognize their particularities and that respond to their demands in relation to the uses they maintain around firewood as energy.

5.4. Analysis of Gender and Health Aspects

To carry out the analysis of gender aspects, it is important to point out the difference between sex and gender. For this purpose, we return to what Ochi Curiel stated in the text *Sex and Gender* (1968) Stoller, where he analyzed the difference between sex and gender to distinguish between sexual identity (gender) and biological sex (sex)261 .

Based on this distinction, it was pointed out that *"sex is shaped by social intervention, therefore, the subordination of women is a product of the relationships that organize and produce sexuality and gender, so the origin of the oppression of women must be placed in the social, not in biology"* 262 .

This clarification is necessary because it allows us to recognize that, if society has constructed this subordination, society itself must work to eliminate it, and undoubtedly designing strategies to replace or reduce the use of firewood as an energy source in the country would contribute

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259 Research Group on Identity Studies. Who are the peasants today in Colombia? School of Human Sciences. University of Rosario. Page 12.

260 Rural Mission Foundation. Analysis of different theoretical conceptions of the peasant and their forms Organizational structure. 2013. Study funded by the Ministry of Agriculture. Page 3.

261 Curiel. Ochi. Gender, race, sexuality: contemporary debates. Conference, Universidad del Rosario. Page 2.

262 Ibid. Page 8.

Machine Translated by Google

to improve the living conditions of women, mainly peasant, indigenous and Afro-descendant women, who correspond to one of the most vulnerable sectors of the population.

Another important topic to include in the analyses is the relationship with the value of domestic work:

> *"Macroeconomic analyses focused on the market and consumption simultaneously portray domestic and care work as an activity that contributes to the country's economy. However, these analyses overlook the responsibility for the living conditions of the women who perform this type of work. Furthermore, there is no systemic view of the economy that integrates subsistence, well-being, the social reproduction of life, and dignified living conditions for women, who are the ones who sustain this reproductive labor. This situation demonstrates the persistence of discrimination that denies women's contributions to the economy and only considers an androcentric view of macroeconomic development. Likewise, this phenomenon persists in the agricultural sector, where inequality and inequity are exacerbated for women, making it one of the sectors with the highest levels of disparity."*
>
> *difficulties for women"* [263]

Using firewood as the main energy source for food preparation is a task
This requires additional effort from families, as it involves more work time due to the need to collect and carry the firewood, an activity supported by men. However, this additional burden falls primarily on women because they are the ones who must transport it home. Analyzing the value of domestic work, cooking with firewood should be valued more highly.

According to data from the Ministry of Health in April 2017, based on a study conducted by the World Bank in 2014, it states that *"The average annual cost of health impacts from indoor air pollution associated with the use of*
Traditional fuels (mainly firewood) in rural areas of Colombia amount to $1.129 billion pesos (0.22% of GDP in 2009). Infant mortality represents 6% of the costs; female mortality represents approximately 78% of the cost. Acute respiratory infection (ARI) in children and adult women and COPD morbidity in adult women represent 16% of the cost." 264

Similarly, it is established that women with chronic domestic exposure to wood smoke in enclosed spaces develop lung disease with clinical and radiological characteristics of chronic bronchitis, obstructive ventilatory impairment, air trapping, severe hypoxemia, and moderate hypercapnia, especially in the

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[263] First Report specifically on rural and peasant women in Colombia. Presented to the 72nd session of the Committee on the Elimination of All Forms of Discrimination against Women – CEDAW. Page 11.

264 Ministry of Health. 2017. The use of efficient stoves and their impact on health promotion in the Colombian context. Page 11.

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Advanced stages of the disease and frequent signs of pulmonary hypertension, even in patients without severe obstruction. The diffusion behavior suggests the absence of significant emphysema, indicating that the flow obstruction is due to airway compromise.

In the study *"Pneumoconiosis due to inhalation of wood smoke,* 1983", it is concluded that *"based on histopathological studies verified by animal experimentation, on clinical findings, on patient biopsies and on the high content of silica and other materials in wood, we verified the hypothesis that the smoke produced by wood, used in cooking, causes in humans a pneumoconiosis-type lung disease265".*

To reinforce the serious health consequences of using firewood, a report from the International Respiratory Societies forum on Biomass Fuels and Respiratory Diseases states that using firewood as the main fuel in homes generates harmful health effects, finding evidence that acute respiratory infections in children and chronic obstructive pulmonary disease (COPD) in women are associated with the use of biomass266 .

Another finding presented in this report is the comparison between COPD related to smoking and COPD related to exposure to biomass. At the Expert Workshop held on October 24, 2019, by the consulting team, pulmonologist Carlos Torres from the Colombian Pulmonology Foundation participated and explained that *"the disease caused by wood smoke is different from that caused by cigarette smoke, but since there is no differentiation between COPD caused by wood and COPD caused by cigarette smoke, the treatment provided to people is the same. They are different diseases because the particle deposition is not the same. In the case of wood smoke, the bronchi of patients appear black, a phenomenon that does not occur in COPD caused by cigarette smoke."*

That is why thinking about replacing firewood in the country is an urgent task, since it not only reveals shortcomings in the rural development process, but also affects the quality of life of people, especially women, children and the elderly, with serious health effects.

5.5. Methodological Recommendations

Taking into account the identified cultural, ethnic, and gender factors, it is recommended to introduce, in addition to technological, financial, regulatory, and economic aspects, these three methodological approaches to community intervention when formulating

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[265] Restrepo. J, Reyes. P and De Ochoa. P. 1983. Pneumoconiosis due to inhalation of wood smoke. Page 13.

[266] Varios Autores. Forum of International Respiratory Societies Report Biomass Fuels and Respiratory Diseases. 2008. Pág. 577.

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and implement the strategies for the progressive replacement of firewood in the country: 1. Intercultural dialogues, 2. Differential approach and 3. Do no harm.

5.5.1. Intercultural Dialogues

The concept put forward by UNESCO is that: *"Equitable exchange, as well as dialogue between civilizations, cultures and peoples, based on mutual understanding and respect, on the equal dignity of cultures, are the sine qua non condition for building social cohesion, reconciliation between peoples and peace between nations267."*
These dialogues become fundamental in Colombia because it has a history of internal armed conflict and social conflict that has generated years of violence and disagreements, which has affected rural areas more cruelly, spaces where the use of firewood as energy is mostly concentrated.

5.5.2. Differential Approach

The mandate of the UN Office for Human Rights states that *"A well-founded conceptualization must begin by saying that the differential approach has a dual meaning: it is both a method of analysis and a guide for action. In the first case, it employs a reading of reality that seeks to make visible the forms of discrimination against those groups or populations considered different by a majority or by a hegemonic group. In the second case, it takes into account this analysis to provide adequate attention and protection of the rights of the population268."*

Regarding the legal framework, Article 13 of the Colombian Constitution establishes that *"all persons are born free and equal before the law, shall receive the same protection and treatment from the authorities, and shall enjoy the same rights, freedoms, and opportunities without any discrimination based on sex, race, national or family origin, language, religion, or political or philosophical opinion."* It also establishes the following specific obligations of the State: *i) to promote conditions that ensure real and effective equality, and ii) to adopt measures in favor of discriminated and marginalized groups, especially those who, due to their economic, physical, or mental condition, find themselves in a situation of manifest vulnerability, and to punish any abuse or mistreatment committed against them.* Therefore, designing a differentiated approach to creating firewood substitution strategies is fundamental, since the target populations have cultural characteristics that determine the use of firewood as an energy source, and these communities are among the most vulnerable and discriminated-against sectors of the population in our country.

The Colombian Institute of Family Welfare, one of the entities that has worked most on the issue in Colombia, understands that: *"The differential approach is a contribution of the law*

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267 http://www.unesco.org/new/es/culture/themes/dialogue/intercultural-dialogue/). Accessed 3 September 2019.

268 https://www.hchr.org.co/index.php/76-boletin/recursos/2470-ique-es-el-enfoque-diferencial. Accessed October 30, 2019.

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International human rights law is applied to the implementation of public policies, seeking to ensure that the focus of social intervention is determined by the characteristics of the social subject and their context. Therefore, it is conceived as a method of analysis and action that recognizes inequalities, risks, and vulnerabilities and values the capacities and diversity of a given subject—individual or collective—in order to influence the design, implementation, monitoring, and evaluation of public policy, with a view to guaranteeing the effective enjoyment of rights, especially the right to equality and non-discrimination. It is implemented through: affirmative action, adaptation of the institutional offering, and development of specialized services.269" Based on these concepts and the characterization carried out above, two main differential approaches should be introduced: ethnic and gender-based.

Ethnic Differential Approach

The inclusion of the ethnic or ethnic and cultural variable is a process that opens opportunities for exchange, innovation, and creativity in a pluralistic Colombian society—culturally and linguistically—that aims for the survival of ethnic groups.270 Thus, the ethnic approach is another component of the differential approach. It relates to ethnic and cultural diversity. This diversity manifests itself in the singularity and, at the same time, in the plurality of identities that characterize the groups and societies that contribute to the richness of humanity. It is a source of innovation, creativity, and the maintenance of necessary biological diversity. In programs that seek to modify daily habits such as food preparation, home heating, and the ritual use of firewood as an energy source, the recognition of ethnic diversity becomes a fundamental axis, since these modifications should not infringe upon their knowledge and, moreover, should be well received in a way that allows for the sustainability of these solutions.

Differential gender approach

In patriarchal and sexist societies like ours, the gender perspective aims to find solutions to problems such as the persistent and growing burden of poverty on women, and their unequal and inadequate access to education and training.
Inappropriate access to health and related services. Violence against women and their limited political participation. The disparity between men and women in the exercise of power.
The persistent discrimination and violation of girls' rights271. This approach is cross-cutting because the situation of women, whether indigenous, Afro-Colombian, or peasant, places them in a high degree of vulnerability in our country. However, this

269 ICBF. 2017. Differential Approach Model. Page 75. 270 National Planning Department. 2012. Guide for the incorporation of the ethnic variable and the differential approach in the formulation and implementation of plans and policies at the national and territorial level. Page 25.

271 https://www.hchr.org.co/index.php/76-boletin/recursos/2470-ique-es-el-enfoque-diferencial. Query performed. October 31, 2019.

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It can also be seen as an opportunity not only to remove women from the use of firewood, but also to empower them and recognize them as active subjects with power in decision-making.

Do No Harm

The concept put forward by Ana Lucia Rodríguez is revisited, who states that *"Do No Harm (DNH) is an ethical approach based on the ancient Hippocratic principle of medicine of "do no harm." Hippocrates affirmed that the first consideration when choosing a treatment is to avoid harm ("Primum non nocere"). From this arises a moral obligation and, in general, a demand for continuous and critical reflection on what is going to be done and on "what has been done," in terms of its principles, consequences, and impacts272."*

This working perspective is nothing more than an ethical reflection on the interventions carried out in communities or population groups and the question of the harm that can be caused, a reflection that starts from some minimum ethical principles that should guide the actions carried out, both by States and by cooperation agencies: dignity, autonomy and freedom.

- Dignity: Every human being is an end in himself; he cannot be reduced to an instrument for other people's ends.
- Autonomy: People are able to define the type and project of life they want to live and also have the ability to provide their own solutions.
  They just need support.
- Freedom: people have the possibility of making decisions for the realization of their life projects.

In conclusion, it can be said that these intervention approaches should be seen as complementary to each other; intercultural dialogues will allow us to move beyond the dichotomy of cultured and uncultured, which has in some cases determined the failure or success of intervention programs, as they end up being rejected or deemed inappropriate.

This is complemented by the idea of recognizing the need to build methodologies that include different references, understanding the existence of particularities that require specific methodological designs.

Finally, we must consider the principle of "do no harm" to continuously evaluate interventions in different areas of the country. This leads us to design strategies that include analyses of the specific contexts related to firewood use, the development of flexible methodologies that allow for adjustments, and the recognition of the need to foster the participation of the beneficiary populations.

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[272] Rodríguez Puentes, Ana Luz. 2009. The do no harm approach. National University of Colombia. Page 7.

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6. Analysis of existing information and figures

According to preliminary information from the *2018 National Population and Housing Census* published by DANE, 96.3% of households in Colombia have access to electricity service and 66.8% of households have access to natural gas service connected to a public network273 .

The 2018 *National Quality of Life Survey* shows that, when asked "What energy or fuel do you primarily use for cooking?" approximately 10.7% of households nationwide responded that they use firewood or wood, and 0.3% that they use charcoal.274 This percentage of households is higher than that reported by the quality of life survey between 2014 and 2016. Therefore, in order to establish benchmarks and standardize information, this section analyzes the existing figures and considers two scenarios of firewood consumption.

6.1. Secondary sources

Among the existing sources in Colombia, two main official sources are identified for determining firewood consumption scenarios in the residential sector: the National Survey of Quality of Life of the DANE; and the Colombian Energy Balance (BECO) of the Mining-Energy Planning Unit.

6.1.1. National Quality of Life Survey

The National Quality of Life Survey (ECV) is a research project carried out by the National Department of Statistics (DANE) in order to collect information "on different aspects and dimensions of the well-being and living conditions of households, including topics such as: access to public, private or community goods and services; health, education, comprehensive care of children under 5 years of age, among others."275

This information is an input for conducting analyses on the factors that explain the living standards that exist in Colombian society.

The first applications of the quality of life survey in Colombia took place in 1991 and 1993, driven by the need to characterize the poor population beyond the instruments used exclusively for poverty measurement. In 1997, the National Quality of Life Survey (ECV) was conducted for the first time, following the methodology for measuring living standards (The Living Standards Measurement Study - LSMS) promoted by the World Bank, with national representation and data from eight regions of the country. Following nationwide applications in 2003 and 2008,

273 DANE (2019). How we live. Retrieved from: https://www.dane.gov.co/index.php/estadisticas-por-tema/demografia-y-poblacion/censo-nacional-de-poblacion-y-vivienda-2018/como-vivimos
274DANE (2019). National Quality of Life Survey 2018. Retrieved from: https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2018#informacion-nacional-regional
275 DANE (2019). General Methodology National Quality of Life Survey - ECV. Retrieved from: https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2018#informacion-nacional-regional

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It was determined that the survey should be applied at intervals of less than five years, so it was carried out in 2010 and from that year onwards it has been carried out annually.

In 2018, the application of the ECV had departmental coverage for the first time, so there are results for the national total, and by departments, capitals and populated centers - dispersed rural.

Over the years, special modules (permanent or periodic) have been included and modifications have been made to the form to address the information needs of multidimensional poverty measurement, gender-sensitive studies, monitoring of the Millennium Development Goals (MDGs), and the Sustainable Development Goals.

Sustainable Development Goals (SDGs), *"as well as other recommendations or requirements of national and international entities"* 276

The objective of the ECV is "to obtain information that allows analysis and comparisons of the socioeconomic conditions of Colombian households, which make it possible to monitor the variables necessary for the design and implementation of public policies." It is made up of 11 permanent modules:

i. Housing details,
ii. Household services,
iii. Characteristics and composition of the household,
iv. Health,
v. Comprehensive care for children under 5 years of age,
vi. Education,
vii. Workforce,
viii. Information and communication technologies (ICTs)
ix. Child labor,
x. Ownership and financing of the dwelling occupied by the household,
xi. Household living conditions and property ownership.

The "Household Services" module includes questions about "the availability of rooms in the home, the quality of services available, the adoption of responsible water and energy consumption practices, among others." Questions that can provide information about firewood and other energy consumption for cooking in Colombian households include:

- What energy or fuel do they *primarily* use for cooking?
  - or Electricity
  - Natural gas connected to the public grid
  - or Petroleum, gasoline, kerosene, alcohol
  - o Propane/LPG gas (in cylinder or pipette)
  - or Mineral coal
  - or Firewood, wood

276 DANE (2019). General Methodology National Quality of Life Survey - ECV. Retrieved from: https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2018#informacion-nacional-regional

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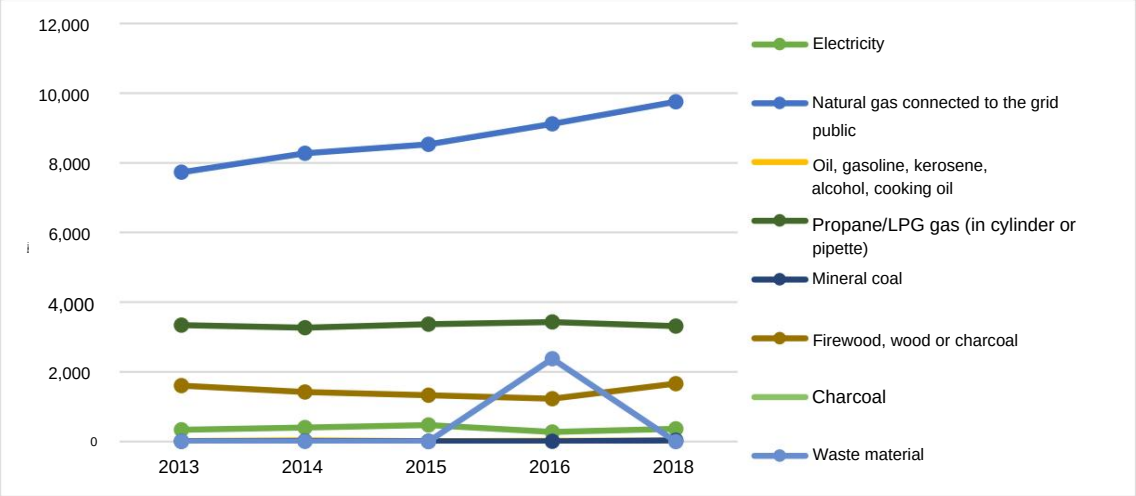
or Charcoal
or Waste material

- Additionally, do they use any other fuel for cooking?

Or Yes: Which one?
or No.

Regarding the question about the energy source primarily used for cooking in Colombian households, a graph with this information between 2013 and 2018 is presented below, according to information from the ECV of DANE.

Figure 14. Energy or fuel used mainly for cooking in households in Colombia between 2013 and 2018 according to the ECV



Source: ECV 2013, 2014, 2015, 2016, 2018.

Overall, energy consumption trends have remained consistent, although there has been an increase in households using firewood in the last year, probably related to the expansion of the household sample this year.

In the 2016 ECV (National Survey of Living Conditions), the sample size was 22,893 households; in 2017, it was 13,034 households; and in 2018, it was 89,522 households. Data from 2017 are not included in the graph because they are only representative of all the country's urban centers, while in 2018, municipalities previously inaccessible were included. Therefore, the sample size allows for the disaggregation of information for the country's 33 departments and between urban centers and rural areas, except for Bogotá and San Andrés, where disaggregation is only available for urban centers.

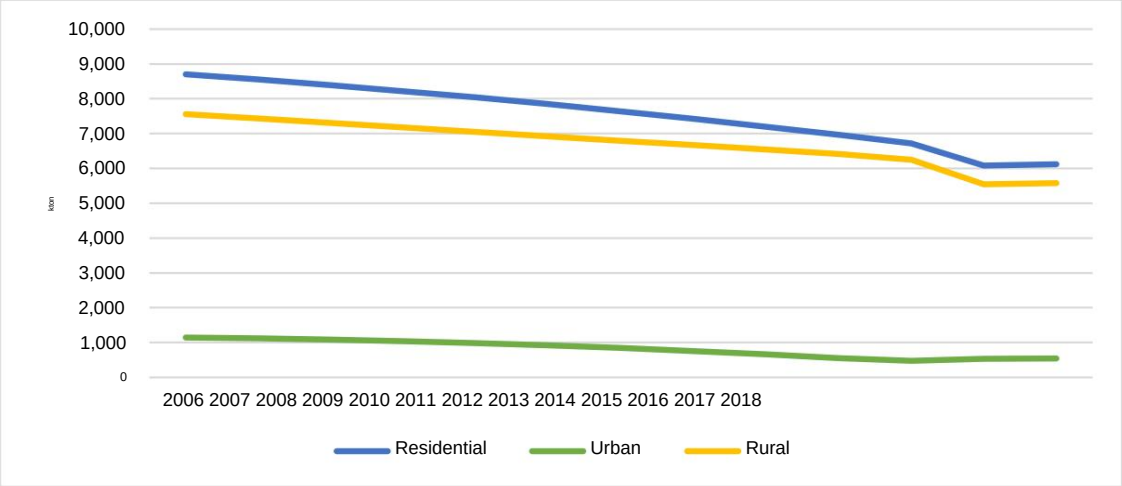
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6.1.2. ALLEY

The Colombian Energy Balance (BECO) "summarizes the information on the production, transformation and consumption of energy in the country, expressed in original physical units and common energy units, corresponding to a particular calendar year"277 .

For the purposes of this study, the BECO (Basic Energy Consumption Index) includes the consumption of each energy source (primary or secondary) by consumption sector. In the case of the residential sector, it is evident that firewood consumption has decreased over time, although the reduction has been greater in the urban residential sector (53% between 2006 and 2018) than in the rural residential sector (26% between 2006 and 2018).

Figure 15. Firewood consumption in the residential sector (urban and rural) in Colombia between 2006 and 2018, according to BECO



Source: BECO 2006 to 2018.

Currently, UPME is conducting a study with the objective of "Designing a methodology and a mathematical model to estimate the firewood consumption of rural households in the country in order to use this information for compiling the Colombian energy balance and to be used to monitor and carry out energy economic policy."

Once this study is completed, products will include: a proposed set of questions to be included in a module on firewood consumption for residential use that can be included in surveys or censuses, a methodology and mathematical model of firewood consumption for the rural residential sector, the results of firewood consumption in the country, and by household, at the national and regional level in the rural area for the years 2010-2019, and a technical document with the analysis of the review of the 2010-2017 historical series of firewood consumption for rural residential use of the Colombian Energy Balance, evaluating its economic consistency.

277 UPME (2019). Colombian Energy Balance – BECO. Retrieved from: https://www1.upme.gov.co/InformacionCifras/Paginas/BalanceEnergetico.aspx

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6.2. Firewood consumption laboratory

6.2.1. Background

In Colombia, several studies have been conducted to determine the per capita consumption of firewood for cooking in rural areas of the country; some have involved field measurements, while others have estimated these values. A study carried out by Ocaña and Linares278 in the rural area of the municipality of Encino (Santander) showed that 79% of the families evaluated used firewood exclusively as cooking fuel, and on average, a family consumed 6.2 tons annually, representing a per capita consumption of 2.9 kg/day; this data was collected through community surveys in 2005.

In Colombia, according to the study conducted by Silva and Nakata in 2008,279 daily consumption per person is assumed to be 11.2 kg of wood in non-interconnected areas with fewer than 50,000 inhabitants. In 2010, Aristizábal280 evaluated the firewood consumption of three types of improved stoves installed in the municipality of Encino (Santander) and compared them with the traditional stove using the Controlled Cooking Test (CCT); the result showed a firewood consumption of 2.6 kg/day for the traditional stove per meal.

A 2014 study in eastern Antioquia281 reported that families consumed approximately 27 kg/day of wood, in about 8 hours that they had their traditional hearth lit; this equates to 5.5 kg/day per person. The data were measured in the field, for which 108 families were selected and the total amount of dry wood consumed by each family for food preparation was weighed.

In the document by Arístizabal, 2014, it is discussed that the region's consumption is high compared to reports for traditional or open stoves, which report 4.64 kg/person/day282 under the Kitchen Performance Test (KPT) protocol, and 4.8 kg/person/day283 found through surveys of the community of San José de Suaita.

278 Ocaña Figueroa, RE, & Linares Castillo, É. L. (2005). Dendroenergetic plant species used by the inhabitants of Encino, Santander, Colombia. Acta Biológica Colombiana Vol. 10, no. 1, 97.

279 Silva Herran, D., & Nakata, T. (2008). Renewable technologies for rural electrification in. International Journal of Energy Sector Management, 139 - 154.

280 Aristizábal, J. (2010). Improved cook stoves and fuelwood lots: an alternative of fuel self-supply for small farmers dependent of oak forests in the Colombia eastern cordillera. Colombia Forestal, 245–256.

281 Ramírez Quirama, JF, & León Tabor, A. (2014). Firewood consumption in traditional stoves in peasant families of eastern Antioquia. Producción + Limpia, 99•114.

282 Aristizábal Hernández, JD (2014). Validation and comparative evaluation of the efficiency of an improved wood stove under controlled conditions and field test. Informador Técnico (Colombia) Volume 78, 12-24.

283 Valderrama, E., & Linares, É. L. (2008). Use and management of firewood by the peasant community of San José de Suaita (Santander). Revista Colombia Forestal Vol. 11, 19-34.

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Table 33. Compilation of studies on per capita firewood consumption in improved stoves in Colombia

Author	Year	Area	Data collection technique	Per capita consumption (kg/da
Ocaña and Linares	2005	Encino (Santander)	Surveys	2,9
Valderrama and Linares	2008	Suaita (Santander)	Surveys	4,8
Silva and Nakata 2008		ZNI (Colombia)	Taken	11,2
Aristizábal	2010	Encino (Santander)	Direct measurement — PCC	2,6
Ramirez and Lion	2014	Eastern Antioquia,	Direct measurement	5,5
Aristizábal	2014	Encino (Santander)	Direct measurement – KPT	4,6

Source: Prepared by the author based on the sources consulted.

Similarly, internationally, consumption in rural communities in Peru in 2010 was reported to be 3 kg/person/day284, and in Mexico, the average consumption of firewood ranges between 2 and 3 kg per person per day285 according to a 2006 study. On the other hand, the World LPG Association (WLPGA) estimated that annually a person in the world consumes 400 kg of firewood for cooking286, which is equivalent to 1 kg/person/day, the lowest of the calculations presented so far.

6.2.2. Measurement protocol

The method used in laboratories to estimate per capita firewood consumption is known as KPT (Kitchen Performance Test); it consists of calculating the consumption of firewood and other fuels used in a stove or hearth, under everyday cooking conditions.
Breakfast, lunch, and dinner for household members. The information is collected through a survey and on-site measurements. The survey is qualitative and gathers information on the user's perception of the wood-burning stove, hearth, and/or improved stove.

The measurement of firewood consumption is carried out using the KPT method, which consists of weighing the cooked ingredients and the firewood, measuring the moisture content of the firewood, counting the number of people who will consume the food cooked on the stove, and adding information such as

284 Muro, HA, Paredes, JN, & Bravo, CA (2010). Environmental impact produced by the use of firewood in the Vilacota-Maure regional conservation area of the Tacna region. IV Latin American Conference on Solar Energy (IV ISES_CLA) and XVII Peruvian Symposium on Solar Energy (XVII-SPES) (pp. 1-5). Cusco: Jorge Basadre Grohmann National University.

285 Masera, CO, & Fuentes, GA (2006). Current state of bioenergy in Mexico. Bioenergy in Mexico. A catalyst for sustainable development. Conafor, Mexican Bioenergy Network, 1-10.
286 WLPGA. (2018). Substituting LPG for Wood: Carbon and Deforestation Impacts. Neuilly-sur-Seine, France: WLPGA.

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Air quality inside the kitchen was measured using a PM 2.5 and 10 particulate matter detector and ambient CO2 levels. Energy losses from wood combustion were measured using a gas analyzer installed in the stove's chimney. Several regions of the country were selected to determine the most commonly used types of firewood, taking into account geographical, technological, and cultural conditions.

6.2.3. Results

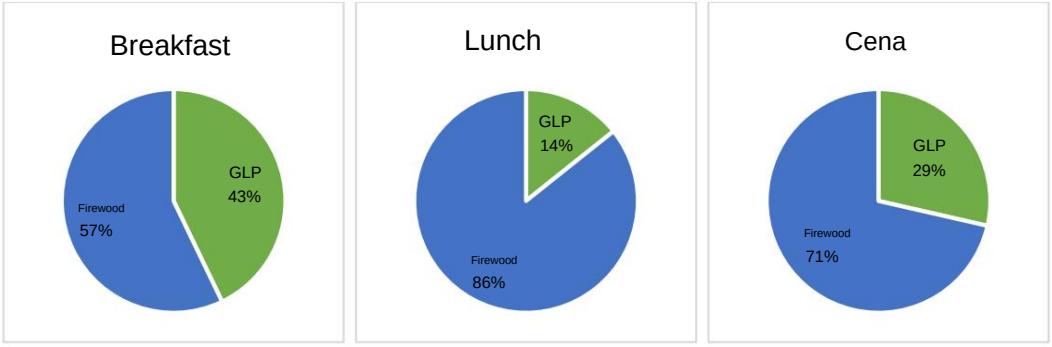
By December 2019, 11 municipalities across the country had been visited, located at different altitudes, in departments such as La Guajira, Caquetá, Tolima, Cauca, Boyacá, Cundinamarca, and the archipelago of San Andrés, Providencia, and Santa Catalina. In these locations, firewood consumption was measured using four three-stone hearth stoves and five improved stoves.

The study included 142 participants, among them farmers and indigenous people from the Pijao and Wayuu ethnic groups. The preparation of lunches was the primary focus of the evaluation. The average moisture content of the firewood was 14.6% on a wet basis. Various types of firewood were found, with no clear preference for any particular species. Among the preferred species in the visited areas were Arrayán, Trupillo, Eucalyptus, and Guácimo. The main criterion for firewood selection was its abundance in the area. Furthermore, 100% of the respondents did not purchase firewood but rather collected it from their farms or the surrounding areas.

The estimated average per capita firewood consumption would be 1.13 kg per person per meal; that is, each meal of the day (breakfast, lunch, or dinner) is considered a meal, **which translates to a daily firewood consumption of 3.39 kg per person.** Graph 16 shows that in the households visited, firewood is primarily used for lunch, even when LPG is available.

The **average weekly use of stoves or cookstoves is 75.5 hours in the homes visited.** Only the two homes visited in La Guajira rely solely on firewood; in the other homes, LPG is used as an alternative for cooking food during the week and for basic meals such as breakfasts and dinners.

Chart 16. Fuel usage for cooking breakfast, lunch and dinner

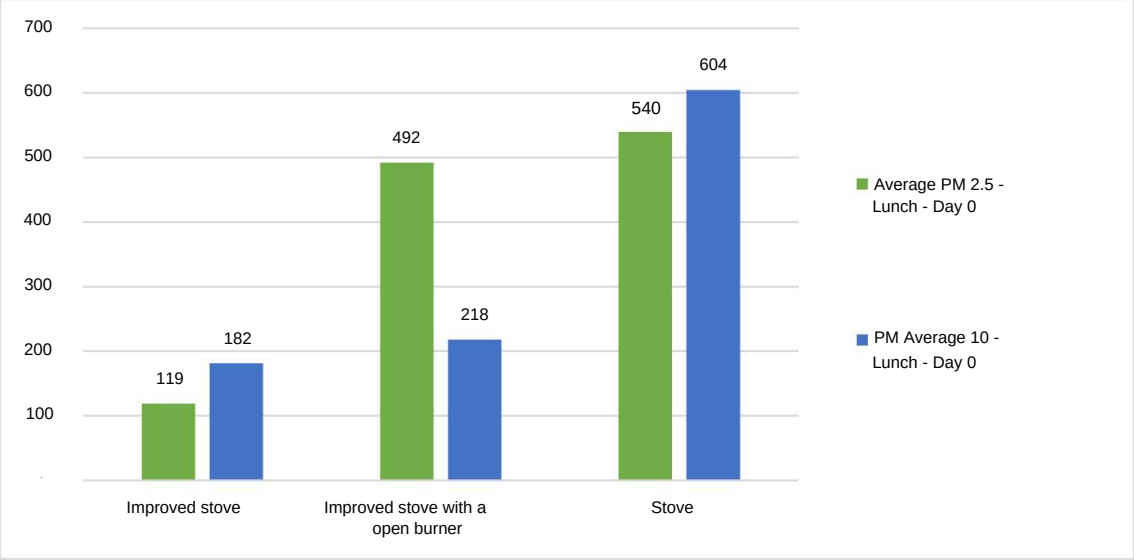


Source: Prepared by the author based on data from the wood laboratories.

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Ambient air measurements around the stoves showed an improvement in air quality inside the kitchen when using an improved stove compared to a traditional cookstove. The particulate matter present in the air around the improved stove is lower than the levels measured around the cookstoves. Both the improved stove with the burner open and the cookstove exceed the maximum permissible levels of 100 µg/m³ for PM10 and 50 µg/m³ for PM2.5287 ; therefore, it is concluded that there is a risk to the health of those cooking.

Figure 17. Average air quality measurements in the different types of stoves visited, PM [µg/m3]



Source: Prepared by the author based on data from the wood laboratories.

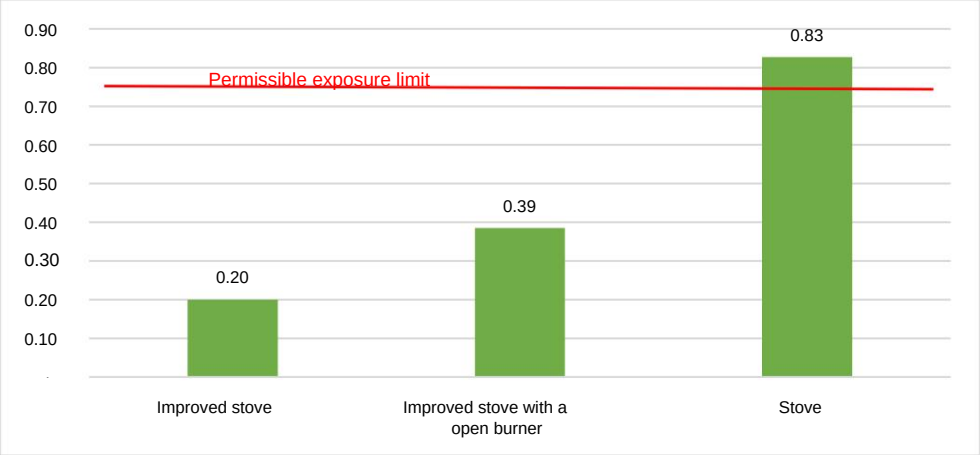
The formaldehyde released into the kitchen environment by the wood-burning stove is higher than the values measured in the surroundings of the improved stove and the same stove with the burner open. The stove is approaching the permissible exposure limit288 (0.75 mg/m3), adding another harm to human health.

287 Ministry of Environment and Sustainable Development (2017). Resolution 2254 of 2017 By which the ambient air quality standard is adopted and other provisions are issued. Retrieved from: http://www.minambiente.gov.co/images/normativa/app/resoluciones/96-res%202254%20de%202017.pdf

288 Ministry of Environment and Sustainable Development (2017). Resolution 2254 of 2017 By which the ambient air quality standard is adopted and other provisions are issued. Retrieved from: http://www.minambiente.gov.co/images/normativa/app/resoluciones/96-res%202254%20de%202017.pdf

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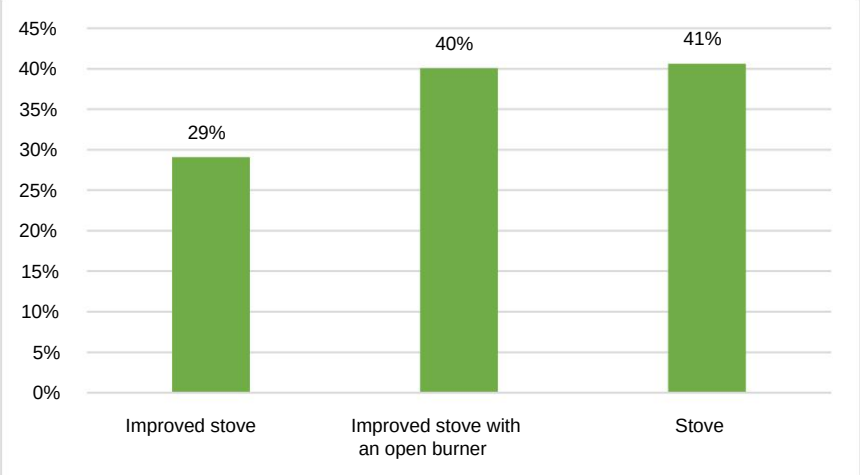
Figure 18. Average air quality measurements in the different types of stoves visited, formaldehyde (HCHO) [mg/m3]



Source: Prepared by the author based on data from the wood laboratories.

The improved stove has less combustion loss than a traditional open fire. Proper use of the improved stove is crucial. This includes cleaning the internal ash, sweeping the chimney, keeping the burners covered, and the stove doors closed. All of this reduces exposure to pollution levels that could harm the health of those cooking and, moreover, increases the efficiency of the energy released by the firewood, resulting in significant energy savings.

Figure 19. Average measurements of energy losses in wood combustion in the different types of stoves visited



Source: Prepared by the author based on data from the wood laboratories.

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6.3. Scenario Estimation

To standardize information, two alternative scenarios for firewood consumption in the residential sector in Colombia have been established. The first scenario (Scenario A) is based on the information contained in the National Quality of Life Survey (ECV) and the Colombian Energy Balance (BECO). The second scenario (Scenario B) incorporates information from the ECV, as well as data from secondary sources and information from the firewood laboratories conducted as part of this study.

6.3.1. Scenario A

The first scenario of firewood consumption in the residential sector in Colombia is based on the number of households that state that firewood is the main energy source used for cooking in the ECV and the total amount of firewood consumed in the residential sector according to the BECO.
According to the figures for this scenario, total firewood consumption for 2018 was 6,123 kilotons, equivalent to 104,046 TJ. Considering that 1,664,000 households in the ECV consume firewood and that there are an average of 3.2 people per household, it is estimated that daily firewood consumption per household is 10.1 kg and consumption per person is approximately 3.1 kg/day.

Table 34. Scenario A of firewood consumption (2018)

Homes	15,493,441 Number of Households
Households that cook with firewood	1,664,229 Number of Households
Percentage of households that cook with firewood	11%
Firewood consumed in the residential sector	6.123 tons
People per household	3.2 Number of People
Firewood consumption per household	3.7 tons/year
Firewood consumption per household	**10.1 kg/day**
Firewood consumption per person	**3.1 kg/day**

Source: Prepared by the author based on ECV 2018 and BECO 2018.

6.3.2. Scenario B

The second scenario proposed to quantify firewood consumption in the residential sector in Colombia is based on the ECV (Colombian Living Conditions Survey) to determine the number of households that consume firewood for cooking. However, it does not start from the total firewood consumption figures for the residential sector presented by BECO (Colombian Forestry Survey), but rather considers the firewood consumption determined by the consulting team during the country's firewood consumption studies. Among the starting data for scenario B, in addition to the number of households that report using firewood primarily for cooking and the number of people per household, the percentage of households that report using another fuel in addition is considered. The firewood consumption indicator obtained from the ECV is also taken into account.

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The laboratory work involved 1.1 kg of firewood consumed for cooking a meal menu, whether breakfast, lunch or dinner.

Table 35. Starting data for scenario B

Total households	15,493,441 Number of Households
Households that cook with firewood	1,664,229 Number of Households
Households that cook exclusively with firewood	70% Percentage
Households that cook with firewood and diversify	30% Percentage
People per household	3.2 Number of People
Factor TJ/kg firewood	17.0 TJ/kg
Firewood consumption per menu	1.1 kg/menu/day

Source: Prepared by the author based on ECV 2018 and wood consumption laboratory.

Based on this data, households that use only firewood for cooking are analyzed separately from those that use another energy source. According to the ECV (National Survey of Living Conditions), 1,164,421 households in the country cooked exclusively with firewood or wood in 2018. Using the estimated firewood consumption per meal obtained from the laboratory studies conducted by the consulting team, it was found that cooking three meals a day would consume approximately 3.4 kg of firewood per person per day, and 11 kg per household, for an estimated total of 4,675 kilotons of firewood consumed annually by households that have no other energy alternative for cooking.

Table 36. Data on households that use firewood or wood as the sole energy source for cooking

Households that cook only with firewood	1,164,421 Number of Households
Firewood consumption per person per day (firewood only)	3.4 kg/person/day
Daily firewood consumption per household (firewood only)	11.0 kg/household/day
Total annual firewood consumption per household (firewood only)	4,675 kton/year

Source: Prepared by the author based on ECV 2018 and wood consumption laboratory.

Likewise, it is estimated that 499,809 households use an energy source other than firewood; of these households, 92.6% use LPG, 2.4% electricity, 1.8% charcoal, 1.4% coal, and the remaining 1.9% use other energy sources such as natural gas, gasoline, or waste material, among others.289 For all of these households, it is assumed that cooking with firewood corresponds to the main meal of the day (lunch), and for other foods that require shorter cooking times or are used for heating, another energy source is used. Therefore, it is estimated that a person in a household that diversifies its energy sources for cooking consumes 1.1 kg of firewood per day, a household 3.7 kg of firewood per day, and that all households in the country that diversify their energy sources by using firewood consume 669 kilotons of firewood per year.

289 Microdata National Survey of Quality of Life 2018.

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Table 37. Data on households that use other energy sources besides firewood

Households that cook with firewood and diversify	499,809 Number of Households
Firewood consumption per person per day (diversify)	1.1 kg/person/day
Daily firewood consumption per household (diversify)	3.7 kg/household/day
Total annual firewood consumption per household (diversified)	669 kton/year

Source: Prepared by the author based on ECV 2018 and wood consumption laboratory.

As a conclusion to scenario B, annual firewood consumption in the residential sector is estimated at 5,344 ktons, which is equivalent to approximately 90,802 kJ consumed annually in this sector. Therefore, considering the total number of households that cook with firewood as their primary energy source (1,664,229), on average a household consumes approximately 8.8 kg of firewood per day, and an individual consumes approximately 2.7 kg of firewood per day.

Table 38. Scenario B of firewood consumption (2018)

Annual firewood consumption in the residential sector	**5,344 Kton/year**
Annual firewood consumption in the residential sector	**90,802 kJ/year**
Firewood consumption per average household	8.8 kg/household/day
Average firewood consumption per person	2.7 kg/person/day
Households that cook with firewood	1,664,229 Number of Households
Households that cook only with firewood	1,164,421 Number of Households
Households that cook with firewood and diversify with other energy sources	499,809 Number of Households

Source: Prepared by the author based on ECV 2018 and wood consumption laboratory.

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7. Estimate of the reduction of and associated with the implementation of program

7.1. The IPCC's global view of the use of fuelwood and other traditional fuels biomass

This section presents the global view of the problems of using biomass for cooking and heating, according to IPCC290 .

Traditional biomass (firewood, charcoal, agricultural waste, animal manure) is used for cooking and heating by approximately 2.8 billion people (38% of the world's population). This accounts for more than half of all bioenergy used globally. Cooking with traditional biomass has multiple negative impacts on human health, particularly that of women, children, young people, and the elderly, and on household productivity because it represents undervalued labor for women and young people.
through harvesting and cooking.

Traditional biomass is intensive in impacts on the land due to the use of open fires, inefficient stoves and overexploitation of firewood, which contribute to land degradation, loss of biodiversity and reduction of ecosystem services291 .

Firewood accounts for 1.9-2.3% of total global GHG emissions292 , particularly in "hot spots" of land degradation and fuelwood depletion in East Africa and South Asia, so that one-third of the world's firewood is harvested unsustainably in developing countries.

Scenarios to significantly reduce dependence on traditional biomass in developing countries present multiple co-benefits (high evidence, high agreement) and include the reduction of black carbon emissions, a short-lived climate forcer that also causes respiratory diseases293 .

Although it would be possible to obtain this biomass more sustainably, the ecosystem impacts and the increased demand for cooking fuel would be reduced through the use of other renewable fuels or, in some cases, non-renewable fuels (LPG), as well as through greater end-use efficiency and improved resource and supply chain management. Integrated response options such as agroforestry and sound governance mechanisms for forest and agricultural management can support the transition to sustainable energy for households and reduce the environmental impacts of traditional biomass.

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290 IPCC – SRCCL - Final Government Distribution - Chapter 4: Land Degradation. Date of draft: 07/08/2019

291 IPCC – SRCCL - Final Government Distribution - Chapter 4: Land Degradation. Date of draft: 07/08/2019

292 IPCC – SRCCL - Final Government Distribution - Chapter 4: Land Degradation. Date of draft: 07/08/2019

293 IPCC – SRCCL - Final Government Distribution - Chapter 4: Land Degradation. Date of draft: 07/08/2019

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As can be understood from the previous IPCC analysis, it is clear that the traditional use of firewood represents a problem for the health of users, but it also raises options to reduce the impacts of firewood use on ecosystems and health, as is done in "hot spots" in Africa and Asia, characterized by intensive use of forests and agricultural waste for cooking and heating, a use that does not occur with that intensity in Colombia.

These options include the use of other renewable fuels, non-renewable fuels such as LPG, greater end-use efficiency, better resource management, integrated responses such as agroforestry, and better governance mechanisms for forest and agricultural management.

7.2. Amount of energy to be delivered for cooking

To compare different fuels, it is necessary to define a common energy output for each cooking system. *Each cooking system must be capable of delivering* ***1000 MJ*** of energy useful to the cooking process in the pot.

(or 1GJ) free of transmission or distribution losses of energy.

7.3. Life cycle analysis

The life cycle assessment (LCA) focuses on the fuels currently used and potential future fuels used for cooking food in rural Colombia, as shown in Figure 11. The following stages are considered:

- Production. All stages are considered, from the extraction or obtaining of the primary energy source in nature to the form that allows its processing into cooking fuel (e.g., cultivation or collection of firewood, extraction of crude oil from an oil well).

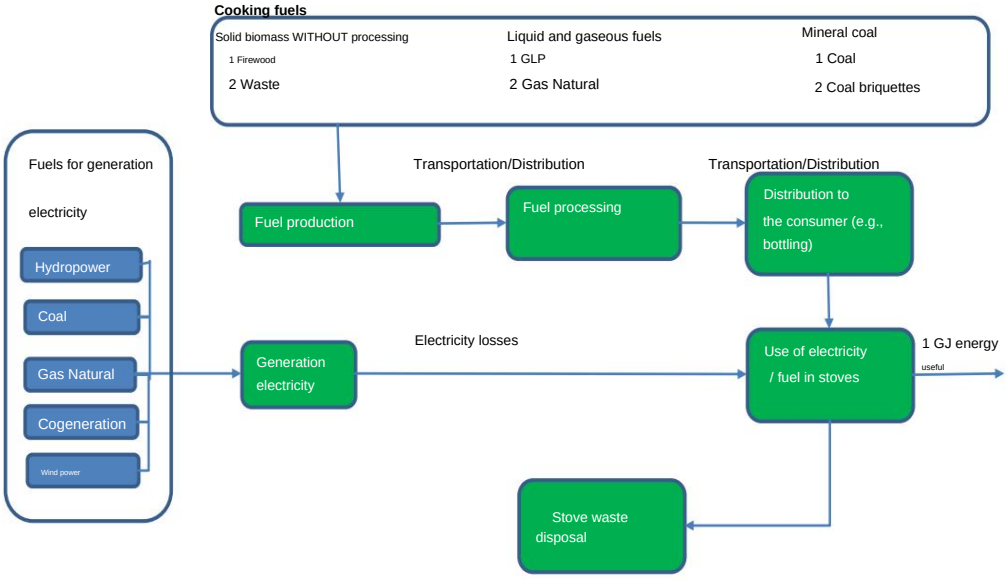
- Processing the fuel into a form ready for use in cooking.
- Distribution of fuels from the production site to the processing plant and their distribution to the consumer. Distribution includes the bottling of fuels, for example, LPG in cylinders.

- Use of fuel via combustion or use of electricity in the stove, including the disposal of combustion residues (e.g., ash)

Illustration 11 shows the flow diagram for calculating the emissions of 2 produced to deliver one GJ of useful energy in the cooking process, considering for the different fuels/energy sources the emissions of 2 for the production of the fuel, **plus** the additional emissions due to the transport/distribution process up to the fuel processing, **plus** those of transport and distribution to reach the final consumer, who finally, depending on the technology of the stove and its efficiency, delivers 1 GJ of useful energy.

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Figure 11. Flowchart for the ACL



Source: Prepared by the consulting team

Outside the system boundary, it is necessary to consider solid waste, which in the case of wood is ash; in the case of coal, unburned waste; and in the case of LPG and Natural Gas, none.

In the case of electricity, the emissions coefficient (factor) of 2 calculated by UPME is considered, which corresponds to the combined generation of the country 's power plants.294 Additionally, it is necessary to take into account the losses due to transmission and distribution in delivering the energy to the user. These losses amount to 11% for the year 2014295. This figure is used to correct the SIN emissions factor.

In the case of firewood, its production, processing (splitting, cracking), and transport are considered to be done using human energy, without emissions of carbon dioxide or other gases into the environment. For this reason, only the calorific value of the firewood and its emissions are considered.

Finally, to LPG and its emissions, it is necessary to add the emissions caused by the transport of the cylinder (round trip by land), from the nearest town

294 Electricity emissions factor SIN: 0.380 tCOÿ/MWh, UPME Resolution 774 Dec 28 of 2018. 1 MWh=3.6 GJ, Therefore, the emissions factor is 0.1056 tCO2/GJ.

295 https://datos.bancomundial.org/indicador/EG.ELC.LOSS.ZS

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up to user296. These emissions are negligible for the example given below, so that the emissions of LPG are 0.0651 t 2/GJ297 .

Finally, the emissions of 2 per GJ of fuel are given in the following table

Table 39. Total CO2 emissions per GJ and per fuel

Combustible	Production, Transport/Distribution (t COÿ/ GJ)	(t COÿ/ GJ)	Total (t COÿ/ GJ)
Generic biogas	0,0844	negligible	0,0844
Generic charcoal	0,0881		0,0881
Electricity	0,1056	0,0116	0,1172
Generic LPG	0,0672	0,0002800028	0,50675
Generic firewood	0,0895		0,0895

Source: http://www.upme.gov.co/calculadora_emisiones/aplicacion/calculadora.html and calculations by the consulting team

If different technologies for each type of fuel are considered and compared against the "3-stone" stove, this stove has the highest emissions, while the other technologies produce lower emissions. Another stove to consider is the traditional coal-fired stove, which is used near coal-producing areas, for example, in the Cundinamarca-Boyacá highlands. This stove is generally closed, not open like the "3-stone" stove, but it is also necessary to consider that it has high emissions of

2, in addition to other gases with negative effects on health.

296 River transport can also be considered
297 Considering a cylinder containing 40 lb (18.14 kg) of LPG, it has a tare weight of 30.5 lb (13.83 kg) and a total weight of 70.5 lb (31.98 kg). The tare weight represents 43.2% of the weight of the filled cylinder. Therefore, it is a significant weight. A small gasoline-powered truck with a 5-ton payload capacity transporting 50 40-lb LPG cylinders (i.e., 1.6 tons of cargo) and traveling 2 x 5 km to deliver and pick up cylinders, with a fuel consumption of 0.2 L/km, also consumes 2 L of gasoline. If it delivers all 50 cylinders, the emissions from transport are 0.04 L of gasoline per cylinder. Gasoline emissions are 2.3 kgCO2/l, and therefore, emissions from transporting 1 cylinder are 0.000092 tCO2. Considering that the energy capacity of a 40 lb LPG cylinder is 18.14 kg * 11,000 MJ/kg = 199.54 GJ, then the

Emissions from transporting one cylinder are 4 x 10^-7 tCO2/GJ. This amount is *negligible* compared to the 0.06508346 tCO2/GJ of LPG.

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Table 40. CO2 emissions per effective GJ delivered to the cooking process

Combustible	Production (tCOÿ/ GJ)	Transport/ Distribution (tCOÿ/ GJ)	Total (tCOÿ/ GJ)	Technology Efficiency	Emissions Effective (tCOÿ / GJ)	Index (%)
Generic biogas	0.0844 negligible	0.0844		Biogas greenhouse	37,2% 0,2268	11
Generic charcoal	0,0881	-	0,0881	Buitron stove	9,5% 0,9277	46
Electricity 0.1056		0,0116	0,1172	Stove resistance	63,0% 0,1860	9
Generic LPG	0,0672	0,0002800028	0,50675	Stove GLP	40,0% 0,1680	8
Generic firewood	0,0895	-	0,0895	3-stone stove	4,4% 2,0347	100
Generic firewood	0,0895	-	0,0895	Improved stove	6,3% 1,4324	70

Source: Calculations by the consulting team

As can be seen, all fuels show a reduction in CO2 emissions compared to firewood in the traditional use of the three-stone stove. Mineral coal, used in stoves in the Cundinamarca and Boyacá regions, also stands out for its high emissions.

7.4. Combustion of firewood

The primary source of energy in firewood is the reaction of the abundant carbon in the wood with oxygen. This combustion reaction releases thermal energy and combustion products.

The most important factors influencing the use of firewood in rural areas have been its availability, free access, traditional use, and calorific value. However, due to the problems associated with the non-renewable use of firewood—including emissions of gases, particulate matter, and even formaldehyde—which negatively impact both the environment and, increasingly, human health, efforts are underway to improve combustion efficiency, heat transfer during cooking, and to vent gas and particulate matter emissions outside of homes.

In this way, improving combustion reduces firewood consumption, the volume of polluting gases produced, and by releasing harmful gaseous emissions outside the home, the inhabitants' exposure to these pollutants and the impact on their health is reduced. Another alternative is to use fuels with a higher calorific value and cleaner combustion processes.

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combustion (lower pollutant and CO2 emissions) such as LPG or electricity generated preferably with renewable energy.

The difference between using one fuel or another in relation to CO2 emissions
The fact is that for both firewood and biogas, the CO2 emitted is endogenous298 , while the CO2 from the other energy sources considered is not.

7.5. Calorific value of firewood

The calorific value of firewood depends on several factors:

- The wood species of the firewood
- The part of the tree from which it comes (trunk, branches or root)
- The age of the tree
- The moisture content of the firewood is a factor that greatly influences its power calorific

7.6. Emissions associated with the combustion of firewood

In the combustion process, whether of firewood or any other carbonaceous fuel, both thermal energy and emissions are released. These two factors, energy and emissions, depend on the type of fuel and the technology used. In order to make the best use of the released energy, the aim is for it to be applied as directly as possible to the process, in this case, cooking food. Cooking food involves using pots or other types of containers that must be exposed to heat. The emissions produced during combustion consist of CO2, as well as gases and particles, including PM2.5 and black carbon.

Information on CO2 emissions and other combustion products from wood is scarce, despite it being a major global issue. In this study, we used data from the Patsari stove. The Patsari stove (in the Purépecha language of Mexico, "the one that stores," referring to its ability to retain heat, as well as

because it preserves health and protects forests); it is a stove resulting from evolution and
Improvements to the initial Lorena-type stove (mud and sand). Improvements to the Lorena
There are several, including the combustion chamber, the air intake, and the fire ducts in the

298 The emissions generated in the anaerobic digestion process of the biogas production plant result from endogenous reactions by microorganisms that produce CO2 and CH4 as products. This methane is considered biogenic in origin; the carbon returns to the atmosphere in the same way it would have occurred naturally (as CO2). Therefore, it is not considered an anthropogenic emission and is not included in the calculation of GHG emissions.

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interior, fireplace, among others. This stove has been developed in Mexico since 2003.
and more than 150,000 of them have been installed.299 This stove has, among other features,
Compared to other developments, the fact that they had consumption measurements of
fuel and emissions that other efficient wood stove developments do not have.

CO2 is the greenhouse gas (GHG) used as a reference for climate change studies. In the combustion of firewood, CO2 *emissions are considered neutral because the combustion process produces as much* CO2 *as was absorbed from the environment during the plant's growth.* Furthermore, firewood requires CO2 for combustion as much as is consumed in its natural decomposition process .

However, depending on the technology used, the incomplete combustion of firewood produces emissions of other greenhouse gases such as CO4, CO2 , and CO, which have a global warming potential many times greater than that of CO2. In addition to these emissions, byproducts such as water vapor, particulate matter, and other gases are produced, some with harmful effects on health and the environment. Finally, combustion residues (ash) remain, which must be disposed of.

Among the emissions mentioned are Total Non-Methane Hydrocarbons (TNMHCs). These are various organic compounds containing only carbon and hydrogen, which, when oxidized in the atmosphere, ultimately form COÿ. While they have a low GWP, they participate in the formation of organic aerosols and in photochemical reactions, which can form, for example, 3-phosphate in the presence of NOx (IPCC 2001b). Also mentioned are Total Suspended Particulates (TSPs), which are solid or liquid materials small enough to remain suspended in the air. They have negative health effects and are not considered a greenhouse gas.

The following table shows the carbon emissions per kg of fuel in two types of stoves: traditional “3 stones” stove and Patsari stove.

Table 41. Emissions from a traditional “3 stones” stove and an improved Patsari-type stove

Emission (g [carbon]/ kg of fuel)			
Gas	**Traditional Stove** 377.9	**Patsari Greenhouse**	**Change**
CO2	31.2	400,3	5,6%
CO		7,6	-310,5%

[299] Berrueta, Victor M., Montserrat Serrano-Medrano, Carlos Garcia-Bustamante, Martha Astier & Omar R. Macera. Promoting sustainable local development of rural communities and mitigating climate change: the case of Mexico's Patsari improved cookstove project. Climatic Change (2017) 140:63–77

300 Berrueta Soriano, Víctor M., Adrián Ghilardi, Omar R. Masera. Mitigation of greenhouse gases through the use of efficient wood-burning stoves. Mexican Bioenergy Network (REMBIO) 2008.

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PTS	9	2	-350,0%
CH4	3,8	0,8	-375,0%
TNMHC	3,5	0,3	-1066,7%
N2O	0,1	0,1	0,0%
Total	**425,5**	**411,1**	**-3,5%**

Source: Adapted from Edwards et al, 2006, Johnson et al, 2007.301 Note: emissions are expressed in grams of carbon, except for Nitrous Oxide N2O.

As these measurements show, carbon emissions are reduced by 3.5% when using the Patsari stove. While this reduction is not very large, the increase in CO2 and the significant decrease in CO are *important ,* indicating *improved combustion.* Other emissions are also significantly reduced, resulting from the improved carbon combustion and, above all, the reduction of particulate matter and volatile organic compounds.

In the case of the Mexico study, the annual firewood consumption when using a traditional hearth is 5.8 tons/year/ family, and when using only the Patsari stove, it is 2,041 kg/year/family (2 tons/year/family), representing a savings of 3,791 kg/year/family, or 65%. In terms of CO2 emissions, there is a reduction of 7.05 tons of CO2/year when considering only gases 2, 4 , and 2 ; and 8.5 tons of CO2/year when considering all gases.

greenhouse gases. The following table shows these reductions. The factors used for GWP (Global Warming Potential) are presented below.

Table 42. Firewood consumption and reduction of consumption and emissions between 3 stone hearth and Patsari stove – Case of Mexico

Technology	Firewood consumption (Ton/year/family)	Gases	Annual emissions (Tons) Spout)	Mitigated emissions (Tons) Spout)	GWP 100 emissions (Ton equivalent/year)
Traditional stove	5,832	CO2	2,204		
		TNMHC	0,021		
		CO	0,182		
		CH4	0,022		
		N2O	0,0003		
		PTS	0,052		
Patsari	2,041	CO2	0,817	1,387	5,087
		TNMHC	0,001	0,02	0,299
		CO	0,018	0,166	1,158
		CH4	0,002	0,02	1,716
		N2O	0,0001	0	0,247
		PTS	0,004	0,048	0

301 Berrueta Soriano, Víctor M., Adrián Ghilardi, Omar R. Masera. Mitigation of greenhouse gases through the use of efficient wood-burning stoves. Mexican Bioenergy Network (REMBIO) 2008.

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Reduction Total 1	3,791				7,050
Reduction Total 2					8,507
Reduction Firewood	3,791				
Reduction Firewood	65%				

Note: Emissions are presented in tons of carbon except for 2 , which is presented in tons of the same gas.
Source: Adapted from Edwards et al, 2006, Johnson et al, 2007.

Total Reduction 1 includes only gases 2, 4, and 2 , which are those considered in the Kyoto Protocol. Total Reduction 2 includes all gases in the Table.

For the calculation of CO2 equivalent emissions, the authors used the table of GWP factors given in Table 43.

Table 43. GWP Factors

Gas	GWP 100
CO2	1
TNMHC	1,9
CO	4,1
CH4	23
N2O	296
PTS	0

Source: Adapted from Edwards et al, 2006, Johnson et al, 2007.

Globally, improved cookstove programs represent an attractive and cost-effective option for reducing carbon emissions, especially considering that nearly 3 billion people worldwide cook with solid fuels. Rather than focusing solely on mitigation, cookstove programs should consider the reduction of greenhouse gas (GHG) emissions as a byproduct of broader sustainable development goals, such as the health benefits resulting from reduced indoor air pollution, as well as the economic and environmental benefits from decreased fuel consumption.

For programs to be effective, it is also essential that they are based on an understanding of local culinary practices and that they encourage the participation of the local population in the different stages of the program, from stove design to monitoring.

Finally, it is also critical that programs go beyond the installation or sale of stoves; measures are needed to ensure the continued use of the stoves in the long term, aimed at guaranteeing sustainability, since almost all programs are aimed at leaving the technology installed but few carry out educational campaigns, follow-up and verification that guarantee appropriation and maintenance of the methodologies by their beneficiaries.

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The factors that directly impact the mitigation potential of improved cookstoves are the reduction in annual fuelwood consumption, GHG emission factors, the renewability of the fuelwood, and the continued use of the stove, which ultimately determines whether the impacts are sustained over time. If the improved wood-burning stove is not used continuously, there is no long-term impact from its use.

7.7. Origin of the carbon in firewood

The carbon in firewood comes from the atmosphere and is incorporated into it through photosynthesis. Since the carbon in firewood, when burned, returns to the biomass carbon cycle, emissions from burning it are considered carbon neutral.

However, it is necessary to consider the origin of the firewood. In the past, it has been argued that firewood comes from deforestation and is therefore a product of logging.

In the Colombian case, although this could happen, it has been found that most of the firewood is collected from the necromass of the forests and therefore the amount of firewood from logging is considered unknown, although very small.

Therefore, the necromass would decompose naturally, returning the carbon to the atmosphere from which it came. Thus, the combustion of firewood at level 2 is simply the same process, but accelerated. However, regarding the combustion process itself, other greenhouse gases are produced, including particulate matter and CO, both harmful which has effects on climate change to health, as well as other compounds and particles such as black carbon, and health. Additionally, in some cases, other compounds such as persistent organic pollutants (dioxins or furans) are produced by the incomplete combustion of wood containing substances like pesticides, which release these POPs, classified as carcinogens, during combustion.

7.8. Firewood and LPG consumption

The firewood consumption measured in the tests carried out by CORPOEMA is 1.22 kg/person/day for lunch (A). If we consider usage mode 2 in the following table, in which lunch (A) is cooked with firewood, and breakfast (D) and dinner (C) using LPG, the monthly firewood consumption is 183.6 kg/month/person, and the LPG consumption is 7.5 kg/month/person (see usage mode 2).

Table 44. Fuel consumption by mode of use

Mode of Use	1	2		3	4
	100% Firewood	50% Firewood	50% LPG	100% LPG	100% Carbon
(kg/mes/f)	367,2	183,6	7.5	15,0	100
(Ton/year/f)	4,41	2,20	0,09	0,18	1,2
Food	D,A,C	A	D,C	D,A,C	D,A,C

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D: Breakfast, A: Lunch, C: Dinner, f: Family

When this family uses only firewood, their consumption doubles (See mode 1). And if this family uses only LPG, LPG consumption doubles (See usage mode 3).

This consultancy has found that 75% of households have Use Mode 2, while 25% of households that are very remote and indigenous consume only firewood, according to Use Mode 1302. These households that consume exclusively firewood may even have a higher consumption of up to **2.5 kg/person/day**
(which would give a higher value than the 4.41Ton/year/f) indicated in usage mode 1.

In the Cundinamarca-Boyacá region, the use of coal has been observed, generally burned in traditional stoves, but usually with a flue and external chimneys for the discharge of unburned material and ashes, and the escape of combustion gases. It was found that these households consume 2 sacks of coal per month, equivalent to 100 kg/month/family (use mode 4).

Firewood consumption also depends on other factors that occur in other countries. In Mexico, for example, firewood consumption for households that use 100% firewood for cooking amounts to 3.4 kg/person/day, which corresponds to a consumption of 6.8 t/year/family. Berrueta et al. (2007), in a case study conducted within the framework of the Patsari Project (Masera et al., 2005), determined savings in firewood consumption of 3,791 kg/family/year for exclusive firewood users, and for mixed users, savings of 2,641 kg/family/year and LP gas of 154 kg/family/year.

7.9. Estimation Analysis for Biogas303

Greenhouse gases (GHGs) are the main cause of climate change; they are derived from human lifestyles and activities. Monitoring them is important to understand their behavior, accumulation, and effects on the climate, the Earth's atmosphere, and the oceans.

In response to the need for climate change mitigation, several countries initiated programs to reduce or prevent GHG emissions. Colombia, for its part, committed to reducing its emissions by 20% compared to the baseline or business-as-usual (BAU) scenario by 2030, through the implementation of renewable energy sources, intensive use of natural gas, and increased energy efficiency, among other actions on the energy demand side. This mitigation involves the use of new technologies or renewable energy sources; in this particular case, the transition from firewood use to

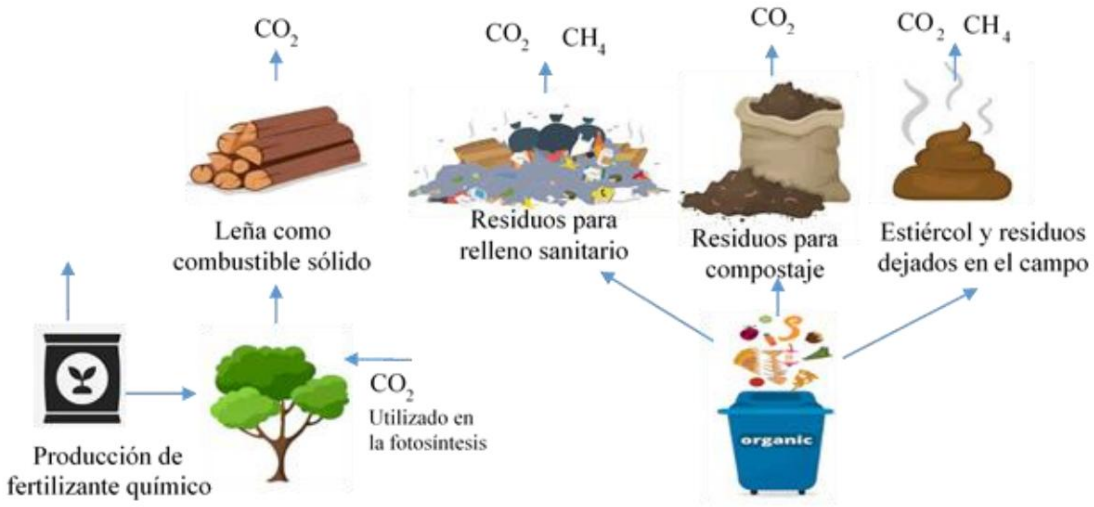
302 Personal communication from Physicist F. González, CORPOEMA.
303 This section was authored by José María Rincón Martínez and Diana Marcela Durán Hernández, from the Industrial Development Center - Tecsol. Bogotá - Colombia
304 B. Del and C. Clim, Means of implementing the UNFCCC in Colombia. 2017.

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Use of biogas and biomethane in rural areas of Colombia, the latter with properties similar to those of natural gas.

In rural areas, the use of firewood promotes forest degradation and CO2 emissions, including those from the production of chemical fertilizers. The same occurs with agricultural waste left in the fields, used for composting, or sent to landfills, as shown in the following illustration.

Figure 12. CO2 and CH4 emissions **produced by activities and waste disposal in the field**

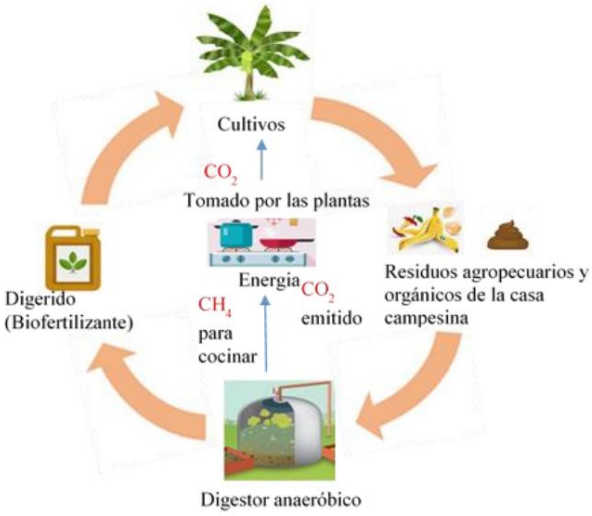


Source: Prepared by the authors

However, when waste in the field is properly utilized and another energy source such as biogas, which is considered a second-generation fuel, is available, the carbon balance is neutral.

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Figure 13. Emissions avoided in the biofertilizer and energy production cycle through the anaerobic digestion process



Source: Prepared by the authors

A carbon footprint is defined as the total greenhouse gas (GHG) emissions resulting directly and indirectly from an individual or product, based on a life cycle analysis. Indirect emissions are associated with all emissions produced along the value chain of a product or service.

In the case of constructing biogas plants for rural areas, there are a number of emissions related to the production, transport, and waste generated from inputs such as the piping for transporting the biogas, inputs for cleaning the biogas, the biobag for biogas storage, and all the materials used to construct the biodigester. Similarly, there are emissions associated with the disposal of these materials at the end of their useful life.

The following shows the emissions avoided through the use of biogas and biomethane. This includes emissions avoided at each stage, from source to use.

- Emissions from biomass collection and transport: One of the biggest drawbacks of utilizing organic waste is the logistics required for its collection and transport. Therefore, it is essential to structure an appropriate and relevant logistics system. In the case of small biogas plants for cooking food, the work done for the sanitary maintenance of the barn or pens is equivalent to the work done for manure collection; for this reason, there is no increase in costs or emissions in this operation.

- Emissions from anaerobic digestion activities: Normally the reactor is heated with the biogas produced; however, when another source is used

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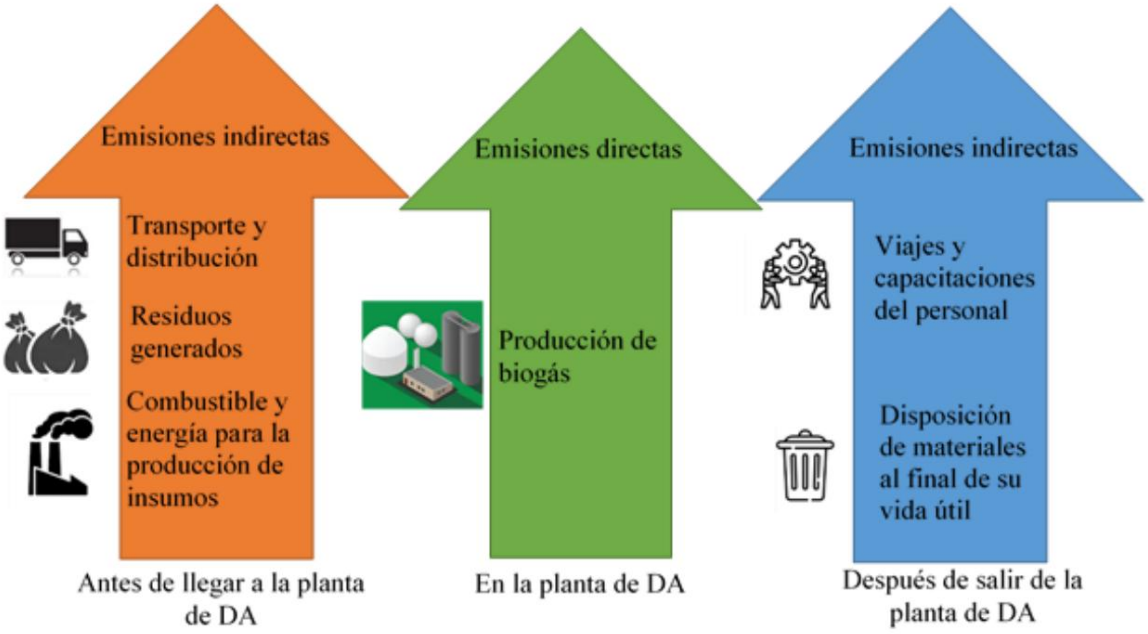
As with solar siphons, the respective calculation of their GHG emissions must be performed. In tropical zones, and for biodigesters smaller than 10 m3
When used in rural areas, it is not necessary to heat the digester.

- Biogas cleaning emissions: Biogas impurities can be removed by simple methods such as passing the gas stream through iron sponge, activated carbon, or caustic soda solutions. There are no emissions during the process itself; however, indirect emissions related to the production of the sponge, activated carbon, or caustic soda must be considered.

- Emissions from the transport and use of biogas and/or biomethane: In rural areas, the biodigester is installed near its point of use, using piping to transport the biogas free of H2S and moisture. When the system is completely airtight and there are no leaks, there are no emissions from this operation.

- Emissions from the use of digestate (biofertilizer): Digest, a byproduct of the anaerobic digestion process, has a high nutrient content. There are no associated emissions at this stage; the irrigation time for the biofertilizer is equivalent to that of the chemical fertilizer, resulting in no increase in costs or emissions.

Illustration 14. Indirect (before and after the Anaerobic Digestion DA plant) and direct emissions associated with biogas production and cleaning in rural areas.



Source: Prepared by the authors

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Under the concept of carbon footprint, the following table presents the activities, direct emissions, indirect emissions and avoided emissions of the corresponding activities.

Table 45. Description of direct, indirect and avoided emissions according to the activities related to the anaerobic digestion process.

Activities	Direct emissions	Indirect emissions	Emissions avoided
Collection and transport of waste	Fuel consumption	Electricity consumption	CH4 avoided by utilizing waste instead of leaving it in the countryside or in landfills.
Anaerobic digestion	Fuel consumption, methanization process	Electricity consumption	Heat production from biogas. Electricity production from biogas. Biofertilizer production.
Biogas cleaning	Not applicable	Electricity consumption	Not applicable
Transport and use of biogas	Fossil fuel consumption	Electricity consumption	CO2 avoided by the use of biogas processed on site, taking into account the gases avoided by the transport of fossil fuels per unit length
Biofertilizer	Consumption of chemical products	Electricity consumption. Fuel consumption.	Production of biofertilizers, in exchange for chemical fertilizers such as ammonium nitrate, urea, phosphorus products and potassium salts.

Source: Prepared by the authors

7.9.1. Methodology for calculating GHG emissions

Currently, several methodologies exist for calculating carbon footprints; among them, those described in ISO 14064 and ISO 14065 stand out as the most impactful ISO standards related to carbon footprints. Other well-known methods include the Greenhouse Gas Protocol (GHG Protocol), Bilan Carbone, PAS 2050, and PAS 2060. These latter standards offer free calculation platforms.

The first step in estimating GHG reduction is deciding on the calculation period; one year is suggested. Next, the boundaries are established, that is, considering the activities over which control is exercised to ensure that the carbon footprint includes all plant activities. Once the boundaries are defined, the

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Sources of GHG generation and finally the scope of the carbon footprint calculation is established; three scopes are known and each is related to different activities.

- Scope 1: Direct GHG emissions; these are produced by sources that are owned or controlled by the company or plant.
- Scope 2: Indirect GHG emissions; associated with the acquisition of electricity, these emissions are generated in the plant that produces electricity, but it is the energy consumed in the facilities and processes of the company or plant.
- Scope 3: These emissions result from the company's activities and occur at sources that are neither owned nor controlled by the company. These activities include waste management, transportation, and distribution.

The methodology used in this case to calculate GHG emissions is by applying emission factors (See Eq 1).

$$_{2} = \quad \dot{y} \quad _{\text{from}} \qquad \text{Ec. 1}$$

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8. Valuation of positive environmental and social externalities

8.1. Access to energy

Access to energy is a concept that allows us to establish guidelines on the satisfaction of energy needs, both for individuals and for them as a whole in a society.

The data and projections presented in the World Energy Outlook 2010305 focus on two elements of energy access: *a household that has access to electricity* and *clean cooking facilities.*

For its part, the International Energy Agency (IEA) defines ***access to energy*** as *"a household that has reliable and affordable access to* ***clean cooking facilities***
such as ***electricity,*** *which is sufficient to initially supply a basic package of energy services, and then an increasing level of electricity over time, to reach the regional average"* 306. This definition of **access to energy** serves as a benchmark for measuring progress on the SDGs, towards SDG target 7.1 (Sustainable Development Goal 7.1) and as a metric for our forward-looking analysis.

Access ***to electricity*** means that a household has initial access to enough electricity to power a basic package of energy services, at a minimum, several light bulbs, lighting for tasks (such as a flashlight), charging a phone and a radio, with a level of service capable of growing over time.

According to projections made by the consultancy, the average household with access to electricity has enough power to run four light bulbs for five hours a day; a refrigerator; a fan for six hours a day; a mobile phone charger; and a television for four hours a day. This equates to an electricity consumption of 1,250 kWh/year/household with standard appliances and 420 kWh/year/household with energy-efficient appliances (energy-efficient appliances consume one-third of the consumption of standard appliances). This definition is only a guideline and should be adjusted to the specific conditions of each country.

Access to ***clean cooking facilities*** means access to (and primary use of) modern fuels and technologies, including natural gas, liquefied petroleum gas (LPG), electricity, and biogas, or improved biomass stoves (ICS) that have considerably lower emissions and higher efficiencies than three-stone open-fire stoves. *Currently, very few ICS models achieve this lower emissions target, particularly under real-world cooking conditions.*

305 IEA. 2010. Energy Poverty: How to make modern energy Access universal.

306 International Agency, downloaded 10/11/2019 https://www.iea.org/energyaccess/methodology/

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Therefore, our *clean cooking* access database refers to households that rely primarily on fuels other than biomass (such as firewood, charcoal, tree leaves, crop residues and animal manure), coal or kerosene for cooking.

The main sources are the World Health Organization (WHO) Household Energy Database and the IEA Energy Balances.

Globally, updated data from the World Health Organization and the IEA Energy Data Center also show that the number of people without access to clean cooking facilities has begun to gradually decline, partly due to increased ***reliance on liquefied petroleum gas (LPG) and improved biomass stoves.*** This is very welcome news, as it will result in fewer premature deaths related to household air pollution.

These efforts have resulted in progress worldwide. For example, the development of China's natural gas infrastructure is helping to reduce the use of biomass and kerosene. This has resulted in 15% fewer people lacking access to clean cooking compared to 2010. In India, 50 million free LPG stoves and initial refills have been provided to poor households through the high-profile Pradhan Mantri Ujjwala Yojana scheme since 2015, and the government has set a target of providing LPG connections to 80 million households by 2020. In Sudan, around half of the urban population now uses LPG for cooking nationwide. In Cameroon, where LPG penetration rates reach 21%, the government has published the region's first "master plan" to increase the proportion of households using LPG to 58% by 2030. Finally, in Kenya, 24% of urban households now use LPG.

Gains in access to modern cooking facilities can sometimes go hand in hand with gains in access to electricity. For example, in Myanmar, where policymakers are motivated to *promote clean cooking in part because of the high rate of deforestation, reliance on biomass has declined from 94% of the population in 2009 to 76% in 2015, largely replaced by electricity.* In Ethiopia, almost a quarter of urban households now cook with electricity, compared to less than 5% in 2011. Meanwhile, in South Africa, electricity is now the primary clean energy source for cooking, used by three-quarters of households nationwide.

8.2. Impacts of the use of firewood

8.2.1. Impact on health

One of the main sources of air pollution in homes in developing countries is the *burning of solid fuels.* Nearly 3 billion of the world's poorest people still *rely on solid fuels (wood,* animal dung, charcoal, crop residues, and coal) *burned on inefficient and highly polluting stoves.*

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Pollutants from cooking and heating. The resulting household air pollution caused more than 4 million premature deaths among children and adults in 2012.307

In terms of health, of the 4.3 million people who die annually from exposure to household air pollutants, the majority die from stroke (34%), ischemic heart disease (26%), and chronic obstructive pulmonary disease (22%). Pneumonia and lung cancer account for 12% and 6% of deaths, respectively.308

Women *and young children,* who spend most of their time near the home, are *particularly vulnerable.* More than 50% of pneumonia deaths in children under 5 are linked to indoor air pollution.

8.2.2. Impacts on the planet's environmental health

Beyond the harmful impact of burning these cooking fuels on human health, these combustion products have effects on the environmental health of the planet.

Polluting domestic fuels also endanger the health of the planet.
Black carbon and methane emissions from inefficient stoves contribute to *outdoor air pollution and increase the rate of climate change.*

8.2.3. Impact on human activities

In many regions, collecting fuel for traditional stoves consumes a considerable amount of time for women and children, limiting other productive activities such as income generation, keeping children out of school, or reducing free time for other human development activities. Safe and efficient cooking stoves would reduce the workload for women and children and decrease the demand for scarce natural resources (e.g., forests).

8.2.4. Impact on natural resources

The use of firewood, when it is taken from trees felled for that purpose, increases pressure on the forest. Firewood should not cause deforestation for the purpose of providing firewood to users.

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307 World Health Organization. Downloaded 10/11/2019 from https://www.who.int/features/qa/indoor-air-pollution/en/

308 https://www.who.int/en/news-room/fact-sheets/detail/household-air-pollution-and-health

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8.3. Combustion of firewood

Smoke is formed when wood or other organic matter is burned. Wood smoke is composed of a complex mixture of gases and fine particles (also called particulate matter pollution, or PM). In addition to particulate pollution, wood smoke contains several toxic pollutants that are harmful to the air, human health, and ecosystems, including benzene, formaldehyde, acrolein, and polycyclic aromatic hydrocarbons (PAHs).

Furthermore, when wood does not burn completely, it produces carbon monoxide (CO), an odorless, colorless, and toxic gas that causes poisoning and is responsible for deaths annually. When burning wood in a wood stove or fireplace, it is advisable to install a digital carbon monoxide (CO) detector, which will sound an alarm when CO levels rise.

Therefore, the more efficiently wood is burned (for example, using an efficient wood stove309 and well-dried wood), the less smoke is generated.

8.3.1. Effects of wood smoke on health

The biggest health threat from smoke is fine particulate matter, also called PM2.5. These microscopic particles can enter the eyes and respiratory system, where they can cause burning eyes, runny nose, and illnesses such as bronchitis.

Fine particulate matter can worsen asthma symptoms and trigger asthma attacks. It can also trigger heart attacks, strokes, irregular heart rhythms, and heart failure, especially in people already at risk for these conditions.

Wood smoke can affect everyone, but children, teenagers, older adults, people with lung diseases, including asthma and COPD, or people with heart disease are the most vulnerable. Research indicates that obesity or diabetes may also increase the risk. New mothers or pregnant women should also take precautions to protect the health of their babies, because some studies indicate that they may be at greater risk.

8.4. Valuation of positive environmental externalities

Given that of the 1,600,000 households that reported cooking with firewood in the 2018 DANE Quality of Life Survey, 70% use firewood without diversifying with other energy sources for cooking food, and taking into account the

309 In the United States, the Environmental Protection Agency (EPA) certifies wood-burning stoves and other wood-burning equipment.

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Based on the information gathered on the cultural, gender, and health aspects of deliverable 2 of this consultancy, we can say that cooking with this energy source in stoves known as three stones or similar adaptations produces:

- Indoor air pollution: It releases smoke into the kitchen environment that directly and indirectly affects family members exposed to it.

- Forest degradation: By collecting firewood, pressure on forests is increased because it is necromass that the forest needs to maintain and diversify existing species.

- Heat loss: Since there is no accumulation of heat, the stove loses energy produced in the combustion of the wood and makes the consumption of wood inefficient.

Thus, the **positive externalities of the energy sources identified as alternatives to replace firewood for cooking are presented below,** with the objective of identifying the contribution that the implementation of each technology would have throughout the national territory.

8.4.1. Improved Stoves

Among the positive externalities identified in improved stoves are:

Table 46. Environmental externalities and their effect Improved stoves

Externality	Effect
Reduction of indoor air pollution	Relief for the health of people in the home who have direct and indirect contact with the smoke produced in cooking or food preparation, given that the amounts of smoke in the environment are reduced.
Decrease in forest degradation	Installing improved stoves saves firewood when preparing food, which reduces energy consumption in the home, thus reducing pressure on the forest.
Improvement of the combustion process	Because the heat from the stove is concentrated more evenly, the energy it produces is used more efficiently than if you had a three-stone stove. It reduces the cooking time of food.
Cleanliness and comfort	Improves cleanliness and comfort conditions during food preparation.
Conservation of fauna and flora	When forest degradation is controlled, the fauna and flora of the habitat are preserved.

Source: Prepared by the authors

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8.4.2. Electrical Energy

The following are the positive environmental externalities for induction and resistance stoves:

Table 47. Environmental externalities and their effect on electrical energy

Externality	Effect
Improves energy efficiency	These types of kitchens make use of almost all the energy they consume because their technology prevents heat leaks and concentrates it only on the cooking utensils.
Shorter cooking times	They reach high temperatures in a short time, which reduces food cooking times.
Less environmental pollution and forest degradation	The environmental impact of using these stoves It is minimal. Likewise, it reduces the impact and pressure on forests.

Source: Prepared by the authors

8.4.3. Liquefied Petroleum Gas

Table 48. Environmental externalities and their effect on LPG

Externality	Effect
Less environmental pollution	No smoke, soot, or suspended particles are produced that could affect the local environment. Carbon dioxide emissions are lower than those produced by burning wood.
Fast heating and shorter cooking times	The concentrated heat from its flame allows cooking vessels to heat evenly and more quickly, thus reducing cooking times.
Reduction of emissions	It does not produce emissions into the environment during combustion, thus reducing the health effects on people who use this energy source for cooking. It does not contain pollutants such as sulfur or lead.
No forest degradation	In rural areas, the use of firewood and its collection causes forest degradation, which is avoided by using LPG for cooking food.

Source: Prepared by the authors

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8.4.4. Biogas310

Table 49 presents the positive environmental externalities obtained by installing a biogas plant for cooking food in the rural area.

Table 49. Environmental externalities and their effect.

Externality	Effect
Odor control	The microorganisms present in biogas production mineralize the organic matter; therefore, the nauseating odors from the decomposition of organic matter are eliminated.
Decrease in the dispersion of microorganisms	Manure is normally removed by washing, causing the dispersal of microorganisms downstream. When manure is used in biogas production, a large part of these microorganisms are eliminated.
Utilization of organic waste	In the countryside, when organic waste accumulates, greenhouse gases such as CH4 and NOx are released into the environment during decomposition. In biogas plants, the CH4 is stored for later use.
Negative CO2 balance	The utilization of biomass through anaerobic digestion is CO2 neutral. However, during the anaerobic digestion process, nitrogen, phosphorus, and potassium are mineralized, thus preventing the formation of N2O , which is released into the atmosphere during combustion. When the digestate is used as a biofertilizer instead of chemical fertilizer, the CO2 released during its production by thermal processes using fossil fuels is avoided; as an overall result, the CO2 balance is negative.
Deforestation control	In rural areas, the use of firewood causes forest degradation, which is avoided by using biogas for cooking food.
Conservation of fauna and flora	When forest degradation is controlled, the fauna and flora of the habitat are preserved.
Improving water quality	During anaerobic digestion, the water undergoes biological treatment, resulting in water free of some pathogenic microorganisms. Furthermore, replacing chemical fertilizers with biofertilizers prevents contamination of nearby surface and groundwater sources.
Production and use of biofertilizers	The use of organic fertilizers improves soil properties and crop quality. Furthermore, it presents a business opportunity for exporting organic products.

Source: Prepared by the authors

310 This section was authored by José María Rincón Martínez and Diana Marcela Durán Hernández, from the Industrial Development Center - Tecsol. Bogotá - Colombia

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8.5. Valuation of positive social externalities

Similarly, in the case of social externalities, the use of firewood for cooking in traditional stoves produces:

- Instability in the pot support with risk of burns.
- An improper posture while cooking that causes back pain.
- Risk of burns from the open flame.
- Respiratory and eye problems caused by the smoke are evident from the combustion of wood.
- An unhygienic environment where soot is evident in the surroundings the kitchen.
- Increased fuel consumption and longer travel times for people to obtain energy.
- Delay in food preparation: resulting in energy inefficiency.

The following presents the positive externalities of the energy sources identified as alternatives to replace firewood for cooking, with the aim of identifying the contribution that the implementation of each technology would have throughout the national territory.

8.5.1. Improved Stoves

Table 50. Social externalities and their effect Improved stoves

Externality	Effect
It reduces respiratory and eye problems.	It prevents women from having bad posture when preparing food, reduces eye burning and congestion, as well as the smell of smoke in women's clothes.
Lower risk of burns	Because there is a platform and clearly defined cooking areas, exposure to the flame is less, thus reducing the risk of suffer burns on the hands.
Improved kitchen hygiene	Kitchen utensils are preserved for a longer time, and by extracting the smoke through the chimney, the kitchen does not accumulate soot and tar.
It makes household chores easier.	Greater convenience when preparing food, since people using the stove will have space to cook comfortably and perform other preparations on it.
New jobs	The construction, installation, and maintenance of the Improved stoves are made with local labor. They create new jobs.

Source: Prepared by the authors

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8.5.2. Electrical Energy

Table 51. Social externalities and their effect on electricity

Externality	Effect
Energy efficiency	It is the most efficient technology available at the residential level for cooking food.
Ease of cleaning and improved quality of life	The burners are easy to use; when food boils, there is no worry about the energy going out; cleaning is quick and safe since the stove does not overheat.
Automatic container detection	The heat is transferred directly to the special cooking containers, so there is no considerable energy loss.
Reduced cooking times	Because the heat is concentrated evenly, the preparations take less time to cook.
No emissions are generated in the kitchen	Combustion in stoves does not generate fumes or emissions that can harm people's health.

Source: Prepared by the authors

8.5.3. Liquefied Petroleum Gas (LPG)

Table 52. Social externalities and their GLP effect

Externality	Effect
Easy to clean	It has very efficient combustion, leaves no marks on kitchenware, and makes cleaning easier.
Reduced cooking times	The heat produced by LPG responds to the regulators and is distributed evenly across the base of the cooking vessels.
No emissions are generated in the kitchen	The combustion of LPG in stoves does not generate fumes or emissions that can harm people's health.
Low costs and versatility	This type of gas is considered the most economical fuel for long distances, it is portable, has great flexibility to be used in various applications and provides greater confidence to the user since it is easy to handle and transport.
It improves the quality of life.	It improves the quality of life and, more importantly, allows women and children to spend less time collecting fuel, giving them more time to go to school or to carry out value-added economic activities within their community.
Easy transport	LPG does not require extensive pipeline infrastructure for transport. It is often the only fuel that

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	It reaches island communities or communities located at high altitudes and distances.

Source: Prepared by the authors

8.5.4. Biogas

Table 53 shows the positive social externalities obtained by installing a biogas plant for cooking food in the rural area.

Table 53. Social externalities and their effect

Externality	Effect
Control of pulmonary diseases	The combustion of biogas does not generate particulate matter or unburned particles, which are the main causes of lung diseases.
Decreased migration of farmers	When there are job opportunities in rural areas and an improved quality of life, farmers see no need to migrate to the city. Likewise, the use of biofertilizers improves crop yields, and the energy produced from biogas promotes and enhances farmers' production processes.
New jobs	The construction, installation, and maintenance of biogas plants are carried out with local labor; consequently, new jobs are created.
Improving the quality of life	Biogas eliminates the need to collect firewood, produces electricity, and can power cooling systems for food preservation.

Source: Prepared by the authors

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9. Cost-benefit analysis of the various available sources and technologies

Cost-benefit analyses are a tool frequently used in pre-implementation project evaluation, from both a financial and a public policy perspective. This type of analysis considers that the positive and negative factors of a project can be quantified, allowing for an assessment of whether a project's costs are less than its potential benefits. Based on the information gathered from the valuation of externalities and the estimated reduction associated with the program's implementation, a preliminary cost-benefit analysis of the technologies and energy sources identified in the first phase of the study is conducted.

[2]

9.1. Monthly cooking costs with different energy sources

To advance the analysis, a preliminary estimate of the monthly costs per household for each cooking technology is presented below. These costs include, on the one hand, the investment costs for the cooking equipment and the energy required for cooking per month, and on the other hand, the health and environmental costs associated with cooking using each energy source per household over the same period. This section describes how the costs for each type of cost were calculated and the total estimated costs per technology.

9.1.1. Initial investment costs

Based on information provided on the websites of various supermarket chains in the country and studies reviewed in section 3 of this document, average investment costs were estimated for different cooking technologies according to the energy source they use. This information is presented in the following table:

Table 54. Investment costs in stoves

Energy used	Type of stove	Total cost	Monthly cost financed over 10 years
Improved wood or coal stove	With fireplace near urban centers	1.500.000	12.500
	With a fireplace in remote homes	2.000.000	**16.667**
LPG stove	LPG for built-in	300.000	2.500
	LPG oven	700.000	5.833
	Average	500.000	**4.167**
Biogas	Biodigester and gas stove	1.500.000	**12.500**
Electric stove	Conventional electric resistance (4 positions)	300.000	**2.500**

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	Induction cooktop with special set of pots	1.500.000	**12.500**

Source: Prepared by the consulting team. Average costs available on the market.

Therefore, we have:

- For improved wood or coal stoves, the average construction cost is $1,500,000 near populated centers. Since the majority of the study population resides in rural areas, the estimated average cost per stove is $2,000,000, with a monthly financed cost of $16,667 over 10 years.

On the other hand, for LPG, an average was calculated between the built-in stove prototypes and the prototype with an oven. The estimated total average cost of a stove that uses LPG for cooking is $500,000, financed at $4,167 per month for 10 years.

- The Biodigester, together with the gas stove, has an average value of $1,500,000 taking into account the average costs of a Taiwanese or sausage-type artisanal biogas generation plant and a geotextile311 .

- In the case of the electric stove, the average cost of a 4-burner electric resistance stove of $300,000 and an induction stove with a special set of pots of $1,500,000 were taken into account separately.

9.1.2. Energy cost

In order to unify the information in the study, the monthly costs associated with the use of different energy sources were estimated from the data collected by the consulting team.

Table 55. Monthly cost of energy for cooking

Energetic	Monthly cost ($COP)
Firewood	160.000
Coal	40.000
GLP	60.000
Biogas	160.000
Electricity conventional	66.000
Induction electricity	59.400

Source: Prepared by the consulting team

Thus, in the case of firewood, the energy cost corresponds to one hour per day in the process of searching, collecting and storing, which is calculated as 4 days of collection work per month, at a rate of $40,000/day, for a total of $160,000.

311 As explained in section 8.2 Biogas

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Currently, these costs are not factored into household budgets, but it is important to quantify the daily effort families make to use firewood for cooking. Similarly, in the case of biogas, approximately one hour of work per day is invested in system maintenance, resulting in an estimated cost of $160,000 per month, based on a daily wage of $40,000.

The cost of charcoal corresponds to the monthly purchase of two 50kg bags of charcoal at $20,000 each. For LPG, the estimated cost is $60,000 for a 15kg or 33lb cylinder in populated areas, with transportation costs ranging from $15,000 to $25,000. Finally, the monthly cost of electricity is estimated at $66,000 for a conventional stove, assuming a monthly consumption of 132 kWh per cooking session, and $59,400 for an induction stove, assuming a 10% reduction in monthly consumption equivalent to 118.8 kWh, at an average price of $500 per kWh.

9.1.3. Health costs

One of the most relevant factors to consider in the cost-benefit analysis of alternative technologies to cooking with firewood is the health costs the country faces due to the health problems experienced by firewood users. According to the National Planning Department (DNP), the economic value of morbidity and mortality resulting from the degradation of indoor air quality in 2015 amounted to $3.14 trillion pesos. 98.7% of that value comes from mortality and 1.3% from morbidity . ³¹²

Table 56. Health costs from consumption of firewood or charcoal

	Cost of intramural pollution per year (Millions of pesos)	Number of households	Annual cost per household per year (Pesos)	Monthly cost per household (Pesos)
Cost per morbidity	40.820	1.664.229	24.528	2.044
Cost per mortality	3.099.180	1.664.229	1.862.231	155.186
Total / household / month				157.230

Source: Own elaboration based on DNP (2018)313 .

Considering that the estimated total health costs in households that consume firewood are approximately $157,230 pesos per month per household, it is considered necessary to differentiate between the health costs of a household that only consumes firewood as fuel for cooking, and those of a household that also consumes another energy source.

312 DNP - National Planning Department - (2018). Economic Valuation of Environmental Degradation. Retrieved from: https://colaboracion.dnp.gov.co/CDT/Prensa/Valoraci%C3%B3n%20econ%C3%B3mica%20de%20 in Colombia 2015

environmental degradation.pdf
313 Ibid.

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as well as considering the difference between cooking with firewood in an open fire and in one with a smoke outlet to the outside of the house (chimney).

Table 57. Health costs from consumption of firewood or charcoal

Cooking type	Monthly health cost per household (Pesos)
Only firewood	184.976
Firewood and other	92.488
Only firewood in an improved stove	25.897
Firewood and other items in an improved stove	12.948

Source: Own elaboration based on DNP (2018)314 .

According to data from scenario B, based on the number of households that reported having an additional energy source for cooking in the quality of life survey, it is estimated that 70% of households cook exclusively with firewood, while 30% diversify their energy consumption, although they cook at least half of their food with firewood. This results in a monthly healthcare cost of $184,976 pesos for households that cook solely with firewood, while a monthly healthcare cost of $92,488 pesos for households that use another energy source in addition to firewood.

Likewise, taking into account that in these homes where cooking with wood or coal is done in improved stoves with smoke outlets to the outside, the degradation of indoor air quality can be reduced by approximately 86%315, which yields
an approximate monthly health cost of $25,897 pesos for households that cook only with firewood but in an improved stove, and $12,948 pesos in households that cook with firewood in an improved stove and diversify with another energy source.

9.1.4. Environmental cost

To determine the environmental cost of using the different technologies proposed for the replacement plan, it is necessary to identify the calculation methodology. The emission factor is estimated in kg² /kg of firewood. The emission factors used are developed in section 7 and presented in the following table.

314 Ibid.

315 According to the results obtained in the wood laboratories presented in section 6.2 of this document.

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Table 58. Emission factors of the energy sources considered

Energy Emission factor
Firewood 1.52 kgCO2/kg
Coal 2.53 kgCO2/kg
GLP 3,05 kgCO2/kg
Biogas 2.06 kgCO2/m3
Electricity 0.42 kgCO2/kWh

Source: Prepared by the consulting team, taking into account the section on estimating CO2 reduction .

As input for calculating the energy costs associated with firewood consumption in homes, the following monthly energy consumptions are considered for each energy option:

Table 59. Energy consumption per household (monthly)

Energetic	Energy only	Mix con GLP316
Firewood (kg)	367	184
Improved stove firewood (kg)	257	
Coal (kg)	100	50
GLP (kg)	15	8
Biogas (m3)	0	
Conventional electricity (kWh)	132	
Induction electricity (kWh)	119	

Source: Own elaboration.

Subsequently, by multiplying the emissions factor of each energy source by its consumption, as presented in Table 59, it is found that firewood has a higher value than the other cooking options: 558 kg CO2 per month in open fires and 391 kg CO2 in improved stoves, above coal with 253 kg CO2. On the other hand, cooking with electricity in a conventional stove has emissions of 56 kg CO2 and in an induction stove emissions of 50 kg CO2, while cooking with LPG generates emissions of 46 kg CO2.

316 Scenario B, section 6.3.2

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Table 60. Monthly emissions by energy source (kgCO2)

Energetic	Energy emissions [kgCO2/mes]	Emissions Mix with LPG [kgCO2/mes]
Firewood	558	302
Improved stove wood	391	
Coal	253	150
GLP	46	
Biogas	0	
Conventional electricity	56	
Induction electricity	50	

Source: Own elaboration.

Now, according to Law 1819 of 2016, the average value of a ton of CO2 emissions
In the country, it is $16,000 pesos. Thus, it is estimated that the monthly environmental opportunity cost of not mitigating CO2 emissions that could be avoided by burning firewood per household is $8,935 for a traditional stove and $6,255 for an improved stove, followed by coal at $4,054, LPG at $732, and electricity at $891 for a conventional stove and $802 for an induction stove.

Table 61. Opportunity cost of not mitigating tons of CO2 and selling these reductions to interested parties

Energetic	Energy only [$/month]	Mix firewood or charcoal with LPG [$/month]
Firewood	8.935	4.834
Improved stove wood	6.255	
Coal	4.054	2.393
GLP	732	
Biogas	0	
Conventional electricity	891	
Induction electricity	802	

Source: Own elaboration.

9.1.5. Total monthly cost

Based on the previously considered costs, several technological options are proposed, along with their respective associated costs for initial investment, energy consumption, health, and the environment. The summary of costs by factor and technology, as well as the total monthly cost, is presented in the following table.

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Table 62. Total monthly cooking cost per household

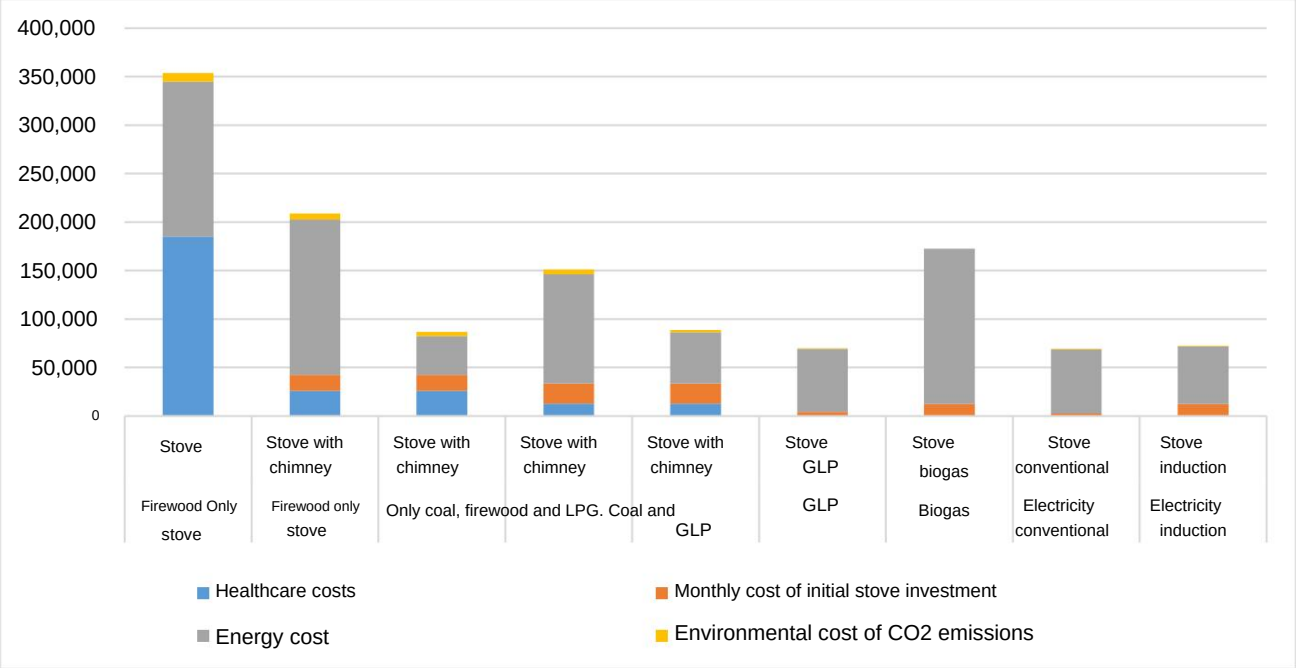
Cooking costs in rural and urban households using firewood, coal, and LPG						
Energy used	Type of stove	Monthly cost of initial stove investment	Cost of the energetic	Cost in health	Environmental opportunity cost for not mitigating CO2 emissions	Total monthly cos
Firewood only	Stove	0	160.000	184.976	8.935	353.912
Wood-burning stove only. Improved stove with chimney.		16.667	160.000	25.897	6.255	208.818
Only coal stove improved	Stove with chimney	16.667	40.000	25.897	4.054	86.617
Firewood and LPG	Stove with chimney	20.833	112.500	12.948	4.834	151.116
Coal and LPG	Stove with chimney	20.833	52.500	12.948	2.393	88.675
GLP	LPG stove	4.167	65.000	0	732	69.899
Biogas	Biogas greenhouse	12.500	160.000	0	0	172.500
Conventional electricity. Conventional stove.		2.500	66.000	0	891	69.391
Induction electricity	Induction stove	12.500	59.400	0	802	72.702

Source: Own elaboration.

As shown in Table 62, the traditional wood-burning stove is the only technology that has no initial investment cost, as stated in chapter 9.1.1. Likewise, the health costs of cooking with LPG, biogas and electricity are considered to be zero, as well as the environmental cost of cooking with biogas.

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Chart 20. Total monthly cooking cost per household by cost type



Source: Own elaboration.

Figure 20 shows that electric and LPG stoves have the lowest total cooking costs in a Colombian household. Wood-burning stoves have the highest costs, primarily due to the health costs associated with their use, presented in section 9.1.3, and the energy costs related to the time spent collecting firewood, presented in section 9.1.2.

Although cooking with biogas does not represent environmental or health costs, the costs of obtaining the energy (estimated from the time required) and initial investment increase its total cost.

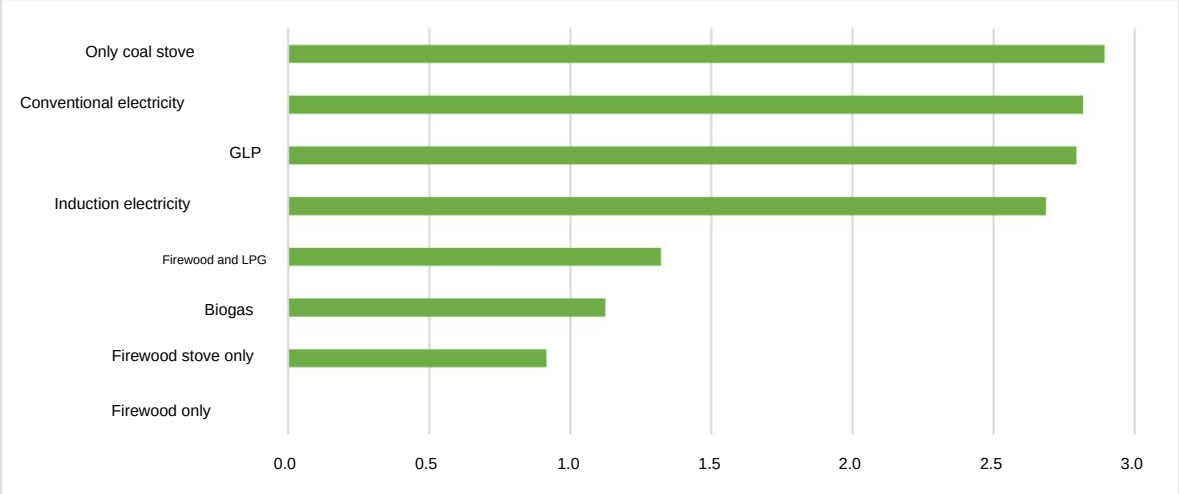
9.2. Cost-benefit

Based on the information presented in section 9.1, a cost-benefit analysis is performed on the technological alternatives identified and considered throughout this document, for the progressive replacement of firewood for cooking in the residential sector in Colombia.

This analysis considers, on the one hand, the costs of the proposed technologies and the energy required for their use, as presented in sections 9.1.1 and 9.1.2, and on the other hand, the environmental and health savings that would be generated by using these technologies compared to a traditional wood-burning stove. These environmental and health savings are estimated by comparing the cost difference between each proposed technology and cooking with wood in a traditional stove.

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Figure 21. Cost-benefit indicator of the technologies considered



Source: Own elaboration.

When estimating the benefit-cost ratio (B/C) by dividing the benefits of each technology by its costs, it is found that a higher B/C ratio indicates that the benefits outweigh the costs. As can be seen in Figure 21, cooking with electricity, coal, and LPG presents the best indicators, followed by cooking with diversified firewood and LPG; then by cooking with biogas; and finally, as expected, by cooking with improved wood-burning stoves.

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10. Feasibility analysis for the installation of viable solutions

Conducting a feasibility analysis helps guide decision-making in evaluating alternatives to firewood, taking into account the specific conditions of each region. This involves a market study for each energy source, reviewing current findings and verifying the existence of a potential market or unmet need, which, for the purposes of this study, is defined by prioritizing health aspects.

Subsequently, the information presented here will assume that technical feasibility and the availability of resources, both human and material, must be guaranteed.
administrative and logistical aspects for the implementation of the most viable solutions for the installation of the technologies.

This will allow the advantages to be verified from a financial, economic, social, cultural and environmental point of view, in order to allocate resources in a sustainable way over time, in the development of the plan for the progressive replacement of firewood as an energy source in the residential sector in Colombia, with the necessary components for its implementation.

For the Firewood Replacement Program, four energy sources have been proposed: Improved Stoves, Biogas, Electricity, and LPG. The analysis of each energy source is presented below, prioritizing what is considered relevant to the study, as shown in Table 63.

Table 63. Monthly cooking costs in rural and urban households

associated with each alternative ($COP)

Energy used	Type of stove	Monthly cost of initial stove investment	Energy cost	Healthcare costs	Environmental opportunity cost why not mitigate CO2 emissions	Total monthly cost
Firewood only	Stove	$0	$160.000	$184.976	$8.935	$353.912
Firewood stove only	Stove with chimney	$16.667	$160.000	$25.897	$ 6.255	$208.818
GLP	Stove GLP	$4.167	$65.000	$0	$732	$69.899
Biogas	Biogas greenhouse	$12.500	$160.000	$0	$0	$172.500
Induction electricity	Induction stove	$12.500	$59.400	$0	$802	$72.702

Source: Prepared by the consulting team

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If the $160,000 equivalent to the time spent by each family per month collecting firewood were not taken into account, the total monthly cost for each energy source considered would be:

- $193,912 for the case of the stove with only a burner.
- $72,702 for the case of electricity.
- $69,899 for the case of LPG.
- $48,818 in the case of the improved stove.
- $12,500 for the case of Biogas

Thus, although wood-burning stoves remain the most expensive technology, alternatives such as electric and LPG stoves would be more costly than improved wood-burning stoves. This analysis indicates that biogas would be the least expensive in economic terms and that the implementation of induction or resistance stoves, along with LPG stoves, should be accompanied by a subsidy to make them affordable for households identified for the gradual replacement of firewood nationwide.

For the consulting case, the analysis is considered taking into account the cost of $160,000 per month for firewood collection per household.

10.1. Improved stoves

In accordance with the above, the feasibility of implementing a Firewood Replacement program in the country and its sustainability over time for the supply of improved stoves as a replacement alternative in some regions of the national territory is analyzed.

10.1.1. Technical Feasibility

Globally, the implementation of improved cookstoves is accompanied by the design of technologies adapted to local circumstances and cooking methods. As observed in deliverable 1 of this consultancy, the success of improved cookstoves is linked to the cooking preferences and dietary habits of different communities, respecting their traditions. Therefore, cookstoves must be evaluated and designed to ensure that households prefer them over traditional or open fires, which have proven to be highly detrimental to the health of those who inhale the smoke, given the associated effects.

The addition of a flue to wood-burning stoves allows the gases produced by burning wood to travel through a duct and out of the house, reducing the concentration of harmful particles and other noxious substances in the home environment. This led to the development of various stove designs aimed at addressing the environmental, social, economic, and health problems that were occurring in homes worldwide.

From there emerged prototype stoves such as the Patsari in Mexico, the Inkawasi in Peru, and the Huellas in Colombia, among others, which have become established as mitigation alternatives in the communities where they have been implemented and have been able to meet the needs of households related to the daily preparation of traditional dishes. An example of this is the

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tortillas in Mexico, where they are prepared daily and a special stove is needed to make them properly.

Currently, the technical standard NTC 6358 is approved, which establishes the Requirements and test methods for evaluating the safety, energy efficiency and control of pollutant emissions of food cooking stoves that use biomass (organic matter such as wood or agricultural surpluses), in which it is expected that the needs of each region will be taken into account with respect to the cooking method, climatic conditions and the cultural and gender aspects of the different communities.

According to information from the Ministry of the Environment, this regulation will be the basis for the development of the national plan for improved stoves in 2020.

Improved cookstoves can be seen as a replacement alternative for homes in hard-to-reach areas where implementing other cooking methods or technologies is impractical and access is limited. This would ensure that households that haven't been able to switch from firewood to other cooking fuels can use it more efficiently, addressing a health-related problem that is likely to become more serious in the medium to long term.

An improved stove technology has been identified, constructed with brick and block, which is fixed and suitable for home kitchens, and has a lifespan of over 10 years. The materials for a fixed stove are: brick, cement, block, sand, metal fittings (burners, oven doors, and other cavities), and rebar.

10.1.2. Commercial Feasibility

Since the 1980s, improved cookstove installation projects have been implemented in the country. According to the national cookstove survey conducted by the Ministry of Environment and Sustainable Development in July 2014, “28,238 efficient cookstoves have been installed nationwide”317 at an average cost of $1,200,000 each. Cornare plans to build an additional 2,000 cookstoves in 2020-2021, on top of the approximately 31,000 it had already built in its jurisdiction by 2019, for a total of 33,000, including support from the Ministry of Environment and Sustainable Development and the Natura Foundation.

The initiatives undertaken by the various regional autonomous corporations are estimated to cover a total of 50,000 stoves. Thus, it can be said that a risk has been identified and attempts have been made to mitigate it through technological change in the communities. However, these have not been massive replacement programs but rather individual initiatives by local governments, supported by regional corporations and private entities.

Table 64. Development of improved wood-burning stoves in Colombia

Region	Number of stoves installed	Installed stove prototypes	Unit cost

317 Ministry of Environment and Sustainable Development. National Stove Survey 2014

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Atlantic	624	Footprints	721.500
		Corantioquia	530.000
Central	2.030	Footprints (UNDP)	1.350.000
		Lorena (Natural Heritage)	1.230.000
		Two positions (Heritage Natural)	1.200.000
		Rocket (Natural Heritage)	380.000
Pacific	1.336	Footprints (Natural Heritage)	1.230.000
		Footprints (Corponariño)	1.200.000
		Triangular hybrid (Patrimonio Natural)	462.300
		Triangular block (Heritage Natural)	932.840
		Linear hybrid with two burners (Natural Heritage)	491.950
Oriental	3.069	Footprints (Corpoguavio)	718.398
		Footprints (Car)	1.720.840
		Natura Foundation	700.000
		ICA-1791 (Corporate)	1.720.840
Antioquia	21.179	Footprints (Corpocaldas)	900.000
		Footprints (CAM)	931.060
		Footprints (Carder)	1.333.700
Total	**28.238**	**14 Prototypes**	

Source: National Survey of Improved Stoves, Ministry of Environment and Sustainable Development, 2014

The potential demand would be in those departments where access to other energy sources or alternative technologies is not possible due to the geographical conditions of the territories, and where the maintenance of the technologies can be carried out by household members without associated transportation costs or the need to pay a technician. The useful life of an improved stove is approximately 10 years, although in the workshop held in Sogamoso on October 17, 2019, by the consulting firm, the mothers in attendance stated that an improved stove can last from 15 to 20 years, depending on the maintenance performed on the firebox and the ash chamber.

Currently, there are metalworking companies and hardware stores in the country whose production processes include the design and creation of improved portable metal stoves, or that sell the hardware for manufacturing improved fixed stoves made of brick and block. Among the companies identified are: Induestufas, Metrococinas, and Alf Soluciones Metálicas in Bogotá; Fundiciones Industriales SAS in Medellín; and El Nuevo Cóndor in Santander, among others. Prices range from $750,000 to $1,000,000, not including transportation costs for improved portable stoves.318 Similarly, in regional workshops, it was found that there are individuals who build improved fixed stoves, charging between $300,000 and $400,000 per stove.

318Consulted on: Mercado Libre, link: https://listado.mercadolibre.com.co/estufa-eficiente-de-le%C3%B1a consulted on 02/12/2019

Machine Translated by Google

installation/assembly plus the surplus of materials, which depends on their availability in the region.

There is a market for stove manufacturing that caters to the specific needs of households, which has led to the development of new accessories for water heating, grills, and the addition of "tamale-cooking" stoves319 or larger ovens. For the purposes of this consultancy, the conventional fixed improved stove, constructed with brick and block, will be used as a reference. Its price ranges from $2,000,000 to $2,500,000, including transportation costs.

10.1.3. Economic Feasibility

While it is true that using a traditional wood-burning stove in homes does not entail a monetary cost associated with obtaining firewood (since it is collected in the forest for free), the replacement of any energy source implies assuming new costs. This consultancy has found that the implementation of the traditional stove has repercussions for health in the medium and long term, such as respiratory problems and diseases mainly associated with COPD (Chronic Obstructive Pulmonary Disease).

By calculating the estimated monthly energy cost described in Table 63, and comparing the costs between the traditional stove and the improved cookstove, it is found that implementing the traditional stove in homes results in significant costs that primarily affect health. It is important to remember that the costs assessed for each technology included components such as: the average value of daily work hours used to collect firewood, equivalent to $160,000 per month; estimated health costs based on the budget allocation designated to treat illnesses associated with indoor air pollution from cooking with firewood, which are valued at $184,976 per month per family; and an environmental cost equivalent to the opportunity cost of the unavoidable emissions from using the stove.

valued at $8,935 per month.

Table 65. Monthly savings associated with the costs of implementing improved stoves per household

Technology	Monthly cost of initial stove investment	Energy cost	Healthcare costs	Environmental cost	Total Cost
Traditional stove	$0	$160.000	$ 184.976	$ 25.897	$ 8.935 $ 353.912
Improved Stove	$16.667	**-$0**	**$**	$160.000	$ 6.255 $ 208.818
Saving	**$16.667**	**159.080**		**$ 2.681**	**$ 145.094**

319 Refers to stoves or burners with a larger diameter than conventional ones, with the aim of being able to mount larger pots on the stovetop.

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Source: Prepared by the consulting team

By adding up the total monthly cost of the improved stoves, it can be seen that their implementation would be equivalent to a decrease of $145,094 per month per family.
Although households are not the ones directly assuming the costs, the State
It would be investing in a technology that would help mitigate them, since it would be preventing a problem that in some years could be considered a burden on health systems due to the care of diseases that could be avoided through alternatives such as those proposed by this consultancy.

In the case of improved stoves, the government's investment should cover 100% of the technology cost, which is an estimated $2,500,000 per family. If we consider that the savings in healthcare and environmental costs amount to $161,761
monthly, the investment would be returned in 12.6 months.

10.1.4. Environmental Feasibility

Among the factors to consider in the environmental feasibility of the consultancy are cultural, social, political, territorial and environmental characteristics prevailing at the time of identifying which alternative is most suitable for each territory.

Thus, by assigning an environmental cost to each technology, in this case the implementation of improved stoves to replace traditional hearths, the monetary savings would be approximately $2,681 per month, which is equivalent to the environmental savings from the opportunity cost of not mitigating emissions.

2.

Table 66. Monthly savings per family in environmental costs with the implementation of improved stoves

Technology	Environmental cost
Traditional stove	$ 8.935
Improved Stove	$ 6.255
Saving	**$ 2.681**

Source: Prepared by the consulting team

10.1.5. Feasibility in Health

As confirmed in section 10.1.2, the failure to implement measures to replace and improve the efficiency of firewood use for cooking represents an expense or liability that the State must assume to address the health problems of the households under study in the medium and long term. The most affected are housewives and children, who spend the most time in the kitchen preparing food at home. The health system would save approximately $159,080 per month per household if firewood were replaced with improved stoves.

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Table 67. Monthly savings per family in health costs with the implementation of improved stoves

Technology	Healthcare costs
Traditional stove	$ 184.976
Improved Stove	$ 25.897
Saving	**$ 159.080**

On the other hand, Source: Prepared by the consulting team

10.1.6. Social Feasibility

Installing an improved stove in the home ensures that families can optimize their use of energy from the hearth, as the heat is distributed in a closed system within the stove. From the perspective of the time spent collecting firewood, this technology still entails travel and collection costs which, although shorter, remain significant for the household. The time freed up can result in greater economic productivity or improved family well-being through quality time with children or for other family, educational, or leisure activities. It is also socially viable in that, with the presence of a chimney, families are not directly exposed to particulate matter, leading to a perceived improvement in quality of life and a positive perception because clothes are not permeated with the smell of smoke and the kitchen does not become a soot-filled enclosure.

and tar.

10.2. GLP

The following section analyzes the feasibility of using LPG in the Firewood Replacement Program and the long-term sustainability of supplying this energy source to the households included in the Plan. It should be noted that regulatory and organizational issues within the sector are addressed in the dedicated chapter on this topic.

10.2.1. Technical Feasibility

LPG is a fuel derived primarily from two hydrocarbons: butane and propane. When subjected to moderate pressure or cooled, it is in a liquid state. In its liquid state, it is versatile, allowing for easy transport and storage, similar to other petroleum products, via propane pipelines, multi-product pipelines, barges, and tank trucks. This enables LPG to reach remote and difficult-to-access locations, given the country's unique geography.

In Colombia, it is transported to terminals or supply plants, where it is usually stored in steel or aluminum containers, from where it is distributed wholesale by tank trucks and then packaged and sold to residential consumers, mainly in cylinders.

Among the existing technical regulations for LPG is resolution MME 40245 of March 7, 2016, which regulates cylinders and stationary tanks; the resolution

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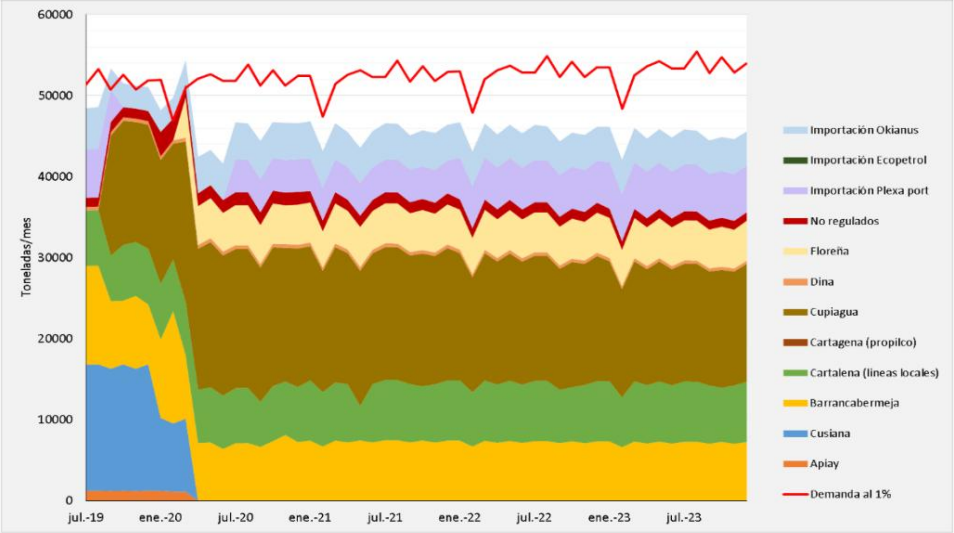
MME 40246 of March 7, 2016, amended by Resolution 40867 of September 8, 2016 on LPG storage and distribution; Resolution MME 40247 of March 7, 2016, amended by Resolution 40868 of 2016 on bottling plants;

Resolution MME 40 248 of March 7, 2016, as amended by Resolution 40869 of 2016 concerning storage facilities, outlets, and points of sale, and Resolution 90902 of 2013 concerning internal combustible gas installations, establishes the regulatory framework. This framework indicates that LPG is a viable alternative to firewood for cooking in regions where it is feasible, according to certain parameters and criteria.

10.2.2. Commercial Feasibility

Both in Colombia and worldwide, LPG is produced in refineries and also as a byproduct of the separation and treatment of natural gas streams associated with crude oil in oil and gas production fields. Additionally, shortfalls to meet national demand are obtained through imports via the privately owned ports in Cartagena and ECOPETROL's facilities. The commercial chain thus begins with the production and importation of LPG, whether at refineries or oil and gas production fields. The following graph shows the current and projected supply and demand composition for the period 2019-2023:

Chart 22. Composition of current and projected LPG supply and demand to 2023



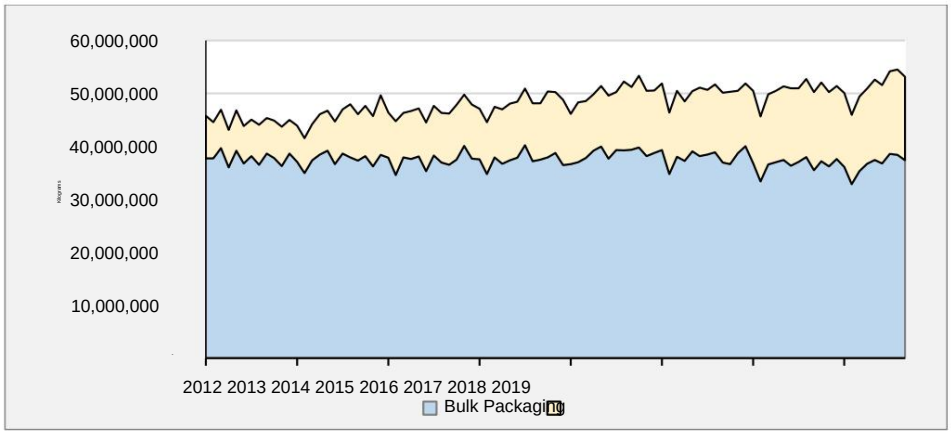
Source: GASNOVA – Statistical Report October 2019.

As shown in the chart, the production sources are listed on the right and include the Barrancabermeja and Cartagena refineries and ECOPETROL's Dina, Cupiagua, Cusiana, Floreña, Apiay, and other fields. Import options for ECOPETROL and private agents are also indicated. In 2020, the LPG from the Cusiana treatment plant is expected to be released for use in co-dilution of heavy crude oils; this necessitates LPG imports to meet 100% of the projected demand.

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In the marketing and distribution of LPG, the service is provided through two channels, which are cylinders and bulk, as shown in the following graph:

Chart 23. National demand – sales channel



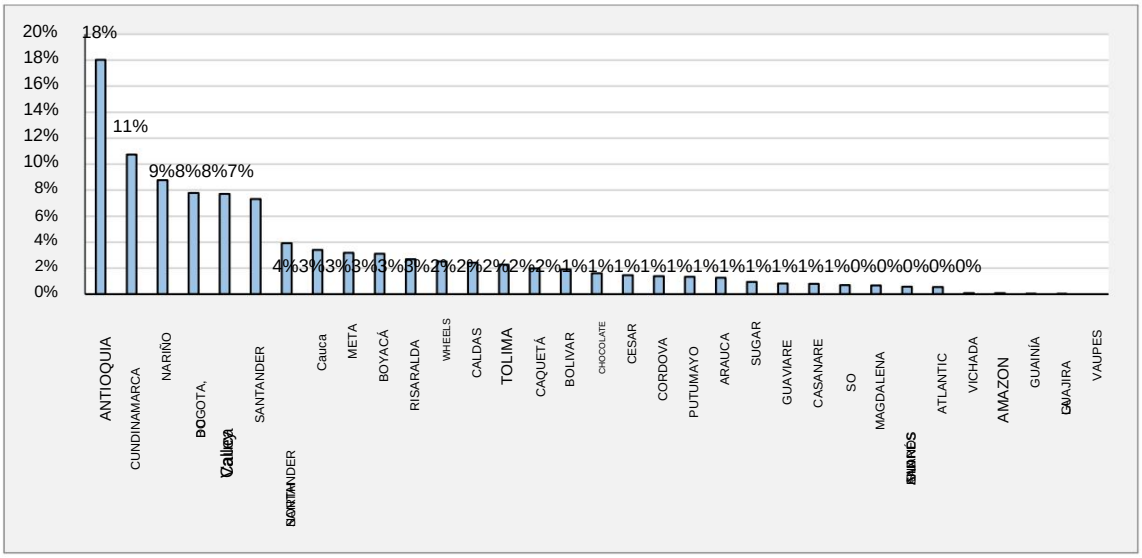
Source: GASNOVA – Statistical Report October 2019.

It is observed, then, that sales in cylinders represent the largest percentage of LPG distribution and marketing, reaching 72% of the total volume, while bulk sales account for the remaining 28%. Distribution and marketing activities are carried out through 50 retail cylinder distributors and a total of 37 companies that distribute LPG via stationary tanks.320 Thanks to this distribution and marketing scheme, LPG reaches each and every one of the country's 32 departments, as shown in the following graph, which also indicates the share of LPG demand in each department relative to the national total.321

320 UPME (2017). Liquefied petroleum gas (LPG) chain.
321 GASNOVA – Statistical Report October 2019.

Machine Translated by Google

Chart 24. Share of LPG demand by department



Source: GASNOVA – Statistical Report October 2019.

LPG is transported partially via pipelines for the product from the Barrancabermeja Refinery and by road in other cases, as is the case with gas from the Cartagena Refinery and the Cusiana and Cupiagua fields. When road transport is not feasible, LPG must be transported by river or even air, as in the case of Mitú and other isolated communities. Therefore, while coverage is nationwide, in some areas transportation costs increase the price of the product. It can be said, then, that the supply, marketing, and distribution of LPG is an organized activity, with a robust commercial system and a multitude of distribution and marketing agents reaching every department in the country.

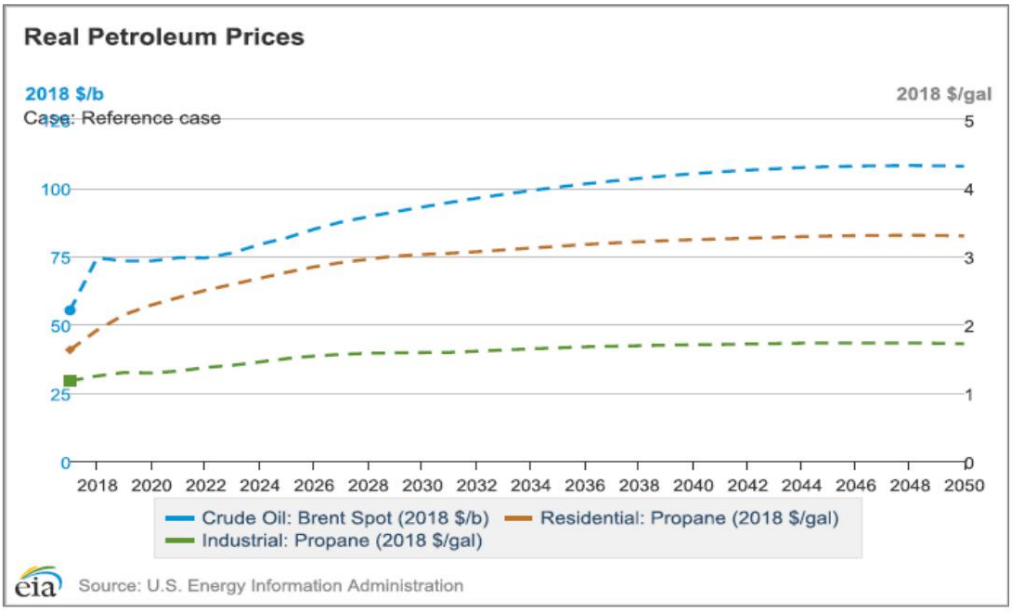
In the medium and long term, the greatest uncertainty facing LPG in the country relates to prices, considering its impact on the residential market, regarding which there are two considerations:

1) Dependence on the international market due to the local supply deficit to meet demand. A graph is presented showing projected LPG prices in the United States market for residential and industrial use, providing an initial approximation of the evolution of import price indicators.[322] It can be observed that for residential use, the price in 2022 reaches USD 2.5/gallon, compared to USD 2.15/gallon in 2019, representing a 16% increase. Consequently, it will be necessary to closely monitor price trends in the United States market, considering their impact on the Colombian market.

[322] Energy Information Admnistration. www.eia.doe.gov

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Figure 25. Projection of LPG prices in the United States market at the residential and industrial levels



Fuente: Energy Information Admnistration. www.eia.doe.gov

2) Furthermore, the national regulation of LPG prices is under review, and we will have to wait for the results of the study that Fedesarrollo is conducting for the UPME regarding price recommendations for setting the producer's income, in this case, ECOPETROL. It should be noted that the current price regulation establishes a maximum price for all of ECOPETROL's LPG production points, while the production of other agents has a free market price.

10.2.3. Economic Feasibility

On the other hand, according to the 2018323 Quality of Life Survey , the average income of households that reported using firewood as their main cooking energy source is $793,549; an LPG cylinder cost of $65,000 was assumed, equivalent to 8.1% of the average income of households that reported using firewood as their main cooking energy source.

According to the 2014 Quality of Life Survey, which asks about household expenses on LPG324 services, the household has a subsistence expenditure of $37,542, equivalent to 4.7% of the average household income.

In this way, the State, to guarantee the use of energy and its replacement over time, must provide a monthly subsidy of approximately between $22,000 and

323 Quality of Life Survey, 2018
324 The estimate of minimum or subsistence expenses is calculated for the year 2014 and based on the consumer price index these values are expressed at constant 2018 prices

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$30,000 per month, taking into account a variation in the price of LPG for dispersed areas of $20,000.

If we consider that the total health and environmental costs saved by using LPG compared to a traditional wood-burning stove are $284,012 per month, compared to $193,179 per month, and considering that the average price of an LPG stove was estimated at $500,000, with an average useful life of 10 years, it is concluded that the estimated health and environmental savings would recoup the investment in 2.6 months.

Technology	Monthly cost initial investment stove	Energy cost	Healthcare costs	Environmental cost	Cost Total
Traditional stove	$0	$160.000	$ 184.976	$ 8.935	$ 353.962
GLP	$4.167 -	$65.000	$0	$732	$ 69.899
Saving	**$4.167**	**$95.000**	**$ 184.976**	**$ 8.203**	**$ 284.012**

Source: Prepared by the consulting team

10.2.4. Environmental Feasibility

Giving a cost-environmental assessment to each technology, in this case the implementation of LPG to replace the traditional stove, the monetary savings would amount to $8,203 per month, equivalent to the environmental savings in emissions of 2.

Table 68. Monthly savings per family in environmental costs with the implementation of LPG

Technology	Environmental cost
Traditional stove	$ 8.935
GLP	$ 732
Saving	**$ 8.203**

Source: Prepared by the consulting team

10.2.5. Feasibility in Health

The failure to implement measures to replace firewood as a cooking fuel implies an expense or liability that the State must assume to address the health problems of the households under study in the medium and long term. Those most affected are housewives and children, who spend the most time in the kitchen preparing food at home. According to the study conducted by the National Planning Department (DNP), "Economic Valuation of Environmental Degradation (2015)," the health system would save $184,976 per household served if firewood were replaced with LPG.

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monthly, which includes direct costs for morbidity and mortality and where
Days of restricted activity are also taken into account. Lost productivity due to disability

Table 69. Monthly savings per family in health costs with the implementation of GLP

Technology	Healthcare costs
Traditional stove	$ 184.976
GLP	$ 0
Saving	**$ 184.976**

Source: Prepared by the consulting team

10.2.6. Social Feasibility

LPG is an easy-to-transport technology that does not emit particulate matter into the environment during combustion, ensuring that household members, and especially women, no longer have to worry about keeping their home walls clean or inhaling substances harmful to their health.

On the other hand, it will no longer be necessary to collect firewood from the forest, since this energy source is delivered in cylinders and is easy to use. Likewise, the implementation of this technology is seen as an improvement in the quality of life in homes and provides the possibility of greater comfort. The only limitation to accessing the technology relates to the energy costs, so it would be essential to ensure the sustainability of the technology in the home by using a subsidy.

10.3. Electricity: Induction stoves

10.3.1. Technical Feasibility

Induction cooking is a type of ceramic cooktop that directly heats the cookware using an electromagnetic field, rather than the heat produced by the stovetop. This process results in less energy loss compared to the other alternatives proposed in this consultancy. The main reason is that the cookware is heated directly. This contributes to energy savings, which are increasingly valued in today's society.

Similarly, the induction cooker can detect through a sensor system whether or not there are containers on its surface, and if there are none, it does not emit energy.
On the other hand, the cooking time is very short since it takes less time to reach the right temperature.

Internationally, technology has evolved enough to offer a wide range of options that suit the particular needs of the home.
It should also be considered that it has the limitation of only being able to use certain materials for kitchenware and the condition that in domestic environments it cannot be

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used at a distance of less than approximately 60 cm by people who have pacemakers325 .

El estudio suizo *"Exposure of the Human Body to Professional and Domestic Induction Cooktops Compared to the Basic Restrictions"* published in the scientific journal Bioelectromagnetics in 2012 concludes that:

- Although most of the induction cookers measured comply with the public exposure limits to the magnetic field at the distance specified by the International Electrotechnical Commission (IEC 62233 standard), 30 cm from the edge of the cooker, most exceed these limits at shorter distances.

- For a cooker that complies with IEC 62233, the current density may exceed the basic restrictions of the International Commission on Non-Ionizing Radiation Protection (ICNIRP) of 1998 by a factor of 16.

- The brain tissue of young children may be overexposed by a factor of two with respect to basic restrictions for the general public if they go near the kitchen.

- Exposure to fetal central nervous system tissue may exceed limits for the general public if the mother is exposed at an occupational level.
- The IEC 62233 standard does not sufficiently protect exposed individuals according to the basic restrictions defined by ICNIRP 1998.

However, according to the Swiss Office of Public Health, it is not currently known whether the magnetic fields generated by an induction cooktop pose a health risk, and there are regulations in place that allow the use of this technology in the country.
In addition, chain stores and appliance companies distribute and sell the technology.

10.3.2. Commercial Feasibility

The public residential electricity service is one of the services with the greatest coverage nationwide, reaching a level of 97.02%326 in December 2016. The largest proportion of users is located in urban centers; however, in rural areas a significant effort has been made to increase the supply of this service, which, given the geographical conditions, has not had access to the network.

Electricity is produced locally, and the country has sufficient installed capacity to generate the energy required to meet demand. From the perspective of expanding supply, there are specific mechanisms in the current regulations (such as the reliability charge) that ensure the installation of new capacity when the country requires it. And, from the perspective of demand, the pattern is stable, and it can be stated that the country is not an energy-intensive user.

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325 Spanish Agency for Medicines and Health Products, and Spanish Society of Cardiology. "Guide for patients with pacemakers."

326 Taken from the coverage indicators for the year 2016, the most recent published by UPME in the Colombian Electrical Information System (SIEL).

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When analyzing the per capita consumption indicator, Colombia registered a per capita consumption in 2014 of 1,312 kWh/per capita, compared to 7,819 and 12,994 kWh/per capita for Japan and the United States, respectively, for the year 2014.

In Colombia, electricity for cooking and water heating was the most widely used energy source until the latter part of the last century. However, high costs for the end user, coupled with substantial tariff subsidies, suggested the need to switch to other energy sources such as natural gas and LPG.

This policy was defined in the Gas Massification Plan, which has been successfully implemented since the 1990s. Although induction cooktops are not yet widely known in the country, as their use becomes more widespread and they are promoted, they may become more affordable, since production volumes are expected to increase. In this regard, domestic manufacturers such as Haceb and chain stores like Homecenter, Éxito, Alkosto, Falabella, and others have the necessary technology to implement a program for the gradual replacement of firewood.

The Study has suggested that the power demand of an induction cooker places demands on the network from users who use electricity for cooking, who would normally be located at the end of a distribution system, typically radial.

In these areas where electricity is primarily used for lighting and refrigeration, the increased demand for cooking could overload the system, leading to outages and service interruptions. Therefore, while not ruling out the use of electricity, a program to replace firewood with this energy source requires a careful evaluation of the distribution network to ensure that the radial system can handle the new loads.

On the other hand, policies aimed at improving both the quality and quantity of energy services must be supported by policies that promote investment, growth, and productive employment.327 In rural areas, this includes the development of rural infrastructure, training, and support through microcredit programs,

which are possible to the extent that the general legal and institutional framework is strengthened.

10.3.3. Economic Feasibility

Currently, the technology available in the country is sold in chain stores at an estimated cost of $1,500,000 for a set of pots and pans. As noted in section 9.1.2, an average monthly electricity cost for cooking of $66,000 was assumed, equivalent to 132 kWh at $500 per kWh (8.3% of the average household income that uses firewood as its primary energy source for cooking – $793,549). Households are willing to pay no more than 6% of their income for energy. Therefore, similar to LPG, electricity for cooking should be subsidized by an average of $33,000, equivalent to 50% of the estimated monthly consumption.

327 UNDP, 2005

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The subsidy is given to low-income users in areas of the country where natural gas service is not offered by network, that is, the departments of Caquetá, Nariño, Putumayo and San Andrés, Providencia and Santa Catalina and the indigenous communities, indigenous communities that are currently potential consumers of firewood in the national territory.

This energy source has not been widely adopted as a cooking alternative in Colombian homes because it requires higher power output, which would necessitate changes to the grid and increased subsidies for the technology to be accessible to the target population. However, considering the estimated economic savings in health, environmental, and social areas, the return on investment for implementing this technology would be 7.8 months.

Table 70. Monthly savings per household from the implementation of induction cooktops

Technology	Monthly cost of initial stove investment	Energy cost	Healthcare costs	Environmental cost	Total Cost
Traditional stove	$0	$160.000	$ 184.976	$ 8.935	$ 353.912
Electric stove	$12.500	$59.400	$ 0	$ 802	$ 72.702
Saving	**-$12.500**	**$100.600**	**$ 184.976**	**$ 8.133**	**$ 281.210**

Source: Prepared by the consulting team

10.3.4. Environmental Feasibility

By assigning an environmental cost to each technology, in this case the implementation of induction stoves to replace traditional stoves, the monetary savings would amount to $8,133 per month, equivalent to the environmental savings from the opportunity cost of not mitigating emissions.

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Table 71. Monthly savings in environmental costs (opportunity cost of not mitigating emissions) with the implementation of induction stoves

Technology	Environmental cost
Traditional stove	$ 8.935
Induction stove	$ 802
Saving	**$ 8,133**

Source: Prepared by the consulting team

10.3.5. Feasibility in Health

The health system would save $184,976 per month for each household served in the case of the replacement with induction stoves.

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Table 72. Savings in health costs per household with the implementation of induction stoves

Technology	Healthcare costs
Traditional stove	$ 184.976
Induction stove	$ 0
Saving	**$ 184.976**

Source: Prepared by the consulting team

10.3.6. Social Feasibility

Induction cooking eliminates direct contact with the heat produced by a traditional stovetop, preventing burns, particulate matter emissions, and smoky odors that linger on clothing. Since cooking time is significantly reduced, it frees up time for other household chores or activities that enhance well-being. With no risk of accidental explosions, it improves the safety and comfort of homes that use this technology.

10.4. Biogas

10.4.1. Technical Feasibility

China produced 13 million cubic meters of biogas from digester installations for cooking in 2016, and India 2 million cubic meters.328 Rural household digesters operate primarily using livestock manure, food waste, and agricultural residues. Biogas is used for cooking and heating, thus replacing firewood, coal, and other solid biomass fuels that negatively impact air quality, health, and quality of life. Furthermore, biogas can be stored for later use, acting as a "battery" to supplement intermittent renewable energy sources such as wind and solar power.

A wide range of organic matter can be treated for biogas production, such as domestic and commercial food waste, municipal and industrial wastewater, agricultural material, and livestock manure; in such digestion plants, biogas is used to produce energy, both on a small scale in homes and farms, and on a large scale for entire cities.

In Colombia, many farmers, due to low soil productivity, are forced to move to cities; however, under Decree 926 of 2017, they are certified as 'carbon neutral' when they neutralize the Greenhouse Gas (GHG) emissions generated by the use of fossil fuels, for which they are required to pay the national carbon tax, Law 1819 of 2016.

328 Sarika Jain (2019). Global potential of biogas.

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This tax can be avoided when the transaction is made using the carbon avoided by using biogas as a thermal fuel instead of fossil fuels. Under this concept, the agricultural sector can be made more profitable by selling carbon-neutral certificates for biogas production and use, grouping the various users together.

Furthermore, individual incentives exist, such as VAT exemption for equipment purchases and other benefits under Law 1715 of 2014, which can provide farmers with additional income. Therefore, in the agricultural sector, the implementation of anaerobic digestion technologies can yield significant economic, environmental, social, and energy benefits; it also allows for improved soil management by utilizing the biofertilizer (a residual product of the process) instead of more expensive chemical fertilizers.

Farm-scale biogas plants come in various sizes, designs, and technologies.
Some are very small and technologically simple, while others are quite large and complex, similar to centralized co-digestion plants. However, they all share a common design principle: manure is collected in a pre-storage tank near the digester and pumped into the digester, which is a gas-tight, insulated tank to maintain a constant temperature. Digesters can be horizontal or vertical, generally equipped with agitation systems to mix and homogenize the substrate and minimize the risk of scum buildup and sediment formation.

The size of the digesters determines the scale of biogas plants and varies from a few cubic meters, in the case of small domestic installations, to several thousand cubic meters, as in the case of large commercial plants, often with several digesters that use agro-industrial waste as raw material329 .

The most commonly used small-scale biogas reactors in rural areas are dome-type and floating drum reactors, which have seen little design innovation in recent years.330 Furthermore, these reactors require considerable technical skill to construct and are prone to leaks. Tubular reactors constructed with geomembranes are currently among the most widely used on an individual scale in the field.

Before designing a biogas plant for rural areas, it is necessary to estimate the production potential and biogas needs to determine its feasibility, considering the livestock management practices and, therefore, the minimum amount of waste required to produce the biogas needed to meet the stated requirements.

Table 73. Biogas needs for a family of 5 people

Request	Factor	Consumption (m3 /day)
Cooking time (5 hours) 0.30	* 5	1,50
Lighting (3 lamps, 3 hours) 0.15	* 3 * 3	1,35

329 M. Dumitru and M. Bibu, “Researches on the digesters and reactors which can be used in a farm scale biogas plant,” vol. 14, no. 3, pp. 63–68, 2014.
330 M. Dumitru and M. Bibu, “Researches on the digesters and reactors which can be used in a farm scale biogas plant,” vol. 14, no. 3, pp. 63–68, 2014.